



US 20250281601A1

(19) **United States**(12) **Patent Application Publication**
Watson et al.(10) **Pub. No.: US 2025/0281601 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **VACCINATION AGAINST BACTERIAL AND
BETACORONAVIRUS INFECTIONS**on May 3, 2021, provisional application No. 63/325,
126, filed on Mar. 29, 2022.(71) Applicant: **PFIZER INC.**, NEW YORK, NY (US)**Publication Classification**(72) Inventors: **Wendy Jo Watson**, Blue Bell, PA (US);
Annaliesa Sybil Anderson, Upper
Saddle River, NJ (US); **Alejandro**
David Cane, Berwyn, PA (US);
William Carl Gruber, Warwick, NY
(US); **Kathrin Ute Jansen**, New York,
NY (US); **Luis Pascual Jodar**
Martin-Montalvo, Villanova, PA (US);
Stephen Paul Lockhart, Sandwich
(GB); **Daniel Alfred Scott**, Dagsboro,
DE (US); **Kari Ann Yacisin**, King of
Prussia, PA (US)(51) **Int. Cl.****A61K 39/215** (2006.01)
A61K 39/00 (2006.01)
A61K 39/095 (2006.01)
A61K 39/108 (2006.01)
A61K 39/385 (2006.01)
A61K 47/02 (2006.01)
A61K 47/22 (2006.01)
A61K 47/26 (2006.01)
A61P 31/14 (2006.01)
C12N 7/00 (2006.01)(52) **U.S. Cl.**CPC **A61K 39/215** (2013.01); **A61K 39/0258**
(2013.01); **A61K 39/095** (2013.01); **A61K**
39/385 (2013.01); **A61K 47/02** (2013.01);
A61K 47/22 (2013.01); **A61K 47/26** (2013.01);
A61P 31/14 (2018.01); **C12N 7/00** (2013.01);
A61K 2039/53 (2013.01); **A61K 2039/545**
(2013.01); **A61K 2039/55505** (2013.01); **A61K**
2039/6037 (2013.01); **A61K 2039/70** (2013.01)(73) Assignee: **PFIZER INC.**, NEW YORK, NY (US)(21) Appl. No.: **18/557,987**(22) PCT Filed: **Apr. 28, 2022**(86) PCT No.: **PCT/IB2022/053951**

§ 371 (c)(1),

(2) Date: **Oct. 30, 2023****Related U.S. Application Data**(60) Provisional application No. 63/183,254, filed on May
3, 2021, provisional application No. 63/183,630, filed

(57)

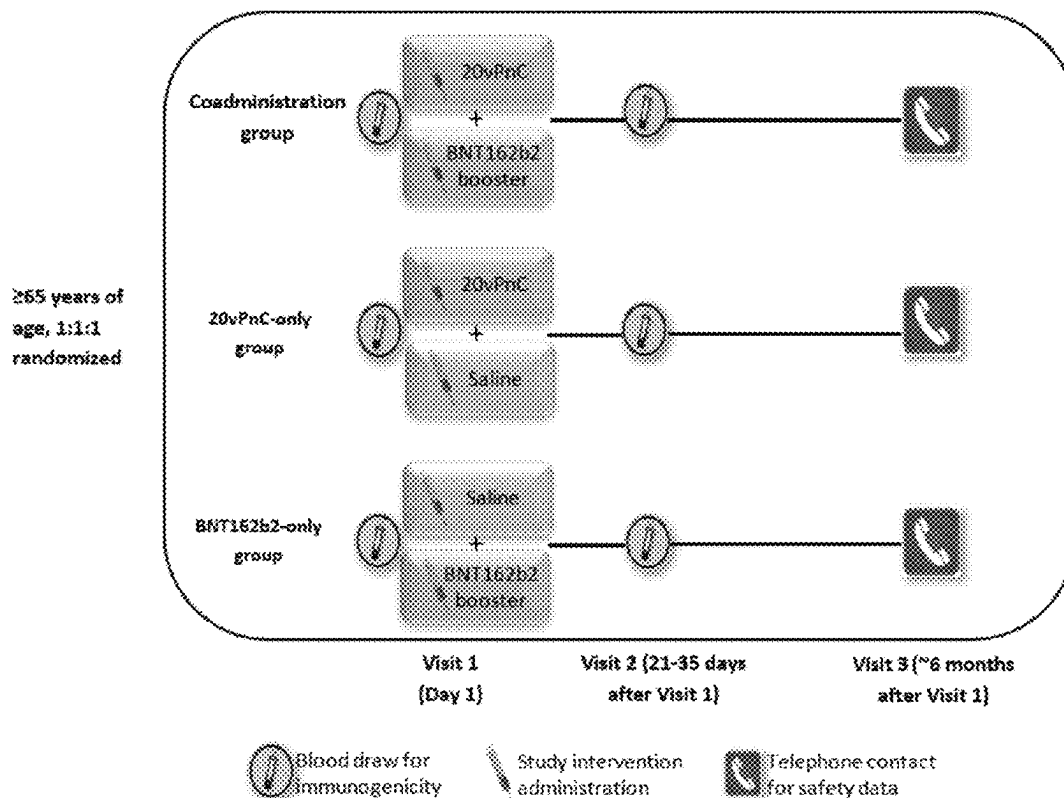
ABSTRACTThe invention relates to vaccination of human subjects, in
particular elderly, against bacterial infections, wherein the
bacterial infection is not pneumococcal, and COVID-19
infections.**Specification includes a Sequence Listing.**

Fig.1

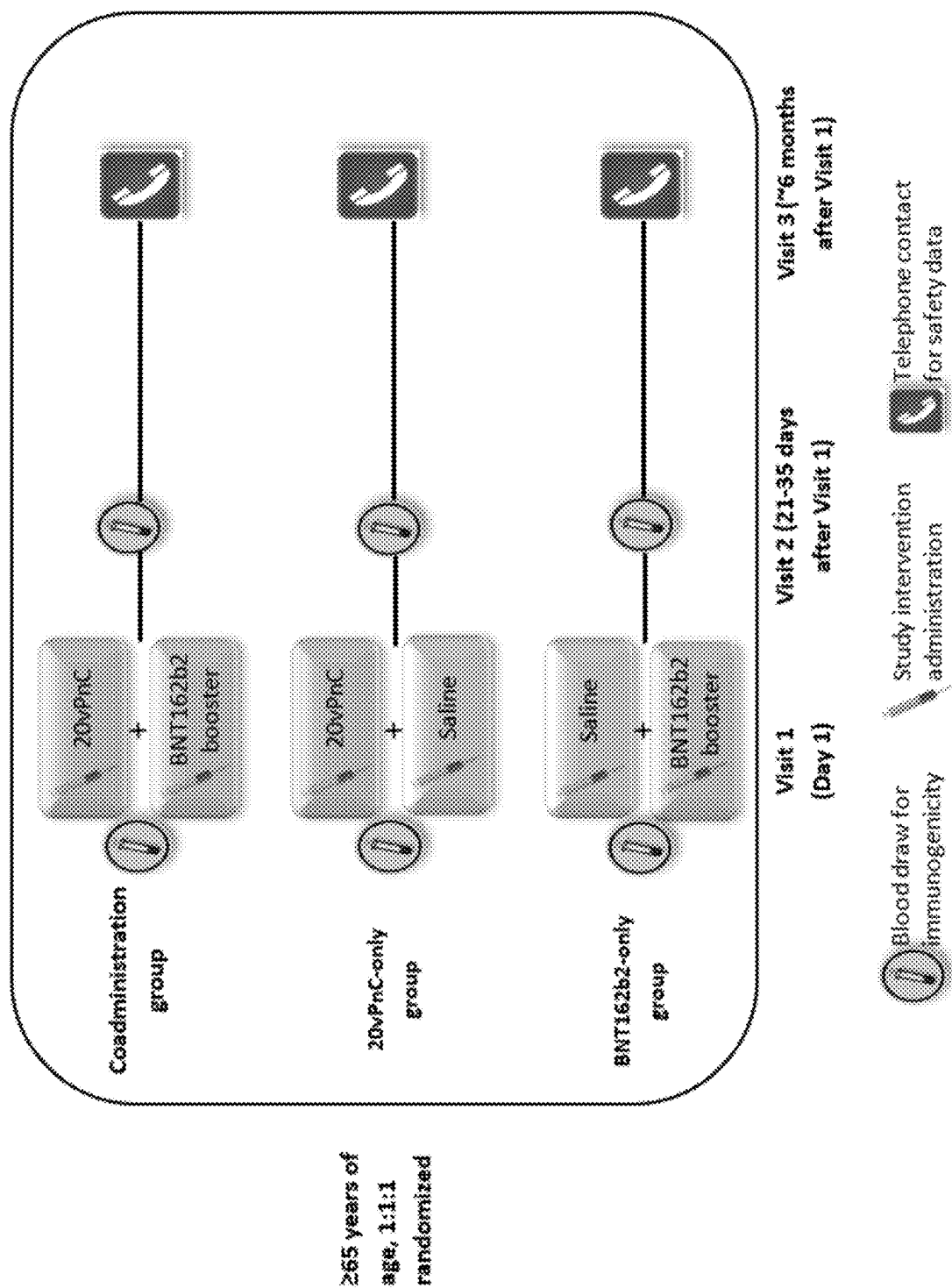
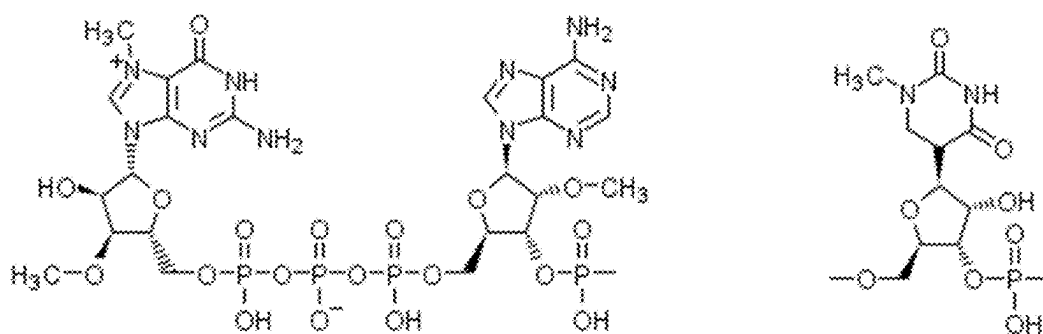
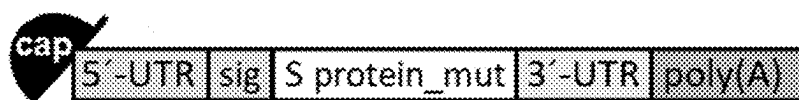


Fig. 2



5'-capping structure

cap G¹A² = m⁷G⁺m^{3'}-5'-ppp-5'-Am^{2'}-3'-p-

[m⁷ = 7-CH₃; m^{3'} = 3'-O-CH₃; m^{2'} = 2'-O-CH₃;

-ppp- = -PO₂H-O-PO₂H-O-PO₂H)-; -p- = -PO₂H-]

m¹Ψ = 1-methyl-3'-pseudouridylyl

Sequence (SEQ ID NO: 1) :

GAGAAΨAAAC	ΨAGΨAΨΨCΨΨ	CΨGGΨCCCCA	CAGACTCAGA	GAGAACCCGC	50
CACCAΨGWΨC	GWΨΨCΨVGG	ΨGCΨGCΨGCC	ΨCΨGGΨΨCC	AGCCAGΨGWG	100
ΨGAACCΨGAC	CACCAGAACA	CAGCΨGCCΨC	CAGCCΨACAC	CAACAGCΨΨΨ	150
ACCAGAGGCG	ΨGWACΨACCC	CGACAAGGΨG	ΨΨCAGAVCCA	GCGΨGCΨGCA	200
CΨCΨACCCAG	GACCCΨΨΨCC	ΨGCCΨΨΨCΨΨ	CAGCAACGΨG	ACCΨGGΨΨCC	250
ACGCCAΨCCA	CGΨGΨCCGGC	ACCAAΨGGCA	CCAAGAGAVΨ	CGACAACCCC	300
GΨGCΨGCCCCΨ	ΨCAACGACGG	GGΨGWACΨΨΨ	GCCAGCACCG	AGAAGΨCCAA	350
CAΨCAΨCAGA	GGCΨGGAΨCΨ	ΨCGGCACCAC	ACΨGGACAGC	AAGACCCAGA	400
GCCΨGCΨGAV	CGΨGAACAAC	GCCACCAACG	ΨGGΨCAΨCAA	AGΨGWGCGAG	450
ΨΨCCAGΨΨCΨ	GCAACGACCC	CΨΨCCΨGGGC	GΨCΨACΨACC	ACAAGAACAA	500
CAAGAGCΨGG	AΨGGAAAGCG	AGΨΨCCGGGΨ	GΨACAGCAGC	GCCAACAACΨ	550
GCACCΨΨCGA	GΨACGΨGΨCC	CAGCCΨΨΨCC	ΨGAΨGGACCΨ	GGAAGGCAAG	600
CAGGGCAACΨ	ΨCAAGAAACΨ	GCGCGAGΨΨC	GΨGΨΨΨAAGA	ACAΨCGACGG	650
CΨACΨΨCAAG	AΨCΨACAGCA	AGCACACCCC	ΨAΨCAACCΨC	GΨGCGGGAVC	700
ΨGCCΨCAGGG	CΨΨCΨCΨGCΨ	CΨGGAACCCC	ΨGGΨGGAΨCΨ	GCCCAΨCGGC	750
AΨCAACAΨCA	CCCCGGΨΨΨCA	GACACΨGCΨG	GCCCΨGCACA	GAAGCΨACCΨ	800
GACACCΨVGGC	GAVAGCAGCA	GCGGAVGGAC	AGCΨGGΨGCC	GCCGCΨΨACΨ	850
AΨGWGGGCWA	CCΨGCAGCCΨ	AGAACCΨΨCC	ΨGCΨGAAGΨA	CAACGAGAAC	900
GGCACCAΨCA	CCGACGCCGΨ	GGAΨΨGWGCV	CΨGGAΨCCΨC	ΨGAGCGAGAC	950
AAAGΨGCACC	CΨGAAGΨCCΨ	ΨCACCΨVGGGA	AAAGGGCAVC	ΨACCAGACCA	1000
GCAACΨΨCCG	GGΨGCAGCCC	ACCGAΨΨCCA	ΨCGΨGCCGGΨΨ	CCCCAAΨAΨC	1050
ACCAAΨCΨGW	GCCCCΨΨCGG	CGAGGΨGWΨC	AAΨGCCACCA	GAVΨCGCCΨC	1100
ΨGWΨVACGCC	ΨGGAACCCGGA	AGCGGAVCAG	CAAΨΨGCGΨG	GCCGACΨACΨ	1150
CCGΨGCΨGWA	CAACΨCCGCC	AGCΨΨCAGCA	CCΨΨCAAGΨG	CΨACGGCGΨG	1200
ΨCCCCΨACCA	AGCΨGAACGA	CCΨGWGCΨΨC	ACAAACGΨGW	ACGCCGACAG	1250
CΨΨCGΨGAVC	CGGGGAGAVG	AAGΨGCGGCA	GAVΨGCCCCΨ	GGACAGACAG	1300
GCAAGAVCGC	CGACΨACAAC	ΨACAAGCΨGC	CCGACGACΨΨ	CACCGGCΨGW	1350
GWGAVΨGCCΨ	GGAACAGCAA	CAACCΨGGAC	ΨCCAAAGΨCG	GCGGCAACΨA	1400
CAAΨΨACCΨG	ΨACCGGCΨGW	ΨCCGGAAGΨC	CAAΨCΨGAAG	CCCΨΨCGAGC	1450
GGGACAΨCΨC	CACCGAGAVC	ΨAΨCAGGCCG	GCAGCACCCC	ΨΨGΨAACGGC	1500
GWGGAAGGCV	ΨCAACΨGCΨA	CΨΨCCCACΨG	CAGΨCCΨACG	GCΨΨΨCAGCC	1550
CACAAΨVGGC	GΨGGGCΨAΨC	AGCCCΨACAG	AGΨGGΨGGΨG	CΨGAGCΨΨCG	1600
AACΨGCΨGCA	ΨGCCCCΨGCC	ACAGΨGWGCG	GCCCΨAAGAA	AAGCACCAAΨ	1650
CΨCGΨGAAGA	ACAAAVGCGΨ	GAACΨΨCAAC	ΨΨCAACGGCC	ΨGACCGGCAC	1700
CGGCΨΨGCΨG	ACAGAGAGCA	ACAAGAAGΨΨ	CCΨGCCAΨΨC	CAGCAGΨΨΨG	1750
GCCGGGAVAΨ	CGCCGAVACC	ACAGACGCCG	ΨΨAGAGAVCC	CCAGACACΨG	1800
GAAAVCCΨGG	ACAΨCACCCC	ΨΨGCAGCΨΨC	GGCGGAGΨGW	CΨGWGAΨCAC	1850
CCCΨGGCACC	AACACCAGCA	AΨCAGGΨGGC	AGΨGCΨGWAC	CAGGACGΨGA	1900
ACΨGΨACCGA	AGΨGCCCGΨG	GCCAVΨCAGC	CCGAVCAGCΨ	GACACCΨACA	1950
ΨGGCGGGΨGW	ACΨCCAACCGG	CAGCAAVGΨG	ΨΨΨCAGACCA	GAGCCGGCΨG	2000
ΨCΨGAVCGGA	GCCGAGCAGC	ΨGAACAAΨAG	CΨACGAGΨGC	GACAΨCCCCA	2050
ΨCGGCGCΨGG	AAΨCΨGCGCC	AGCΨACCAGA	CACAGACAAA	CAGCCΨCGG	2100
AGAGCCAGAA	GCGΨGGCCAG	CCAGACAVC	AΨΨGCCΨACA	CAAΨGWCVΨCΨ	2150
GGGCGCCGAG	ACGAGCGΨWG	CCΨACΨCCAA	CAACΨCΨAVC	CGΨAVCCCCA	2200
CCAACΨΨCAC	CAΨCAGCGΨG	ACCACAGAGA	ΨCCΨGCCΨGW	GWCCAVGACC	2250
AAGACCAGCG	ΨGGACΨGCAC	CAΨGWACAΨC	ΨGCGGCGAVΨ	CCACCAGΨG	2300
CΨCCAACCΨG	CΨGCΨGCAGΨ	ACGGCAGCΨΨ	CΨGCACCCAG	CΨGAAVAGAG	2350
CCCΨGACAGG	GAΨCGCCGΨG	GAACAGGACA	AGAACACCCA	AGAGGWGΨΨC	2400
GCCAAAGΨGA	AGCAGAVCΨA	CAAGACCCΨ	CCΨAΨCAAGG	ACΨΨCGGCGG	2450
CΨΨCAAΨΨΨC	AGCCAGAVΨC	ΨGCCCCAVCC	ΨAGCAAGCCC	AGCAAGCGGA	2500
GCΨΨCAΨCGA	GGACCΨGCΨG	ΨΨCAACAAAG	ΨGACACΨGGC	CGACGCCGGC	2550
ΨΨCAΨCAAGC	AGΨAVGGCGA	ΨΨGWCVGGGC	GACAΨΨGCCG	CCAGGGAVCΨ	2600

GAWΨΨGCGCC	CAGAAGΨΨΨA	ACGGACΨGAC	AGΨGCΨGCCΨ	CCΨCΨGCΨGA	2650
CCGAWGAGAW	GAΨCGCCCAG	ΨACACAΨCΨG	CCCΨGCΨGGC	CGGCACAAΨC	2700
ACAAGCGGCΨ	GGACAΨΨΨGG	AGCAGGCGCC	GCΨCΨGCAGA	ΨCCCCΨΨΨGC	2750
ΨAWGCAGAWG	GCCΨACCGGΨ	ΨCAACGGCAΨ	CGGAGΨGACC	CAGAAΨGΨGC	2800
ΨGΨACGAGAA	CCAGAAGCΨG	AΨCGCCAACC	AGΨΨCAACAG	CGCCAΨCGGC	2850
AAGAWCCAGG	ACAGCCΨGAG	CAGCACAGCA	AGCGCCCΨGG	GAAAGCΨGCA	2900
GGACGΨGGΨC	AACCAGAAΨG	CCCAGGCACΨ	GAACACCCΨG	GΨCAAGCAGC	2950
ΨGΨCCΨCCAA	CΨΨCGGCGCC	AΨCAGCΨCΨG	ΨGCΨGAACGA	ΨAWCCΨGAGC	3000
AGACΨGGACC	CΨCCΨGAGGC	CGAGGΨGCAG	AΨCGACAGAC	ΨGAΨCACAGG	3050
CAGACΨGCAG	AGCCΨCCAGA	CAΨACGΨGAC	CCAGCAGCΨG	AΨCAGAGCCG	3100
CCGAGAWΨWAG	AGCCΨCΨGCC	AAΨCΨGGCCG	CCACCAAGAΨ	GΨCΨGAGΨGΨ	3150
GΨGCΨGGGCC	AGAGCAAGAG	AGΨGGACΨΨΨ	ΨGCGGCAAGG	GCΨACCACCΨ	3200
GAΨGAGCΨΨC	CCΨCAGΨCΨG	CCCCΨCACGG	CGΨGGΨGΨΨΨ	CΨGCACGΨGA	3250
CAΨAWGΨGCC	CGCΨCAAGAG	AAGAΨΨΨΨCA	CCACCGCΨCC	AGCCAΨCΨGC	3300
CACGACGGCA	AAGCCACΨΨ	ΨCCΨAGAGAA	GGCGΨGΨΨCG	ΨGΨCCAACGG	3350
CACCCAWΨGG	ΨΨCGΨGACAC	AGCGGAACΨΨ	CΨACGAGCCC	CAGAWCAΨCA	3400
CCACCGACAA	CACCΨΨCGΨG	ΨCΨGGCAACΨ	GCGACGΨCGΨ	GAΨCGGCAΨΨ	3450
GΨGAACAAΨA	CCGΨGWACGA	CCCΨCΨGCAG	CCCGAGCΨGG	ACAGCΨΨCAA	3500
AGAGGAACΨG	GACAAGΨACΨ	ΨΨAAGAACCA	CACAAGCCCC	GACGΨGGACC	3550
ΨGGGCGAWAW	CAGCGGAAΨC	AAΨGCCAGCG	ΨCGΨGAACAW	CCAGAAAGAG	3600
AΨCGACCGGC	ΨGAACGAGGΨ	GGCCAAGAAΨ	CΨGAACGAGA	GCCΨGAΨCGA	3650
CCΨGCAAGAA	CΨGGGGAAGΨ	ACGAGCAGΨA	CAΨCAAGΨGG	CCCΨGGΨACA	3700
ΨCΨGGCΨGGG	CΨΨΨAWCGCC	GGACΨGAWΨG	CCAΨCGΨGAW	GGΨCACAAΨC	3750
AΨGCΨGΨGΨΨ	GCAΨGACCAG	CΨGCΨGWAGC	ΨGCCΨGAAGG	GCΨGΨΨGΨAG	3800
CΨGΨGGCAGC	ΨGCΨGCAAGΨ	ΨCGACGAGGA	CGAWΨCΨGAG	CCCGΨGCΨGA	3850
AGGGCGΨGAA	ACΨGCACΨAC	<u>ACAΨGAWGAC</u>	ΨCGAGCΨGGΨ	ACΨGCAΨGCA	3900
CGCAAWGCΨA	GCPGCCCCΨΨ	ΨCCCCΨCCΨG	GGΨACCCCGA	GΨCΨCCCCCG	3950
ACCΨCGGGΨC	CCAGGΨAΨGC	ΨCCCACCΨCC	ACCΨGCCCCA	CΨCACCACCΨ	4000
CΨGCΨAGΨΨC	CAGACACCΨC	CCAAGCACGC	AGCAAWGCAG	CΨCAAAACGC	4050
ΨΨAGCCΨAGC	CACACCCCCA	CGGGAAACAG	CAGΨGAΨΨAA	CCΨΨΨAGCAA	4100
ΨAAACGAAAG	ΨΨΨAACΨAAG	CΨAΨACΨAAC	CCCAGGGΨΨG	GΨCAAAΨΨΨCG	4150
ΨGCCAGCCAC	ACCCΨGGAGC	ΨAGCAAAAAA	AAAAAAAAAA	AAAAAAAAAA	4200
AAAAGCAΨAW	GACΨAAAAAA	AAAAAAAAAA	AAAAAAAAAA	AAAAAAAAAA	4250
AAAAAAAAAA	AAAAAAAAAA	AAAAAAAAAA	AAAA (SEQ ID NO: 1)		4284

Ψ = 1-methyl-3'-pseudouridylyl

Fig. 3

GGGAAATAAGAGAGAAAAGAAGAGTAAGAAGAAATATAAGACCCCGGCGCCGCCACCATG
TTCGTGTTCCCTGGTGCTGCTGCCCCCTGGTGAGCAGCCAGTGCGTGAACCTGACCACCCGG
ACCCAGCTGCCACCAGCCTACACCAACAGCTTCACCCGGGGCGTCTACTACCCCGACAAG
GTGTTCCGGAGCAGCGTCCCTGCACAGCACCCAGGACCTGTTCCCTGCCCCCTCTTCAGCAAC
GTGACCTGGTTCCACGCCATCCACGTGAGCGGCACCAACGGCACCAGCGGTTGACAAAC
CCCGTGCTGCCCCCTCAACGACGGCGTGTACTTCGCCAGCACCCGAGAAGAGCAACATCATC
CGGGGCTGGATCTTCGGCACCCACCTGGACAGCAAGACCCAGAGCCTGCTGATCGTGAAT
AACGCCACCAACGTGGTGATCAAGGTGTGCGAGTTCCAGTTCTGCAACGACCCCTTCCTG
GGCGTGTAATAACACAAGAACAACAAGAGCTGGATGGAGAGCGAGTTCGGGGTGTAACAGC
AGCGCCAACAACCTGCACCTTCGAGTACGTAGAGCCAGCCCTTCCTGATGGACCTGGAGGG
CAAGCAGGGCAACTTCAAGAACCTGCGGGAGTTTCGTGTTCAAGAACATCGACGGCTACTT
CAAGATCTACAGCAAGCACACCCCAATCAACCTGGTGCGGGATCTGCCCCAGGGCTTCTC
AGCCCTGGAGCCCCCTGGTGACCTGCCCATCGGCATCAACATCACCCGGTTCCAGACCCT
GCTGGCCCTGCACCCGAGCTACCTGACCCCAAGGCGACAGCAGCAGCGGGTGACAGCAGG
CGCGGCTGCTTACTACGTGGGCTACCTGCAGCCCCGGACCTTCCTGCTGAAGTACAACGA
GAACGGCACCATCACCGACGCCGTGGACTGCGCCCTGGACCTTCCTGAGCGAGACCAAGTG
CACCTGAAGAGCTTCACCGTGGAGAAGGGCATCTACCAGACCAGCAACTTCGGGGTGCA
GCCCCAGGAGCATCGTGCGGTTCCCAACATCACCAACCTGTGCCCTTCGGCGAGGT
GTTCAACGCCACCCGGTTTCGCCAGCGTGACGCTGGAACCGGAAGCGGATCAGCAACTG
CGTGGCCGACTACAGCGTGCTGTACAACAGCGCCAGCTTCAGCACCTTCAAGTGCTACGG
CGTGAGCCCCACCAAGCTGAACGACCTGTCTTCACCAACGTGTACGCCGACAGTTTCGTGA
TCCGTGGCGACGAGGTGCGGCAGATCGCACCCGGCCAGACAGGCAAGATCGCCGACTACA
ACTACAAGCTGCCCCGACGACTTCACCGGCTGCGTGATCGCCTGGAACAGCAACAACCTCG
ACAGCAAGGTGGGCGCAACTACAACCTACCTGTACCGGCTGTTCCGGAAGAGCAACCTGA
AGCCCTTCGAGCGGGACATCAGCACCGAGATCTACCAAGCCGGCTCCACCCCTTGCAACG
GCGTGGAGGGCTTCAACTGCTACTTCCCTCTGCAGAGCTACGGCTTCAGCCCCACCAACG
GCGTGGGCTTACCAGCCCCTACCGGGTGGTGGTGCTGAGCTTCGAGCTGCTGCACGCCCCA
GCCACCGTGTGTGGCCCCAAGAAGAGCACCAACCTGGTGAAGAACAAGTGCGTGAACCTTC
AACTTCAACGGCCTTACCGGCACCGCGTGCTGACCGAGAGCAACAAGAAATTCCTGCCC
TTTCAGCAGTTCGGCCGGGACATCGCCGACACCCAGCGCTGTGCGGGATCCCCAGACC
CTGGAGATCTTGGACATCACCCCTTGCAGCTTCGGCGGCGTGAGCGTGATCACCCAGGC
ACCAACACCAGCAACCAGGTGGCCGTGTGTACCAGGACGTGAACCTGCACCGAGGTGCCC
GTGGCCATCCACGCCGACCAGCTGACACCCACCTGGCGGGTCTACAGCACCGGCAGCAAC
GTGTTCCAGACCCGGGCGCGGTTGCCTGATCGGCGCCGAGCACGTGAACAACAGCTACGAG
TGCGACATCCCCATCGGCGCCGGCATCTGTGCCAGCTACCAGACCCAGACCAATTCACCC
CGGAGGGCAAGGAGCGTGGCCAGCCAGAGCATCATCGCTACACCATGAGCCTGGGCGCC
GAGAACAGCGTGCCCTACAGCAACAACAGCATCGCCATCCCCACCAACTTCACCATCAGC
GTGACCACCGAGATTCTGCCCGTGAGCATGACCAAGACCAGCGTGGACTGCACCATGTAC
ATCTGCGGCGACAGCACCGAGTGACAGCAACCTGCTGCTGCAGTACGGCAGCTTCTGCACC
CAGCTGAACCGGGCCCTGACCGGCATCGCCGTGGAGCAGGACAAGAACACCCAGGAGGTG
TTCGCCCAGGTGAAGCAGATCTACAAGACCCCTCCCATCAAGGACTTCGGCGGCTTCAAT
TCAGCCAGATCCTGCCCCGACCCAGCAAGCCCAGCAAGCGGAGCTTCATCGAGGACCTGC
TGTTCAACAAGGTGACCCCTAGCCGACGCCGGCTTCATCAAGCAGTACGGCGACTGCCTCG
GCGACATAGCCGCCCGGGACCTGATCTGCGCCAGAAGTTCAACGGCCTGACCGTGCTGC
CTCCCTGCTGACCGACGAGATGATCGCCAGTACACCAGCGCTGTAGCCGGAACCAT
CACCAGCGGCTGGACTTTCGGCGCTGGAGCCGCTCTGCAGATCCCCCTTCGCCATGCAGAT
GGCCTACCGGTTCAACGGCATCGGCGTGACCCAGAACGTGCTGTACGAGAACCAGAAGCT
GATCGCCAACCAAGTTCAACAGCGCCATCGGCAAGATCCAGGACAGCCTGAGCAGCACCGC
TAGCGCCCTGGGCAAGCTGCAGGACGTGGTGAACCAGAACGCCCAGGCCCTGAACACCT
GGTGAAGCAGCTGAGCAGCAACTTCGGCGCCATCAGCAGCGTGCTGAACGACATCCTGAG

CCGGCTGGACCCCTCCCGAGGCCGAGGTGCAGATCGACCGGCTGATCACTGGCCGGCTGCA
GAGCCTGCAGACCTACGTGACCCAGCAGCTGATCCGGGCCGCGGAGATTCCGGGCCAGCGC
CAACCTGGCCGCCACCAAGATGAGCGAGTGCGTGCTGGGCCAGAGCAAGCGGGTGGACTT
CTGCGGCAAGGGCTACCACCTGATGAGCTTTCCCCAGAGCGCACCCACGGAGTGGTGTT
CCTGCACGTGACCTACGTGCCCCGCCAGGAGAAGAACTTCACCACCGCCCCAGCCATCTG
CCACGACGGCAAGGCCCACTTTTCCCCGGGAGGGCGTGTTTCGTGAGCAACGGCACCCACTG
GTTTCGTGACCCAGCGGAACTTCTACGAGCCCCAGATCATCACCACCGACAACACCTTCGT
GAGCGGCAACTGCGACGTGGTGATCGGCATCGTGAACAACACCGTGTACGATCCCCTGCA
GCCCCAGCTGGACAGCTTCAAGGAGGAGCTGGACAAGTACTTCAAGAATCACACCAGCCC
CGACGTGGACCTGGGCGACATCAGCGGCATCAACGCCAGCGTGGTGAACATCCAGAAGGA
GATCGATCGGCTGAACGAGGTGGCCAAGAACCTGAACGAGAGCCTGATCGACCTGCAGGA
GCTGGGCAAGTACGAGCAGTACATCAAGTGGCCCTGGTACATCTGGCTGGGCTTCATCGC
CGGCCTGATCGCCATCGTGATGGTGACCATCATGCTGTGCTGCATGACCAGCTGCTGCAG
CTGCCTGAAGGGCTGTTGCAGCTGCGGCAGCTGCTGCAAGTTCGACGAGGACGACAGCGA
GCCCCGTGCTGAAGGGCGTGAAGCTGCACTACACCTGATAATAGGCTGGAGCCTCGGTGGC
CTAGCTTCTTGCCCCCTTGGGCCTCCCCCAGCCCCCTCCTCCCCCTTCCTGCACCCGTACCC
CCGTGGTCTTTGAATAAAGTCTGAGTGGGCGGCAAAAAAAAAA

(SEQ ID NO: 2)

Fig. 4

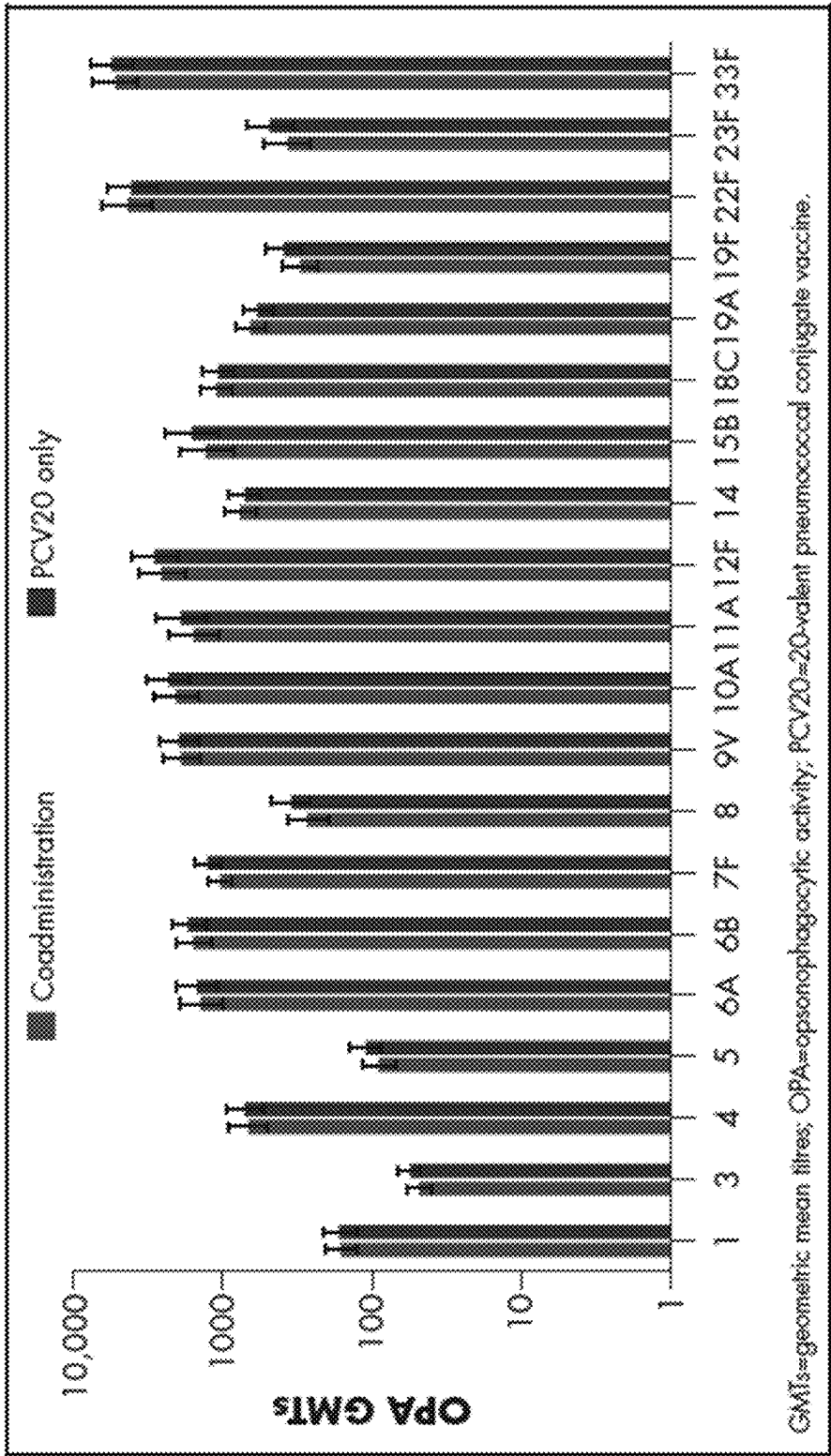


Fig. 5

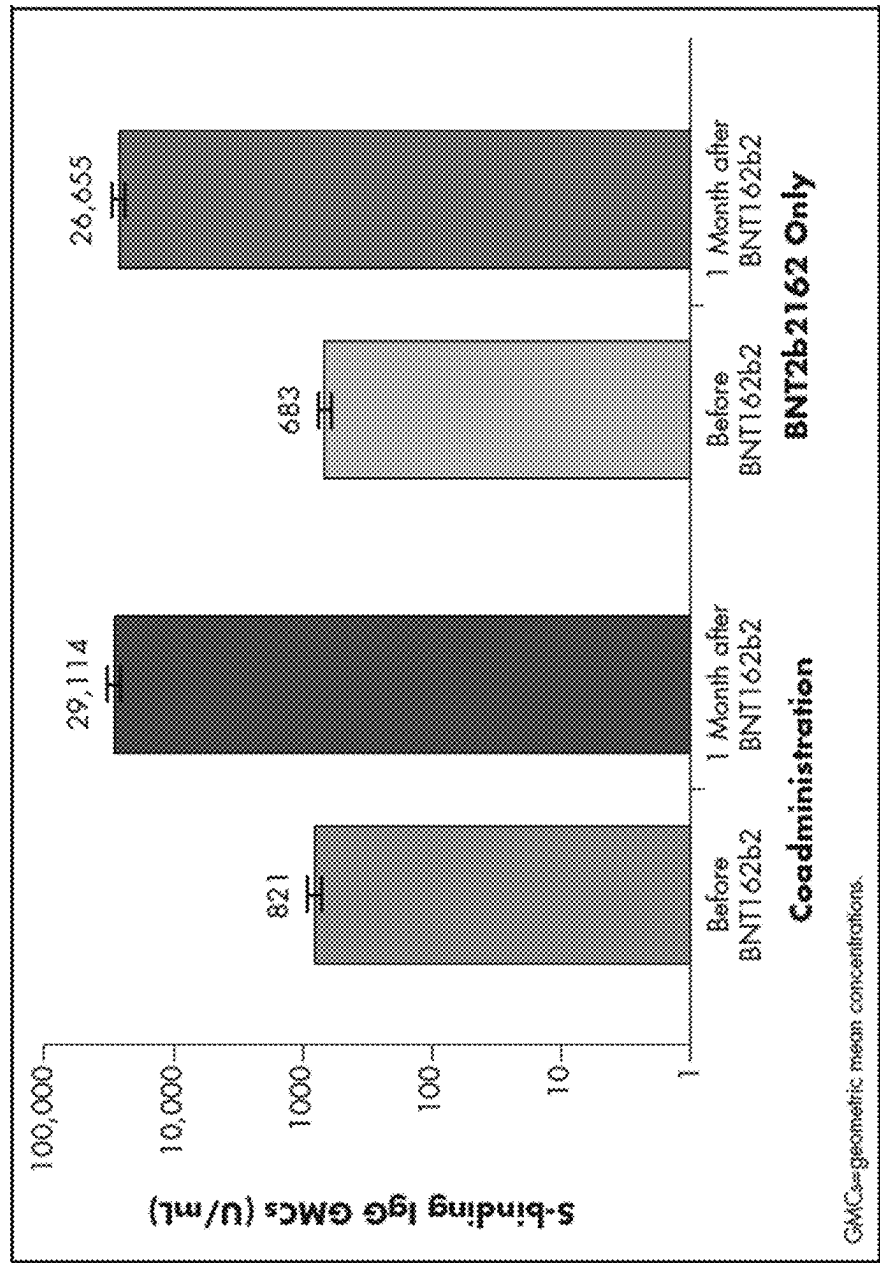
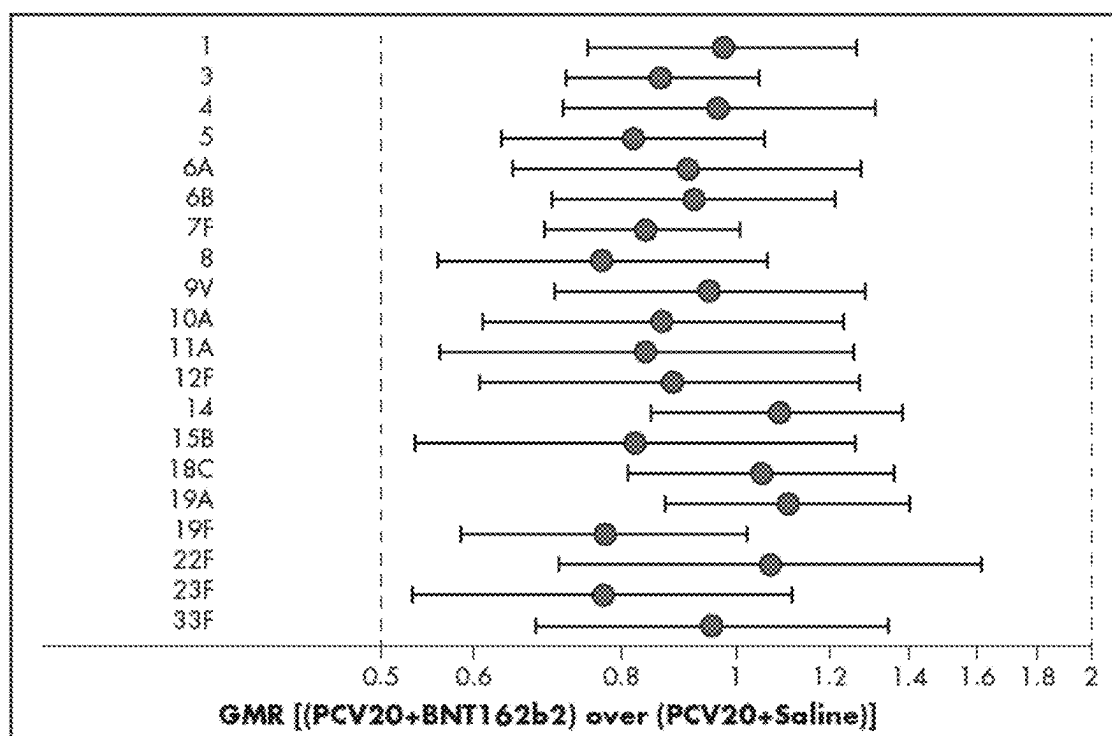


Fig. 6



VACCINATION AGAINST BACTERIAL AND BETACORONAVIRUS INFECTIONS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of U.S. Provisional Patent Application 63/183,254, filed on May 3, 2021, U.S. Provisional Patent Application 63/183,630, filed on May 3, 2021, and U.S. Provisional Patent Application 63/325,126, filed Mar. 29, 2022. The entire contents of the aforementioned applications are herein incorporated by reference in their entireties.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

[0002] The instant application includes an electronically submitted sequence listing in .txt format. The .txt file contains a sequence listing entitled "PC72752A_SEQListing_ST25.txt" created on May 22, 2024, and having a size of 78 KB. The sequence listing contained in this .txt file is part of the specification and is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0003] The invention relates to vaccination of human subjects, in particular elderly, adolescent, and infant subjects, with bacterial vaccines in combination with mRNA vaccines. In particular, the present invention relates to vaccinations against bacterial and COVID-19 infections, preferably wherein the bacterial infection is not pneumococcal.

BACKGROUND OF THE INVENTION

[0004] Protein and/or polysaccharide antigens from pathogens have long been used in vaccines, designed to elicit neutralizing antibody and/or cell-mediated immune responses in the recipient, specific for the antigen. Cell-mediated immune responses, particularly the generation of effector T-cells (including cytotoxic T-cells), may be a desirable component of the immune response elicited from vaccines having a polypeptide and/or polysaccharide component. Antibodies may also be a desirable component of the protective immune response for pathogens particularly bacteria and certain viruses such as the influenza viruses.

[0005] Nucleic acid-based vaccines may elicit cell-mediated immunity (e.g., involving effector T-cells, such as interferon- γ secreting antigen-specific T-cells and antigen-specific cytotoxic T-cells). Generating antibodies against the antigen that is encoded and expressed by the nucleic acid component may also be a desirable component of the immune response elicited from nucleic acid-based vaccines.

[0006] There is a need to improve immunization regimens such that immune responses observed with coadministration of a first immunogenic composition having a polypeptide and/or polysaccharide component and a second immunogenic composition having a nucleic acid component are non-inferior, or preferably enhanced, for the respective antigens compared to the immune response against the respective antigens when the immunogenic compositions are not coadministered. For example, the first immunogenic composition may include a polypeptide, a toxoid, a polysaccharide, and/or a polysaccharide-conjugate. The compositions may be useful for generating an immune response,

for example, to reduce the likelihood of infection, by an infectious agent, such as pneumococci.

[0007] Infections caused by pneumococci are a major cause of morbidity and mortality all over the world. Pneumonia, febrile bacteraemia and meningitis are the most common manifestations of invasive pneumococcal disease, whereas bacterial spread within the respiratory tract may result in middle-ear infection, sinusitis or recurrent bronchitis. Compared with invasive disease, the non-invasive manifestations are usually less severe, but considerably more common.

[0008] In Europe and the United States, pneumococcal pneumonia is the most common community-acquired bacterial pneumonia, estimated to affect approximately 100 per 100,000 adults each year. The corresponding figures for febrile bacteraemia and meningitis are 15-19 per 100,000 and 1-2 per 100,000, respectively. The risk for one or more of these manifestations is much higher in infants and elderly people, as well as immune compromised persons of any age. Even in economically developed regions, invasive pneumococcal disease carries high mortality; for adults with pneumococcal pneumonia the mortality rate averages 10%-20%, whilst it may exceed 50% in the high-risk groups. Pneumonia is by far the most common cause of pneumococcal death worldwide.

[0009] Pneumococcal conjugate vaccines (PCVs) are pneumococcal vaccines used to protect against disease caused by *S. pneumoniae* (pneumococcus). There are currently three PCV vaccines available on the global market: PREVNAR® (PREVENAR® in some countries) (heptavalent vaccine), SYNFLORIX® (a decavalent vaccine) and PREVNAR 13® (PREVENAR 13® in some countries) (tridecavalent vaccine).

[0010] Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) is the virus that causes COVID-19 (coronavirus disease 2019), the respiratory illness responsible for the COVID-19 pandemic. SARS-CoV-2 is a positive-sense single-stranded RNA virus. The virus primarily spreads between people through close contact and via respiratory droplets produced from coughs or sneezes. It mainly enters human cells by binding to the angiotensin converting enzyme 2 (ACE2).

[0011] There is a need to effectively protect patients against both bacterial (e.g., pneumococcal) infections and/or bacterial (e.g., pneumococcal) disease as well as against coronavirus disease.

[0012] An object of the schedules of administration of the present invention is to provide for appropriate protection against bacterial (e.g., *S. pneumoniae*) and COVID-19.

SUMMARY OF THE INVENTION

[0013] The invention relates generally to methods and uses for eliciting an immune response in a human subject against an infectious disease-causing bacterium and a betacoronavirus, the method includes co-administering to the human subject an effective dose of a first immunogenic composition comprising an antigen derived from the bacterium and an immunogenic composition comprising mRNA encoding an antigen derived from the betacoronavirus. For example, disclosed herein is administration of a first immunogenic composition that includes an RNA component and a second immunogenic composition that includes a polypeptide and/or polysaccharide component. Administration of the first and second immunogenic compositions may

enhance the immune response to the respective pathogen(s), as compared to administration of an immunogenic composition including RNA alone, or polypeptide and/or polysaccharide alone. In some embodiments, the first immunogenic composition and the second immunogenic composition individually elicit an immune response against the same epitope. In some embodiments, the first immunogenic composition and the second immunogenic composition individually elicit an immune response against epitopes that are not the same between the first and second compositions.

[0014] In one aspect the invention relates to a method for eliciting an immunoprotective response in a human subject against an infectious disease-causing bacterium (e.g., selected from any one of *S. pneumoniae*, *N. meningitidis*, *C. difficile*, and *E. coli*) and betacoronavirus (e.g., SARS-CoV-2), the method includes co-administering to the human subject an effective dose of a first immunogenic composition including an antigen selected from any one of a polypeptide, toxoid, polysaccharide, and polysaccharide conjugate; and a second immunogenic composition including mRNA against a betacoronavirus. In an aspect, said first immunogenic composition against the bacterium and said second immunogenic composition mRNA vaccine against betacoronavirus are administered concurrently or concomitantly. In a preferred embodiment, the antigen selected from any one of a polypeptide, toxoid, polysaccharide, and polysaccharide conjugate, is derived from the infectious disease-causing bacterium.

[0015] In an aspect the invention is directed to a method for eliciting an immunoprotective response in a human subject against *S. pneumoniae* and Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), the method comprising co-administering to the human subject an effective dose of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2. In an aspect, said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are administered concurrently or concomitantly.

[0016] The invention further relates to a pneumococcal conjugate vaccine and an mRNA vaccine against SARS-CoV-2 for use in a method for eliciting an immunoprotective response in a human subject against *S. pneumoniae* and Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), said method comprising co-administering to the human subject said vaccines.

[0017] Another aspect of the invention relates to the use of the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 as a booster dose of an mRNA vaccine against SARS-CoV-2.

[0018] The invention further relates to the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for use as a booster dose of an mRNA vaccine against SARS-CoV-2 and to the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for use in a method of boosting an mRNA vaccine against SARS-CoV-2.

[0019] In an aspect the invention is directed to a method for eliciting an immunoprotective response in a human subject against *Neisseria meningitidis* and betacoronavirus (e.g., SARS-CoV-2), the method comprising co-administering to the human subject an effective dose of a first immunogenic composition including an antigen derived from *Neisseria meningitidis* and of an mRNA vaccine against betacoronavirus. In an aspect, said first immunogenic com-

position and said mRNA vaccine against betacoronavirus are administered concurrently or concomitantly.

[0020] The invention further relates to a first immunogenic composition including an antigen derived from *Neisseria meningitidis* and an mRNA vaccine against SARS-CoV-2 for use in a method for eliciting an immunoprotective response in a human subject against *Neisseria meningitidis* and betacoronavirus, said method comprising co-administering to the human subject said compositions.

[0021] Another aspect of the invention relates to the use of the co-administration of a first immunogenic composition including an antigen derived from *Neisseria meningitidis* and of an mRNA vaccine against betacoronavirus as a booster dose of an mRNA vaccine against the betacoronavirus.

[0022] The invention further relates to the co-administration of a first immunogenic composition including an antigen derived from *Neisseria meningitidis* and of an mRNA vaccine against betacoronavirus for use as a booster dose of an mRNA vaccine against betacoronavirus and to the co-administration of a first immunogenic composition including an antigen derived from *Neisseria meningitidis* and of an mRNA vaccine against betacoronavirus for use in a method of boosting an mRNA vaccine against betacoronavirus.

[0023] In an aspect the invention is directed to a method for eliciting an immunoprotective response in a human subject against *Clostridium difficile*, now *Clostridioides difficile* (*C. difficile*) and betacoronavirus (e.g., SARS-CoV-2), the method comprising co-administering to the human subject an effective dose of a first immunogenic composition including an antigen derived from *C. difficile* and of an mRNA vaccine against betacoronavirus. In an aspect, said first immunogenic composition and said mRNA vaccine against betacoronavirus are administered concurrently or concomitantly. The invention further relates to a first immunogenic composition including an antigen derived from *C. difficile* and an mRNA vaccine against SARS-CoV-2 for use in a method for eliciting an immunoprotective response in a human subject against *C. difficile* and betacoronavirus, said method comprising co-administering to the human subject said compositions.

[0024] Another aspect of the invention relates to the use of the co-administration of a first immunogenic composition including an antigen derived from *C. difficile* and of an mRNA vaccine against betacoronavirus as a booster dose of an mRNA vaccine against the betacoronavirus.

[0025] The invention further relates to the co-administration of a first immunogenic composition including an antigen derived from *C. difficile* and of an mRNA vaccine against betacoronavirus for use as a booster dose of an mRNA vaccine against betacoronavirus and to the co-administration of a first immunogenic composition including an antigen derived from *C. difficile* and of an mRNA vaccine against betacoronavirus for use in a method of boosting an mRNA vaccine against betacoronavirus.

[0026] In an aspect the invention is directed to a method for eliciting an immunoprotective response in a human subject against *Escherichia coli* (*E. coli*) and betacoronavirus (e.g., SARS-CoV-2), the method comprising co-administering to the human subject an effective dose of a first immunogenic composition including an antigen derived from *E. coli* and of an mRNA vaccine against betacoronavirus. In an aspect, said first immunogenic composition and

said mRNA vaccine against betacoronavirus are administered concurrently or concomitantly.

[0027] The invention further relates to a first immunogenic composition including an antigen derived from *E. coli* and an mRNA vaccine against SARS-CoV-2 for use in a method for eliciting an immunoprotective response in a human subject against *E. coli* and betacoronavirus, said method comprising co-administering to the human subject said compositions.

[0028] Another aspect of the invention relates to the use of the co-administration of a first immunogenic composition including an antigen derived from *E. coli* and of an mRNA vaccine against betacoronavirus as a booster dose of an mRNA vaccine against the betacoronavirus.

[0029] The invention further relates to the co-administration of a first immunogenic composition including an antigen derived from *E. coli* and of an mRNA vaccine against betacoronavirus for use as a booster dose of an mRNA vaccine against betacoronavirus and to the co-administration of a first immunogenic composition including an antigen derived from *E. coli* and of an mRNA vaccine against betacoronavirus for use in a method of boosting an mRNA vaccine against betacoronavirus.

FIGURES

[0030] FIG. 1 Schematic representation of the design of a study to describe the safety and immunogenicity of co-administration of a 20-valent Pneumococcal Conjugate Vaccine (20vPnC) when Coadministered with an mRNA vaccine to prevent infection with SARS-CoV-2 (BNT162b2), together at the same visit compared to each of the vaccines given alone in adults 65 years of age.

[0031] FIG. 2 Schematic and sequence relating to the nucleoside-modified mRNA (modRNA) sequence of the vaccine BNT162b2 (Comirnaty®; INN: tozinameran); Description: Messenger RNA encoding the full-length SARS-CoV-2 spike glycoprotein. UTR=Untranslated region; sig=extended signal sequence of the S glycoprotein; S protein_mut=S glycoprotein sequence containing mutations K986P and V987P; poly(A)=polyadenylate signal tail.

[0032] FIG. 3 The putative sequence of the vaccine mRNA-1273 (SEQ ID NO: 2)

[0033] FIG. 4 OPA GMTs (With 2-Sided 95% CIs) 1 Month After Vaccination by Group and Vaccine Serotype; GMTs=geometric mean titres; OPA=opsonophagocytic activity; PCV20=20-valent pneumococcal conjugate vaccine.

[0034] FIG. 5. SARS-CoV-2 Full-length S-binding IgG GMCs (With 2-Sided 95% CIs) Before and 1 Month After BNT162b2 by Group; GMCs=geometric mean concentrations.

[0035] FIG. 6. Model-Based OPA GMRs (With 95% CIs) 1 Month After PCV20 (Evaluable Immunogenicity Population); BLQ=below limit of quantitation; BMI=body mass index; GMR=geometric mean ratio; LLOQ=lower limit of quantitation; LS=least squares; OPA=opsonophagocytic activity. Assay results below the LLOQ or denoted as BLQ were set to 0.5×LLOQ in the analysis. 1 Month after PCV20 refers to 1 month after vaccination with PCV20 for both groups. GMRs (PCV20+BNT162b2 to PCV20+Saline) and 2-sided CIs were calculated by exponentiating the difference of LS means for the OPA titres and the corresponding CIs based on the regression model adjusted with vaccine group,

sex, smoking status, age at vaccination in years, baseline log-transformed OPA titres, prior pneumococcal vaccination status, and BMI group.

DETAILED DESCRIPTION OF THE INVENTION

[0036] The present invention relates to vaccinations with immunogenic compositions against infectious disease-causing bacterium and mRNA vaccines.

[0037] The present invention combines vaccinations with PCVs and mRNA vaccines.

1. mRNA Vaccines of the Invention The present invention relates to mRNA vaccines in general. To our best knowledge, the present invention is the first to combine an immunogenic composition against an infectious disease-causing bacterium (e.g., PCVs) and mRNA vaccines.

[0038] A number of mRNA vaccine platforms are available in the prior art. The basic structure of in vitro transcribed (IVT) mRNA closely resembles “mature” eukaryotic mRNA, and consists of (i) a protein-encoding open reading frame (ORF), flanked by (ii) 5' and 3' untranslated regions (UTRs), and at the end sides (iii) a 7-methyl guanosine 5' cap structure and (iv) a 3' poly(A) tail. The non-coding structural features play important roles in the pharmacology of mRNA and can be individually optimized to modulate the mRNA stability, translation efficiency, and immunogenicity.

[0039] By incorporating modified nucleosides, mRNA transcripts referred to as “nucleoside-modified mRNA” can be produced with reduced immunostimulatory activity, and therefore an improved safety profile can be obtained. In addition, modified nucleosides allow the design of mRNA vaccines with strongly enhanced stability and translation capacity, as they can avoid the direct antiviral pathways that are induced by type IFNs and are programmed to degrade and inhibit invading mRNA. For instance, the replacement of uridine with pseudouridine in IVT mRNA reduces the activity of 2'-5'-oligoadenylate synthetase, which regulates the mRNA cleavage by RNase L. In addition, lower activities are measured for protein kinase R, an enzyme that is associated with the inhibition of the mRNA translation process.

[0040] Besides the incorporation of modified nucleotides, other approaches have been validated to increase the translation capacity and stability of mRNA. One example is the development of “sequence-engineered mRNA”. Here, mRNA expression can be strongly increased by sequence optimizations in the ORF and UTRs of mRNA, for instance by enriching the GC content, or by selecting the UTRs of natural long-lived mRNA molecules. Another approach is the design of “self-amplifying mRNA” constructs. These are mostly derived from alphaviruses, and contain an ORF that is replaced by the antigen of interest together with an additional ORF encoding viral replicase. The latter drives the intracellular amplification of mRNA, and can therefore significantly increase the antigen expression capacity.

[0041] Also, several modifications have been implemented at the end structures of mRNA. Anti-reverse cap (ARCA) modifications can ensure the correct cap orientation at the 5' end, which yields almost complete fractions of mRNA that can efficiently bind the ribosomes. Other cap modifications, such as phosphorothioate cap analogs, can further improve the affinity towards the eukaryotic translation initiation factor 4E, and increase the resistance against the RNA decapping complex.

[0042] Conversely, by modifying its structure, the potency of mRNA to trigger innate immune responses can be further improved, but to the detriment of translation capacity. By stabilizing the mRNA with either a phosphorothioate backbone, or by its precipitation with the cationic protein protamine, antigen expression can be diminished, but stronger immune-stimulating capacities can be obtained.

[0043] Preferably, the mRNA vaccine of the present invention is a vaccine directed against infectious disease, preferably against viral infectious disease, preferably coronavirus disease, preferably against Covid-19 disease.

[0044] In one aspect the invention relates to a method for eliciting an immunoprotective response in a human subject against a bacterium and betacoronavirus, the method includes co-administering to the human subject an effective dose of a first immunogenic composition including an antigen selected from any one of a polypeptide, toxoid, polysaccharide, and polysaccharide conjugate; and a second immunogenic composition including mRNA against a betacoronavirus. In an aspect, said first immunogenic composition against the bacterium and said second immunogenic composition mRNA vaccine against betacoronavirus are administered concurrently or concomitantly. One particularly preferred embodiment of the invention combines a PCV of the invention with the mRNA vaccine BNT162b2 (Comirnaty®).

[0045] Another particularly preferred embodiment of the invention combines an immunogenic composition including an antigen derived from *Neisseria meningitidis* with the mRNA vaccine BNT162b2 (Comirnaty®).

[0046] Another particularly preferred embodiment of the invention combines an immunogenic composition including an antigen derived from *C. difficile* with the mRNA vaccine BNT162b2 (Comirnaty®).

[0047] Another particularly preferred embodiment of the invention combines an immunogenic composition including an antigen derived from *E. coli* with the mRNA vaccine BNT162b2 (Comirnaty®).

[0048] In another embodiment, the mRNA vaccine includes a sequence having residues 1-102 of SEQ ID NO: 1 (see FIG. 2) and residues 103-4284 of SEQ ID NO: 1, wherein the sequence for the SARS-CoV-2 antigen of SEQ ID NO: 1 is replaced with SARS-CoV-2 antigen of a variant strain.

[0049] Another particularly preferred embodiment of the invention combines a PCV of the invention with the mRNA vaccine "mRNA-1273".

[0050] Further mRNA vaccines directed against Covid-19 disease currently undergoing clinical trials include:

[0051] 1) MRT5500 Sanofi and Translate Bio Preclinical

[0052] 2) HGC019 Gennova Biopharmaceuticals and HDT Bio Phase I/II

[0053] 3) ARCoV Academy of Military Medical Sciences, Suzhou Abogen Biosciences, Walvax Biotechnology Phase I

[0054] 4) ChulaCoV19 Chula Vaccine Research Centre Phase I

[0055] 5) PTX-COVID19-B Providence Therapeutics Phase I

[0056] 6) ARCT-021 (LUNAR-COV19) Duke-NUS/Arcturus Therapeutics Phase II

[0057] 7) CVnCoV CureVac Phase III

[0058] The mRNA vaccines of the invention comprise mRNA and preferably nucleoside-modified mRNA.

[0059] mRNA useful in the disclosure typically include a first region of linked nucleosides encoding a polypeptide of interest (e.g., a coding region), a first flanking region located at the 5'-terminus of the first region (e.g., a 5'-UTR), a second flanking region located at the 3'-terminus of the first region (e.g., a 3'-UTR), at least one 5'-cap region, and a 3'-stabilizing region. In some embodiments, the mRNA of the invention further includes a poly-A region or a Kozak sequence (e.g., in the 5'-UTR). In some cases, mRNA of the invention may contain one or more intronic nucleotide sequences capable of being excised from the polynucleotide. In some embodiments, mRNA of the invention may include a 5' cap structure, a chain terminating nucleotide, a stem loop, a poly A sequence, and/or a polyadenylation signal. Any one of the regions of a nucleic acid may include one or more alternative components (e.g., an alternative nucleoside). For example, the 3'-stabilizing region may contain an alternative nucleoside such as an L-nucleoside, an inverted thymidine, or a 2'-O-methyl nucleoside and/or the coding region, 5'-UTR, 3'-UTR, or cap region may include an alternative nucleoside such as a 5-substituted uridine (e.g., 5-methoxyuridine), a 1-substituted pseudouridine (e.g., 1-methyl-pseudouridine), and/or a 5-substituted cytidine (e.g., 5-methyl-cytidine). In some embodiments, a LNP includes one or more RNAs, and the one or more RNAs, lipids, and amounts thereof may be selected to provide a specific N:P ratio. The N:P ratio of the composition refers to the molar ratio of nitrogen atoms in one or more lipids to the number of phosphate groups in an RNA. In general, a lower N:P ratio is preferred. The one or more RNA, lipids, and amounts thereof may be selected to provide an N:P ratio from about 2:1 to about 30:1, such as 2:1, 3:1, 4:1, 5:1, 6:1, 7:1, 8:1, 9:1, 10:1, 12:1, 14:1, 16:1, 18:1, 20:1, 22:1, 24:1, 26:1, 28:1, or 30:1. In certain embodiments, the N:P ratio may be from about 2:1 to about 8:1. In other embodiments, the N:P ratio is from about 5:1 to about 8:1. For example, the N:P ratio may be about 5.0:1, about 5.5:1, about 5.67:1, about 6.0:1, about 6.5:1, or about 7.0:1. For example, the N:P ratio may be about 5.67:1.

[0060] mRNA of the invention may include one or more naturally occurring components, including any of the canonical nucleotides A (adenosine), G (guanosine), C (cytosine), U (uridine), or T (thymidine). In one embodiment, all or substantially all of the nucleotides comprising (a) the 5'-UTR, (b) the open reading frame (ORF), (c) the 3'-UTR, (d) the poly A tail, and any combination of (a, b, c, or d above) comprise naturally occurring canonical nucleotides A (adenosine), G (guanosine), C (cytosine), U (uridine), or T (thymidine).

[0061] mRNA of the invention may include one or more alternative components, as described herein, which impart useful properties including increased stability and/or the lack of a substantial induction of the innate immune response of a cell into which the polynucleotide is introduced. For example, a modRNA may exhibit reduced degradation in a cell into which the modRNA is introduced, relative to a corresponding unaltered mRNA. These alternative species may enhance the efficiency of protein production, intracellular retention of the polynucleotides, and/or viability of contacted cells, as well as possess reduced immunogenicity.

[0062] mRNA of the invention may include one or more modified (e.g., altered or alternative) nucleobases, nucleosides, nucleotides, or combinations thereof. The mRNA useful in a LNP can include any useful modification or alteration, such as to the nucleobase, the sugar, or the internucleoside linkage (e.g., to a linking phosphate/to a phosphodiester linkage/to the phosphodiester backbone). In certain embodiments, alterations (e.g., one or more alterations) are present in each of the nucleobase, the sugar, and the internucleoside linkage. Alterations according to the present disclosure may be alterations of ribonucleic acids (RNAs), e.g., the substitution of the 2'-OH of the ribofuranosyl ring to 2'-H, threose nucleic acids (TNAs), glycol nucleic acids (GNAs), peptide nucleic acids (PNAs), locked nucleic acids (LNAs), or hybrids thereof. Additional alterations are described herein.

[0063] mRNA of the invention may or may not be uniformly altered along the entire length of the molecule. For example, one or more or all types of nucleotide (e.g., purine or pyrimidine, or any one or more or all of A, G, U, C) may or may not be uniformly altered in a mRNA, or in a given predetermined sequence region thereof. In some instances, all nucleotides X in a mRNA (or in a given sequence region thereof) are altered, wherein X may any one of nucleotides A, G, U, C, or any one of the combinations A+G, A+U, A+C, G+U, G+C, U+C, A+G+U, A+G+C, G+U+C or A+G+C.

[0064] Different sugar alterations and/or internucleoside linkages (e.g., backbone structures) may exist at various positions in a polynucleotide. One of ordinary skill in the art will appreciate that the nucleotide analogs or other alteration (s) may be located at any position(s) of a polynucleotide such that the function of the polynucleotide is not substantially decreased. An alteration may also be a 5'- or 3'-terminal alteration. In some embodiments, the polynucleotide includes an alteration at the 3'-terminus. The polynucleotide may contain from about 1% to about 100% alternative nucleotides (either in relation to overall nucleotide content, or in relation to one or more types of nucleotide, i.e., any one or more of A, G, U or C) or any intervening percentage (e.g., from 1% to 20%, from 1% to 25%, from 1% to 50%, from 1% to 60%, from 1% to 70%, from 1% to 80%, from 1% to 90%, from 1% to 95%, from 10% to 20%, from 10% to 25%, from 10% to 50%, from 10% to 60%, from 10% to 70%, from 10% to 80%, from 10% to 90%, from 10% to 95%, from 10% to 100%, from 20% to 25%, from 20% to 50%, from 20% to 60%, from 20% to 70%, from 20% to 80%, from 20% to 90%, from 20% to 95%, from 20% to 100%, from 50% to 60%, from 50% to 70%, from 50% to 80%, from 50% to 90%, from 50% to 95%, from 50% to 100%, from 70% to 80%, from 70% to 90%, from 70% to 95%, from 70% to 100%, from 80% to 90%, from 80% to 95%, from 80% to 100%, from 90% to 95%, from 90% to 100%, and from 95% to 100%). It will be understood that any remaining percentage is accounted for by the presence of a canonical nucleotide (e.g., A, G, U, or C).

[0065] Polynucleotides may contain at a minimum zero and at maximum 100% alternative nucleotides, or any intervening percentage, such as at least 5% alternative nucleotides, at least 10% alternative nucleotides, at least 25% alternative nucleotides, at least 50% alternative nucleotides, at least 80% alternative nucleotides, or at least 90% alternative nucleotides. For example, polynucleotides may contain an alternative pyrimidine such as an alternative uracil or cytosine. In some embodiments, at least 5%, at least

10%, at least 25%, at least 50%, at least 80%, at least 90% or 100% of the uracil in a polynucleotide is replaced with an alternative uracil (e.g., a 5-substituted uracil). The alternative uracil can be replaced by a compound having a single unique structure or can be replaced by a plurality of compounds having different structures (e.g., 2, 3, 4 or more unique structures). In some instances, at least 5%, at least 10%, at least 25%, at least 50%, at least 80%, at least 90% or 100% of the cytosine in the polynucleotide is replaced with an alternative cytosine (e.g., a 5-substituted cytosine). The alternative cytosine can be replaced by a compound having a single unique structure or can be replaced by a plurality of compounds having different structures (e.g., 2, 3, 4 or more unique structures).

[0066] In some instances, nucleic acids do not substantially induce an innate immune response of a cell into which the polynucleotide (e.g., mRNA) is introduced. Features of an induced innate immune response include 1) increased expression of pro-inflammatory cytokines, 2) activation of intracellular PRRs (RIG-I, MDA5, etc., and/or 3) termination or reduction in protein translation.

[0067] In some embodiments, the mRNA comprises one or more alternative nucleoside or nucleotides. The alternative nucleosides and nucleotides can include an alternative nucleobase. A nucleobase of a nucleic acid is an organic base such as a purine or pyrimidine or a derivative thereof. A nucleobase may be a canonical base (e.g., adenine, guanine, uracil, thymine, and cytosine). These nucleobases can be altered or wholly replaced to provide polynucleotide molecules having enhanced properties, e.g., increased stability such as resistance to nucleases. Non-canonical or modified bases may include, for example, one or more substitutions or modifications including but not limited to alkyl, aryl, halo, oxo, hydroxyl, alkyloxy, and/or thio substitutions; one or more fused or open rings; oxidation; and/or reduction.

[0068] In some embodiments, the nucleobase is an alternative uracil. Exemplary nucleobases and nucleosides having an alternative uracil include pseudouridine (ψ), pyridin-4-one ribonucleoside, 5-aza-uracil, 6-aza-uracil, 2-thio-5-aza-uracil, 2-thio-uracil (s2U), 4-thio-uracil (s4U), 4-thio-pseudouridine, 2-thio-pseudouridine, 5-hydroxy-uracil (ho5U), 5-aminoallyl-uracil, 5-halo-uracil (e.g., 5-iodo-uracil or 5-bromo-uracil), 3-methyl-uracil (m U), 5-methoxy-uracil (mo5U), uracil 5-oxyacetic acid (cmo5U), uracil 5-oxyacetic acid methyl ester (mcmo5U), 5-carboxymethyl-uracil (cm5U), 1-carboxymethyl-pseudouridine, 5-carboxy-hydroxymethyl-uracil (chm5U), 5-carboxyhydroxymethyl-uracil methyl ester (mchm5U), 5-methoxycarbonylmethyl-uracil (mcm5U), 5-methoxycarbonylmethyl-2-thio-uracil (mcm5s2U), 5-aminomethyl-2-thio-uracil (nmVu), 5-methylaminomethyl-uracil (mnm5U), 5-methylaminomethyl-2-thio-uracil (mnmVu), 5-methylaminomethyl-2-seleno-uracil (mnm5se2U), 5-carbamoylmethyl-uracil (ncm5U), 5-carboxymethylaminomethyl-uracil (cmnm5U), 5-carboxymethylaminomethyl-2-thio-uracil (cmnmVu), 5-propynyl-uracil, 1-propynyl-pseudouracil, 5-taurinomethyl-uracil (xm5U), 1-taurinomethyl-pseudouridine, 5-taurinomethyl-2-thio-uracil (xm5s2U), 1-taurinomethyl-4-thio-pseudouridine, 5-methyl-uracil (m5U, i.e., having the nucleobase deoxythymine), 1-methyl-pseudouridine (mV), 5-methyl-2-thio-uracil (m5s2U), 1-methyl-4-thio-pseudouridine (m xjl), 4-thio-1-methyl-pseudouridine, 3-methyl-pseudouridine (m \//), 2-thio-1-methyl-pseudouridine, 1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-1-deaza-pseudouridine,

dihydrouracil (D), dihydropseudouridine, 5,6-dihydrouracil, 5-methyl-dihydrouracil (m5D), 2-thio-dihydrouracil, 2-thio-dihydropseudouridine, 2-methoxy-uracil, 2-methoxy-4-thio-uracil, 4-methoxy-pseudouridine, 4-methoxy-2-thio-pseudouridine, NI-methyl-pseudouridine, 3-(3-amino-3-carboxypropyl)uracil (acp U), 1-methyl-3-(3-amino-3-carboxypropyl)pseudouridine (acp ψ), 5-(isopentenylaminomethyl)uracil (imm5U), 5-(isopentenylaminomethyl)-2-thio-uracil (imm5s2U), 5,2'-O-dimethyl-uridine (m5Um), 2-thio-2'-O-methyl-uridine (s2Um), 5-methoxycarbonylmethyl-2'-O-methyl-uridine (mem Um), 5-carbamoylmethyl-2'-O-methyl-uridine (ncm5Um), 5-carboxymethylaminomethyl-2'-O-methyl-uridine (cmnm5Um), 3,2'-O-dimethyl-uridine (m Um), and 5-(isopentenylaminomethyl)-2'-O-methyl-uridine (imm5Um), 1-thio-uracil, deoxythymidine, 5-(2-carbomethoxyvinyl)-uracil, 5-(carbamoylhydroxymethyl)-uracil, 5-carbamoylmethyl-2-thio-uracil, 5-carboxymethyl-2-thio-uracil, 5-cyanomethyl-uracil, 5-methoxy-2-thio-uracil, and 5-[3-(1-E-propenylamino)] uracil.

[0069] In some embodiments, the nucleobase is an alternative cytosine. Exemplary nucleobases and nucleosides having an alternative cytosine include 5-aza-cytosine, 6-aza-cytosine, pseudoisocytidine, 3-methyl-cytosine (m3C), N4-acetyl-cytosine (ac4C), 5-formyl-cytosine (f5C), N4-methyl-cytosine (m4C), 5-methyl-cytosine (m5C), 5-halo-cytosine (e.g., 5-iodo-cytosine), 5-hydroxymethyl-cytosine (hm5C), 1-methyl-pseudoisocytidine, pyrrolo-cytosine, pyrrolo-pseudoisocytidine, 2-thio-cytosine (s2C), 2-thio-5-methyl-cytosine, 4-thio-pseudoisocytidine, 4-thio-1-methyl-1-pseudoisocytidine, 4-thio-1-methyl-1-deaza-pseudoisocytidine, 1-methyl-1-deaza-pseudoisocytidine, zebularine, 5-aza-zebularine, 5-methyl-zebularine, 5-aza-2-thio-zebularine, 2-thio-zebularine, 2-methoxy-cytosine, 2-methoxy-5-methyl-cytosine, 4-methoxy-pseudoisocytidine, 4-methoxy-1-methyl-pseudoisocytidine, lysidine (k2C), 5,2'-O-dimethyl-cytidine (m5Cm), N4-acetyl-2'-O-methyl-cytidine (ac4Cm), N4,2'-O-dimethyl-cytidine (m4Cm), 5-formyl-2'-O-methyl-cytidine (f5Cm), N4,N4,2'-O-trimethyl-cytidine (m42Cm), 1-thio-cytosine, 5-hydroxycytosine, 5-(3-azidopropyl)-cytosine, and 5-(2-azidoethyl)-cytosine.

[0070] In some embodiments, the nucleobase is an alternative adenine. Exemplary nucleobases and nucleosides having an alternative adenine include 2-amino-purine, 2,6-diaminopurine, 2-amino-6-halo-purine (e.g., 2-amino-6-chloro-purine), 6-halo-purine (e.g., 6-chloro-purine), 2-amino-6-methyl-purine, 8-azido-adenine, 7-deaza-adenine, 7-deaza-8-aza-adenine, 7-deaza-2-amino-purine, 7-deaza-8-aza-2-amino-purine, 7-deaza-2,6-diaminopurine, 7-deaza-8-aza-2,6-diaminopurine, 1-methyl-adenine (m1A), 2-methyl-adenine (m2A), N6-methyl-adenine (m6A), 2-methylthio-N6-methyl-adenine (ms2m6A), N6-isopentenyl-adenine (i6A), 2-methylthio-N6-isopentenyl-adenine (ms2i6A), N6-(cis-hydroxyisopentenyl)adenine (io6A), 2-methylthio-N6-(cis-hydroxyisopentenyl)adenine (ms2io6A), N6-glycylcarbamoyl-adenine (g6A), N6-threonylcarbamoyl-adenine (t6A), N6-methyl-N6-threonylcarbamoyl-adenine (m6t6A), 2-methylthio-N6-threonylcarbamoyl-adenine (ms2g6A), N6,N6-dimethyl-adenine (m62A), N6-hydroxynorvalylcarbamoyl-adenine (hn6A), 2-methylthio-N6-hydroxynorvalylcarbamoyl-adenine (ms2hn6A), N6-acetyl-adenine (ac6A), 7-methyl-adenine, 2-methylthio-adenine, 2-methoxy-adenine, N6,2'-O-dimethyl-adenosine (m6Am), N6,N6,2'-O-trimethyl-adenosine (m62Am), 1,2'-O-dimethyl-adenosine (m1Am), 2-amino-N6-methyl-purine, 1-thio-adenine, 8-azido-adenine, N6-(19-amino-pentaoxonadecyl)-adenine, 2,8-dimethyl-adenine, N6-formyl-adenine, and N6-hydroxymethyl-adenine.

ethyl-adenosine (m6Am), N6,N6,2'-O-trimethyl-adenosine (m62Am), 1,2'-O-dimethyl-adenosine (m1Am), 2-amino-N6-methyl-purine, 1-thio-adenine, 8-azido-adenine, N6-(19-amino-pentaoxonadecyl)-adenine, 2,8-dimethyl-adenine, N6-formyl-adenine, and N6-hydroxymethyl-adenine.

[0071] In some embodiments, the nucleobase is an alternative guanine. Exemplary nucleobases and nucleosides having an alternative guanine include inosine (I), 1-methyl-inosine (m1I), wyosine (imG), methylwyosine (mimG), 4-demethyl-wyosine (imG-14), isowyosine (imG2), wybutosine (yW), peroxywybutosine (o2yW), hydroxywybutosine (OHyW), undermodified hydroxywybutosine (OHyW*), 7-deaza-guanine, queuosine (Q), epoxyqueuosine (oQ), galactosyl-queuosine (galQ), mannosyl-queuosine (manQ), 7-cyano-7-deaza-guanine (preQO), 7-aminomethyl-7-deaza-guanine (preQI), archaeosine (G+), 7-deaza-8-aza-guanine, 6-thio-guanine, 6-thio-7-deaza-guanine, 6-thio-7-deaza-8-aza-guanine, 7-methyl-guanine (m7G), 6-thio-7-methyl-guanine, 7-methyl-inosine, 6-methoxy-guanine, 1-methyl-guanine (m1G), N2-methyl-guanine (m2G), N2,N2-dimethyl-guanine (m22G), N2,7-dimethyl-guanine (m2,7G), N2, N2,7-dimethyl-guanine (m2,2,7G), 8-oxo-guanine, 7-methyl-8-oxo-guanine, 1-methyl-6-thio-guanine, N2-methyl-6-thio-guanine, N2,N2-dimethyl-6-thio-guanine, N2-methyl-2'-O-methyl-guanosine (m2Gm), N2,N2-dimethyl-2'-O-methyl-guanosine (m22Gm), 1-methyl-2'-O-methyl-guanosine (m1Gm), N2,7-dimethyl-2'-O-methyl-guanosine (m2,7Gm), 2'-O-methyl-inosine (Im), 1,2'-O-dimethyl-inosine (m1Im), 1-thio-guanine, and O-6-methyl-guanine.

[0072] The alternative nucleobase of a nucleotide can be independently a purine, a pyrimidine, a purine or pyrimidine analog. For example, the nucleobase can be an alternative to adenine, cytosine, guanine, uracil, or hypoxanthine. In another embodiment, the nucleobase can also include, for example, naturally-occurring and synthetic derivatives of a base, including pyrazolo[3,4-d]pyrimidines, 5-methylcytosine (5-me-C), 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine and 2-thiocytosine, 5-propynyl uracil and cytosine, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 8-halo (e.g., 8-bromo), 8-amino, 8-thiol, 8-thio-alkyl, 8-hydroxy and other 8-substituted adenines and guanines, 5-halo particularly 5-bromo, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine and 7-methyladenine, 8-azaguanine and 8-azaadenine, deazaguanine, 7-deazaguanine, 3-deazaguanine, deazaadenine, 7-deazaadenine, 3-deazaadenine, pyrazolo[3,4-d]pyrimidine, imidazo[1,5-a]1,3,5 triazinones, 9-deazapurines, imidazo[4,5-d]pyrazines, thiazolo[4,5-d]pyrimidines, pyrazin-2-ones, 1,2,4-triazine, pyridazine; or 1,3,5 triazine. When the nucleotides are depicted using the shorthand A, G, C, T or U, each letter refers to the representative base and/or derivatives thereof, e.g., A includes adenine or adenine analogs, e.g., 7-deaza adenine).

[0073] The mRNA may include a 5'-cap structure. The 5'-cap structure of a polynucleotide is involved in nuclear export and increasing polynucleotide stability and binds the mRNA Cap Binding Protein (CBP), which is responsible for polynucleotide stability in the cell and translation competency through the association of CBP with poly-A binding protein to form the mature cyclic mRNA species. The cap

further assists the removal of 5'-proximal introns removal during mRNA splicing. Endogenous polynucleotide molecules may be 5'-end capped generating a 5'-ppp-5'-triphosphate linkage between a terminal guanosine cap residue and the 5'-terminal transcribed sense nucleotide of the polynucleotide. This 5'-guanylate cap may then be methylated to generate an N7-methyl-guanylate residue. The ribose sugars of the terminal and/or antiterminal transcribed nucleotides of the 5' end of the polynucleotide may optionally also be 2'-O-methylated. 5'-decapping through hydrolysis and cleavage of the guanylate cap structure may target a polynucleotide molecule, such as an mRNA molecule, for degradation.

[0074] Alterations to polynucleotides may generate a non-hydrolyzable cap structure preventing decapping and thus increasing polynucleotide half-life. Because cap structure hydrolysis requires cleavage of 5'-ppp-5' phosphodiester linkages, alternative nucleotides may be used during the capping reaction. For example, a Vaccinia Capping Enzyme from New England Biolabs (Ipswich, MA) may be used with a-thio-guanosine nucleotides according to the manufacturer's instructions to create a phosphorothioate linkage in the 5'-ppp-5' cap.

[0075] Additional alternative guanosine nucleotides may be used such as a-methyl-phosphonate and seleno-phosphate nucleotides. Additional alterations include, but are not limited to, 2'-O-methylation of the ribose sugars of 5'-terminal and/or 5'-antiterminal nucleotides of the polynucleotide (as mentioned above) on the 2'-hydroxy group of the sugar. Multiple distinct 5'-cap structures can be used to generate the 5'-cap of an mRNA molecule.

[0076] Cap analogs, which herein are also referred to as synthetic cap analogs, chemical caps, chemical cap analogs, or structural or functional cap analogs, differ from natural (i.e., endogenous, wild-type, or physiological) 5'-caps in their chemical structure, while retaining cap function. Cap analogs may be chemically (i.e., non-enzymatically) or enzymatically synthesized and/linked to a polynucleotide. For example, the Anti-Reverse Cap Analog (ARCA) cap contains two guanoses linked by a 5'-5'-triphosphate group, wherein one guanosine contains an N7-methyl group as well as a 3-O-methyl group (i.e., N7,'0-dimethyl-guanosine-5'-triphosphate-5'-guanosine, m7G-3'mppp-G, which may equivalently be designated 3' 0-Me-m7G(5')ppp(5')G). The 3'-O atom of the other, unaltered, guanosine becomes linked to the 5'-terminal nucleotide of the capped polynucleotide (e.g., an mRNA). The N7- and 3'-O-methylated guanosine provides the terminal moiety of the capped polynucleotide (e.g., mRNA). Another exemplary cap is mCAP, which is similar to ARCA but has a 2'-O-methyl group on guanosine (i.e., N7,2'-O-dimethyl-guanosine-5'-triphosphate-5'-guanosine, m7Gm-ppp-G).

[0077] A cap may be a dinucleotide cap analog. As a non-limiting example, the dinucleotide cap analog may be modified at different phosphate positions with a boranophosphate group or a phosphoselenoate group such as the dinucleotide cap analogs described in U.S. Pat. No. 8,519, 110, the cap structures of which are herein incorporated by reference.

[0078] Alternatively, a cap analog may be a N7-(4-chlorophenoxy ethyl) substituted dinucleotide cap analog known in the art and/or described herein. Non-limiting examples of N7-(4-chlorophenoxy ethyl) substituted dinucleotide cap analogs include a N7-(4-chlorophenoxyethyl)-G(5)ppp(5')G

and a N7-(4-chlorophenoxyethyl)-m3'-OG(5)ppp(5')G cap analog (see, e.g., the various cap analogs and the methods of synthesizing cap analogs described in Kore et al. Bioorganic & Medicinal Chemistry 2013 21:4570-4574; the cap structures of which are herein incorporated by reference). In other instances, a cap analog useful in the polynucleotides of the present disclosure is a 4-chloro/bromophenoxy ethyl analog.

[0079] While cap analogs allow for the concomitant capping of a polynucleotide in an in vitro transcription reaction, up to 20% of transcripts remain uncapped. This, as well as the structural differences of a cap analog from endogenous 5'-cap structures of polynucleotides produced by the endogenous, cellular transcription machinery, may lead to reduced translational competency and reduced cellular stability.

[0080] Alternative polynucleotides may also be capped post-transcriptionally, using enzymes, in order to generate more authentic 5'-cap structures. As used herein, the phrase "more authentic" refers to a feature that closely mirrors or mimics, either structurally or functionally, an endogenous or wild type feature. That is, a "more authentic" feature is better representative of an endogenous, wild-type, natural or physiological cellular function, and/or structure as compared to synthetic features or analogs of the prior art, or which outperforms the corresponding endogenous, wild-type, natural, or physiological feature in one or more respects. Non-limiting examples of more authentic 5'-cap structures useful in the polynucleotides of the present disclosure are those which, among other things, have enhanced binding of cap binding proteins, increased half-life, reduced susceptibility to 5'-endonucleases, and/or reduced 5'-decapping, as compared to synthetic 5'-cap structures known in the art (or to a wild-type, natural or physiological 5'-cap structure). For example, recombinant Vaccinia Virus Capping Enzyme and recombinant 2'-O-methyltransferase enzyme can create a canonical 5'-5'-triphosphate linkage between the 5'-terminal nucleotide of a polynucleotide and a guanosine cap nucleotide wherein the cap guanosine contains an N7-methylation and the 5'-terminal nucleotide of the polynucleotide contains a 2'-O-methyl. Such a structure is termed the CapI structure. This cap results in a higher translational-competency, cellular stability, and a reduced activation of cellular pro-inflammatory cytokines, as compared, e.g., to other 5' cap analog structures known in the art. Other exemplary cap structures include 7mG(5')ppp(5')N.pN2p (Cap 0), 7mG(5')ppp(5')NlmpNp (Cap 1), 7mG(5')-ppp(5')NlmpN2mp (Cap 2), and m(7)Gpppm(3)(6,6,2')Apm(2')Apm(2')Cpm(2)(3,2')Up (Cap 4).

[0081] Because the alternative polynucleotides may be capped post-transcriptionally, and because this process is more efficient, nearly 100% of the mRNA may be capped. This is in contrast to ~80% when a cap analog is linked to a polynucleotide in the course of an in vitro transcription reaction. 5'-terminal caps may include endogenous caps or cap analogs. A 5'-terminal cap may include a guanosine analog. Useful guanosine analogs include inosine, NI-methyl-guanosine, 2'-fluoro-guanosine, 7-deaza-guanosine, 8-oxo-guanosine, 2-amino-guanosine, LNA-guanosine, and 2-azido-guanosine. In some cases, a polynucleotide contains a modified 5'-cap. A modification on the 5'-cap may increase the stability of polynucleotide, increase the half-life of the polynucleotide, and could increase the polynucleotide translational efficiency. The modified 5'-cap may include, but is not limited to, one or more of the following modifications: modification at the 2'- and/or 3'-position of a capped

guanosine triphosphate (GTP), a replacement of the sugar ring oxygen (that produced the carbocyclic ring) with a methylene moiety (CH₂), a modification at the triphosphate bridge moiety of the cap structure, or a modification at the nucleobase (G) moiety.

[0082] A 5'-UTR may be provided as a flanking region to the mRNA. A 5'-UTR may be homologous or heterologous to the coding region found in a polynucleotide. Multiple 5'-UTRs may be included in the flanking region and may be the same or of different sequences. Any portion of the flanking regions, including none, may be codon optimized and any may independently contain one or more different structural or chemical alterations, before and/or after codon optimization.

[0083] To alter one or more properties of an mRNA, 5'-UTRs which are heterologous to the coding region of an mRNA may be engineered. The mRNA may then be administered to cells, tissue or organisms and outcomes such as protein level, localization, and/or half-life may be measured to evaluate the beneficial effects the heterologous 5'-UTR may have on the mRNA. Variants of the 5'-UTRs may be utilized wherein one or more nucleotides are added or removed to the termini, including A, T, C or G. 5'-UTRs may also be codon-optimized, or altered in any manner described herein. mRNAs may include a stem loop such as, but not limited to, a histone stem loop. The stem loop may be a nucleotide sequence that is about 25 or about 26 nucleotides in length. The histone stem loop may be located 3'-relative to the coding region (e.g., at the 3'-terminus of the coding region). As a non-limiting example, the stem loop may be located at the 3'-end of a polynucleotide described herein. In some cases, an mRNA includes more than one stem loop (e.g., two stem loops). A stem loop may be located in a second terminal region of a polynucleotide. As a non-limiting example, the stem loop may be located within an untranslated region (e.g., 3'-UTR) in a second terminal region. In some cases, a mRNA which includes the histone stem loop may be stabilized by the addition of a 3'-stabilizing region (e.g., a 3'-stabilizing region including at least one chain terminating nucleoside). Not wishing to be bound by theory, the addition of at least one chain terminating nucleoside may slow the degradation of a polynucleotide and thus can increase the half-life of the polynucleotide. In other cases, a mRNA, which includes the histone stem loop may be stabilized by an alteration to the 3'-region of the polynucleotide that can prevent and/or inhibit the addition of oligio(U). In yet other cases, a mRNA, which includes the histone stem loop may be stabilized by the addition of an oligonucleotide that terminates in a 3'-deoxynucleoside, 2',3'-dideoxynucleoside 3'-O-methylnucleosides, 3-O-ethyl-nucleosides, 3'-arabinosides, and other alternative nucleosides known in the art and/or described herein. In some instances, the mRNA of the present disclosure may include a histone stem loop, a poly-A region, and/or a 5'-cap structure. The histone stem loop may be before and/or after the poly-A region. The polynucleotides including the histone stem loop and a poly-A region sequence may include a chain terminating nucleoside described herein. In other instances, the polynucleotides of the present disclosure may include a histone stem loop and a 5'-cap structure. The 5'-cap structure may include, but is not limited to, those described herein and/or known in the art. In some cases, the conserved stem loop region may include a miR sequence described herein. As a non-limiting example, the stem loop region may

include the seed sequence of a miR sequence described herein. In another non-limiting example, the stem loop region may include a miR-122 seed sequence.

[0084] mRNA may include at least one histone stem-loop and a poly-A region or polyadenylation signal. In certain cases, the polynucleotide encoding for a histone stem loop and a poly-A region or a polyadenylation signal may code for a pathogen antigen or fragment thereof. In other cases, the polynucleotide encoding for a histone stem loop and a poly-A region or a polyadenylation signal may code for a therapeutic protein. In some cases, the polynucleotide encoding for a histone stem loop and a poly-A region or a polyadenylation signal may code for a tumor antigen or fragment thereof. In other cases, the polynucleotide encoding for a histone stem loop and a poly-A region or a polyadenylation signal may code for an allergenic antigen or an autoimmune self-antigen.

[0085] An mRNA may include a polyA sequence and/or polyadenylation signal. A polyA sequence may be comprised entirely or mostly of adenine nucleotides or analogs or derivatives thereof. A polyA sequence may be a tail located adjacent to a 3' untranslated region of a nucleic acid. During RNA processing, a long chain of adenosine nucleotides (poly-A region) is normally added to messenger RNA (mRNA) molecules to increase the stability of the molecule. Immediately after transcription, the 3'-end of the transcript is cleaved to free a 3-hydroxy. Then poly-A polymerase adds a chain of adenosine nucleotides to the RNA. The process, called polyadenylation, adds a poly-A region that is between 100 and 250 residues long. Unique poly-A region lengths may provide certain advantages to the alternative polynucleotides of the present disclosure. Generally, the length of a poly-A region of the present disclosure is at least 30 nucleotides in length. In another embodiment, the poly-A region is at least 35 nucleotides in length. In another embodiment, the length is at least 40 nucleotides. In another embodiment, the length is at least 45 nucleotides. In another embodiment, the length is at least 55 nucleotides. In another embodiment, the length is at least 60 nucleotides. In another embodiment, the length is at least 70 nucleotides. In another embodiment, the length is at least 80 nucleotides. In another embodiment, the length is at least 90 nucleotides. In another embodiment, the length is at least 100 nucleotides. In another embodiment, the length is at least 120 nucleotides. In another embodiment, the length is at least 140 nucleotides. In another embodiment, the length is at least 160 nucleotides. In another embodiment, the length is at least 180 nucleotides. In another embodiment, the length is at least 200 nucleotides. In another embodiment, the length is at least 250 nucleotides. In another embodiment, the length is at least 300 nucleotides. In another embodiment, the length is at least 350 nucleotides. In another embodiment, the length is at least 400 nucleotides. In another embodiment, the length is at least 450 nucleotides. In another embodiment, the length is at least 500 nucleotides. In another embodiment, the length is at least 600 nucleotides. In another embodiment, the length is at least 700 nucleotides. In another embodiment, the length is at least 800 nucleotides. In another embodiment, the length is at least 900 nucleotides. In another embodiment, the length is at least 1000 nucleotides. In another embodiment, the length is at least 1100 nucleotides. In another embodiment, the length is at least 1200 nucleotides. In another embodiment, the length is at least 1300 nucleotides. In another embodiment, the length is

at least 1400 nucleotides. In another embodiment, the length is at least 1500 nucleotides. In another embodiment, the length is at least 1600 nucleotides. In another embodiment, the length is at least 1700 nucleotides. In another embodiment, the length is at least 1800 nucleotides. In another embodiment, the length is at least 1900 nucleotides. In another embodiment, the length is at least 2000 nucleotides. In another embodiment, the length is at least 2500 nucleotides. In another embodiment, the length is at least 3000 nucleotides. In some instances, the poly-A region may be 80 nucleotides, 120 nucleotides, 160 nucleotides in length on an alternative polynucleotide molecule described herein. In other instances, the poly-A region may be 20, 40, 80, 100, 120, 140 or 160 nucleotides in length on an alternative polynucleotide molecule described herein. In some cases, the poly-A region is designed relative to the length of the overall alternative polynucleotide. This design may be based on the length of the coding region of the alternative polynucleotide, the length of a particular feature or region of the alternative polynucleotide (such as mRNA) or based on the length of the ultimate product expressed from the alternative polynucleotide. When relative to any feature of the alternative polynucleotide (e.g., other than the mRNA portion which includes the poly-A region) the poly-A region may be 10, 20, 30, 40, 50, 60, 70, 80, 90 or 100% greater in length than the additional feature. The poly-A region may also be designed as a fraction of the alternative polynucleotide to which it belongs. In this context, the poly-A region may be 10, 20, 30, 40, 50, 60, 70, 80, or 90% or more of the total length of the construct or the total length of the construct minus the poly-A region.

[0086] In certain cases, engineered binding sites and/or the conjugation of mRNA for poly-A binding protein may be used to enhance expression. The engineered binding sites may be sensor sequences which can operate as binding sites for ligands of the local microenvironment of the mRNA. As a non-limiting example, the mRNA may include at least one engineered binding site to alter the binding affinity of poly-A binding protein (PABP) and analogs thereof. The incorporation of at least one engineered binding site may increase the binding affinity of the PABP and analogs thereof.

[0087] Additionally, multiple distinct mRNA may be linked together to the PABP (poly-A binding protein) through the 3'-end using alternative nucleotides at the 3'-terminus of the poly-A region. Transfection experiments can be conducted in relevant cell lines at and protein production can be assayed by ELISA at 12 hours, 24 hours, 48 hours, 72 hours, and day 7 post-transfection. As a non-limiting example, the transfection experiments may be used to evaluate the effect on PABP or analogs thereof binding affinity as a result of the addition of at least one engineered binding site. In certain cases, a poly-A region may be used to modulate translation initiation. While not wishing to be bound by theory, the poly-A region recruits PABP which in turn can interact with translation initiation complex and thus may be essential for protein synthesis. In some cases, a poly-A region may also be used in the present disclosure to protect against 3'-5'-exonuclease digestion. In some instances, an mRNA may include a polyA-G Quartet. The G-quartet is a cyclic hydrogen bonded array of four guanosine nucleotides that can be formed by G-rich sequences in both DNA and RNA. In this embodiment, the G-quartet is incorporated at the end of the poly-A region. The resultant mRNA may be assayed for stability, protein production and

other parameters including half-life at various time points. It has been discovered that the polyA-G quartet results in protein production equivalent to at least 75% of that seen using a poly-A region of 120 nucleotides alone. In some cases, mRNA may include a poly-A region and may be stabilized by the addition of a 3'-stabilizing region. The mRNA with a poly-A region may further include a 5'-cap structure. In other cases, mRNA may include a poly-A-G Quartet. The mRNA with a poly-A-G Quartet may further include a 5'-cap structure. In some cases, the 3'-stabilizing region which may be used to stabilize mRNA includes a poly-A region or poly-A-G Quartet. In other cases, the 3'-stabilizing region which may be used with the present disclosure include a chain termination nucleoside such as 3'-deoxyadenosine (cordycepin), 3'-deoxyuridine, 3'-deoxycytosine, 3'-deoxyguanosine, 3'-deoxy thymine, 2',3'-dideoxynucleosides, such as 2',3'-dideoxyadenosine, 2',3'-dideoxyuridine, 2',3'-dideoxycytosine, 2',3'-dideoxyguanosine, 2',3'-dideoxythymine, a 2'-deoxynucleoside, or an O-methylnucleoside. In other cases, mRNA which includes a polyA region or a poly-A-G Quartet may be stabilized by an alteration to the 3'-region of the polynucleotide that can prevent and/or inhibit the addition of oligio(U). In yet other instances, mRNA which includes a poly-A region or a poly-A-G Quartet may be stabilized by the addition of an oligonucleotide that terminates in a 3'-deoxynucleoside, 2',3'-dideoxynucleoside 3-O-methylnucleosides, 3'-O-ethylnucleosides, 3'-arabinosides, and other alternative nucleosides known in the art and/or described herein.

[0088] In an embodiment, the mRNA vaccines of the invention comprise lipids. The lipids and modRNA can together form nanoparticles. The lipids can encapsulate the mRNA in the form of a lipid nanoparticle (LNP) to aid cell entry and stability of the RNA/lipid nanoparticles.

[0089] Lipid nanoparticles may include a lipid component and one or more additional components, such as a therapeutic and/or prophylactic. A LNP may be designed for one or more specific applications or targets. The elements of a LNP may be selected based on a particular application or target, and/or based on the efficacy, toxicity, expense, ease of use, availability, or other feature of one or more elements. Similarly, the particular formulation of a LNP may be selected for a particular application or target according to, for example, the efficacy and toxicity of particular combinations of elements. The efficacy and tolerability of a LNP formulation may be affected by the stability of the formulation.

[0090] Lipid nanoparticles may be designed for one or more specific applications or targets. For example, a LNP may be designed to deliver a therapeutic and/or prophylactic such as an RNA to a particular cell, tissue, organ, or system or group thereof in a mammal's body.

[0091] Physiochemical properties of lipid nanoparticles may be altered in order to increase selectivity for particular bodily targets. For instance, particle sizes may be adjusted based on the fenestration sizes of different organs. The therapeutic and/or prophylactic included in a LNP may also be selected based on the desired delivery target or targets. For example, a therapeutic and/or prophylactic may be selected for a particular indication, condition, disease, or disorder and/or for delivery to a particular cell, tissue, organ, or system or group thereof (e.g., localized or specific delivery). In certain embodiments, a LNP may include an mRNA

encoding a polypeptide of interest capable of being translated within a cell to produce the polypeptide of interest. Such a composition may be designed to be specifically delivered to a particular organ. In some embodiments, a composition may be designed to be specifically delivered to a mammalian liver. In some embodiments, a composition may be designed to be specifically delivered to a lymph node. In some embodiments, a composition may be designed to be specifically delivered to a mammalian spleen.

[0092] A LNP may include one or more components described herein. In some embodiments, the LNP formulation of the disclosure includes at least one lipid nanoparticle component. Lipid nanoparticles may include a lipid component and one or more additional components, such as a therapeutic and/or prophylactic, such as a nucleic acid. A LNP may be designed for one or more specific applications or targets. The elements of a LNP may be selected based on a particular application or target, and/or based on the efficacy, toxicity, expense, ease of use, availability, or other feature of one or more elements. Similarly, the particular formulation of a LNP may be selected for a particular application or target according to, for example, the efficacy and toxicity of particular combination of elements. The efficacy and tolerability of a LNP formulation may be affected by the stability of the formulation.

[0093] In some embodiments, for example, a polymer may be included in and/or used to encapsulate or partially encapsulate a LNP. A polymer may be biodegradable and/or biocompatible. A polymer may be selected from, but is not limited to, polyamines, polyethers, polyamides, polystyrenes, poly carbamates, polyureas, polycarbonates, polystyrenes, polyimides, polysulfones, polyurethanes, polyacetylenes, polyethylenes, polyethyleneimines, polyisocyanates, polyacrylates, polymethacrylates, polyacrylonitriles, and polyarylates. For example, a polymer may include poly(caprolactone) (PCL), ethylene vinyl acetate polymer (EVA), poly(lactic acid) (PLA), poly(L-lactic acid) (PLLA), poly(glycolic acid) (PGA), poly(lactic acid-co-glycolic acid) (PLGA), poly(L-lactic acid-co-glycolic acid) (PLLGA), poly(D,L-lactide) (PDLA), poly(L-lactide) (PLLA), poly(D,L-lactide-co-caprolactone), poly(D,L-lactide-co-caprolactone-co-glycolide), poly(D,L-lactide-co-PEO-co-D,L-lactide), poly(D,L-lactide-co-PPO-co-D,L-lactide), polyalkyl cyanoacrylate, polyurethane, poly-L-lysine (PLL), hydroxypropyl methacrylate (HPMA), polyethyleneglycol, poly-L-glutamic acid, poly(hydroxy acids), polyanhydrides, polyorthoesters, poly(ester amides), polyamides, poly(ester ethers), polycarbonates, polyalkylenes such as polyethylene and polypropylene, polyalkylene glycols such as poly(ethylene glycol) (PEG), polyalkylene oxides (PEO), polyalkylene terephthalates such as poly(ethylene terephthalate), polyvinyl alcohols (PVA), polyvinyl ethers, polyvinyl esters such as poly(vinyl acetate), polyvinyl halides such as poly(vinyl chloride) (PVC), polyvinylpyrrolidone (PVP), polysiloxanes, polystyrene, polyurethanes, derivatized celluloses such as alkyl celluloses, hydroxyalkyl celluloses, cellulose ethers, cellulose esters, nitro celluloses, hydroxypropylcellulose, carboxymethylcellulose, polymers of acrylic acids, such as poly(methyl(meth)acrylate) (PMMA), poly(ethyl(meth)acrylate), poly(butyl(meth)acrylate), poly(isobutyl(meth)acrylate), poly(hexyl(meth)acrylate), poly(isodecyl(meth)acrylate), poly(lauryl(meth)acrylate), poly(phenyl(meth)acrylate), poly(methyl acrylate), poly(isopropyl acrylate), poly(isobutyl acrylate), poly(octadecyl acrylate)

and copolymers and mixtures thereof, polydioxanone and its copolymers, polyhydroxyalkanoates, polypropylene fumarate, polyoxymethylene, poloxamers, poloxamines, poly(ortho)esters, poly(butyric acid), poly(valeric acid), poly(lactide-co-caprolactone), trimethylene carbonate, poly(N-acryloylmorpholine) (PACM), poly(2-methyl-2-oxazoline) (PMOX), poly(2-ethyl-2-oxazoline) (PEOZ), and polyglycerol.

[0094] Surface altering agents may include, but are not limited to, anionic proteins (e.g., bovine serum albumin), surfactants (e.g., cationic surfactants such as dimethyldioctadecyl-ammonium bromide), sugars or sugar derivatives (e.g., cyclodextrin), nucleic acids, polymers (e.g., heparin, polyethylene glycol, and poloxamer), mucolytic agents (e.g., acetylcysteine, mugwort, bromelain, papain, clerodendrum, bromhexine, carbocysteine, eprazinone, mesna, ambroxol, sobrerol, domiodol, letosteine, stepronin, tiopronin, gelsolin, thymosin β 4, dornase alfa, neltexine, and erdosteine), and DNases (e.g., rhDNase). A surface altering agent may be disposed within a nanoparticle and/or on the surface of a LNP (e.g., by coating, adsorption, covalent linkage, or other process). A LNP may also comprise one or more functionalized lipids. For example, a lipid may be functionalized with an alkyne group that, when exposed to an azide under appropriate reaction conditions, may undergo a cycloaddition reaction. In particular, a lipid bilayer may be functionalized in this fashion with one or more groups useful in facilitating membrane permeation, cellular recognition, or imaging. The surface of a LNP may also be conjugated with one or more useful antibodies. Functional groups and conjugates useful in targeted cell delivery, imaging, and membrane permeation are well known in the art.

[0095] In addition to these components, lipid nanoparticles may include any substance useful in pharmaceutical compositions. For example, the lipid nanoparticle may include one or more pharmaceutically acceptable excipients or accessory ingredients such as, but not limited to, one or more solvents, dispersion media, diluents, dispersion aids, suspension aids, surface active agents, buffering agents, preservatives, and other species. Surface active agents and/or emulsifiers may include, but are not limited to, natural emulsifiers (e.g., acacia, alginic acid, sodium alginate, cholesterol, and lecithin), sorbitan fatty acid esters (e.g., polyoxyethylene sorbitan monolaurate [TWEEN®20], polyoxyethylene sorbitan [TWEEN®60], polyoxyethylene sorbitan monooleate [TWEEN®80], sorbitan monopalmitate [SPAN®40], sorbitan monostearate [SPAN®60], sorbitan tristearate [SPAN®65], glyceryl monooleate, sorbitan monooleate [SPAN®80]), polyoxyethylene esters (e.g., polyoxyethylene monostearate [MYRJ®45], polyoxyethylene hydrogenated castor oil, polyethoxylated castor oil, polyoxymethylene stearate, and SOLUTOL®), sucrose fatty acid esters, polyethylene glycol fatty acid esters (e.g., CREMOPHOR®), polyoxyethylene ethers, (e.g., polyoxyethylene lauryl ether [BRIJ®30]), poly(vinyl-pyrrolidone), diethylene glycol monolaurate, triethanolamine oleate, sodium oleate, potassium oleate, ethyl oleate, oleic acid, ethyl laurate, sodium lauryl sulfate, PLURONIC®F 68, POLOXAMER® 188, cetrimonium bromide, cetylpyridinium chloride, benzalkonium chloride, docusate sodium, and/or combinations thereof.

[0096] Examples of preservatives may include, but are not limited to, antioxidants, chelating agents, free radical scavengers, antimicrobial preservatives, antifungal preserva-

tives, alcohol preservatives, acidic preservatives, and/or other preservatives. Examples of antioxidants include, but are not limited to, alpha tocopherol, ascorbic acid, ascorbyl palmitate, butylated hydroxyanisole, butylated hydroxy toluene, monothioglycerol, potassium metabisulfite, propionic acid, propyl gallate, sodium ascorbate, sodium bisulfite, sodium metabisulfite, and/or sodium sulfite. Examples of chelating agents include ethylenediaminetetraacetic acid (EDTA), citric acid monohydrate, disodium edetate, dipotassium edetate, edetic acid, fumaric acid, malic acid, phosphoric acid, sodium edetate, tartaric acid, and/or trisodium edetate. Examples of antimicrobial preservatives include, but are not limited to, benzalkonium chloride, benzethonium chloride, benzyl alcohol, bronopol, cetrimide, cetylpyridinium chloride, chlorhexidine, chlorobutanol, chlorocresol, chloroxylenol, cresol, ethyl alcohol, glycerin, hexetidine, imidurea, phenol, phenoxyethanol, phenylethyl alcohol, phenylmercuric nitrate, propylene glycol, and/or thimerosal. Examples of antifungal preservatives include, but are not limited to, butyl paraben, methyl paraben, ethyl paraben, propyl paraben, benzoic acid, hydroxybenzoic acid, potassium benzoate, potassium sorbate, sodium benzoate, sodium propionate, and/or sorbic acid. Examples of alcohol preservatives include, but are not limited to, ethanol, polyethylene glycol, benzyl alcohol, phenol, phenolic compounds, bisphenol, chlorobutanol, hydroxybenzoate, and/or phenylethyl alcohol. Examples of acidic preservatives include, but are not limited to, vitamin A, vitamin C, vitamin E, beta-carotene, citric acid, acetic acid, dehydroascorbic acid, ascorbic acid, sorbic acid, and/or phytic acid. Other preservatives include, but are not limited to, tocopherol, tocopherol acetate, deteroxime mesylate, cetrimide, butylated hydroxyanisole (BHA), butylated hydroxy toluene (BHT), ethylenediamine, sodium lauryl sulfate (SLS), sodium lauryl ether sulfate (SLES), sodium bisulfite, sodium metabisulfite, potassium sulfite, potassium metabisulfite, GLYDANT PLUS®, PHENONIP®, methylparaben, GERMALL® 115, GERMABEN® II, NEOLONE™, KATHON™, and/or EUXYL®. An exemplary free radical scavenger includes butylated hydroxytoluene (BHT or butylhydroxytoluene) or deferoxamine.

[0097] Examples of buffering agents include, but are not limited to, citrate buffer solutions, acetate buffer solutions, phosphate buffer solutions, ammonium chloride, calcium carbonate, calcium chloride, calcium citrate, calcium gluconate, calcium gluceptate, calcium gluconate, d-gluconic acid, calcium glycerophosphate, calcium lactate, calcium lactobionate, propanoic acid, calcium levulinate, pentanoic acid, dibasic calcium phosphate, phosphoric acid, tribasic calcium phosphate, calcium hydroxide phosphate, potassium acetate, potassium chloride, potassium gluconate, potassium mixtures, dibasic potassium phosphate, monobasic potassium phosphate, potassium phosphate mixtures, sodium acetate, sodium bicarbonate, sodium chloride, sodium citrate, sodium lactate, dibasic sodium phosphate, monobasic sodium phosphate, sodium phosphate mixtures, tromethamine, amino-sulfonate buffers (e.g., HEPES), magnesium hydroxide, aluminum hydroxide, alginic acid, pyrogen-free water, isotonic saline, Ringer's solution, ethyl alcohol, and/or combinations thereof.

[0098] In some embodiments, the formulation including a LNP may further include a salt, such as a chloride salt. In some embodiments, the formulation including a LNP may further include a sugar such as a disaccharide. In some

embodiments, the formulation further includes a sugar but not a salt, such as a chloride salt. In some embodiments, a LNP may further include one or more small hydrophobic molecules such as a vitamin (e.g., vitamin A or vitamin E) or a sterol. Carbohydrates may include simple sugars (e.g., glucose) and polysaccharides (e.g., glycogen and derivatives and analogs thereof). The characteristics of a LNP may depend on the components thereof. For example, a LNP including cholesterol as a structural lipid may have different characteristics than a LNP that includes a different structural lipid. As used herein, the term "structural lipid" refers to sterols and also to lipids containing sterol moieties. As defined herein, "sterols" are a subgroup of steroids consisting of steroid alcohols. In some embodiments, the structural lipid is a steroid. In some embodiments, the structural lipid is cholesterol. In some embodiments, the structural lipid is an analog of cholesterol. In some embodiments, the structural lipid is alpha-tocopherol.

[0099] In some embodiments, the characteristics of a LNP may depend on the absolute or relative amounts of its components. For instance, a LNP including a higher molar fraction of a phospholipid may have different characteristics than a LNP including a lower molar fraction of a phospholipid. Characteristics may also vary depending on the method and conditions of preparation of the lipid nanoparticle. In general, phospholipids comprise a phospholipid moiety and one or more fatty acid moieties.

[0100] A phospholipid moiety can be selected, for example, from the non-limiting group consisting of phosphatidyl choline, phosphatidyl ethanolamine, phosphatidyl glycerol, phosphatidyl serine, phosphatidic acid, 2-lyso-phosphatidyl choline, and a sphingomyelin. A fatty acid moiety can be selected, for example, from the non-limiting group consisting of lauric acid, myristic acid, myristoleic acid, palmitic acid, palmitoleic acid, stearic acid, oleic acid, linoleic acid, alpha-linolenic acid, erucic acid, phytanoic acid, arachidic acid, arachidonic acid, eicosapentaenoic acid, behenic acid, docosapentaenoic acid, and docosahexaenoic acid. Particular phospholipids can facilitate fusion to a membrane. In some embodiments, a cationic phospholipid can interact with one or more negatively charged phospholipids of a membrane (e.g., a cellular or intracellular membrane). Fusion of a phospholipid to a membrane can allow one or more elements (e.g., a therapeutic agent) of a lipid-containing composition (e.g., LNPs) to pass through the membrane permitting, e.g., delivery of the one or more elements to a target tissue. Non-natural phospholipid species including natural species with modifications and substitutions including branching, oxidation, cyclization, and alkynes are also contemplated. In some embodiments, a phospholipid can be functionalized with or cross-linked to one or more alkynes (e.g., an alkenyl group in which one or more double bonds is replaced with a triple bond). Under appropriate reaction conditions, an alkyne group can undergo a copper-catalyzed cycloaddition upon exposure to an azide. Such reactions can be useful in functionalizing a lipid bilayer of a nanoparticle composition to facilitate membrane permeation or cellular recognition or in conjugating a nanoparticle composition to a useful component such as a targeting or imaging moiety (e.g., a dye). Phospholipids include, but are not limited to, glycerophospholipids such as phosphatidylcholines, phosphatidyl-ethanolamines, phosphatidylserines, phosphatidylinositols, phosphatidyl glycerols, and phosphatidic acids. Phospholip-

ids also include phosphosphingolipid, such as sphingomyelin. In some embodiments, a phospholipid useful or potentially useful in the present invention is an analog or variant of DSPC.

[0101] Lipid nanoparticles may be characterized by a variety of methods. For example, microscopy (e.g., transmission electron microscopy or scanning electron microscopy) may be used to examine the morphology and size distribution of a LNP. Dynamic light scattering or potentiometry (e.g., potentiometric titrations) may be used to measure zeta potentials. Dynamic light scattering may also be utilized to determine particle sizes. Instruments such as the Zetasizer Nano ZS (Malvern Instruments Ltd, Malvern, Worcestershire, UK) may also be used to measure multiple characteristics of a LNP, such as particle size, polydispersity index, and zeta potential.

[0102] The mean size of a LNP may be between 10s of nm and 100s of nm, e.g., measured by dynamic light scattering (DLS). For example, the mean size may be from about 40 nm to about 150 nm, such as about 40 nm, 45 nm, 50 nm, 55 nm, 60 nm, 65 nm, 70 nm, 75 nm, 80 nm, 85 nm, 90 nm, 95 nm, 100 nm, 105 nm, 110 nm, 115 nm, 120 nm, 125 nm, 130 nm, 135 nm, 140 nm, 145 nm, or 150 nm. In some embodiments, the mean size of a LNP may be from about 50 nm to about 100 nm, from about 50 nm to about 90 nm, from about 50 nm to about 80 nm, from about 50 nm to about 70 nm, from about 50 nm to about 60 nm, from about 60 nm to about 100 nm, from about 60 nm to about 90 nm, from about 60 nm to about 80 nm, from about 60 nm to about 70 nm, from about 70 nm to about 100 nm, from about 70 nm to about 90 nm, from about 70 nm to about 80 nm, from about 80 nm to about 100 nm, from about 80 nm to about 90 nm, or from about 90 nm to about 100 nm. In certain embodiments, the mean size of a LNP may be from about 70 nm to about 100 nm. In a particular embodiment, the mean size may be about 80 nm. In other embodiments, the mean size may be about 100 nm.

[0103] A LNP may be relatively homogenous. A polydispersity index may be used to indicate the homogeneity of a LNP, e.g., the particle size distribution of the lipid nanoparticles. A small (e.g., less than 0.3) polydispersity index generally indicates a narrow particle size distribution. A LNP may have a polydispersity index from about 0 to about 0.25, such as 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.10, 0.11, 0.12, 0.13, 0.14, 0.15, 0.16, 0.17, 0.18, 0.19, 0.20, 0.21, 0.22, 0.23, 0.24, or 0.25. In some embodiments, the polydispersity index of a LNP may be from about 0.10 to about 0.20.

[0104] The zeta potential of a LNP may be used to indicate the electrokinetic potential of the composition. For example, the zeta potential may describe the surface charge of a LNP. Lipid nanoparticles with relatively low charges, positive or negative, are generally desirable, as more highly charged species may interact undesirably with cells, tissues, and other elements in the body. In some embodiments, the zeta potential of a LNP may be from about -10 mV to about +20 mV, from about -10 mV to about +15 mV, from about -10 mV to about +10 mV, from about -10 mV to about +5 mV, from about -10 mV to about 0 mV, from about -10 mV to about +20 mV, from about -5 mV to about +15 mV, from about -5 mV to about +10 mV, from about -5 mV to about +5 mV, from about -5 mV to about 0 mV, from about 0 mV to about +20 mV, from about 0 mV to about +15 mV, from about 0 mV to about +10

mV, from about 0 mV to about +5 mV, from about +5 mV to about +20 mV, from about +5 mV to about +15 mV, or from about +5 mV to about +10 mV.

[0105] The efficiency of encapsulation of a therapeutic and/or prophylactic describes the amount of therapeutic and/or prophylactic that is encapsulated or otherwise associated with a LNP after preparation, relative to the initial amount provided. The encapsulation efficiency is desirably high (e.g., close to 100%). The encapsulation efficiency may be measured, for example, by comparing the amount of therapeutic and/or prophylactic in a solution containing the lipid nanoparticle before and after breaking up the lipid nanoparticle with one or more organic solvents or detergents. Fluorescence may be used to measure the amount of free therapeutic and/or prophylactic (e.g., RNA) in a solution. For the lipid nanoparticles described herein, the encapsulation efficiency of a therapeutic and/or prophylactic may be at least 50%, for example 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%. In some embodiments, the encapsulation efficiency may be at least 80%. In certain embodiments, the encapsulation efficiency may be at least 90%.

[0106] A LNP may optionally comprise one or more coatings. For example, a LNP may be formulated in a capsule, film, or tablet having a coating. A capsule, film, or tablet including a composition described herein may have any useful size, tensile strength, hardness, or density.

[0107] Formulations comprising amphiphilic polymers and lipid nanoparticles may be formulated in whole or in part as pharmaceutical compositions. Pharmaceutical compositions may include one or more amphiphilic polymers and one or more lipid nanoparticles. For example, a pharmaceutical composition may include one or more amphiphilic polymers and one or more lipid nanoparticles including one or more different therapeutics and/or prophylactics. Pharmaceutical compositions may further include one or more pharmaceutically acceptable excipients or accessory ingredients such as those described herein. General guidelines for the formulation and manufacture of pharmaceutical compositions and agents are available, for example, in Remington's The Science and Practice of Pharmacy, 21st Edition, A. R. Gennaro; Lippincott, Williams & Wilkins, Baltimore, MD, 2006. Conventional excipients and accessory ingredients may be used in any pharmaceutical composition, except insofar as any conventional excipient or accessory ingredient may be incompatible with one or more components of a LNP or the one or more amphiphilic polymers in the formulation of the disclosure. An excipient or accessory ingredient may be incompatible with a component of a LNP or the amphiphilic polymer of the formulation if its combination with the component or amphiphilic polymer may result in any undesirable biological effect or otherwise deleterious effect.

[0108] In some embodiments, one or more excipients or accessory ingredients may make up greater than 50% of the total mass or volume of a pharmaceutical composition including a LNP. For example, the one or more excipients or accessory ingredients may make up 50%, 60%, 70%, 80%, 90%, or more of a pharmaceutical convention. In some embodiments, a pharmaceutically acceptable excipient is at least 95%, at least 96%, at least 97%, at least 98%, at least 99%, or 100% pure. In some embodiments, an excipient is approved for use in humans and for veterinary use. In some

embodiments, an excipient is approved by United States Food and Drug Administration. In some embodiments, an excipient is pharmaceutical grade. In some embodiments, an excipient meets the standards of the United States Pharmacopoeia (USP), the European Pharmacopoeia (EP), the British Pharmacopoeia, and/or the International Pharmacopoeia. Relative amounts of the one or more amphiphilic polymers, the one or more lipid nanoparticles, the one or more pharmaceutically acceptable excipients, and/or any additional ingredients in a pharmaceutical composition in accordance with the present disclosure will vary, depending upon the identity, size, and/or condition of the subject treated and further depending upon the route by which the composition is to be administered. By way of example, a pharmaceutical composition may comprise between 0.1% and 100% (wt wt) of one or more lipid nanoparticles. As another example, a pharmaceutical composition may comprise between 0.1% and 15% (wt/vol) of one or more amphiphilic polymers (e.g., 0.5%, 1%, 2.5%, 5%, 10%, or 12.5% w/v). In certain embodiments, the lipid nanoparticles and/or pharmaceutical compositions of the disclosure are refrigerated or frozen for storage and/or shipment (e.g., being stored at a temperature of 4° C. or lower, such as a temperature between about -150° C. and about 0° C. or between about -80° C. and about -20° C. (e.g., about -5° C., -10° C., -15° C., -20° C., -25° C., -30° C., -40° C., -50° C., -60° C., -70° C., -80° C., -90° C., -130° C. or -150° C.)). For example, the pharmaceutical composition comprising one or more amphiphilic polymers and one or more lipid nanoparticles is a solution or solid (e.g., via lyophilization) that is refrigerated for storage and/or shipment at, for example, about -20° C., -30° C., -40° C., -50° C., -60° C., -70° C., or -80° C. In certain embodiments, the disclosure also relates to a method of increasing stability of the lipid nanoparticles by adding an effective amount of an amphiphilic polymer and by storing the lipid nanoparticles and/or pharmaceutical compositions thereof at a temperature of 4° C. or lower, such as a temperature between about -150° C. and about 0° C. or between about -80° C. and about -20° C., e.g., about -5° C., -10° C., -15° C., -20° C., -25° C., -30° C., -40° C., -50° C., -60° C., -70° C., -80° C., -90° C., -130° C. or -150° C.).

[0109] The chemical properties of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation of the present disclosure may be characterized by a variety of methods. In some embodiments, electrophoresis (e.g., capillary electrophoresis) or chromatography (e.g., reverse phase liquid chromatography) may be used to examine the mRNA integrity.

[0110] In some embodiments, the LNP integrity of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation of the present disclosure is about 20% or higher, about 25% or higher, about 30% or higher, about 35% or higher, about 40% or higher, about 45% or higher, about 50% or higher, about 55% or higher, about 60% or higher, about 65% or higher, about 70% or higher, about 75% or higher, about 80% or higher, about 85% or higher, about 90% or higher, about 95% or higher, about 96% or higher, about 97% or higher, about 98% or higher, or about 99% or higher.

[0111] In some embodiments, the LNP integrity of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation of the present disclosure is higher than the LNP integrity of the LNP, LNP suspension, lyophilized LNP

composition, or LNP formulation produced by a comparable method by about 5% or higher, about 10% or more, about 15% or more, about 20% or more, about 30% or more, about 40% or more, about 50% or more, about 60% or more, about 70% or more, about 80% or more, about 90% or more, about 1 folds or more, about 2 folds or more, about 3 folds or more, about 4 folds or more, about 5 folds or more, about 10 folds or more, about 20 folds or more, about 30 folds or more, about 40 folds or more, about 50 folds or more, about 100 folds or more, about 200 folds or more, about 300 folds or more, about 400 folds or more, about 500 folds or more, about 1000 folds or more, about 2000 folds or more, about 3000 folds or more, about 4000 folds or more, about 5000 folds or more, or about 10000 folds or more.

[0112] In some embodiments, the T₅₀ % of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation of the present disclosure is about 12 months or longer, about 15 months or longer, about 18 months or longer, about 21 months or longer, about 24 months or longer, about 27 months or longer, about 30 months or longer, about 33 months or longer, about 36 months or longer, about 48 months or longer, about 60 months or longer, about 72 months or longer, about 84 months or longer, about 96 months or longer, about 108 months or longer, about 120 months or longer.

[0113] In some embodiments, the T₅₀ % of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation of the present disclosure is longer than the T₅₀ % of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation produced by a comparable method by about 5% or higher, about 10% or more, about 15% or more, about 20% or more, about 30% or more, about 40% or more, about 50% or more, about 60% or more, about 70% or more, about 80% or more, about 90% or more, about 1 folds or more, about 2 folds or more, about 3 folds or more, about 4 folds or more, about 5 folds or more.

[0114] In some embodiments, the T_{1/2} of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation of the present disclosure is about 12 months or longer, about 15 months or longer, about 18 months or longer, about 21 months or longer, about 24 months or longer, about 27 months or longer, about 30 months or longer, about 33 months or longer, about 36 months or longer, about 48 months or longer, about 60 months or longer, about 72 months or longer, about 84 months or longer, about 96 months or longer, about 108 months or longer, about 120 months or longer.

[0115] In some embodiments, the T_{1/2} of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation of the present disclosure is longer than the T_{1/2} of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation produced by a comparable method by about 5% or higher, about 10% or more, about 15% or more, about 20% or more, about 30% or more, about 40% or more, about 50% or more, about 60% or more, about 70% or more, about 80% or more, about 90% or more, about 1 folds or more, about 2 folds or more, about 3 folds or more, about 4 folds or more, about 5 folds or more.

[0116] As used herein, “Tx” refers to the amount of time lasted for the nucleic acid integrity (e.g., mRNA integrity) of a LNP, LNP suspension, lyophilized LNP composition, or LNP formulation to degrade to about X of the initial integrity of the nucleic acid (e.g., mRNA) used for the preparation of the LNP, LNP suspension, lyophilized LNP

composition, or LNP formulation. For example, "T80 %" refers to the amount of time lasted for the nucleic acid integrity (e.g., mRNA integrity) of a LNP, LNP suspension, lyophilized LNP composition, or LNP formulation to degrade to about 80% of the initial integrity of the nucleic acid (e.g., mRNA) used for the preparation of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation. For another example, "T_{1/2}" refers to the amount of time lasted for the nucleic acid integrity (e.g., mRNA integrity) of a LNP, LNP suspension, lyophilized LNP composition, or LNP formulation to degrade to about 1/2 of the initial integrity of the nucleic acid (e.g., mRNA) used for the preparation of the LNP, LNP suspension, lyophilized LNP composition, or LNP formulation.

[0117] Lipid nanoparticles may include a lipid component and one or more additional components, such as a therapeutic and/or prophylactic, such as a nucleic acid. A LNP may be designed for one or more specific applications or targets. The elements of a LNP may be selected based on a particular application or target, and/or based on the efficacy, toxicity, expense, ease of use, availability, or other feature of one or more elements. Similarly, the particular formulation of a LNP may be selected for a particular application or target according to, for example, the efficacy and toxicity of particular combination of elements. The efficacy and tolerability of a LNP formulation may be affected by the stability of the formulation.

[0118] The lipid component of a LNP may include, for example, a cationic lipid, a phospholipid (such as an unsaturated lipid, e.g., DOPE or DSPC), a PEG lipid, and a structural lipid. The elements of the lipid component may be provided in specific fractions.

[0119] In some embodiments, the LNP further comprises a phospholipid, a PEG lipid, a structural lipid, or any combination thereof. Suitable phospholipids, PEG lipids, and structural lipids for the methods of the present disclosure are further disclosed herein. In some embodiments, the lipid component of a LNP includes a cationic lipid, a phospholipid, a PEG lipid, and a structural lipid. In certain embodiments, the lipid component of the lipid nanoparticle includes about 30 mol % to about 60 mol % cationic lipid, about 0 mol % to about 30 mol % phospholipid, about 18.5 mol % to about 48.5 mol % structural lipid, and about 0 mol % to about 10 mol % of PEG lipid, provided that the total mol % does not exceed 100%. In some embodiments, the lipid component of the lipid nanoparticle includes about 35 mol % to about 55 mol % compound of cationic lipid, about 5 mol % to about 25 mol % phospholipid, about 30 mol % to about 40 mol % structural lipid, and about 0 mol % to about 10 mol % of PEG lipid. In a particular embodiment, the lipid component includes about 50 mol % said cationic lipid, about 10 mol % phospholipid, about 38.5 mol % structural lipid, and about 1.5 mol % of PEG lipid. In another particular embodiment, the lipid component includes about 40 mol % said cationic lipid, about 20 mol % phospholipid, about 38.5 mol % structural lipid, and about 1.5 mol % of PEG lipid. In some embodiments, the phospholipid may be DOPE or DSPC. In other embodiments, the PEG lipid may be PEG-DMG and/or the structural lipid may be cholesterol.

[0120] The amount of a therapeutic and/or prophylactic in a LNP may depend on the size, composition, desired target and/or application, or other properties of the lipid nanoparticle as well as on the properties of the therapeutic and/or prophylactic. For example, the amount of an RNA useful in

a LNP may depend on the size, sequence, and other characteristics of the RNA. The relative amounts of a therapeutic and/or prophylactic (i.e. pharmaceutical substance) and other elements (e.g., lipids) in a LNP may also vary. In some embodiments, the wt/wt ratio of the lipid component to a therapeutic and/or prophylactic in a LNP may be from about 5:1 to about 60:1, such as 5:1, 6:1, 7:1, 8:1, 9:1, 10:1, 11:1, 12:1, 13:1, 14:1, 15:1, 16:1, 17:1, 18:1, 19:1, 20:1, 25:1, 30:1, 35:1, 40:1, 45:1, 50:1, and 60:1. For example, the wt/wt ratio of the lipid component to a therapeutic and/or prophylactic may be from about 10:1 to about 40:1.

[0121] In certain embodiments, the wt/wt ratio is about 20:1. The amount of a therapeutic and/or prophylactic in a LNP may, for example, be measured using absorption spectroscopy (e.g., ultraviolet-visible spectroscopy).

[0122] In some embodiments, the ionizable lipid is a compound of Formula (IL-1): or their N-oxides, or salts or isomers thereof, wherein:

[0123] Ri is selected from the group consisting of C5-30 alkyl, C5-20 alkenyl, —R*YR", —YR", and —R"M'R'; R2 and R3 are independently selected from the group consisting of H, C1-14 alkyl, C2-14 alkenyl, —R*YR", —YR", and —R*OR", or R2 and R3, together with the atom to which they are attached, form a heterocycle or carbocycle; R4 is selected from the group consisting of hydrogen, a C3-6 carbocycle, —(CH₂)_nQ, —(CH₂)_nCHQR, —CHQR, —CQ(R)₂, and unsubstituted C1-6 alkyl, where Q is selected from a carbocycle, heterocycle, —OR, —O(CH₂)_n(R)₂, —C(O)OR, —OC(O)R, —CX₃, —CX₂H, —CXH₂, —CN, —N(R)₂, —C(O)N(R)₂, —N(R)C(O)R, —N(R)S(O)₂R, —N(R)C(O)N(R)₂, —N(R)C(S)N(R)₂, —N(R)Re, N(R)S(O)₂R₈, —O(CH₂)_nOR, —N(R)C(=NR₉)N(R)₂, —N(R)C(=CHR₉)N(R)₂, —OC(O)N(R)₂J, —N(R)C(O)OR, —N(OR)C(O)R, —N(OR)S(O)₂R, —N(OR)C(O)OR, —N(OR)C(O)N(R)₂, —N(OR)C(S)N(R)₂, —N(OR)C(=NR₉)N(R)₂, —N(OR)C(=CHR₉)N(R)₂, —C(=NR₉)N(R)₂, —C(=NR₉)R, —C(O)N(R)OR, and —C(R)N(R)₂C(O)OR, and each n is independently selected from 1, 2, 3, 4, and 5; each R5 is independently selected from the group consisting of C1-3 alkyl, C2-3 alkenyl, and H; each Re is independently selected from the group consisting of C1-3 alkyl, C2-3 alkenyl, and H; M and M' are independently selected from —C(O)O—, —OC(O)—, —OOC(O)—M"—C(O)O—, —C(O)N(R')—, —N(R')C(O)—, —C(O)—, —C(S)—, —C(S)S—, —SC(S)—, —CH(OH)—, —P(O)(OR')O—, —S(O)₂—, —S—S—, an aryl group, and a heteroaryl group, in which M" is a bond, C1-13 alkyl or C2-13 alkenyl; R7 is selected from the group consisting of C1-3 alkyl, C2-3 alkenyl, and H; Re is selected from the group consisting of C3-6 carbocycle and heterocycle; R9 is selected from the group consisting of H, CN, NO₂, C1-6 alkyl, —OR, —S(O)₂R, —S(O)₂N(R)₂, C2-6 alkenyl, C3-6 carbocycle and heterocycle; each R is independently selected from the group consisting of C1-3 alkyl, C2-3 alkenyl, and H; each R' is independently selected from the group consisting of C1-3 alkyl, C2-3 alkenyl, —R*YR", —YR", and H; each R" is independently selected from the group consisting of C3-15 alkyl and C3-15 alkenyl; each R* is independently selected from the group consisting of C1-12 alkyl and C2-12 alkenyl; each Y is independently a C3-6 carbocycle; each X is independently selected from the group consisting of F, Cl, Br, and I; and m is selected from 5, 6, 7, 8, 9, 10, 11, 12, and 13; and wherein when R4 is —(CH₂)_nQ, —(CH₂)_nCHQR, —CHQR, or —CQ(R)₂, then (i) Q is

10-nucleotide linker sequence, and 70 other adenosine residues (bases 4175-4284). The sequence contains no uridine residues; they are replaced by 1-methyl-3'-pseudouridylyl. The 2P proline substitutions in the spike proteins were originally developed for a MERS vaccine by researchers at the National Institute of Allergy and Infectious Diseases' Vaccine Research Center, Scripps Research, and Jason McLellan's team (at the University of Texas at Austin, previously at Dartmouth College).

Composition of BNT162b2

[0128] In addition to the mRNA molecule, the vaccine contains the following inactive ingredients (excipients):

- [0129] ALC-0315, ((4-hydroxybutyl)azanediyl)bis(hexane-6,1-diyl)bis(2-hexyldecanoate)
- [0130] ALC-0159, 2-[(polyethylene glycol)-2000]-N,N-ditetradecylacetamide
- [0131] 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC)
- [0132] cholesterol
- [0133] dibasic sodium phosphate dihydrate
- [0134] monobasic potassium phosphate
- [0135] potassium chloride
- [0136] sodium chloride
- [0137] sucrose
- [0138] water for injection

[0139] The first four of these are lipids. The lipids are intended to encapsulate the mRNA in the form of a lipid nanoparticle to aid cell entry and stability of the RNA/lipid nanoparticles. ALC-0315 is the functional cationic lipid component of the drug product. When incorporated in lipid nanoparticles, it helps regulate the endosomal release of the RNA.

[0140] During drug product manufacturing, introduction of an aqueous RNA solution to an ethanolic lipid mixture containing ALC-0315 at a specific pH leads to an electrostatic interaction between the negatively charged RNA backbone and the positively charged cationic lipid. This electrostatic interaction leads to encapsulation of RNA drug substance resulting with particle formation.

[0141] The primary function of the PEGylated lipid ALC-0159 is to form a protective hydrophilic layer that sterically stabilises the lipid nanoparticle which contributes to storage stability and reduces non-specific binding to proteins.

[0142] Cholesterol is included in the formulation to support bilayer structures in the lipid nanoparticle and to provide mobility of the lipid components within the lipid nanoparticle structure.

[0143] DSPC is a phospholipid component intended to provide a stable bilayer-forming structure to balance the non-bilayer propensity of the cationic lipid. DSPC is a non-pharmacopeial excipient and an adequate specification has been provided. The lipids and modRNA together form nanoparticles.

[0144] In some embodiments, the BNT162b2 composition includes 30 mcg of the nucleoside-modified messenger RNA encoding a mutated viral spike (S) glycoprotein of SARS-CoV-2. The BNT162b2 composition for each dose includes the modRNA and the following: lipids (0.43 mg (4-hydroxybutyl)azanediyl)bis(hexane-6,1-diyl)bis(2-hexyldecanoate), 0.05 mg 2[(polyethylene glycol)-2000]-N,N-ditetradecylacetamide, 0.09 mg 1,2-distearoyl-sn-glycero-3-phosphocholine, and 0.2 mg cholesterol), 0.01 mg potassium chloride, 0.01 mg monobasic potassium phosphate, 0.36 mg sodium

chloride, 0.07 mg dibasic sodium phosphate dihydrate, and 6 mg sucrose. The diluent (0.9% Sodium Chloride Injection) contributes an additional 2.16 mg sodium chloride per dose. The BNT162b2 Vaccine can have a dosing regimen that includes two doses of 0.3 mL each, 3 weeks apart. The BNT162b2 vaccine includes mRNA having the sequence as set forth in SEQ ID NO: 1 (see FIG. 2).

[0145] The vaccine is supplied in a multidose vial as "a white to off-white, sterile, preservative-free, frozen suspension for intramuscular injection". It must be thawed to room temperature and diluted with normal saline before administration.

The Vaccine mRNA-1273

[0146] The mRNA-1273 vaccine composition includes 100 mcg of the nucleoside-modified messenger RNA encoding a mutated viral spike (S) glycoprotein of SARS-CoV-2 (see FIG. 3). The vaccine composition includes the following: lipids (SM-102; 1,2-dimyristoyl-rac-glycero-3-methoxypolyethylene glycol-2000 [PEG2000-DMG]; cholesterol; and 1,2-distearoyl-sn-glycero-3-phosphocholine [DSPC]), tromethamine, tromethamine hydrochloride, acetic acid, sodium acetate, and sucrose. The mRNA-1273 vaccine may have a dosing regimen that is two doses of 0.5 mL each, one month apart. The mRNA-1273 vaccine includes mRNA having the sequence shown in FIG. 3 (SEQ ID NO: 2). In another embodiment, the mRNA vaccine includes a sequence as shown in FIG. 3 (SEQ ID NO: 2), wherein the "spike encoding region" is replaced with a SARS-CoV-2 S-antigen of a variant strain.

2. Immunogenic Composition Against a Bacterium

2.1 Pneumococcal Conjugate Vaccines (PCVs) of the Invention

[0147] Pneumococcal conjugate vaccines of the present invention will typically comprise conjugated capsular saccharide antigens (also named herein conjugates or glycoconjugates), wherein the saccharides are derived from serotypes of *S. pneumoniae*. In a preferred embodiment, the saccharides are each individually conjugated to different molecules of the protein carrier (each molecule of protein carrier only having one type of saccharide conjugated to it). In said embodiment, the capsular saccharides are said to be individually conjugated to the carrier protein.

[0148] Preferably, the number of *S. pneumoniae* capsular saccharides can range from 13 serotypes (or "v", valences) to 20 different serotypes (25v). In one embodiment there are 12 different serotypes. In one embodiment there are 13 different serotypes. In one embodiment there are 14 different serotypes. In one embodiment there are 15 different serotypes. In one embodiment there are 16 different serotypes. In an embodiment there are 17 different serotypes. In an embodiment there are 18 different serotypes. In an embodiment there are 19 different serotypes. In an embodiment there are 20 different serotypes. The capsular saccharides are conjugated to a carrier protein to form glycoconjugates as described here below.

[0149] If the protein carrier is the same for 2 or more saccharides in the composition, the saccharides could be conjugated to the same molecule of the protein carrier (carrier molecules having 2 or more different saccharides conjugated to it) [see for instance WO 2004/083251].

[0150] In a preferred embodiment though, the saccharides are each individually conjugated to different molecules of

the protein carrier (each molecule of protein carrier only having one type of saccharide conjugated to it). In said embodiment, the capsular saccharides are said to be individually conjugated to the carrier protein.

[0151] For the purposes of the invention the term ‘glycoconjugate’ or ‘conjugate’ indicates a capsular saccharide linked covalently to a carrier protein. In one embodiment a capsular saccharide is linked directly to a carrier protein. In a second embodiment a bacterial saccharide is linked to a protein through a spacer/linker.

2.1.1 Capsular Saccharide of the Invention

[0152] The term “saccharide” throughout this specification may indicate polysaccharide or oligosaccharide and includes both. In frequent embodiments, the saccharide is a polysaccharide, in particular a *S. pneumoniae* capsular polysaccharide.

[0153] Capsular polysaccharides are prepared by standard techniques known to those of ordinary skill in the art.

[0154] Typically, capsular polysaccharides are produced by growing each *S. pneumoniae* serotype in a medium (e.g., in a soy-based medium), the polysaccharides are then prepared from the bacteria culture. Bacterial strains of *S. pneumoniae* used to make the respective polysaccharides that are used in the glycoconjugates of the invention may be obtained from established culture collections (such as for example the Streptococcal Reference Laboratory (Centers for Disease Control and Prevention, Atlanta, GA)) or clinical specimens.

[0155] The population of the organism (each *S. pneumoniae* serotype) is often scaled up from a seed vial to seed bottles and passaged through one or more seed fermentors of increasing volume until production scale fermentation volumes are reached. At the end of the growth cycle the cells are lysed and the lysate broth is then harvested for downstream (purification) processing (see for example WO 2006/110381, WO 2008/118752, and U.S. Patent App. Pub. Nos. 2006/0228380, 2006/0228381, 2008/0102498 and 2008/0286838).

[0156] The individual polysaccharides are typically purified through centrifugation, precipitation, ultra-filtration, and/or column chromatography (see for example WO 2006/110352 and WO 2008/118752).

[0157] Purified polysaccharides may be activated (e.g., chemically activated) to make them capable of reacting (e.g., either directly to the carrier protein or via a linker such as an eTEC spacer) and then incorporated into glycoconjugates of the invention, as further described herein.

[0158] *S. pneumoniae* capsular polysaccharides comprise repeating oligosaccharide units which may contain up to 8 sugar residues.

[0159] In an embodiment, capsular saccharide of the invention may be one oligosaccharide unit, or a shorter than native length saccharide chain of repeating oligosaccharide units. In an embodiment, capsular saccharide of the invention is one repeating oligosaccharide unit of the relevant serotype.

[0160] In an embodiment, capsular saccharide of the invention may be oligosaccharides. Oligosaccharides have a low number of repeat units (typically 5-15 repeat units) and are typically derived synthetically or by hydrolysis of polysaccharides.

[0161] In an embodiment, all of the capsular saccharides of the present invention and in the immunogenic composi-

tions of the present invention are polysaccharides. High molecular weight capsular polysaccharides are able to induce certain antibody immune responses due to the epitopes present on the antigenic surface. The isolation and purification of high molecular weight capsular polysaccharides is preferably contemplated for use in the conjugates, compositions and methods of the present invention.

[0162] In some embodiments, the purified polysaccharides before conjugation have a molecular weight of between 5 kDa and 4,000 kDa. In other such embodiments, the polysaccharide has a molecular weight of between 10 kDa and 4,000 kDa; between 50 kDa and 4,000 kDa; between 50 kDa and 3,000 kDa; between 50 kDa and 2,000 kDa; between 50 kDa and 1,500 kDa; between 50 kDa and 1,000 kDa; between 50 kDa and 750 kDa; between 50 kDa and 500 kDa; between 100 kDa and 4,000 kDa; between 100 kDa and 3,000 kDa; 100 kDa and 2,000 kDa; between 100 kDa and 1,500 kDa; between 100 kDa and 1,000 kDa; between 100 kDa and 750 kDa; between 100 kDa and 500 kDa; between 100 and 400 kDa; between 200 kDa and 4,000 kDa; between 200 kDa and 3,000 kDa; between 200 kDa and 2,000 kDa; between 200 kDa and 1,500 kDa; between 200 kDa and 1,000 kDa. In an embodiment, the capsular polysaccharide has a molecular weight of between or between 200 kDa and 500 kDa. In another embodiment, the capsular polysaccharide has a molecular weight of between 100 kDa to 500 kDa.

[0163] In further embodiments, the capsular polysaccharide has a molecular weight of between 5 kDa to 100 kDa; 7 kDa to 100 kDa; 10 kDa to 100 kDa; 20 kDa to 100 kDa; 30 kDa to 100 kDa; 40 kDa to 100 kDa; 50 kDa to 100 kDa; 60 kDa to 100 kDa; 70 kDa to 100 kDa; 80 kDa to 100 kDa; 90 kDa to 100 kDa; 5 kDa to 90 kDa; 5 kDa to 80 kDa; 5 kDa to 70 kDa; 5 kDa to 60 kDa; 5 kDa to 50 kDa; 5 kDa to 40 kDa; 5 kDa to 30 kDa; 5 kDa to 20 kDa or 5 kDa to 10 kDa. Any whole number integer within any of the above ranges is contemplated as an embodiment of the disclosure.

[0164] A polysaccharide can become slightly reduced in size during normal purification procedures. Additionally, polysaccharide can be subjected to sizing techniques before conjugation. Mechanical or chemical sizing maybe employed.

[0165] In a preferred embodiment the purified polysaccharides, are capsular polysaccharide from serotypes 11, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 22F, 23F or 33F of *S. pneumoniae*, wherein the capsular polysaccharide has a molecular weight falling within one of the molecular weight ranges as described here above.

[0166] As used herein, the term “molecular weight” of polysaccharide or of carrier protein-polysaccharide conjugate refers to the weight average molecular weight (Mw) which can be measured by size exclusion chromatography (SEC) combined with multiangle laser light scattering detector (MALLS).

[0167] In some embodiments, the pneumococcal saccharides from serotypes 9V, 18C, 11A, 15B, 22F and/or 33F of the invention are O-acetylated. In some embodiments, the pneumococcal saccharides from serotypes 9V, 11A, 15B, 22F and/or 33F of the invention are O-acetylated. In a preferred embodiment, the pneumococcal saccharide from serotype 18C of the invention is de-O-acetylated. For example, saccharides of serotype 18C can be de-O-acetylated by acidic treatment (see e.g. WO2006/110381, page 37 lines 1-4).

[0168] The degree of O-acetylation of the polysaccharide can be determined by any method known in the art, for example, by proton NMR (see for example Lemercinier et al. (1996) Carbohydrate Research 296:83-96, Jones et al. (2002) J. Pharmaceutical and Biomedical Analysis 30:1233-1247, WO 2005/033148 and WO 00/56357). Another commonly used method is described in Hestrin (1949) J. Biol. Chem. 180:249-261. Preferably, the presence of O-acetyl groups is determined by ion-HPLC analysis.

[0169] The purified polysaccharides described herein are chemically activated to make the saccharides capable of reacting with the carrier protein. These pneumococcal conjugates are prepared by separate processes and formulated into a single dosage formulation as described below.

2.1.2 Glycoconjugates of the Invention

[0170] The purified saccharides are chemically activated to make the saccharides capable of reacting with the carrier protein (i.e., activated saccharides), either directly or via a linker. Once activated, each capsular saccharide is separately conjugated to a carrier protein to form a glycoconjugate. In one embodiment, each capsular saccharide is conjugated to the same carrier protein. The chemical activation of the saccharides and subsequent conjugation to the carrier protein can be achieved by the activation and conjugation methods.

[0171] Capsular polysaccharides from *S. pneumoniae* can be prepared as disclosed above.

[0172] In a preferred embodiment, at least one of capsular polysaccharides from serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 22F, 23F and 33F of *S. pneumoniae* is conjugated to the carrier protein by reductive amination (such as described in U.S. Patent Appl. Pub. Nos. 2006/0228380, 2007/184072, 2007/0231340 and 2007/0184071, WO 2006/110381, WO 2008/079653, and WO 2008/143709).

[0173] In a preferred embodiment of the present invention, the glycoconjugate from *S. pneumoniae* serotype 15B is prepared by reductive amination.

[0174] In a preferred embodiment of the present invention, the glycoconjugate from *S. pneumoniae* serotype 18C is prepared by reductive amination. In a preferred embodiment of the present invention, the glycoconjugate from *S. pneumoniae* serotype 6A is prepared by reductive amination. In a preferred embodiment of the present invention, the glycoconjugate from *S. pneumoniae* serotype 19A is prepared by reductive amination. In a preferred embodiment of the present invention, the glycoconjugate from *S. pneumoniae* serotype 3 is prepared by reductive amination. In a preferred embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 6A and 19A are prepared by reductive amination. In a preferred embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 3, 6A and 19A are prepared by reductive amination. In a preferred embodiment of the present invention, the glycoconjugate from *S. pneumoniae* serotype 33F is prepared using the eTEC conjugation. In a preferred embodiment of the present invention, the glycoconjugate from *S. pneumoniae* serotype 12F is prepared using TEMPO/NCS-reductive amination.

[0175] In an embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 4, 6B, 9V, 14, 18C, 19F and 23F are prepared by reductive amination.

[0176] In an embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 4, 6B, 9V, 14, 18C, 19F and 23F are prepared by reductive amination. In a preferred embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 4, 5, 6B, 9V, 14, 18C, 19F and 23F are prepared by reductive amination. In an embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 4, 5, 6B, 7F, 9V, 14, 18C, 19F and 23F are prepared by reductive amination. In an embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19F and 23F are prepared by reductive amination. In an embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F are prepared by reductive amination. In a preferred embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F are all prepared by reductive amination. In a preferred embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 9V, 14 and 18C are prepared by reductive amination in aqueous solvent, the glycoconjugates from *S. pneumoniae* serotypes 6A, 6B, 7F, 19A, 19F and 23F are prepared by reductive amination in aprotic solvent. In a preferred embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 9V, 14 and 18C are prepared by reductive amination in aqueous solvent, the glycoconjugates from *S. pneumoniae* serotypes 6A, 6B, 7F, 19A, 19F and 23F are prepared by reductive amination in DMSO. In another preferred embodiment, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 22F and 23F are all prepared by reductive amination.

[0177] In another embodiment, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F and 33F are all prepared by reductive amination. In another embodiment, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 22F, 23F and 33F are all prepared by reductive amination. In a preferred embodiment when the vaccine is a 15-valent vaccine, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 9V, 14, 22F and 33F are prepared by reductive amination in aqueous solvent and the glycoconjugates from *S. pneumoniae* serotypes 6A, 6B, 7F, 18C, 19A, 19F and 23F are prepared by reductive amination in aprotic solvent. In a preferred embodiment when the vaccine is a 15-valent vaccine, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 9V, 14, 22F and 33F are prepared by reductive amination in aqueous solvent and the glycoconjugates from *S. pneumoniae* serotypes 6A, 6B, 7F, 18C, 19A, 19F and 23F are prepared by reductive amination in DMSO.

[0178] In another embodiment, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 15B, 18C, 19A, 19F, 22F and 23F are all prepared by reductive amination.

[0179] In another preferred embodiment, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 22F and 23F are all prepared by reductive amination.

[0180] In another embodiment, the glycoconjugates from *S. pneumoniae* serotypes 1, 4, 5, 6A, 6B, 7F, 9V, 12F, 14, 18C, 19A, 19F, 22F, 23F and 33F are all prepared by reductive amination.

[0181] In another preferred embodiment, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 14, 15B, 18C, 19A, 19F, 22F and 23F are all prepared by reductive amination.

[0182] In another embodiment, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 22F, 23F and 33F are all prepared by reductive amination.

[0183] In a preferred embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 14, 15B, 18C, 19A, 19F, 22F and 23F are prepared by reductive amination, the glycoconjugate from *S. pneumoniae* serotype 33F is prepared using the eTEC conjugation and the glycoconjugate from *S. pneumoniae* serotype 12F is prepared using TEMPO/NCS-reductive amination.

[0184] In a preferred embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 9V, 11A, 14 and 18C are prepared by reductive amination in aqueous solvent, the glycoconjugates from *S. pneumoniae* serotypes 6A, 6B, 7F, 8, 10A, 15B, 19A, 19F, 22F and 23F are prepared by reductive amination in aprotic solvent, the glycoconjugate from *S. pneumoniae* serotype 33F is prepared using the eTEC conjugation and the glycoconjugate from *S. pneumoniae* serotype 12F is prepared using TEMPO/NCS-reductive amination in aqueous solvent.

[0185] In a preferred embodiment of the present invention, the glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 9V, 11A, 14 and 18C are prepared by reductive amination in aqueous solvent, the glycoconjugates from *S. pneumoniae* serotypes 6A, 6B, 7F, 8, 10A, 15B, 19A, 19F, 22F and 23F are prepared by reductive amination in DMSO, the glycoconjugate from *S. pneumoniae* serotype 33F is prepared using the eTEC conjugation and the glycoconjugate from *S. pneumoniae* serotype 12F is prepared using TEMPO/NCS-reductive amination in aqueous solvent.

[0186] Reductive amination involves two steps, (1) oxidation of the polysaccharide, (2) reduction of the activated polysaccharide and a carrier protein to form a conjugate. Before oxidation, the polysaccharide is optionally hydrolyzed. Mechanical or chemical hydrolysis may be employed. Chemical hydrolysis may be conducted using acetic acid.

[0187] The oxidation step may involve reaction with periodate. For the purpose of the present invention, the term "periodate" includes both periodate and periodic acid; the term also includes both metaperiodate (IO_4^-) and orthoperiodate (IO_6^{5-}) and includes the various salts of periodate (e.g., sodium periodate and potassium periodate). In an embodiment the capsular polysaccharide is oxidized in the presence of metaperiodate, preferably in the presence of sodium periodate (NaIO_4). In another embodiment the capsular polysaccharide is oxidized in the presence of orthoperiodate, preferably in the presence of periodic acid.

[0188] In an embodiment, the oxidizing agent is a stable nitroxyl or nitroxide radical compound, such as piperidine-N-oxyl or pyrrolidine-N-oxyl compounds, in the presence of an oxidant to selectively oxidize primary hydroxyls (as described in WO 2014/097099). In said reaction, the actual oxidant is the N-oxoammonium salt, in a catalytic cycle. In an aspect, said stable nitroxyl or nitroxide radical compound are piperidine-N-oxyl or pyrrolidine-N-oxyl compounds. In an aspect, said stable nitroxyl or nitroxide radical compound bears a TEMPO (2,2,6,6-tetramethyl-1-piperidinyloxy) or a PROXYL (2,2,5,5-tetramethyl-1-pyrrolidinyloxy) moiety.

In an aspect, said stable nitroxyl radical compound is TEMPO or a derivative thereof. In an aspect, said oxidant is a molecule bearing a N-halo moiety. In an aspect, said oxidant is selected from the group consisting of N-Chlorosuccinimide, N-Bromosuccinimide, N-Iodosuccinimide, Dichloroisocyanuric acid, 1,3,5-trichloro-1,3,5-triazinane-2,4,6-trione, Dibromoisocyanuric acid, 1,3,5-tribromo-1,3,5-triazinane-2,4,6-trione, Diiodoisocyanuric acid and 1,3,5-triiodo-1,3,5-triazinane-2,4,6-trione. Preferably said oxidant is N-Chlorosuccinimide.

[0189] In a preferred embodiment, capsular polysaccharides from serotypes 12F *S. pneumoniae* are conjugated to the carrier protein by reductive amination, wherein the oxidizing agent is 2,2,6,6-Tetramethyl-1-piperidinyloxy (TEMPO) free radical and N-Chlorosuccinimide (NCS) as the cooxidant (as described in WO 2014/097099). Therefore in one aspect, the glycoconjugates from *S. pneumoniae* serotype 12F are obtainable by a method comprising the steps of: a) reacting a 12F saccharide with 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO) and N-chlorosuccinimide (NCS) in an aqueous solvent to produce an activated saccharide; and b) reacting the activated saccharide with a carrier protein comprising one or more amine groups (said method is designated "TEMPO/NCS-reductive amination").

[0190] Optionally the oxidation reaction is quenched by addition of a quenching agent. The quenching agent may be selected from vicinal diols, 1,2-aminoalcohols, amino acids, glutathione, sulfite, bisulfate, dithionite, metabisulfite, thiosulfate, phosphites, hypophosphites or phosphorous acid (such as glycerol, ethylene glycol, propan-1,2-diol, butan-1,2-diol or butan-2,3-diol, ascorbic acid).

[0191] Following the oxidation step of the polysaccharide, the polysaccharide is said to be activated and is referred to an "activated polysaccharide" here below. The activated polysaccharide and the carrier protein may be lyophilized (freeze-dried), either independently (discrete lyophilization) or together (co-lyophilized). In one embodiment the activated polysaccharide and the carrier protein are co-lyophilized. In another embodiment the activated polysaccharide and the carrier protein are lyophilized independently.

[0192] In one embodiment the lyophilization takes place in the presence of a non-reducing sugar, possible non-reducing sugars include sucrose, trehalose, raffinose, stachyose, melezitose, dextran, mannitol, lactitol and palatinin.

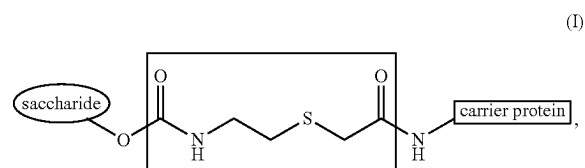
[0193] The second step of the conjugation process is the reduction of the activated polysaccharide and a carrier protein to form a conjugate (so-called reductive amination), using a reducing agent. Reducing agents which are suitable include the cyanoborohydrides (such as sodium cyanoborohydride, sodium triacetoxymethylborohydride or sodium or zinc borohydride in the presence of Bronsted or Lewis acids), amine boranes such as pyridine borane, 2-Picoline Borane, 2,6-diborane-methanol, dimethylamine-borane, t-BuMePrN— BH_3 , benzylamine- BH_3 or 5-ethyl-2-methylpyridine borane (PEMB) or borohydride exchange resin. In one embodiment the reducing agent is sodium cyanoborohydride.

[0194] In an embodiment, the reduction reaction is carried out in aqueous solvent (e.g., selected from PBS, MES, HEPES, Bis-tris, ADA, PIPES, MOPSO, BES, MOPS, DIPSO, MOBS, HEPPSO, POPSO, TEA, EPPS, Bicine or HEPB, at a pH between 6.0 and 8.5, 7.0 and 8.0, or 7.0 and 7.5), in another embodiment the reaction is carried out in

aprotic solvent. In an embodiment, the reduction reaction is carried out in DMSO (dimethylsulfoxide) or in DMF (dimethylformamide) solvent. The DMSO or DMF solvent may be used to reconstitute the activated polysaccharide and carrier protein which has been lyophilized. At the end of the reduction reaction, there may be unreacted aldehyde groups remaining in the conjugates, these may be capped using a suitable capping agent. In one embodiment this capping agent is sodium borohydride (NaBH_4). Following the conjugation (the reduction reaction and optionally the capping), the glycoconjugates may be purified (enriched with respect to the amount of polysaccharide-protein conjugate) by a variety of techniques known to the skilled person. These techniques include dialysis, concentration/diafiltration operations, tangential flow filtration precipitation/elution, column chromatography (DEAE or hydrophobic interaction chromatography), and depth filtration. In an embodiment, the glycoconjugates are purified by diafiltration or ion exchange chromatography or size exclusion chromatography.

[0195] In one embodiment the glycoconjugates are sterile filtered.

[0196] In an embodiment, the glycoconjugates of the invention are prepared using the eTEC conjugation, such as described in WO 2014/027302. Said glycoconjugates comprise a saccharide covalently conjugated to a carrier protein through one or more eTEC spacers, wherein the saccharide is covalently conjugated to the eTEC spacer through a carbamate linkage, and wherein the carrier protein is covalently conjugated to the eTEC spacer through an amide linkage. The eTEC linked glycoconjugates of the invention may be represented by the general formula (I):



where the atoms that comprise the eTEC spacer are contained in the central box.

[0197] The eTEC spacer includes seven linear atoms (i.e., $-\text{C}(\text{O})\text{NH}(\text{CH}_2)_2\text{SCH}_2\text{C}(\text{O})-$) and provides stable thioether and amide bonds between the saccharide and carrier protein. Synthesis of the eTEC linked glycoconjugate involves reaction of an activated hydroxyl group of the saccharide with the amino group of a thioalkylamine reagent, e.g., cystamine or cysteamine or a salt thereof, forming a carbamate linkage to the saccharide to provide a thiolated saccharide. Generation of one or more free sulfhydryl groups is accomplished by reaction with a reducing agent to provide an activated thiolated saccharide. Reaction of the free sulfhydryl groups of the activated thiolated saccharide with an activated carrier protein having one or more α -haloacetamide groups on amine containing residues generates a thioether bond to form the conjugate, wherein the carrier protein is attached to the eTEC spacer through an amide bond.

[0198] In said glycoconjugates of the invention, the saccharide may be a polysaccharide or an oligosaccharide. The carrier protein may be selected from any suitable carrier as

described herein or known to those of skill in the art. In frequent embodiments, the saccharide is a polysaccharide. In some such embodiments, the carrier protein is CRM₁₉₇. In some such embodiments, the eTEC linked glycoconjugate comprises a *S. pneumoniae* serotype 33F capsular polysaccharide.

[0199] In particularly preferred embodiments, the eTEC linked glycoconjugate comprises a pneumococcal serotype 33F (Pn33F) capsular polysaccharide, which is covalently conjugated to CRM₁₉₇ through an eTEC spacer (serotype 33F eTEC linked glycoconjugates).

[0200] In some embodiments, the glycoconjugate from *S. pneumoniae* serotypes 1, 7F, 9V and/or 18C of the invention are O-acetylated. In some embodiments, the glycoconjugate from *S. pneumoniae* serotypes 1, 7F and 9V is O-acetylated and the glycoconjugate from *S. pneumoniae* serotype 18C is de-O-acetylated.

[0201] In some embodiments, the glycoconjugates of the present invention comprise a saccharide having a molecular weight of between 5 kDa and 2,000 kDa. In other such embodiments, the saccharide has a molecular weight of between 50 kDa and 1,000 kDa. In other such embodiments, the saccharide has a molecular weight of between 70 kDa and 900 kDa. In other such embodiments, the saccharide has a molecular weight of between 100 kDa and 800 kDa. In other such embodiments, the saccharide has a molecular weight of between 200 kDa and 600 kDa. In other such embodiments, the saccharide has a molecular weight of between 100 kDa and 500 kDa. In other such embodiments, the saccharide has a molecular weight of between 100 kDa and 400 kDa. In other such embodiments, the saccharide has a molecular weight of between 150 kDa and 300 kDa. In further embodiments, the saccharide has a molecular weight of between 5 kDa to 100 kDa; 10 kDa to 100 kDa; 20 kDa to 100 kDa; 30 kDa to 100 kDa; 40 kDa to 100 kDa; 50 kDa to 100 kDa; 60 kDa to 100 kDa; 70 kDa to 100 kDa; 80 kDa to 100 kDa; 90 kDa to 100 kDa; 5 kDa to 90 kDa; 5 kDa to 80 kDa; 5 kDa to 70 kDa; 5 kDa to 60 kDa; 5 kDa to 50 kDa; 5 kDa to 40 kDa; 5 kDa to 30 kDa; 5 kDa to 20 kDa or 5 kDa to 10 kDa. Any whole number integer within any of the above ranges is contemplated as an embodiment of the disclosure. In some such embodiments, the glycoconjugate is prepared using reductive amination.

[0202] In some embodiments, the glycoconjugate of the invention has a molecular weight of between 100 kDa and 15,000 kDa. In some embodiments, the glycoconjugate of the invention has a molecular weight of between 500 kDa and 10,000 kDa. In some embodiments, the glycoconjugate of the invention has a molecular weight of between 2,000 kDa and 10,000 kDa. In some embodiments, the glycoconjugate of the invention has a molecular weight of between 3,000 kDa and 8,000 kDa. In some embodiments, the glycoconjugate of the invention has a molecular weight of between 3,000 kDa and 5,000 kDa. In other embodiments, the glycoconjugate has a molecular weight of between 500 kDa and 10,000 kDa. In other embodiments, glycoconjugate has a molecular weight of between 1,000 kDa and 8,000 kDa. In still other embodiments, the glycoconjugate has a molecular weight of between 2,000 kDa and 8,000 kDa or between 3,000 kDa and 7,000 kDa.

[0203] The molecular weight of the glycoconjugate is measured by SEC-MALLS. Any whole number integer within any of the above ranges is contemplated as an embodiment of the disclosure.

[0204] Another way to characterize the glycoconjugates of the invention is by the number of lysine residues in the carrier protein (e.g., CRM₁₉₇) that become conjugated to the saccharide which can be characterized as a range of conjugated lysines (degree of conjugation). The evidence for lysine modification of the carrier protein, due to covalent linkages to the polysaccharides, can be obtained by amino acid analysis using routine methods known to those of skill in the art. Conjugation results in a reduction in the number of lysine residues recovered, compared to the carrier protein starting material used to generate the conjugate materials. In a preferred embodiment, the degree of conjugation of the glycoconjugates of the invention is between 2 and 15. In an embodiment, the degree of conjugation of the glycoconjugates of the invention is between 2 and 13. In an embodiment, the degree of conjugation of the glycoconjugates of the invention is between 2 and 10. In an embodiment, the degree of conjugation of the glycoconjugates of the invention is between 2 and 8. In an embodiment, the degree of conjugation of the glycoconjugates of the invention is between 2 and 6. In an embodiment, the degree of conjugation of the glycoconjugates of the invention is between 3 and 10. In an embodiment, the degree of conjugation of the glycoconjugates of the invention is between 3 and 6. In an embodiment, the degree of conjugation of the glycoconjugates of the invention is between 5 and 10. In an embodiment, the degree of conjugation of the glycoconjugates of the invention is between 8 and 12. In an embodiment, the degree of conjugation of the glycoconjugate of the invention is about 2. In an embodiment, the degree of conjugation of the glycoconjugate of the invention is about 3. In an embodiment, the degree of conjugation of the glycoconjugate of the invention is about 4. In an embodiment, the degree of conjugation of the glycoconjugate of the invention is about 5.

[0205] In an embodiment, the degree of conjugation of the glycoconjugate of the invention is about 6. In an embodiment, the degree of conjugation of the glycoconjugate of the invention is about 8. In an embodiment, the degree of conjugation of the glycoconjugate of the invention is about 10. In an embodiment, the degree of conjugation of the glycoconjugate of the invention is about 12. In an embodiment, the degree of conjugation of the glycoconjugate of the invention is about 15. In a preferred embodiment, the degree of conjugation of the glycoconjugate of the invention is between 4 and 7. In some such embodiments, the carrier protein is CRM₁₉₇.

[0206] The glycoconjugates of the invention may also be characterized by the ratio (weight/weight) of saccharide to carrier protein. In some embodiments, the ratio of polysaccharide to carrier protein in the glycoconjugate (w/w) is between 0.5 and 3. In some embodiments, the ratio of polysaccharide to carrier protein in the glycoconjugate (w/w) is about 0.8. In some embodiments, the ratio of polysaccharide to carrier protein in the glycoconjugate (w/w) is about 0.9. In some embodiments, the ratio of polysaccharide to carrier protein in the glycoconjugate (w/w) is about 1.0. In some embodiments, the ratio of polysaccharide to carrier protein in the glycoconjugate (w/w) is about 1.2. In some embodiments, the ratio of polysaccharide to carrier protein in the glycoconjugate (w/w) is about 1.5. In some embodiments, the ratio of polysaccharide to carrier protein in the glycoconjugate (w/w) is about 1.8. In some embodiments, the ratio of

polysaccharide to carrier protein in the glycoconjugate (w/w) is about 2.0. In some embodiments, the ratio of polysaccharide to carrier protein in the glycoconjugate (w/w) is about 2.5. In some embodiments, the ratio of polysaccharide to carrier protein in the glycoconjugate (w/w) is about 3.0. In other embodiments, the saccharide to carrier protein ratio (w/w) is between 0.5 and 2.0. In other embodiments, the saccharide to carrier protein ratio (w/w) is between 0.5 and 1.5. In further embodiments, the saccharide to carrier protein ratio (w/w) is between 0.8 and 1.2. In a preferred embodiment, the ratio of capsular polysaccharide to carrier protein in the conjugate is between 0.9 and 1.1. In some such embodiments, the carrier protein is CRM₁₉₇.

[0207] The glycoconjugates and immunogenic compositions of the invention may contain free saccharide that is not covalently conjugated to the carrier protein, but is nevertheless present in the glycoconjugate composition. The free saccharide may be non-covalently associated with (i.e., non-covalently bound to, adsorbed to, or entrapped in or with) the glycoconjugate.

[0208] In a preferred embodiment, the glycoconjugate comprises less than about 50%, 45%, 40%, 35%, 30%, 25%, 20% or 15% of free polysaccharide compared to the total amount of polysaccharide. In a preferred embodiment the glycoconjugate comprises less than about 25% of free polysaccharide compared to the total amount of polysaccharide. In a preferred embodiment the glycoconjugate comprises less than about 20% of free polysaccharide compared to the total amount of polysaccharide. In a preferred embodiment the glycoconjugate comprises less than about 15% of free polysaccharide compared to the total amount of polysaccharide.

[0209] The glycoconjugates may also be characterized by their molecular size distribution (K_d). Size exclusion chromatography media (CL-4B) can be used to determine the relative molecular size distribution of the conjugate. Size Exclusion Chromatography (SEC) is used in gravity fed columns to profile the molecular size distribution of conjugates. Large molecules excluded from the pores in the media elute more quickly than small molecules. Fraction collectors are used to collect the column eluate. The fractions are tested colorimetrically by saccharide assay. For the determination of K_d , columns are calibrated to establish the fraction at which molecules are fully excluded (V_0), ($K_d=0$), and the fraction representing the maximum retention (V_i), ($K_d=1$). The fraction at which a specified sample attribute is reached (V_e), is related to K_d by the expression, $K_d=(V_e-V_0)/(V_i-V_0)$.

[0210] In a preferred embodiment, at least 30% of the glycoconjugate has a K_d below or equal to 0.3 in a CL-4B column. In a preferred embodiment, at least 40% of the glycoconjugate has a K_d below or equal to 0.3 in a CL-4B column. In a preferred embodiment, at least 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, or 85% of the glycoconjugate has a K_d below or equal to 0.3 in a CL-4B column. In a preferred embodiment, at least 60% of the glycoconjugate has a K_d below or equal to 0.3 in a CL-4B column. In a preferred embodiment, between 50% and 80% of the glycoconjugate has a K_d below or equal to 0.3 in a CL-4B column. In a preferred embodiment, between 65% and 80% of the glycoconjugate has a K_d below or equal to 0.3 in a CL-4B column.

[0211] The frequency of attachment of the saccharide chain to a lysine on the carrier protein is another parameter

for characterizing the glycoconjugates of the invention. For example, in some embodiments, at least one covalent linkage between the carrier protein and the polysaccharide occurs for every 4 saccharide repeat units of the polysaccharide. In another embodiment, the covalent linkage between the carrier protein and the polysaccharide occurs at least once in every 10 saccharide repeat units of the polysaccharide. In another embodiment, the covalent linkage between the carrier protein and the polysaccharide occurs at least once in every 15 saccharide repeat units of the polysaccharide. In a further embodiment, the covalent linkage between the carrier protein and the polysaccharide occurs at least once in every 25 saccharide repeat units of the polysaccharide.

[0212] In frequent embodiments, the carrier protein is CRM₁₉₇ and the covalent linkage via an eTEC spacer between the CRM₁₉₇ and the polysaccharide occurs at least once in every 4, 10, 15 or 25 saccharide repeat units of the polysaccharide.

[0213] In other embodiments, the conjugate comprises at least one covalent linkage between the carrier protein and saccharide for every 5 to 10 saccharide repeat units. In other embodiments, the conjugate comprises at least one covalent linkage between the carrier protein and saccharide every 2 to 7 saccharide repeat units. In other embodiments, the conjugate comprises at least one covalent linkage between the carrier protein and saccharide for every 7 to 12 saccharide repeat units. In other embodiments, the conjugate comprises at least one covalent linkage between the carrier protein and saccharide for every 10 to 15 saccharide repeat units. In other embodiments, the conjugate comprises at least one covalent linkage between the carrier protein and saccharide for every 4 to 8 saccharide repeat units. In other embodiments, the conjugate comprises at least one covalent linkage between the carrier protein and saccharide for every 10 to 20 saccharide repeat units. In other embodiments, the conjugate comprises at least one covalent linkage between the carrier protein and saccharide for every 2 to 25 saccharide repeat units. In frequent embodiments, the carrier protein is CRM₁₉₇.

[0214] In another embodiment, at least one linkage between carrier protein and saccharide occurs for every 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 or 25 saccharide repeat units of the polysaccharide. In an embodiment, the carrier protein is CRM₁₉₇. Any whole number integer within any of the above ranges is contemplated as an embodiment of the disclosure.

2.1.3 Carrier Protein of the Invention

[0215] A component of the conjugate of the invention is a carrier protein to which the pneumococcal saccharide is conjugated. The terms “protein carrier” or “carrier protein” or “carrier” may be used interchangeably herein. Carrier proteins should be amenable to standard conjugation procedures.

[0216] In a preferred embodiment, the carrier protein of the conjugates is selected in the group consisting of: DT (Diphtheria toxin), TT (tetanus toxoid) or fragment C of TT, CRM₁₉₇ (a nontoxic but antigenically identical variant of diphtheria toxin), other DT mutants (such as CRM176, CRM228, CRM45 (Uchida et al. (1973) J. Biol. Chem. 218:3838-3844), CRM9, CRM102, CRM103 or CRM107; and other mutations described by Nicholls and Youle in Genetically Engineered Toxins, Ed: Frankel, Maecel Dekker

Inc. (1992); deletion or mutation of Glu-148 to Asp, Gln or Ser and/or Ala 158 to Gly and other mutations disclosed in U.S. Pat. Nos. 4,709,017 and 4,950,740; mutation of at least one or more residues Lys 516, Lys 526, Phe 530 and/or Lys 534 and other mutations disclosed in U.S. Pat. Nos. 5,917, 017 and 6,455,673; or fragment disclosed in U.S. Pat. No. 5,843,711, pneumococcal pneumolysin (ply) (Kuo et al. (1995) Infect Immun 63:2706-2713) including ply detoxified in some fashion, for example dPLY-GMBS (WO 2004/081515, WO 2006/032499) or dPLY-formol, PhtX, including PhtA, PhtB, PhtD, PhtE (sequences of PhtA, PhtB, PhtD or PhtE are disclosed in WO 00/37105 and WO 00/39299) and fusions of Pht proteins, for example PhtDE fusions, PhtBE fusions, Pht A-E (WO 01/98334, WO 03/054007, WO 2009/000826), OMPC (meningococcal outer membrane protein), which is usually extracted from *Neisseria meningitidis* serogroup B (EP0372501), PorB (from *N. meningitidis*), PD (*Haemophilus influenzae* protein D; see, e.g., EP0594610 B), or immunologically functional equivalents thereof, synthetic peptides (EP0378881, EP0427347), heat shock proteins (WO 93/17712, WO 94/03208), pertussis proteins (WO 98/58668, EP0471177), cytokines, lymphokines, growth factors or hormones (WO 91/01146), artificial proteins comprising multiple human CD4+ T cell epitopes from various pathogen derived antigens (Falugi et al. (2001) Eur J Immunol 31:3816-3824) such as N19 protein (Baraldoi et al. (2004) Infect Immun 72:4884-4887) pneumococcal surface protein PspA (WO 02/091998), iron uptake proteins (WO 01/72337), toxin A or B of *Clostridium difficile* (WO 00/61761), transferrin binding proteins, pneumococcal adhesion protein (PsaA), recombinant *Pseudomonas aeruginosa* exotoxin A (in particular non-toxic mutants thereof (such as exotoxin A bearing a substitution at glutamic acid 553 (Douglas et al. (1987) J. Bacteriol. 169(11): 4967-4971)). Other proteins, such as ovalbumin, keyhole limpet hemocyanin (KLH), bovine serum albumin (BSA) or purified protein derivative of tuberculin (PPD) also can be used as carrier proteins. Other suitable carrier proteins include inactivated bacterial toxins such as cholera toxoid (e.g., as described in WO 2004/083251), *Escherichia coli* LT, *E. coli* ST, and exotoxin A from *P. aeruginosa*.

[0217] In a preferred embodiment, the carrier protein of the conjugates is independently selected from the group consisting of TT, DT, DT mutants (such as CRM₁₉₇), *H. influenzae* protein D, PhtX, PhtD, PhtDE fusions (particularly those described in WO 01/98334 and WO 03/054007), detoxified pneumolysin, PorB, N19 protein, PspA, OMPC, toxin A or B of *C. difficile* and PsaA.

[0218] In an embodiment, the carrier protein of the conjugates of the invention is DT (Diphtheria toxoid). In another embodiment, the carrier protein of the conjugates of the invention is TT (tetanus toxoid).

[0219] In another embodiment, the carrier protein of the conjugates of the invention is PD (*H. influenzae* protein D; see, e.g., EP0594610 B).

[0220] In a preferred embodiment, the pneumococcal capsular saccharides of the invention are conjugated to CRM₁₉₇ protein. The CRM₁₉₇ protein is a nontoxic form of diphtheria toxin but is immunologically indistinguishable from the diphtheria toxin. CRM₁₉₇ is produced by *Corynebacterium diphtheriae* infected by the nontoxigenic phage β 197^{tox-} created by nitrosoguanidine mutagenesis of the toxigenic corynebacter phage beta (Uchida et al. (1971) Nature New Biology 233:8-11). The CRM₁₉₇ protein has the same molecular

weight as the diphtheria toxin but differs therefrom by a single base change (guanine to adenine) in the structural gene. This single base change causes an amino acid substitution (glutamic acid for glycine) in the mature protein and eliminates the toxic properties of diphtheria toxin. The CRM₁₉₇ protein is a safe and effective T-cell dependent carrier for saccharides. Further details about CRM₁₉₇ and production thereof can be found, e.g., in U.S. Pat. No. 5,614,382. In a preferred embodiment, all the pneumococcal capsular saccharides of the invention are individually conjugated to CRM₁₉₇ protein.

[0221] In an embodiment, the pneumococcal capsular saccharides of the invention are conjugated to CRM₁₉₇ protein or the A chain of CRM₁₉₇ (see CN103495161). In an embodiment, the pneumococcal capsular saccharides of the invention are conjugated to the A chain of CRM₁₉₇ obtained via expression by genetically recombinant *E. coli* (see CN103495161). In an embodiment, the capsular saccharides of the invention are all conjugated to CRM₁₉₇. In an embodiment, the capsular saccharides of the invention are all conjugated to the A chain of CRM₁₉₇.

[0222] Accordingly, in frequent embodiments, the glycoconjugates of the invention comprise CRM₁₉₇ as the carrier protein, wherein the pneumococcal capsular polysaccharide is covalently linked to CRM₁₉₇.

2.1.4 Pneumococcal Conjugate Vaccines (PCV) of the Invention

[0223] In an embodiment, the number of different *S. pneumoniae* capsular saccharide serotypes of the pneumococcal conjugate vaccines can range from 13 serotypes (or “v”, valence) to 20 different serotypes (from 13v to 20v). In one embodiment the pneumococcal conjugate vaccine of the invention is a 13-valent pneumococcal vaccine. In one embodiment the pneumococcal conjugate vaccine of the invention is a 14-valent pneumococcal vaccine. In one embodiment the pneumococcal conjugate vaccine of the invention is a 15-valent pneumococcal vaccine. In one embodiment the pneumococcal conjugate vaccine of the invention is a 16-valent pneumococcal vaccine. In one embodiment the pneumococcal conjugate vaccine of the invention is a 17-valent pneumococcal vaccine. In one embodiment the pneumococcal conjugate vaccine of the invention is a 18-valent pneumococcal vaccine. In one embodiment the pneumococcal conjugate vaccine of the invention is a 19-valent pneumococcal vaccine. In one embodiment the pneumococcal conjugate vaccine of the invention is a 20-valent pneumococcal vaccine.

[0224] The capsular saccharides are conjugated to a carrier protein to form glycoconjugates as described here above.

[0225] Preferably, all the glycoconjugates of the above pneumococcal conjugate vaccines are individually conjugated to the carrier protein.

[0226] In an embodiment, the glycoconjugates from *S. pneumoniae* are all individually conjugated to CRM₁₉₇.

[0227] In one embodiment the pneumococcal conjugate vaccine of the invention comprises 13 glycoconjugates from a *Streptococcus pneumoniae* serotype selected from the group consisting of serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 23F, 22F and 33F. Preferably, said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0228] In an embodiment, the pneumococcal conjugate vaccine of the invention is a 13-valent pneumococcal con-

jugate vaccine wherein said 13 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F. Preferably, said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0229] In an embodiment, the pneumococcal conjugate vaccine of the invention is a 14-valent pneumococcal conjugate vaccine wherein said 14 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F. Preferably, said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0230] In an embodiment, the pneumococcal conjugate vaccine of the invention is a 14-valent pneumococcal conjugate vaccine wherein said 14 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F and 33F. Preferably, said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0231] In an embodiment, the pneumococcal conjugate vaccine of the invention is a 15-valent pneumococcal conjugate vaccine wherein said 15 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F, 22F and 33F. Preferably, said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0232] In an embodiment, the pneumococcal conjugate vaccine of the invention is a 16-valent pneumococcal conjugate vaccine wherein said 16 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 15B, 18C, 19A, 19F, 23F, 22F and 33F. Preferably, said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0233] In an embodiment, the pneumococcal conjugate vaccine of the invention is a 17-valent pneumococcal conjugate vaccine wherein said 17 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 11A, 14, 15B, 18C, 19A, 19F, 23F, 22F and 33F. Preferably, said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0234] In an embodiment, the pneumococcal conjugate vaccine of the invention is a 18-valent pneumococcal conjugate vaccine wherein said 18 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 11A, 12F, 14, 15B, 18C, 19A, 19F, 23F, 22F and 33F. Preferably, said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0235] In an embodiment, the pneumococcal conjugate vaccine of the invention is a 19-valent pneumococcal conjugate vaccine wherein said 19 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 11A, 12F, 14, 15B, 18C, 19A, 19F, 23F, 22F and 33F. Preferably, said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0236] In a preferred embodiment, the pneumococcal conjugate vaccine of the invention is a 20-valent pneumococcal conjugate vaccine wherein said 20 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 23F, 22F and 33F. Preferably, said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0237] In an embodiment, the pneumococcal conjugate vaccine of the invention is PREVNAR 13® (PREVENAR 13® in some countries). PREVNAR 13® is a 13-valent PCV where the 13 conjugates consist of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C,

19A, 19F and 23F all individually conjugated to CRM₁₉₇. The glycoconjugates are prepared by reductive amination.

[0238] In an embodiment, the pneumococcal conjugate vaccine of the invention is V114 developed by Merck. V114 is a 15-valent PCV where the 15 conjugates consist of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 22F, 23F and 33F all individually conjugated to CRM₁₉₇. The glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 9V, 14, 22F and 33F are prepared by reductive amination in aqueous solvent and the glycoconjugates from *S. pneumoniae* serotypes 6A, 6B, 7F, 18C, 19A, 19F and 23F are prepared by reductive amination in DMSO.

[0239] In an embodiment, the pneumococcal conjugate vaccine of the invention is 20vPnC. 20vPnC is a 20-valent PCV where the 20 conjugates consist of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 22F, 23F and 33F all individually conjugated to CRM₁₉₇.

2.2. Immunogenic Composition Including Antigen Derived from *Neisseria meningitidis*

[0240] *Neisseria meningitidis* is a Gram-negative encapsulated bacterium that can cause sepsis, meningitis, and death. *N. meningitidis* can be classified into at least 12 serogroups (including serogroups A, B, C, 29E, H, I, K, L, W-135 (mostly now referred to as W), X, Y and Z) based on chemically and antigenically distinctive polysaccharide capsules. Strains with five of the serogroups (A, B, C, Y, and W135) are responsible for the majority of disease.

[0241] Meningococcal meningitis is a devastating disease that can kill children and young adults within hours despite the availability of antibiotics. There is a need for improved immunogenic compositions against meningococcal serogroups A, B, C, Y, and W135 and/or X. Currently, a cross-protective vaccine or composition effective against a wide range of MnB and meningococcal serogroups A, C, Y, and W and/or X isolates is not yet commercially available. Accordingly, a cross-protective vaccine or composition effective against diverse MnB and meningococcal serogroups A, C, Y, and W and/or X isolates is needed.

[0242] In a preferred embodiment, the first immunogenic composition includes an polypeptide including the amino acid sequence set forth in SEQ ID NO: 3. In some embodiments, the polypeptide is lipidated. In some embodiments, the polypeptide is non-lipidated. In some embodiments, the polypeptide is immunogenic. In some embodiments the polypeptide includes the sequence set forth in SEQ ID NO: 3, wherein the cysteine at position 1 is deleted. In some embodiments, the polypeptide does not exhibit a mass shift of about +70 Da compared to the corresponding non-lipidated polypeptide as measured by mass spectrometry. In some embodiments, the first immunogenic composition further includes an adjuvant.

[0243] In some embodiments, the first immunogenic composition includes a) a first lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 3, and b) a second lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 4. In some embodiments, the composition further includes any one of polysorbate-80, aluminum, histidine, and sodium chloride, or a combination thereof.

[0244] In some embodiments, the first immunogenic composition includes: a) a first polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 3; (b) a second

polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 4; (c) a *Neisseria meningitidis* serogroup A capsular saccharide conjugated to an adipic acid dihydrazide (ADH) linker by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, wherein the linker is conjugated to tetanus toxoid by carbodiimide chemistry; (d) a *Neisseria meningitidis* serogroup C capsular saccharide conjugated to an ADH linker by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, wherein the linker is conjugated to tetanus toxoid by carbodiimide chemistry; (e) a *Neisseria meningitidis* serogroup W capsular saccharide directly conjugated to tetanus toxoid by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, in the absence of a linker; and (f) a *Neisseria meningitidis* serogroup Y capsular saccharide directly conjugated to tetanus toxoid by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, in the absence of a linker.

[0245] In some embodiments, the first immunogenic composition includes: a) a liquid composition comprising (i) a first lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 3; and (ii) a second lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 4 and aluminum; and b) a lyophilized composition comprising i) a *Neisseria meningitidis* serogroup A (MenA) capsular saccharide conjugated to an adipic acid dihydrazide (ADH) linker by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, wherein the linker is conjugated to tetanus toxoid (TT) by carbodiimide chemistry (MenAAH-TT conjugate); ii) a *Neisseria meningitidis* serogroup C (MenC) capsular saccharide conjugated to an ADH linker by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, wherein the linker is conjugated to tetanus toxoid (TT) by carbodiimide chemistry (MenCAH-TT conjugate); iii) a *Neisseria meningitidis* serogroup W135 (MenW) capsular saccharide directly conjugated to tetanus toxoid (TT) by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, in the absence of a linker (MenW-TT conjugate); and iv) a *Neisseria meningitidis* serogroup Y (MenY) capsular saccharide directly conjugated to tetanus toxoid (TT) by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, in the absence of a linker (MenY-TT conjugate) wherein the lyophilized composition is reconstituted with the liquid composition. In some embodiments, the immunogenic composition includes aluminum phosphate. In some embodiments, the immunogenic composition further includes polysorbate-80. In some embodiments, the lyophilized composition does not contain aluminum. In some embodiments, the first polypeptide and the second polypeptide are bound to the aluminum. In some embodiments, the immunogenic composition further includes any one of Tris-HCl; sodium chloride; sucrose; histidine; polysorbate 80; and aluminum phosphate. In some embodiments, the immunogenic composition further includes Tris-HCl; sodium chloride; sucrose; histidine; polysorbate 80; and aluminum phosphate.

2.3. Immunogenic Composition Including Antigen Derived from *C. difficile*

[0246] *Clostridium difficile*, now *Clostridioides difficile* (*C. difficile*) is a Gram-positive anaerobic bacterium that is associated with gastrointestinal disease in humans. Colonization of *C. difficile* usually occurs in the colon if the natural gut flora is diminished by treatment with antibiotics. An infection can lead to antibiotic-associated diarrhea and sometimes pseudomembranous colitis through the secretion

of the glucosylating toxins, toxin A and toxin B (approximately 308 and 270 kDa, respectively), which are the primary virulence factors of *C. difficile*.

[0247] In the last decade, the numbers and severity of *C. difficile* outbreaks in hospitals, nursing homes, and other long-term care facilities increased dramatically. Key factors in this escalation include emergence of hypervirulent pathogenic strains, increased use of antibiotics, improved detection methods, and increased exposure to airborne spores in health care facilities.

[0248] In some embodiments, the first immunogenic composition includes a polypeptide including the amino acid sequence set forth in SEQ ID NO: 5, wherein a side chain of a lysine residue of the polypeptide is crosslinked to a beta-alanine moiety. In some embodiments, the polypeptide further includes a glycine moiety crosslinked to a side chain of an aspartic acid residue of the polypeptide or to a side chain of a glutamic acid residue of the polypeptide. In some embodiments, the polypeptide further includes a crosslink between a second lysine residue of the polypeptide and a side chain of an aspartic acid residue of the polypeptide. In some embodiments, the polypeptide further includes a crosslink between a second lysine residue of the polypeptide and a side chain of a glutamic acid residue of the polypeptide. In some embodiments, the polypeptide has an EC50 of at least about 100 µg/ml, as measured by an in vitro cytotoxicity assay. In some embodiments, the polypeptide has an EC50 of at least about 1000 µg/ml, as measured by an in vitro cytotoxicity assay. In some embodiments, the first immunogenic composition further includes a second polypeptide having the amino acid sequence set forth in SEQ ID NO: 6, wherein a side chain of a lysine residue of the second polypeptide is crosslinked to a beta-alanine moiety. In some embodiments, the first polypeptide further includes a crosslink between a side chain of a second lysine residue of the first polypeptide and a side chain of an aspartic acid residue. In some embodiments, the second polypeptide further includes a crosslink between a side chain of a second lysine residue of the second polypeptide and a side chain of an aspartic acid residue. In some embodiments, the first polypeptide further includes a crosslink between a side chain of a second lysine residue of the first polypeptide and a side chain of a glutamic acid residue. In some embodiments, the second polypeptide further includes a crosslink between a side chain of a second lysine residue of the second polypeptide and a side chain of a glutamic acid residue. In some embodiments, the first polypeptide further includes a crosslink between a side chain of an aspartic acid residue of the first polypeptide and a glycine moiety. In some embodiments, the second polypeptide further includes a crosslink between a side chain of an aspartic acid residue of the second polypeptide and a glycine moiety. In some embodiments, the first polypeptide further includes a crosslink between a side chain of a glutamic acid residue of the first polypeptide and a glycine moiety. In some embodiments, the second polypeptide includes a crosslink between a side chain of a glutamic acid residue of the second polypeptide and a glycine moiety. In some embodiments, each polypeptide has an EC50 of at least about 1000 µg/ml, as measured by an in vitro cytotoxicity assay. In some embodiments, the first immunogenic composition further includes an adjuvant.

[0249] In some embodiments, the first immunogenic composition includes: (a) a first polypeptide, which includes the amino acid sequence set forth in SEQ ID NO: 5, wherein a

side chain of a lysine residue of the first polypeptide is crosslinked to a beta-alanine moiety, and wherein the first polypeptide further comprises a crosslink between a side chain of an aspartic acid residue of the first polypeptide and a glycine moiety, and a crosslink between a side chain of a glutamic acid residue of the first polypeptide and a glycine moiety; and (b) a second polypeptide, which includes the amino acid sequence set forth in SEQ ID NO: 6, wherein a side chain of a lysine residue of the second polypeptide is crosslinked to a beta-alanine moiety, and wherein the second polypeptide further comprises a crosslink between a side chain of an aspartic acid residue of the second polypeptide and a glycine moiety, and a crosslink between a side chain of a glutamic acid residue of the second polypeptide and a glycine moiety. In some embodiments, the first immunogenic composition includes: (a) a first polypeptide, which comprises the amino acid sequence set forth in SEQ ID NO: 5, wherein a side chain of a first lysine residue of the first polypeptide is crosslinked to a beta-alanine moiety, and wherein a second lysine residue of the first polypeptide is crosslinked to an aspartic acid residue or to a glutamic acid residue of the first polypeptide; and (b) a second polypeptide, which comprises the amino acid sequence set forth in SEQ ID NO: 6, wherein a side chain of a lysine residue of the second polypeptide is crosslinked to a beta-alanine moiety, and wherein a second lysine residue of the second polypeptide is crosslinked to an aspartic acid residue or to a glutamic acid residue of the second polypeptide. In some embodiments, the second polypeptide further comprises a dehydroalanine moiety. In some embodiments, the composition is lyophilized. In some embodiments, the composition further comprises aluminum hydroxide. In some embodiments, the composition is lyophilized. In some embodiments, the composition further comprises trehalose. In some embodiments, the composition further comprises polysorbate-80.

2.4. Immunogenic Composition Including Antigen Derived from RSV

[0250] Respiratory syncytial virus, or RSV, is a respiratory virus that infects the lungs and breathing passages. RSV is the leading cause of serious viral lower respiratory tract illness in infants worldwide and an important cause of respiratory illness in the elderly.

[0251] RSV is a member of the Paramyxoviridae family. Its genome consists of a single-stranded, negative-sense RNA molecule that encodes 11 proteins, including nine structural proteins (three glycoproteins and six internal proteins) and two non-structural proteins. The structural proteins include three transmembrane surface glycoproteins: the attachment protein G, fusion protein F, and the small hydrophobic SH protein. There are two subtypes of RSV, A and B. They differ primarily in the G glycoprotein, while the sequence of the F glycoprotein is more conserved between the two subtypes.

[0252] The mature F glycoprotein has three general domains: ectodomain (ED), transmembrane domain (TM), and a cytoplasmic tail (CT). CT contains a single palmitoylated cysteine residue.

[0253] The F glycoprotein of human RSV is initially translated from the mRNA as a single 574-amino acid polypeptide precursor (referred to "F0" or "F0 precursor"), which contains a signal peptide sequence (amino acids 1-25) at the N-terminus. Upon translation the signal peptide is removed by a signal peptidase in the endoplasmic reticulum.

The remaining portion of the F0 precursor (i.e., residues 26-574) may be further cleaved at two polybasic sites (a.a. 109/110 and 136/137) by cellular proteases (in particular furin), removing a 27-amino acid intervening sequence designated pep27 (amino acids 110-136) and generating two linked fragments designated F1 (C-terminal portion; amino acids 137-574) and F2 (N-terminal portion; amino acids 26-109). F1 contains a hydrophobic fusion peptide at its N-terminus and two heptad-repeat regions (HRA and HRB). HRA is near the fusion peptide, and HRB is near the TM domain. The F1 and F2 fragments are linked together through two disulfide bonds. Either the uncleaved F0 protein without the signal peptide sequence or a F1-F2 heterodimer can form a RSV F protomer. Three such protomers assemble to form the final RSV F protein complex, which is a homotrimer of the three protomers. The F proteins of subtypes A and B are about 90 percent identical in amino acid sequence. An example sequence of the F0 precursor polypeptide for the A subtype is provided in SEQ ID NO: 7 (A2 strain; GenBank GI: 138251; Swiss Prot P03420), and for the B subtype is provided in SEQ ID NO: 8 (18537 strain; GenBank GI: 138250; Swiss Prot P13843). SEQ ID NO: 7 and SEQ ID NO: 8 are both 574 amino acid sequences. One of the primary antigens explored for RSV subunit vaccines is the F protein. The RSV F protein trimer mediates fusion between the virion membrane and the host cellular membrane and also promotes the formation of syncytia. During RSV entry into cells, the F protein rearranges from the pre-fusion state (which may be referred to herein as "pre-F"), through an intermediate extended structure, to a post-fusion state ("post-F"). To prevent viral entry, F-specific neutralizing antibodies presumably must bind the pre-fusion conformation of F on the virion, or potentially the extended intermediate, before the viral envelope fuses with a cellular membrane. Thus, the pre-fusion form of the F protein is considered the preferred conformation as the desired vaccine antigen.

[0254] In some embodiments, the first immunogenic composition includes a mutant of a wildtype RSV F protein as described in U.S. Pat. Nos. 9,950,058, and 10,821,171, which are herein incorporated by reference.

[0255] In some embodiments, the first immunogenic composition includes a mutant of a wild-type RSV F protein, which mutant comprises a F1 polypeptide and a F2 polypeptide, wherein the mutant comprises at least one introduced amino acid mutation relative to the amino acid sequence of the wild-type RSV F protein, wherein the introduced amino acid mutation is a pair of cysteine mutations selected from the group consisting of: (1) 55C and 188C; (2) 103C and 148C; and (3) 142C and 371C; and (4) 155C and 290C, and wherein amino acid positions are numbered according to SEQ ID NO: 7. In one aspect of this embodiment, the mutant further comprises at least one cavity filling mutation, and at least one electrostatic mutation, wherein the cavity filling mutation is selected from the group consisting of:

- [0256]** (1) substitution of the amino acid at position 62, 155, 190, or 290 with I, Y, L, H, or M;
- [0257]** (2) substitution of the amino acid at position 54, 58, 189, 219, or 397 with I, Y, L, H, or M;
- [0258]** (3) substitution of the amino acid at position 151 with A or H;
- [0259]** (4) substitution of the amino acid at position 147 or 298 with I, L, H, or M; and

[0260] (5) substitution of the amino acid at position 164, 187, 192, 207, 220, 296, 300, or 495 with I, Y, or H, and wherein the electrostatic mutation is selected from the group consisting of:

- [0261]** (1) substitution of the amino acid at position 82, 92, or 487 by D, F, Q, T, S, L, or H;
- [0262]** (2) substitution of the amino acid at position 315, 394, or 399 by F, M, R, S, L, I, Q, or T;
- [0263]** (3) substitution of the amino acid at position 392, 486, or 489 by H, S, N, T, or P; and
- [0264]** (4) substitution of the amino acid at position 106 or 339 by F, Q, N, or W.

[0265] In another aspect of this embodiment, the C-terminus of the F1 polypeptide is linked to a trimerization domain. In another aspect of this embodiment, the F2 polypeptide comprises RSV F amino acid positions 26-109 and F1 polypeptide comprises RSV F amino acid positions 137-513. In another aspect of this embodiment, the wild-type RSV is subtype A, subtype B, strain A2, strain Ontario, or strain Buenos Aires. In another aspect of this embodiment, the pair of cysteine mutations is selected from the group consisting of: (1) 55C and 188C; and (2) 103C and 148C. In another aspect of this embodiment, the cavity filling mutation is selected from the group consisting of:

- [0266]** (1) substitution of the amino acid at position 190 with I, Y, or M;
- [0267]** (2) substitution of the amino acid at position 54 with I or H; and
- [0268]** (3) substitution of the amino acid at position 296 with I.

[0269] In another aspect of this embodiment, the electrostatic mutation is selected from the group consisting of:

- [0270]** (1) substitution of the amino acid at position 487 with D, Q, or H;
- [0271]** (2) substitution of the amino acid at position 489 with H, S, or N; and
- [0272]** (3) substitution of the amino acid at position 486 with H, S, or T.

[0273] In another aspect of this embodiment, wherein:

- [0274]** (i) the pair of cysteine mutations is selected from the group consisting of: (1) 55C and 188C and (2) 103C and 148C;
- [0275]** (ii) the cavity filling mutation is selected from the group consisting of:
- [0276]** (1) substitution of the amino acid at positions 190 with I, Y, or M;
- [0277]** (2) substitution of the amino acid at position 54 with I or H; and
- [0278]** (3) substitution of the amino acid at position 296 with I; and
- [0279]** (iii) the electrostatic mutation is selected from the group consisting of:
- [0280]** (1) substitution of the amino acid at position 487 with D, Q, or H;
- [0281]** (2) substitution of the amino acid at position 486 with H, S, or T; and
- [0282]** (3) substitution of the amino acid at position 489 with H, S, or N.

[0283] In another aspect of this embodiment, wherein the mutant comprises a combination of introduced amino acid mutations selected from the group consisting of:

- [0284]** (1) combination of 103C, 148C, 190I, and 486S;
- [0285]** (2) combination of 54H, 55C, 188C, and 486S;

- [0286] (3) combination of 54H, 103C, 148C, 190I, 296I, and 486S;
- [0287] (4) combination of 54H, 55C, 142C, 188C, 296I, and 371C;
- [0288] (5) combination of 55C, 188C, and 486S;
- [0289] (6) combination of 54H, 55C, 188C, and 190I;
- [0290] (7) combination of 55C, 188C, 190I, and 486S; and
- [0291] (8) combination of 54H, 55C, 188C, 190I, and 486S.
- [0292] In another aspect of this embodiment, wherein the mutant comprises a cysteine (C) at position 103 (103C) and at position 148 (148C), an isoleucine (I) at position 190 (190I), and a serine (S) at position 486 (486S).
- [0293] In another aspect, the mutant comprises a histidine (H) at position 54, a cysteine (C) at positions 103 and 148, an isoleucine (I) at positions 190, and 296, and a serine (S) at position 486.

[0294] In another aspect, wherein the mutant comprises a histidine (H) at position 54, a cysteine (C) at positions 55 and 188, and a serine (S) at position 486.

[0295] In another aspect, wherein the C-terminus of the F1 polypeptide of the mutant is linked to a trimerization domain.

[0296] In another aspect, wherein the trimerization domain is a phage T4 foldon domain.

[0297] In another embodiment, the first immunogenic composition includes a pharmaceutical composition, wherein the F2 polypeptide and F1 polypeptide of the mutant from the F protein of RSV subtype A comprise the amino acid sequence of SEQ ID NO:9 and the amino acid sequence of SEQ ID NO:10, respectively, and wherein the F2 polypeptide and F1 polypeptide of the second mutant from the F protein of RSV subtype B comprise the amino acid sequence of SEQ ID NO:11 and the amino acid sequence of SEQ ID NO:12, respectively. In one aspect, the C-terminus of the F1 polypeptide of the mutant is linked to a trimerization domain. In another aspect, the trimerization domain is a phage T4 foldon domain.

SEQ ID NO: 7. Amino Acid Sequence of the Full Length F0 of Native RSV A2 (GenBank GI: 138251; Swiss Prot P03420)

```
<212>      TYPE: PRT
<213>      ORGANISM: Human respiratory syncytial virus
<223>      OTHER INFORMATION: F protein of human RSV subtype A2
MELLILKANAITTILTAVTFCFASGQNITEEFYQSTCSAVSKGYLSALRTGWYTSVITIE
LSNIKENKCNKGTDAKVKLKQELDKYKNAVTELQLLMQSTPPTNNRRELPRFMNYTLNNAKK
TNVTLSKKRRRFLGFLLVGSAIASGVAISKVLHLEGEVNNIKSALLSTNKAVVSLNGVSVLT
SKVLDLKNYIDKQLLPVIVNKQSCSISNIETVIEFQQKNRLLLEITREFSVNAGVTPVSTYMLTNSE
LLSLINDMPIITNDQKKLMSSNVQIVRQQSYSIMSIIEEVLAYVVQLPLYGVIDTPCWKLHTSPLC
TTNTKEGSNICLTRDRGWYCDNAGSVSFFPQAETCKVQSNRVFCDTMNSLTLPSSEINLCNVDI
FNPKYDCKIMTSKTDVSSSVITSLGAIVSCYGKTKCTASNKNGRIIKTFNSGCDYVSNKGMDTV
VGNTLYYVKNQEGKSLYVKGEPIINFYDPLVFPSPDEFDASISQVNEKINQSLAFIRKSDLLHNVN
AGKSTTNIMITIIIVVILLS LIAVGLLLYCKARSTPVTLSKDQLSGINNIAPFSN
```

SEQ ID NO: 8. Amino Acid Sequence of the Full Length F0 of Native RSV B (18537 strain; GenBank GI: 138250; Swiss Prot P13843)

```
<212>      TYPE: PRT
<213>      ORGANISM: Human respiratory syncytial virus
<223>      OTHER INFORMATION: F protein of human RSV subtype B
MELLIHRSSAIFLTLAVNALYLTSSQNITEEFYQSTCSAVSRGYFSALRTGWYTSVITIE
LSNIKETKCNKGTDTKVLKQELDKYKNAVTELQLLMQNTPAANNRREARQPMNYTINTTKN
LNVISISKKRRRFLGFLLVGSAIASGIAVSKVLHLEGEVNNIKNALLSTNKAVVSLNGVSVLTS
KVLDLKNYINNRLLPVIVNQSCRSISNIETVIEFQQMNSRLLEITREFSVN
AGVTTPLSTYMLTNSELLSLINDMPIITNDQKKLMSSNVQIVRQQSYSIMSIIEEVLAYV
VQLPIYGVIDTPCWKLHTSPLCTTNIKEGSNICLTRDRGWYCDNAGSVSFFPQADTKVQSNR
VFCDTMNSLTLPSSEVSLCNTDIFNSKYDCKIMTSKTDISSSVITSLGAIVSCYGKTKCTASNKNR
GIKTFNSGCDYVSNKGVDTVSGNTLYYVKNLEGNLYVKGEPIINYDPLVFPSPDEFDASISQ
VNEKINQSLAFIRKSDLLHNVNTGKSTTNIMITIIIVVILLS
LIAIGLLLYCKAKNTFVTLTKDQLSGINNIAPFSK
```

SEQ	QNITEEFYQSTCSAV	SEQ ID	FLGFLLVGVSACASGIAVSKVLHLEGEVNNIKSALLS
ID NO:	SKGYLSALRTGWYT	NO: 10	TNKAVVSLNGVSVLT/KVLDLKNYIDKQLLPVIVNKQS
9	SVITIELSNIKCN		CSISNIETVIEFQQKNRLLLEITREFSVNAGVTPVST
	GTDAAVKLIKQELDK		YMLTNSELLSLINDMPIITNDQKKLMSSNVQIVRQQS
	YKNAVTELQLLMQST		YSIMSIIEEVLAYVVQLPLYGVIDTPCWKLHTSPLCT
	PACNSRRAR		TNTKEGSNICLTRDRGWYCDNAGSVSFFPQAETC
			KVQSNRVFCDTMNSLTLPSSEVSLCNTDIFNPKYDCKI
			MTSKTDVSSSVITSLGAIVSCYGKTKCTASNKNGRII
			KTFNSGCDYVSNKGVDTVSGNTLYYVKNQEGKSL
			YVKGEPIINFYDPLVFPSPDEFDASISQVNEKINQSLAF
			IRKSDLL

SEQ	QNITEEFYQSTCSAV	SEQ ID	FLGFLLVGVSACASGIAVSKVLHLEGEVNNIKNALL
ID NO:	SRGYFSALRTGWYT	NO: 12	STNKAVVSLNGVSVLT/KVLDLKNYINNQLLPVIVNQ
11	SVITIELSNIKETKCN		QSCRSISNIETVIEFQQKNRLLLEITREFSVNAGVTP
	GTDTKVLIKQELDK		LSTYMLTNSELLSLINDMPIITNDQKKLMSSNVQIVR

-continued

YKNAVTELQLLMQN	QQSYSIMSIIEEVLAYVVQLPIYGVIDTPCWKLHTS
TPACNNRARR	PLCTTNIKEGSNICLTRDRGWYCDNAGSVSFFPQ
	ADTCKVQSNRVFCDTMNSLTLPSEVSLCNTDIFNS
	KYDCKIMTSKTDISSSVITSLGAIVSCYGKTKCTASN
	KNRGIKTFSGCDYVSNKGVDTVSVGNTLYYVKNL
	EGKNLYVKGEPIINYDPLVFPSEFDASISQVNEKI
	NQSLAFIRRSDELL

2.5. Immunogenic Composition Including Antigen Derived from CMV

[0298] Human cytomegalovirus (HCMV) is a double stranded DNA virus of the β -herpesvirus family.

[0299] HCMV is the leading cause of congenital and neonatal hearing loss resulting from vertical virus transmission following infection or reactivation of latent virus in pregnant women. In addition, HCMV is a common opportunistic pathogen affecting immunosuppressed patients, such as solid organ and stem cell transplant patients, AIDS patients, etc.

[0300] The HCMV genome encodes several envelope glycoproteins, one of which is glycoprotein B (gB). Glycoprotein B is a fusogen that is required for virus entry into cells and an important target for neutralizing antibody (nAb) responses to infection. HCMV vaccines that incorporate gB subunit antigens have been under development. Clinical studies have shown that some gB subunit-based vaccine candidates are safe and immunogenic, though improvements in protective efficacy and durability of protection are desirable.

[0301] In some embodiments, the first immunogenic composition includes a polypeptide comprising any one of the following mutations,

-continued

1) Q98C and G271C	35) T746C and F747C
2) Q98C and I653C	36) K749C and N750C
3) G99C and A267C	37) M96C and D660C
4) T100C and A267C	38) Q98C and N658C
5) T100C and S269C	39) T100C and R258C
6) T100C and L651C	40) T100C and L656C
7) D217C and F584C	41) T100C and N658C
8) Y218C and A585C	42) I117C and T406C
9) S219C and D654C	43) I117C and S407C
10) N220C and D652C	44) Y153C and L712C
11) T221C and D652C	45) L162C and M716C
12) W240C and G718C	46) D217C and S587C
13) Y242C and K710C	47) D217C and Y589C
14) Y242C and D714C	48) S219C and F584C
15) S269C and I653C	49) S219C and A585C
16) G271C and P614C	50) S219C and N586C
17) S367C and L499C	51) N220C and T659C
18) T372C and W506C	52) S223C and T659C
19) F541C and Q669C	53) W240C and A732C
20) L548C and A650C	54) W240C and G735C
21) A549C and I653C	55) Y242C and V728C
22) S550C and D652C	56) Y242C and G731C
23) G604C and F661C	57) R258C and L656C
24) N605C and E665C	58) S269C and L656C
25) R607C and S675C	59) S269C and N658C
26) T608C and D679C	60) D272C and P614C
27) E609C and F678C	61) V273C and V629C
28) R673C and S674C	62) W349C and A650C
29) N676C and V677C	63) S367C and A500C
30) L680C and E681C	64) S367C and A503C
31) I683C and M684C	65) K370C and Q501C
32) F687C and N688C	66) K522C and I683C
33) Y690C and K691C	67) I523C and I683C
34) K695C and K724C	68) I523C and M684C
	69) N524C and M684C
	70) P525C and E681C
	71) R540C and L680C
	72) F541C and L680C
	73) L548C and P655C
	74) A549C and N658C
	75) S550C and P655C
	76) S550C and E657C
	77) Q591C and S668C
	78) L603C and Y667C
	79) G604C and L672C
	80) R607C and N688C
	81) T608C and Q692C
	82) E609C and K691C
	83) E610C and S674C
	84) E610C and S675C
	85) Q612C and V663C
	86) V737C and F755C
	87) V741C and A754C
	88) V741C and F755C
	89) D217C and Y589C
	90) I356C and A500C
	91) S367C and A500C
	92) S367C and A503C
	93) M371C and A505C
	94) M371C and W506C
	95) T374C and A503C
	96) Y160C and Y708C
	97) L162C and M716C
	98) N524C and M684C

-continued

99)	G99C and A267C
100)	T100C and R258C
101)	T221C and E657C
102)	S223C and T659C
103)	F541C and E681C
104)	L603C and Y667C
105)	N605C and K670C
106)	R607C and N688C
107)	E609C and K691C
108)	K670L
109)	K670F
110)	R673L
111)	R673F
112)	K691L
113)	K691F
114)	E686L
115)	R354F
116)	R573F
117)	D101L and K260L
118)	D679S
119)	D679N
120)	E682S
121)	E682Q
122)	E686S
123)	E686Q
124)	N118P
125)	D646P
126)	E686L
127)	E686I
128)	D679A
129)	R354F
130)	R573F
131)	D101L and K260L
132)	D217C and Y589C
133)	D217C, Y589C, D703C, and P704C
134)	D217C, Y589C, Y696C, and V697C
135)	D217C and Y589C
136)	D217C and Y589C
137)	D217C and Y589C
138)	D217C and Y589C
139)	D217C, M371C, W506C, and Y589C
140)	D217C, N524C, Y589C, and M684C
141)	D217C, N524C, Y589C, M684C, D703C, and P704C

or a combination thereof, as compared to SEQ ID NO: 13.

3. Dosage of the Pneumococcal Conjugate Vaccine

[0302] The amount of glycoconjugate(s) in each dose is selected as an amount which induces an immunoprotective response without significant, adverse side effects in typical vaccines. Such amount will vary depending upon which specific immunogen is employed and how it is presented.

[0303] The amount of a particular glycoconjugate in an immunogenic composition can be calculated based on total polysaccharide for that conjugate (conjugated and non-conjugated). For example, a glycoconjugate with 20% free polysaccharide will have about 80 µg of conjugated polysaccharide and about 20 µg of non-conjugated polysaccharide in a 100 µg polysaccharide dose. The amount of glycoconjugate can vary depending upon the streptococcal serotype. The saccharide concentration can be determined by the uronic acid assay.

[0304] The “immunogenic amount” of the different polysaccharide components in the immunogenic composition, may diverge and each may comprise about 1.0 µg, about 2.0 µg, about 3.0 µg, about 4.0 µg, about 5.0 µg, about 6.0 µg, about 7.0 µg, about 8.0 µg, about 9.0 µg, about 10.0 µg, about 15.0 µg, about 20.0 µg, about 30.0 µg, about 40.0 µg,

about 50.0 µg, about 60.0 µg, about 70.0 µg, about 80.0 µg, about 90.0 µg, or about 100.0 µg of any particular polysaccharide antigen.

[0305] Generally, each dose will comprise 0.1 µg to 100 µg of polysaccharide for a given serotype, particularly 0.5 µg to 20 µg, more particularly 1 µg to 10 µg, and even more particularly 2 µg to 5 µg. Any whole number integer within any of the above ranges is contemplated as an embodiment of the disclosure.

[0306] In an embodiment, each dose will comprise 1 µg, 2 µg, 3 µg, 4 µg, 5 µg, 6 µg, 7 µg, 8 µg, 9 µg, 10 µg, 15 µg or 20 µg of polysaccharide for a given serotype.

3.2 Carrier Amount

[0307] Generally, each dose will comprise 5 µg to 150 µg of carrier protein, particularly 10 µg to 100 µg of carrier protein, more particularly 15 µg to 100 µg of carrier protein, more particularly 25 to 75 µg of carrier protein, more particularly 30 µg to 70 µg of carrier protein, more particularly 30 to 60 µg of carrier protein, more particularly 30 µg to 50 µg of carrier protein and even more particularly 40 to 60 µg of carrier protein. In an embodiment, said carrier protein is CRM₁₉₇.

[0308] In an embodiment, each dose will comprise about 25 µg, about 26 µg, about 27 µg, about 28 µg, about 29 µg, about 30 µg, about 31 µg, about 32 µg, about 33 µg, about 34 µg, about 35 µg, about 36 µg, about 37 µg, about 38 µg, about 39 µg, about 40 µg, about 41 µg, about 42 µg, about 43 µg, about 44 µg, about 45 µg, about 46 µg, about 47 µg, about 48 µg, about 49 µg, about 50 µg, about 51 µg, about 52 µg, about 53 µg, about 54 µg, about 55 µg, about 56 µg, about 57 µg, about 58 µg, about 59 µg, about 60 µg, about 61 µg, about 62 µg, about 63 µg, about 64 µg, about 65 µg, about 66 µg, about 67 µg, about 68 µg, about 69 µg, about 70 µg, about 71 µg, about 72 µg, about 73 µg, about 74 µg or about 75 µg of carrier protein. In an embodiment, said carrier protein is CRM₁₉₇.

4. Adjuvant(s) of the Pneumococcal Conjugate Vaccine

[0309] In some embodiments, the pneumococcal conjugate vaccines disclosed herein may further comprise at least one, two or three adjuvants. The term “adjuvant” refers to a compound or mixture that enhances the immune response to an antigen. Antigens may act primarily as a delivery system, primarily as an immune modulator or have strong features of both. Suitable adjuvants include those suitable for use in mammals, including humans.

[0310] Examples of known suitable delivery-system type adjuvants that can be used in humans include, but are not limited to, alum (e.g., aluminum phosphate, aluminum sulfate or aluminum hydroxide), calcium phosphate, liposomes, oil-in-water emulsions such as MF59 (4.3% w/v squalene, 0.5% w/v polysorbate 80 (TWEEN® 80), 0.5% w/v sorbitan trioleate (Span 85)), water-in-oil emulsions such as MON-TANIDE™, and poly(D,L-lactide-co-glycolide) (PLG) microparticles or nanoparticles.

[0311] In an embodiment, the pneumococcal conjugate vaccines disclosed herein comprise aluminum salts (alum) as adjuvant (e.g., aluminum phosphate, aluminum sulfate or aluminum hydroxide). In a preferred embodiment, the pneumococcal conjugate vaccines disclosed herein comprise aluminum phosphate or aluminum hydroxide as adjuvant. In an embodiment, the pneumococcal conjugate vaccines dis-

closed herein comprise from 0.1 mg/mL to 1 mg/mL or from 0.2 mg/mL to 0.3 mg/mL of elemental aluminum in the form of aluminum phosphate. In an embodiment, the pneumococcal conjugate vaccines disclosed herein comprise about 0.25 mg/mL of elemental aluminum in the form of aluminum phosphate.

[0312] In an embodiment the pneumococcal conjugate vaccines of the invention comprises aluminum salt (alum) (e.g., aluminum phosphate, aluminum sulfate or aluminum hydroxide). In an embodiment, the pneumococcal conjugate vaccines of the invention comprise aluminum phosphate or aluminum hydroxide as adjuvant.

5. Formulation of the Pneumococcal Conjugate Vaccine

[0313] The pneumococcal conjugate vaccines of the invention may be formulated in liquid form (i.e., solutions or suspensions) or in a lyophilized form. Liquid formulations may advantageously be administered directly from their packaged form and are thus ideal for injection without the need for reconstitution in aqueous medium as otherwise required for lyophilized compositions of the invention.

[0314] In an embodiment, the pneumococcal conjugate vaccines of the invention is in liquid form, preferably in aqueous liquid form.

[0315] In an embodiment the pneumococcal conjugate vaccines of the invention comprises a buffer. In an embodiment, said buffer has a pKa of about 3.5 to about 7.5. In some embodiments, the buffer is phosphate, succinate, histidine or citrate. In certain embodiments, the buffer is succinate at a final concentration of 1 mM to 10 mM. In one particular embodiment, the final concentration of the succinate buffer is about 5 mM.

[0316] In an embodiment, the pneumococcal conjugate vaccines of the invention comprises a salt. In some embodiments, the salt is selected from the groups consisting of magnesium chloride, potassium chloride, sodium chloride and a combination thereof. In one particular embodiment, the salt is sodium chloride. In one particular embodiment, the pneumococcal conjugate vaccine of the invention comprises sodium chloride at 150 mM.

[0317] In an embodiment, the pneumococcal conjugate vaccines of the invention comprise a surfactant. In an embodiment, the surfactant is selected from the group consisting of polysorbate 20 (TWEENTM20), polysorbate 40 (TWEENTM40), polysorbate 60 (TWEENTM60), polysorbate 65 (TWEENTM65), polysorbate 80 (TWEENTM80), polysorbate 85 (TWEENTM85), TRITONTM N-1 01, TRITONTM X-100, octoxynol 40, nonoxynol-9, triethanolamine, triethanolamine polypeptide oleate, polyoxyethylene-660 hydroxystearate (PEG-15, Solutol H 15), polyoxyethylene-35-ricinoleate (CREMOPHOR[®]EL), soy lecithin and a poloxamer. In one particular embodiment, the surfactant is polysorbate 80. In some said embodiment, the final concentration of polysorbate 80 in the formulation is at least 0.0001% to 10% polysorbate 80 weight to weight (w/w). In some said embodiments, the final concentration of polysorbate 80 in the formulation is at least 0.001% to 1% polysorbate 80 weight to weight (w/w). In some said embodiments, the final concentration of polysorbate 80 in the formulation is at least 0.01% to 1% polysorbate 80 weight to weight (w/w). In other embodiments, the final concentration of polysorbate 80 in the formulation is 0.01%, 0.02%, 0.03%, 0.04%, 0.05%, 0.06%, 0.07%, 0.08%, 0.09% or 0.1% polysorbate 80 (w/w). In another embodiment, the final concentration of

the polysorbate 80 in the formulation is 0.02% polysorbate 80 (w/w). In another embodiment, the final concentration of the polysorbate 80 in the formulation is 0.01% polysorbate 80 (w/w). In another embodiment, the final concentration of the polysorbate 80 in the formulation is 0.03% polysorbate 80 (w/w). In another embodiment, the final concentration of the polysorbate 80 in the formulation is 0.04% polysorbate 80 (w/w). In another embodiment, the final concentration of the polysorbate 80 in the formulation is 0.05% polysorbate 80 (w/w). In another embodiment, the final concentration of the polysorbate 80 in the formulation is 1% polysorbate 80 (w/w). In one particular embodiment, the surfactant is polysorbate 20. In some said embodiment, the final concentration of polysorbate 20 in the formulation is at least 0.0001% to 10% polysorbate 20 weight to weight (w/w). In some said embodiments, the final concentration of polysorbate 20 in the formulation is at least 0.001% to 1% polysorbate 20 weight to weight (w/w). In some said embodiments, the final concentration of polysorbate 20 in the formulation is at least 0.01% to 1% polysorbate 20 weight to weight (w/w). In other embodiments, the final concentration of polysorbate 20 in the formulation is 0.01%, 0.02%, 0.03%, 0.04%, 0.05%, 0.06%, 0.07%, 0.08%, 0.09% or 0.1% polysorbate 20 (w/w). In another embodiment, the final concentration of the polysorbate 20 in the formulation is 0.02% polysorbate 20 (w/w). In another embodiment, the final concentration of the polysorbate 20 in the formulation is 0.01% polysorbate 20 (w/w). In another embodiment, the final concentration of the polysorbate 20 in the formulation is 0.03% polysorbate 20 (w/w). In another embodiment, the final concentration of the polysorbate 20 in the formulation is 0.04% polysorbate 80 (w/w). In another embodiment, the final concentration of the polysorbate 20 in the formulation is 0.05% polysorbate 20 (w/w). In another embodiment, the final concentration of the polysorbate 20 in the formulation is 1% polysorbate 20 (w/w).

[0318] In certain embodiments, the pneumococcal conjugate vaccine of the invention has a pH of 5.5 to 7.5, more preferably a pH of 5.6 to 7.0, even more preferably a pH of 5.8 to 6.0. A typical dose of the pneumococcal conjugate vaccines of the invention for injection has a volume of 0.1 mL to 2 mL, more preferably 0.2 mL to 1 mL, even more preferably a volume of about 0.5 mL.

6. Method for Eliciting an Immunoprotective Response

[0319] In an embodiment the invention relates to a method for eliciting an immunoprotective response in a human against an infectious bacterial antigen (e.g., selected from any one of *S. pneumoniae*, *N. meningitidis*, *C. difficile*, and *E. coli*), and betacoronavirus (e.g., Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2)), the method comprising co-administering (e.g. concomitantly or concurrently) to the human an effective dose of a first immunogenic composition including an antigen selected from any one of a polypeptide, toxoid, polysaccharide, and polysaccharide conjugate derived from the infectious disease-causing bacterium and an mRNA vaccine against betacoronavirus. In an embodiment said first immunogenic composition against the infectious disease-causing bacterium and said mRNA vaccine against betacoronavirus are any of the compositions disclosed herein.

[0320] In an embodiment the invention relates to a method for eliciting an immunoprotective response in a human against *S. pneumoniae* and Severe Acute Respiratory Syn-

drome Coronavirus 2 (SARS-CoV-2), the method comprising co-administering (e.g. concomitantly or concurrently) to the human an effective dose of a pneumococcal conjugate vaccine (PCV) and an mRNA vaccine against SARS-CoV-2. In an embodiment said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are any of the vaccine disclosed herein.

[0321] In a preferred embodiment an immunoprotective response against *S. pneumoniae* can be measured by any method known in the art, such as IgG level, fold rise in IgG level from before to after vaccination, OPA titers and/or fold rise in OPA titers from before to after vaccination (e.g. at least a 4-fold rise in OPA titers).

[0322] For each serotype of *S. pneumoniae* the level of IgG antibodies which are capable of binding *S. pneumoniae* polysaccharide can be determined by ELISA assay.

[0323] In the ELISA (Enzyme-linked Immunosorbent Assay) method, antibodies from the sera of vaccinated subjects are incubated with polysaccharides which have been adsorbed to a solid support. The bound antibodies are detected using enzyme-conjugated secondary detection antibodies. In an embodiment said ELISA assay is the standardized ELISA assay as defined by the WHO in the “Training Manual For Enzyme Linked Immunosorbent Assay For The Quantitation Of *Streptococcus Pneumoniae* Serotype Specific IgG (Pn PS ELISA). (007sp Version)” (available for example at [https://www.vaccine.uab.edu/uploads/mdocs/ELISAProtocol\(007sp\).pdf](https://www.vaccine.uab.edu/uploads/mdocs/ELISAProtocol(007sp).pdf), accessed on May 3, 2021).

[0324] The ELISA measures type specific IgG anti-*S. pneumoniae* capsular polysaccharide (PS) antibodies present in human serum. When dilutions of human sera are added to type-specific capsular PS-coated microtiter plates, antibodies specific for that capsular PS bind to the microtiter plates. The antibodies bound to the plates are detected using a goat anti-human IgG alkaline phosphatase-labeled antibody followed by a p-nitrophenyl phosphate substrate. The optical density of the colored end product is proportional to the amount of anticapsular PS antibody present in the serum.

[0325] In an embodiment an immunoprotective response against *S. pneumoniae* can be measured by IgG level as determined by ELISA assay (such as the standardized ELISA assay as defined by the WHO), where the subject achieves a pre-specified level of pneumococcal IgG concentrations after vaccination for a given serotype. In an embodiment, said level is measured about 1 month after vaccination. In an embodiment the pre-specified levels of IgG concentrations after vaccination are as follows: for serotype 1, 3, 4, 6A, 7F, 9V, 14, 18C, 19F, 23F, 8, 10A, 11A, 12F, 15B, 22F, 33F: at least 0.35 microgram per milliliter, for serotype 5: at least 0.23 microgram per milliliter, for serotype 6B: at least 0.10 microgram per milliliter and for serotype 19A: at least 0.12 microgram per milliliter.

[0326] In a preferred embodiment an immunoprotective response against *S. pneumoniae* can be measured by pneumococcal OPA titers or fold rise in OPA titers from before to after vaccination (e.g. 1 month after vaccination). In such an embodiment, an immunoprotective response against *S. pneumoniae* can be measured by a at least 4-fold rise in OPA titers from before to after vaccination (e.g. 1 month after vaccination).

[0327] The pneumococcal opsonophagocytic assay (OPA), which measures killing of *S. pneumoniae* cells by phagocytic effector cells in the presence of functional anti-

body and complement, is considered to be an important surrogate for evaluating the effectiveness of pneumococcal vaccines.

[0328] In vitro opsonophagocytic assay (OPA) can be conducted by incubating together a mixture of *Streptococcus pneumoniae* cells, a heat inactivated human serum to be tested, differentiated HL-60 cells (phagocytes) and an exogenous complement source (e.g., baby rabbit complement). Opsonophagocytosis proceeds during incubation and bacterial cells that are coated with antibody and complement are killed upon opsonophagocytosis. Colony forming units (cfu) of surviving bacteria that escape from opsonophagocytosis are determined by plating the assay mixture. The OPA titer is defined as the reciprocal dilution that results in a 50% reduction in bacterial count over control wells without test serum. The OPA titer is interpolated from the two dilutions that encompass this 50% killing cut-off.

[0329] An endpoint titer of 1:8 or greater is considered a positive result in these killing type OPA. Therefore, in an embodiment an immunoprotective response against a *S. pneumoniae* serotype can be measured by pneumococcal OPA titer where a result is considered positive when an endpoint titer of 1:8 or greater is measured.

[0330] In some embodiment, the human subjects may have serotype specific OPA titers prior to pneumococcal vaccination due for example to natural exposures to *S. pneumoniae* (e.g., in case of adult subjects). Therefore, comparison of OPA activity of pre- and post-immunization serum with the pneumococcal conjugate vaccine of the invention can be conducted by comparing the potential increase in OPA titers.

[0331] In an embodiment an immunoprotective response against a *S. pneumoniae* serotype can be measured by fold rise in OPA titers from before to after vaccination (e.g. 1 month after vaccination) where a at least 4-fold rise in OPA titers from before to after vaccination is considered positive.

[0332] In a preferred embodiment an immunoprotective response against SARS-CoV-2 can be measured by any method known in the art, such as vaccine-induced antibody response concentrations of S-binding IgG and/or SARS-CoV-2-neutralizing titres.

[0333] In a preferred embodiment an immunoprotective response against SARS-CoV-2 can be measured by full-length S-binding IgG levels (antigen-specific antibodies) and/or by the neutralizing antibody titer produced. In a preferred embodiment an immunoprotective response against SARS-CoV-2 can be measured by full-length S-binding IgG levels. In another preferred embodiment an immunoprotective response against SARS-CoV-2 can be measured by full by the neutralizing antibody titer produced.

[0334] In some embodiments the neutralizing antibody titer is greater than a protein vaccine. In other embodiments the neutralizing antibody titer produced by the mRNA vaccines of the invention is greater than an adjuvanted protein vaccine. In yet other embodiments the neutralizing antibody titer produced by the mRNA vaccines of the invention is 1,000-10,000, 1,200-10,000, 1,400-10,000, 1,500-10,000, 1,000-5,000, 1,000-4,000, 1,800-10,000, 2,000-10,000, 2,000-5,000, 2,000-3,000, 2,000-4,000, 3,000-5,000, 3,000-4,000, or 2,000-2,500. A neutralization titer is typically expressed as the highest serum dilution required to achieve a 50% reduction in the number of plaques. For example, in a clinical trial conducted in 2020, described in *N Engl J Med.* 2020 Dec. 31; 383(27):2603-2615, the 50%

neutralizing geometric mean titers elicited by 30 µg of BNT162b2 in older and younger adults exceeded the geometric mean titer measured in a human convalescent serum panel.

[0335] In exemplary embodiments, the antibody titer (i.e., the amount of antigen-specific antibody (S-binding) produces in a subject) is expressed as the inverse of the greatest dilution (in a serial dilution) that still gives a positive result. In exemplary embodiments, antibody titer is determined or measured by enzyme-linked immunosorbent assay (ELISA). In exemplary embodiments, antibody titer is determined or measured by neutralization assay, e.g., by microneutralization assay. In certain aspects, antibody titer measurement is expressed as a ratio, such as 1:40, 1:100, etc. In exemplary embodiments of the invention, an efficacious vaccine produces an antibody titer of greater than 1:40, greater than 1:100, greater than 1:400, greater than 1:1000, greater than 1:2000, greater than 1:3000, greater than 1:4000, greater than 1:500, greater than 1:6000, greater than 1:7500, greater than 1:10000.

[0336] In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:40. In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:100. In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:400. In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:1000. In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:2000. In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:3000. In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:4000. In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:500. In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:6000. In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:7500. In an embodiment of the invention, an efficacious vaccine produces an antibody titer of greater than 1:10000. In exemplary embodiments, the antibody titer is produced or reached by 10 days following vaccination, by 20 days following vaccination, by 30 days following vaccination, by 40 days following vaccination, or by 50 or more days following vaccination. In an embodiment of the invention, the antibody titer is produced or reached by 10 days following vaccination. In an embodiment of the invention, the antibody titer is produced or reached by 20 days following vaccination. In an embodiment of the invention, the antibody titer is produced or reached by 30 days following vaccination. In an embodiment of the invention, the antibody titer is produced or reached by 40 days following vaccination. In an embodiment of the invention, the antibody titer is produced or reached by 50 or more days following vaccination. In an embodiment of the invention, the antibody titer is produced or reached by 21 to 35 days following vaccination, in exemplary embodiments, the titer is produced or reached following a single dose of vaccine administered to the subject. In other embodiments, the titer is produced or reached following multiple doses, e.g., following a first and a second dose (e.g., a booster dose.). In exemplary embodiments, the titer is produced or reached following three doses of vaccine administered to the subject. In exemplary aspects

of the invention, antigen-specific antibodies are measured in units of µg/ml or are measured in units of IU/L (International Units per liter) or mIU/ml (milli International Units per ml). In exemplary embodiments of the invention, an efficacious vaccine produces >0.05 µg/ml, >0.1 µg/ml, >0.2 µg/ml, >0.35 µg/ml, >0.5 µg/ml, >1 µg/ml, >2 µg/ml, >5 µg/ml or >10 µg/ml. In an embodiment of the invention, an efficacious vaccine produces >0.05 µg/ml of antigen-specific antibodies. In an embodiment of the invention, an efficacious vaccine produces >0.1 µg/ml of antigen-specific antibodies. In an embodiment of the invention, an efficacious vaccine produces >0.2 µg/ml of antigen-specific antibodies. In an embodiment of the invention, an efficacious vaccine produces >0.35 µg/ml of antigen-specific antibodies. In an embodiment of the invention, an efficacious vaccine produces >0.5 µg/ml of antigen-specific antibodies. In an embodiment of the invention, an efficacious vaccine produces >1 µg/ml of antigen-specific antibodies. In an embodiment of the invention, an efficacious vaccine produces >2 µg/ml of antigen-specific antibodies.

[0337] In an embodiment of the invention, an efficacious vaccine produces >5 µg/ml of antigen-specific antibodies. In an embodiment of the invention, an efficacious vaccine produces >10 µg/ml of antigen-specific antibodies.

[0338] In exemplary embodiments of the invention, an efficacious vaccine produces >10 mIU/ml, >20 mIU/ml, >50 mIU/ml, >100 mIU/ml, >200 mIU/ml, >500 mIU/ml or >1000 mIU/ml. In an embodiment of the invention, an efficacious vaccine produces >10 mIU/ml. In an embodiment of the invention, an efficacious vaccine produces >20 mIU/ml. In an embodiment of the invention, an efficacious vaccine produces >50 mIU/ml. In an embodiment of the invention, an efficacious vaccine produces >100 mIU/ml. In an embodiment of the invention, an efficacious vaccine produces >200 mIU/ml. In an embodiment of the invention, an efficacious vaccine produces >500 mIU/ml or >1000 mIU/ml. In exemplary embodiments, the antibody level or concentration is produced or reached by 10 days following vaccination, by 20 days following vaccination, by 30 days following vaccination, by 40 days following vaccination, or by 50 or more days following vaccination. In an embodiment of the invention, the antibody level or concentration is produced or reached by 10 days following vaccination. In an embodiment of the invention, the antibody level or concentration is produced or reached by 20 days following vaccination. In an embodiment of the invention, the antibody level or concentration is produced or reached by 30 days following vaccination. In an embodiment of the invention, the antibody level or concentration is produced or reached by 40 days following vaccination. In an embodiment of the invention, the antibody level or concentration is produced or reached by 50 days following vaccination. In an embodiment of the invention, the antibody level or concentration is produced or reached by 21 to 35 days following vaccination. In exemplary embodiments, the level or concentration is produced or reached following a single dose of vaccine administered to the subject. In other embodiments, the level or concentration is produced or reached following multiple doses, e.g., following a first and a second dose (e.g., a booster dose.). In exemplary embodiments, the titer is produced or reached following three doses of vaccine administered to the subject. In exemplary embodiments, antibody level or concentration is determined or measured by enzyme-linked immunosorbent assay (ELISA). In exem-

plary embodiments, antibody level or concentration is determined or measured by neutralization assay, e.g., by microneutralization assay.

[0339] In another embodiment, the immunoprotective response against SARS-CoV-2 may be measured by CD4+ and CD8+ T-cell responses against SARS-CoV-2 S protein and epitopes thereof. Functionality and polarization of S-specific SARS-CoV-2 T cells induced by the mRNA composition may be assessed by intracellular accumulation of cytokines IFN γ , IL-2, and IL-4 measured after stimulation with overlapping peptide pools representing the full-length sequence of the whole SARS-CoV-2 S protein.

[0340] For example, in a clinical trial conducted in 2020, described in the FDA Briefing Document for the Vaccines and Related Biological Products Advisory Committee Meeting, dated Dec. 10, 2020, most participants who received both doses of BNT162b2 had evidence of SARS-CoV-2 S protein-specific CD4+(39/39, 100%) and CD8+(35/39, 89.7%) T cell responses. These T cell responses were directed against different parts of the antigen, including epitopes in the RBD, indicating the induction of multi-epitope responses by BNT162b2. Functionality and polarization of S-specific BNT162b2-induced SARS-CoV-2 T cells were assessed by intracellular accumulation of cytokines IFN γ , IL-2, and IL-4 measured after stimulation with overlapping peptide pools representing the full-length sequence of the whole SARS-CoV-2 S protein. For benchmarking, PBMC fractions from 15 convalescent patients with virologically confirmed COVID-19 were used. The Th1 polarization of the T helper response was characterized by the IFN γ and IL-2 production, and only minor IL-4, production upon antigen-specific (SARS-CoV-2 S protein peptide pools) re-stimulation. The SARS-CoV-2 neutralizing geometric mean titer (GMTs) increased over baseline after Dose 1, with a boost effect after Dose 2 that was most pronounced at the 30 μ g dose level. Thus, the immunogenicity results from Study BNT162-01 showed evidence of antibody-mediated SARS-CoV-2 neutralization and a Th1 polarization in the cell-mediated cellular immune responses in healthy adults 18 to 55 years of age, which supports the final dose selection and prospect of benefit for the enrollment of larger numbers of participants in Study C4591001.

[0341] In a preferred embodiment the immunoprotective response elicited by the PCV of the invention against *S. pneumoniae* is not decreased by co-administering (e.g. concomitantly or concurrently) a mRNA vaccine of the invention as compared to the administration of the PCV of the invention alone.

[0342] Thus, surprisingly, in these embodiments it has been found that the mRNA vaccine does not immunologically interfere with the patient's response to the PCV, preferably the Prevnar13®, the V114 or the 20vPnC (Prevnar20®) vaccine.

[0343] In a preferred embodiment the immunoprotective response elicited by the PCV of the invention against *S. pneumoniae* is increased by co-administering (e.g. concomitantly or concurrently) a mRNA vaccine of the invention as compared to the administration of the PCV of the invention alone.

[0344] Thus, surprisingly, in these embodiments it has been found that the mRNA vaccine immunologically enhances the patient's response to the PCV, preferably the Prevnar13®, the V114 or the 20vPnC (Prevnar20®) vaccine.

[0345] Preferably such increase is observed for at least one conjugate in a multivalent PCV of the invention. Preferably such increase is observed for at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12 or 13 conjugates in a multivalent PCV of the invention. In some such embodiments, the immunoprotective response is increased for at least one conjugate of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least two conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least three conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least four conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least five conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least six conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least seven conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least eight conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least nine conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least ten conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least eleven conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least three conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least twelve conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least thirteen conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least fourteen conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least fifteen conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least sixteen conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least seventeen conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least eighteen conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for at least nineteen conjugates of the PCV of the invention. In some embodiments, the immunoprotective response is increased for twenty conjugates of the PCV of the invention.

[0346] Preferably such increase is observed for all conjugates of a respective multivalent PCV. For example, in case of a 13-valent PCV (such as Prevnar13®), the immunoprotective response is increased for the thirteen conjugates of the PCV.

[0347] For example, in case of a 15-valent PCV (such as V114), the immunoprotective response is increased for the fifteen conjugates of the PCV.

[0348] For example, in case of a 20-valent PCV (such as 20vPnC (Prevnar20®)), the immunoprotective response is increased for the twenty conjugates of the PCV.

[0349] Preferably, such increase is at least 1.2-fold. In an embodiment, such increase is at least 1.3-fold. In an embodiment, such increase is at least 1.4-fold. In an embodiment,

such increase is at least 1.5-fold. In an embodiment, such increase is at least 1.6-fold. In an embodiment, such increase is at least 1.7-fold. In an embodiment, such increase is at least 1.8-fold. In an embodiment, such increase is at least 1.9-fold. In an embodiment, such increase is at least 2-fold.

[0350] In an embodiment said increase is an increase of the IgG level.

[0351] In an embodiment said increase is an increase of the fold rise in IgG level from before to after vaccination.

[0352] In an embodiment said increase is an increase of the OPA titer.

[0353] In an embodiment said increase is an increase of the fold rise in OPA titers from before to after vaccination (1 month after vaccination).

[0354] Preferably such increase of the invention is statistically significant at a p-value less than 0.05.

[0355] In a preferred embodiment the immunoprotective response elicited by a mRNA vaccine of the invention against SARS-CoV-2 is not decreased by co-administering (e.g. concomitantly or concurrently) a PCV vaccine of the invention as compared to the administration of the mRNA vaccine of the invention alone.

[0356] Thus, surprisingly, in these embodiments it has been found that the PCV does not immunologically interfere with the patient's response to the mRNA vaccine, preferably the BNT162b2 vaccine.

[0357] In a preferred embodiment the immunoprotective response elicited by a mRNA vaccine of the invention against SARS-CoV-2 is increased by co-administering (e.g. concomitantly or concurrently) a PCV vaccine of the invention as compared to the administration of the mRNA vaccine of the invention alone.

[0358] Thus, surprisingly, in these embodiments it has been found that the PCV immunologically enhances the patient's response to the mRNA vaccine, preferably the BNT162b2 vaccine.

[0359] Preferably, such increase is at least 1.2-fold, 1.3-fold, 1.4-fold, 1.5-fold, 1.6-fold, 1.7-fold, 1.8-fold, 1.9-fold or 2-fold of the neutralizing antibody titer. In an embodiment, such increase is at least 1.2-fold, of the neutralizing antibody titer. In an embodiment, such increase is at least 1.3-fold, of the neutralizing antibody titer. In an embodiment, such increase is at least 1.4-fold, of the neutralizing antibody titer. In an embodiment, such increase is at least 1.5-fold, of the neutralizing antibody titer. In an embodiment, such increase is at least 1.6-fold, of the neutralizing antibody titer. In an embodiment, such increase is at least 1.7-fold, of the neutralizing antibody titer. In an embodiment, such increase is at least 1.8-fold, of the neutralizing antibody titer. In an embodiment, such increase is at least 1.9-fold, of the neutralizing antibody titer. In an embodiment, such increase is at least 2-fold, of the neutralizing antibody titer.

[0360] Preferably, such increase is at least 1.2-fold, 1.3-fold, 1.4-fold, 1.5-fold, 1.6-fold, 1.7-fold, 1.8-fold, 1.9-fold or 2-fold of antigen-specific (S-binding) antibody titer. In an embodiment, such increase is at least 1.2-fold of the neutralizing antibody titer. In an embodiment, such increase is at least 1.3-fold, of antigen-specific (S-binding) antibody titer. In an embodiment, such increase is at least 1.4-fold of antigen-specific (S-binding) antibody titer. In an embodiment, such increase is at least 1.5-fold of antigen-specific (S-binding) antibody titer. In an embodiment, such increase is at least 1.6-fold of antigen-specific (S-binding) antibody

titer. In an embodiment, such increase is at least 1.7-fold of antigen-specific (S-binding) antibody titer. In an embodiment, such increase is at least 1.8-fold of antigen-specific (S-binding) antibody titer. In an embodiment, such increase is at least 1.9-fold of antigen-specific (S-binding) antibody titer. In an embodiment, such increase is at least 2-fold of antigen-specific (S-binding) antibody titer.

[0361] Preferably such increase of the invention is statistically significant a p-value less than 0.05.

[0362] In an embodiment, said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are administered concurrently.

[0363] In an embodiment, said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are administered concomitantly.

[0364] By "concurrent administration" is meant the administration of therapeutically effective doses of a first and a second immunogenic compositions through the same access site, but in separate unit dosage forms, within a short period of one another. Concurrent administration is essentially administering the two immunogenic compositions at about the same time but in separate dosage forms, through the same access site. The concurrent administration of the first and the second immunogenic compositions often occurs during the same physician office visit.

[0365] By "concomitant administration" is meant the administration of therapeutically effective doses of a first and a second immunogenic compositions, in separate unit dosage forms within a short period of one another at different anatomic sites. Concomitant administration is essentially administering the two immunogenic compositions at about the same time but in separate dosage forms and at different anatomic sites. The concomitant administration of the first and second immunogenic compositions often occurs during the same physician office visit.

[0366] In some cases, as little as one dose of each of the vaccines according to the invention is administered. In some circumstances however, a second, third or fourth dose may be given. Following an initial vaccination, subjects can receive one or several booster immunizations adequately spaced.

[0367] In an embodiment of the method of the invention, at least 2 doses of mRNA vaccine against SARS-CoV-2 is administered. In an embodiment of the method of the invention, at least 3 doses of mRNA vaccine against SARS-CoV-2 is administered. In an embodiment of the method of the invention, at least 4 doses of mRNA vaccine against SARS-CoV-2 is administered.

[0368] In an embodiment of the method of the invention, 2 doses of mRNA vaccine against SARS-CoV-2 is administered. In an embodiment of the method of the invention, 3 doses of mRNA vaccine against SARS-CoV-2 is administered. In an embodiment of the method of the invention, 4 doses of mRNA vaccine against SARS-CoV-2 is administered. In said embodiments, the doses of mRNA vaccine against SARS-CoV-2 can be separated by an interval of about 2 weeks to about 12 months.

[0369] In an embodiment of the method of the invention, one dose of pneumococcal conjugate vaccine is administered.

[0370] In an embodiment of the method of the invention, at least 2 doses of pneumococcal conjugate vaccine is administered. In an embodiment of the method of the invention, at least 3 doses of pneumococcal conjugate vac-

against SARS-CoV-2 has been administered at least about 3 weeks prior to said co-administration. In an embodiment of the method of the invention, the human subject has already received two mRNA vaccine doses against SARS-CoV-2 prior to said co-administration wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 2 months prior to said co-administration. In an embodiment of the method of the invention, the human subject has already received two mRNA vaccine doses against SARS-CoV-2 prior to said co-administration wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 6 months prior to said co-administration. In an embodiment of the method of the invention, the human subject has already received two mRNA vaccine doses against SARS-CoV-2 prior to said co-administration wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about one year prior to said co-administration. In an embodiment of the method of the invention, the human subject has already received two mRNA vaccine doses against SARS-CoV-2 prior to said co-administration wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about two years prior to said co-administration.

[0396] In an embodiment of the method of the invention, said co-administration is a booster dose of said mRNA vaccine against SARS-CoV-2.

[0397] In an embodiment the invention relates to a pneumococcal conjugate vaccine and an mRNA vaccine against SARS-CoV-2 for use in a method for eliciting an immunoprotective response in a human subject against *S. pneumoniae* and Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), said method comprising co-administering to the human subject said vaccines.

[0398] In an embodiment the invention relates to a pneumococcal conjugate vaccine and an mRNA vaccine against SARS-CoV-2 for use in a method for eliciting an immunoprotective response in a human subject against *S. pneumoniae* and Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), said method comprising co-administering to the human subject said vaccines wherein said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are administered concomitantly.

[0399] In an embodiment the invention relates to the use of the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 as a booster dose of an mRNA vaccine against SARS-CoV-2.

[0400] In an embodiment the invention relates to the use of the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 as a booster dose of said mRNA vaccine against SARS-CoV-2.

[0401] In an embodiment the invention relates to the use of the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for boosting an mRNA vaccine against SARS-CoV-2.

[0402] In an embodiment the invention relates to the use of the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for boosting said mRNA vaccine against SARS-CoV-2.

[0403] In an embodiment the invention relates to the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for use as a booster dose of an mRNA vaccine against SARS-CoV-2.

[0404] In an embodiment the invention relates to the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for use as a booster dose of said mRNA vaccine against SARS-CoV-2.

[0405] In an embodiment the invention relates to the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for use in a method of boosting an mRNA vaccine against SARS-CoV-2.

[0406] In an embodiment the invention relates to the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for use in a method of boosting said mRNA vaccine against SARS-CoV-2.

[0407] Said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 can be as disclosed herein.

[0408] In an embodiment, the vaccines disclosed herein are administered by intramuscular or subcutaneous injection. In an embodiment, the vaccines disclosed herein are administered by intramuscular injection. In an embodiment, the vaccines disclosed herein are administered by subcutaneous injection.

[0409] In an embodiment, the vaccines are administered by intramuscular injection in a thigh or arm. In an embodiment, the injection site is the anterolateral thigh muscle or the deltoid muscle. In an embodiment, the vaccines are administered via intramuscular injection to the deltoid muscle of an arm.

[0410] In an embodiment, the vaccines are administered by subcutaneous injection in a thigh or an arm. In an embodiment, the injection site is the fatty tissue over the anterolateral thigh muscle or the fatty tissue over triceps.

[0411] In case of concomitant administration, the first injection can be made in one thigh and the second in the other thigh (preferably in the anterolateral thigh muscles). Alternatively, the first injection can be made in one arm and the second in the other arm (preferably in the deltoid muscles). The first injection can also be made in a thigh and the second in an arm or the first injection in an arm and the second in a thigh. In case of concomitant administration, the vaccines are preferably administered via intramuscular injection to the deltoid muscle of each arm.

7. Subject to be Treated with the Method of the Invention

[0412] As disclosed herein, the vaccines described herein may be used for eliciting an immunoprotective response in a human against *S. pneumoniae* and Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2).

[0413] In an embodiment of the present invention, the human subject to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 is a human adult 50 years of age or older. More preferably the human subject is a human adult 60 years of age or older. Even more preferably, the human subject is a human adult 65 years of age or older. In an embodiment, the human subject is 70 years of age or older, 75 years of age or older or 80 years of age or older.

[0414] In an embodiment the human subject to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 is an immunocompromised individual. An immunocompromised individual is generally defined as a person who exhibits an attenuated or reduced ability to mount a normal humoral or cellular defense to challenge by infectious agents.

[0415] In an embodiment of the present invention, the immunocompromised human to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 suffers from a disease or condition that impairs the immune system and results in an antibody response that is insufficient to protect against or treat pneumococcal disease.

[0416] In an embodiment, said disease is a primary immunodeficiency disorder. Preferably, said primary immunodeficiency disorder is selected from the group consisting of: combined T- and B-cell immunodeficiencies, antibody deficiencies, well-defined syndromes, immune dysregulation diseases, phagocyte disorders, innate immunity deficiencies, autoinflammatory disorders, and complement deficiencies. In an embodiment, said primary immunodeficiency disorder is selected from the one disclosed on page 24, line 11, to page 25, line 19, of WO 2010/125480.

[0417] In a particular embodiment of the present invention, the immunocompromised human subject to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 suffers from a disease selected from the group consisting of: HIV-infection, acquired immunodeficiency syndrome (AIDS), cancer, chronic heart or lung disorders, congestive heart failure, diabetes mellitus, chronic liver disease, alcoholism, cirrhosis, spinal fluid leaks, cardiomyopathy, chronic bronchitis, emphysema, chronic obstructive pulmonary disease (COPD), spleen dysfunction (such as sickle cell disease), lack of spleen function (asplenia), blood malignancy, leukemia, multiple myeloma, Hodgkin's disease, lymphoma, kidney failure, nephrotic syndrome and asthma.

[0418] In an embodiment of the present invention, the immunocompromised human subject to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 suffers from malnutrition.

[0419] In a particular embodiment of the present invention, the immunocompromised human subject to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 is taking a drug or treatment that lowers the body's resistance to infection. In an embodiment, said drug is selected from the one disclosed on page 26, line 33, to page 26, line 4, of WO 2010/125480.

[0420] In a particular embodiment of the present invention, the immunocompromised human subject to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 is a smoker.

[0421] In a particular embodiment of the present invention, the immunocompromised human subject to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 has a white blood cell count (leukocyte count) below 5×10^9 cells per liter, or below 4×10^9 cells per liter, or below 3×10^9 cells per liter, or below 2×10^9 cells per liter, or below 1×10^9 cells per liter, or below 0.5×10^9 cells per liter, or below 0.3×10^9 cells per liter, or below 0.1×10^9 cells per liter.

[0422] White blood cell count (leukocyte count): The number of white blood cells (WBC) in the blood. The WBC is usually measured as part of the CBC (complete blood count). White blood cells are the infection-fighting cells in the blood and are distinct from the red (oxygen-carrying) blood cells known as erythrocytes. There are different types of white blood cells, including neutrophils (polymorphonuclear leukocytes; PMN), band cells (slightly immature neutrophils), T-type lymphocytes (T-cells), B-type lymphocytes

(B-cells), monocytes, eosinophils, and basophils. All the types of white blood cells are reflected in the white blood cell count. The normal range for the white blood cell count is usually between 4,300 and 10,800 cells per cubic millimeter of blood. This can also be referred to as the leukocyte count and can be expressed in international units as $4.3\text{--}10.8 \times 10^9$ cells per liter.

[0423] In a particular embodiment of the present invention, the immunocompromised human subject to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 suffers from neutropenia. In a particular embodiment of the present invention, the immunocompromised human subject to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 has a neutrophil count below 2×10^9 cells per liter, or below 1×10^9 cells per liter, or below 0.5×10^9 cells per liter, or below 0.1×10^9 cells per liter, or below 0.05×10^9 cells per liter.

[0424] A low white blood cell count or "neutropenia" is a condition characterized by abnormally low levels of neutrophils in the circulating blood. Neutrophils are a specific kind of white blood cell that help to prevent and fight infections. The most common reason that cancer patients experience neutropenia is as a side effect of chemotherapy. Chemotherapy-induced neutropenia increases a patient's risk of infection and disrupts cancer treatment. In a particular embodiment of the present invention, the immunocompromised subject to be vaccinated has a CD4+ cell count below $500/\text{mm}^3$, or CD4+ cell count below $300/\text{mm}^3$ or CD4+ cell count below $200/\text{mm}^3$, CD4+ cell count below $100/\text{mm}^3$, CD4+ cell count below $75/\text{mm}^3$, or CD4+ cell count below $50/\text{mm}^3$.

[0425] CD4 cell tests are normally reported as the number of cells in mm^3 . Normal CD4 counts are between 500 and 1,600, and CD8 counts are between 375 and 1,100. CD4 counts drop dramatically in people with HIV.

[0426] In an embodiment of the invention, any of the immunocompromised human subjects disclosed herein is a human male or a human female.

[0427] In an embodiment of the present invention, the human subject to be co-administered a pneumococcal conjugate vaccine and a mRNA vaccine against SARS-CoV-2 has already received at least one mRNA vaccine dose against SARS-CoV-2 prior to said co-administration. Preferably, said at least one mRNA vaccine dose against SARS-CoV-2 is a dose of BNT162b2.

[0428] In an embodiment of the present invention, the human subject to be co-administered a pneumococcal conjugate vaccine (PCV) and a mRNA vaccine against SARS-CoV-2 has already received at least two mRNA vaccine doses against SARS-CoV-2 prior to said co-administration. Preferably, said at least two mRNA vaccine doses against SARS-CoV-2 each are a dose of BNT162b2. In a preferred embodiment said PCV co-administered with said mRNA vaccine against SARS-CoV-2 is Prevnar13®, V114 or the 20vPnC (Prevnar20®) vaccine.

[0429] Accordingly, it is a preferred embodiment of the present invention to co-administer a dose of a PCV with a booster dose of a mRNA vaccine against SARS-CoV-2. In a preferred embodiment said PCV co-administered with said booster dose is Prevnar13®, V114 or the 20vPnC (Prevnar20®) vaccine.

SEQUENCES

CGSSGGGGVAADIGTGLADALTAPLDHKDKGLKSLTLEDSISQNGTLTLSAQGAECT
FKVGDKNDSLNTGKLNKDKISRFDVQKIEVDGQTI TLASGEFQIYKQDHSVVALQI
EKINNPDKIDSLINQSRFLVSLGGGEHTAFNQLPSGKA EYHGKAFSSDDAGGKLTYYTI
DFAAQGHGKIEHLKTPEQNVELASAEKKADEKSHAVILGDTRYGSEEKGYHLALF
GDRAQEIAGSATVKIREKVHEIGIAGKQ (SEQ ID NO: 3)

CGSSGGGGSGGGGVADIGTGLADALTAPLDHKDKGLKSLTLEDSISQNGTLTLSA
QGAECTYNGDLSLNTGKLNKDKVSRFDQIRQIEVDGQLITLESGEFQVYKQSHSALT
ALQTEQEQQDEHSEKMKVAKRRFRIGDIAGEHTSFDKLPKDVMTYRGTAFGSDDAG
GKLTYYTIDFAAQGHGKIEHLKSPELNVDLAVAYIKPDEKHHAVISGSVLYNQDEKGS
YSLGIFGEKAQEVAGSAE VETANGIHIGLAAKQ (SEQ ID NO: 4)

SLISKEELI KLAYSIRPRE NEYKTILTNL DEYNKLTNN NENKYLQLKK LNESIDVFMN 60
KYKTSSRNRA LSNLKKDILK EVILIKNSNT SPVEKNLHFV WIGGEVSDIA LEYIKQWADI 120
NAEYNIKWLWY DSEAFVNTL KKAIVESST EALQLLEEEI QNPQFDNMKF YKKRMEFIYD 180
RQKRPFINYK SQINKPTVPT IDDIKSHLV SEYNRDETVL ESYRTNSLRK INSNHGDIDIR 240
ANSLFTEQEL LNIYSQELLN RGNLAAASDI VRLALKNFG GVYLAVAMLP GIHSDLFKTI 300
SRPSSIGLDR WEMIKLEAIM KYKKYINNYT SENFDKLDQQ LKDNFKLIIE SKSEKSEIFS 360
KLENLMVSDL EIKIAFALGS VINQALISKQ GSYLTNLVIE QVKNRYQFLN QHLNPAIESD 420
NNFTDTTKIF HDSLFNSATA ENSMFLTKIA PYLQVGFMP E ARSTISLSGP GAYASAYYDF 480
INLQENTIEK TLKASDLIEF KPPENNLSQL TEQEINSLWS FDQASAKYQF EKYVRDYTGG 540
SLSEDNGVDF NKN'TALDKNY LLNNKIPSN VEEAGSKNVV HYIIQLQGDD ISYEATCNLF 600
SKNPKNSIII QRNMNESAKS YFLSDDGESI LELNKYRIPE RLKKNKEKVKV TFIGHGKDEF 660
NTSEFARLSV DLSLSNEISSF LDTIKLDISP KNVEVNLGA NMFSDYENVE ETYPGKLLLS 720
IMDKITSTLP DVNKNSTITIG ANQYEVIRINS EGRKELLAHS GKWINKEEAI MSDLSKEYI 780
FFDSIDNKLK AKSKNIPGLA SISEDIKTLL LDASVSPDTK FILNNLKLNI ESSIGDYIYY 840
EKLEPVKNII HNSIDDLIDE FNLLLENVSE LYLKLLNNL DEKYLISFED ISKNNSTYSV 900
RPINKSNGES VYVTEKEIF SKYSEHITKE ISTIKNSIIT DVNGNLLDNI QLDHTSQVNT 960
LNAAFFIQSL IDYSSNKDVL NDLSTSVKQV LYAQLFSTGL NTIYDSIQLV NLISNAVNDT 1020
INVLPITIEG IPIYSTILDG INLGAAIKEL LDEHDPPLKK ELEAKVGVLA INMSLSIAAT 1080
VASIVGIGAE VTIFLLPIAG ISAGIPSLVN NELILHDKAT SVVNYFNHLS ESKKYGPLKT 1140
EDDKILVPID DLVISEIDEN NNSIKLGTGN ILAMEGGSGH TVTGNIDHFF SSPSISSHIP 1200
SLSIYSAIGI ELVISEIDEN IMMLPNAPSR VFWWETGAVP GLRSLENDGT RLDSIRDLY 1260
PGKFIYWRFA FFDYAITTLK PVYEDTNIKI KLDKDRNFI MPTITTNEIR NKLSYSFDGA 1320
GGTYSLLSS YPISTNINLS KDDLWIFNID NEVREISIEI GTIKKGKLIK DVLSKIDINK 1380
NKLIIQNQTI DFSGDIDNKG RYIFLTCELD DKISLIIIEIN LVAKSYSLLL SGDKNYLISN 1440
LSNIEKINT LGLDSKNIAI NYTDESNNKY FGAISKTSQK SIIHYKDSK NILEFYNDST 1500
LEFNSKDFIA EDINVMKDD INTITGKYV DNNTDKSIDF SISLVSKNQV KVNGLYLNES 1560
VYSSYLDPVK NSDGHNTSN FMNLPLDNIS FWKLPGFENI NFVIDKYFTL VGKTNLGYVE 1620
FICDNNKID IYFGEWTKSS SKSTIFSGNG RNVVVEPIYN PDTGEDISTS LDFSIEPLYG 1680
IDRYINKVLI APDLYTSLIN INTNYSNEY YPEIIVLNPN TFHKKVNINL DSSSFYKKS 1740
TEGSDFILVR YLEESNKKIL QKIRIKGILS NTQSPNMSI DPKDIKKLSL GYIMSNFKSF 1800
NSENELDRDH LGFKIIDNKT YYYDEDSKLV KGLININNSL FYFDPFIEFNL VTGWQTINGK 1860
KYYFDINTGA ALISYKIING KHFYFNNDGV MQLGVFKGPD GFEYFAPANT QNNNIEGQAI 1920
VYQSKFLTIN GKKYFNDNS KAVTGWRIIN NEKYFNPNN AIAAVGLQVI DNNKYFNPND 1980
TAIISKGWQT VNGSRYFDT DTAIAFNGYK TIDGKHFYFD SDCVVKIGVF STSNGFEYFA 2040
PANTYNNIE GQAIYVQSKF LTLNGKKYYP DNNSKAVTGW QTIDSKKYF NTNTAEAAATG 2100
WQTIDGKKYF FQNTNTAAAT GWQTIIDGKY YFNTNTAIA STGYTINGKH FYFNTDQIMQ 2160
IGVFKGPNGF EYFAPANTDA NNIEGQAILY QNEFLTNGK KYFPGSDSKA VTGWRIINN 2220
KYYFNPNNAI AAHLCTINN DKYFYSYDGI LQNGYITIER NNFYFDANNE SKMVTGVFKG 2280
PNGFEYFAPA NTHNNIEGQ AIYVQNKELT LNKKKYFDPN DSKAVTGWQ IDGKKYFNL 2340
NTAEAAATGW TIDGKKYFYN LNTAEAAATG QTIDGKKYF NTNTFIASGT YTSINGKHFY 2400
FNTDQIMQIG VFKGPNGF EYFAPANTHNN IEGQAILYQN KFLTNGKKY YFGSDSKAVT 2460
GLRTIDGKKY YFNTNTAVKE TGMQTINGKK YFNTNTSIA STGYTIISGK HFYFNTDQIM 2520
QIGVFKGPDG FEYFAPANTD ANNIEGQAIR YQNRFLYLHD NIYYFGNNSK AATGWVTIDG 2580
NRYFEPNNTA MGANGYKIID NKNFYFRNGL PQIGVFKGSN GFEYFAPANT DANNIEGQAI 2640
RYQNRFLHLL GKIIYFGNNS KAVTGWQTING KKVYFMPDT AMAAAGLFE IDGVIYFFGV 2700
(SEQ ID NO: 5)

SLVNRKQLE KMANVRFRTO EDEYVAILDA LEEYHNMSN TVVEKYLKLD DINSLTDIYI 60
DTYKKSGRNK ALKKFKEYLV TEVLEKNNN LTPVEKNLHF VWIGGQINDT AINYINQWKD 120
VNSDYNVNVF YDSNAFLINT LKKTVESAI NDTLESFREN LNDPRFDYNK FFRKRMEIY 180
DKQKNFINY KAQREENPEL IIDDIVKTYL SNEYSKEIDE LNTYIEESLN KITQNSGNDV 240
RNFEFFKNGE SFNLYEQELV ERWNLAAASD ILRISALKEI GGMYLAVAML PGIQPDLFES 300
IEKPPSVTVL FWEMTKLEAI MKYKEYIPEY TSEHFDMLDE EVQSSPESVL ASKSDKSEIF 360
SSLGDMESP LKVIAPNSN GIINQGLISV KDSYCSNLIV KQIENRYKI NNSLNPASIE 420
DNDFNNTTNT FIDSIMAEAN ADNGREMMEL GKYLVRGFFP DVKTTINLSG PEAYAAAYQD 480
LLMFKEGSMN IHLIEADLRN FEISKTNISQ STEQEMASLW SEDDARAKA FEEYKRNIFE 540
GSLGEDDNDL FSHQIVVDKE YLLEKISSLA RSSERGIHY IVQLQGDKIS YEACNLFAK 600
TPYDSVLFKQ NIDSEIAYY YNPGDGEIQE IDKYKIPSII SDRPKIKLTF IGHGKDEFNT 660
DIFAGFDVDS LSTEIEAAID LAKEDISPKS IEINLLGAMN PSYSINVEET YPGKLLLVK 720
DKISELMP SI SQDSIIYSAN QYEVIRINSE RRELDDHSGE WINKEESIIK DISSEYISF 780
NPKENKITVK SKNLPSTL LQEIIRNNSNS SDIELEEKVM LTECEINVIS NIDTQIVEER 840
IEEAKNLTSD SINYIKDEFK LIESISDALC DLKQONELED SHFISFEDIS ETDEGFSIRF 900
INKETGESIF VETEKTFSE YANHITEEIS KIKGTIFDTV NGKLVKKVNL DTTHEVNTLN 960
AAFFIQSLIE YNSSKESLSN LSVAMKVQVY AQLFSTGLNT ITDAKVVVEL VSTALDETID 1020

-continued

SEQUENCES						
LLPTLSEGLP	IIATIIDGVS	LGAAIKELSE	TSDPLLQREI	EAKIGIMAVN	LTTATTAIIT	1080
SSLGIASGFS	ILLVPLAGIS	AGIPSLVNNE	LVLDRKATKV	VDYFKHVSIV	ETEGVPTLLD	1140
DKIMMPQDDL	VISEIDFNNN	SIVLGKCEIW	RMEGGSGHTV	TDDIDHFFSA	PSITYREPHL	1200
SIYDVLEVQK	EELDLSDKLM	VLPNAPNRVF	AWETGWTPGL	RSLENDGTKL	LDRIRDNYEG	1260
EFYWRYPAFI	ADALITTLKP	RYEDTNIRIN	LDSNTRSFIV	PIITTEYIRE	KLSYSFYGSG	1320
GTVALSLSQY	NMGINIELSE	SDVWIIDVDN	VVRDVTIESD	KIKKGDLEIG	ILSTLSIEEN	1380
KIILNSHEIN	FSGEVNGSNG	FVSLTFSILE	GINAIEVDL	LSKSYKLLIS	GELKILMLNS	1440
NHIQQKIDYI	GFNSELQKNI	PYSFVDSEKG	ENGFINGSTK	EGLFVSELDP	VVLISKVYMD	1500
DSKPSFGYYS	NNLKDVKVIT	KDNVNILTYG	YLKDDIKISL	SLTLQDEKTI	KLNSVHLDSE	1560
GVAEILKPMN	RKGNNTNTSDS	LMSFLESMNI	KSIFVNFLQS	NIKFILDANF	IISGTTSIGQ	1620
FEFICDENND	IQPYFIKNT	LETNTYTLVG	NRQNMIVEPN	YDLDDSGDIS	STVINFSQKY	1680
LYGIDSCVNK	VVISPNITYD	EINITPVYET	NNTYPEVIVL	DANYINEKIN	VNINDLSIRY	1740
VWSNDGNDFI	LMSTSEENKV	SQVKIRFNVV	FKDKTLANKL	SFNFSDKQDV	PVSEIILSFT	1800
PSYYEDGLIG	YDLGLVSLYN	EKPYINNFGM	MVSGLIYIND	SLYYFKPPVN	NLITGFVTVG	1860
DDKYYFNPIN	GGAASIGETI	IDDKNYFFNQ	SGVLQTVFVS	TEDGFKYFAP	ANTLDENLEG	1920
EAIDFTGKLI	IDENIYYFDD	NYRGAVEWKE	LDGEMHYFSP	ETGKAFKGLN	QIGDYKYYPN	1980
SDGVMQKGFV	SINDNKHYFD	DSGVMKVGYT	EIDGKHFYFA	ENGEMQIGVF	NTEDGPKYFA	2040
HHNEDLGNEE	GEEISYSGIL	NFNKKIYYFD	DSFTAVVGWK	DLEDGSKYFF	DEDTAEAYIG	2100
LSLINDGQYY	FNDDGIMQVG	FVTINDKVFY	FSDSGIIESG	VQNIDDNIFY	IDDNGIVQIG	2160
VFDTSBGYKY	FAPANTVNDN	IYQGAWEYSG	LVRVGEDVYV	FGETYTIETG	WIYDMENESD	2220
KYYFNPETKK	ACKGINLIDD	IKYYPDEKGI	MRTGLISPEN	NNYYFNENGE	MQPGYINIED	2280
KMFYFGEDGV	MQIGVENTPD	GPKYFAHQNT	LDENFEGESI	NYTGWLDDLE	KRYYFTDEYI	2340
AATGSVIIDG	BEYFYDPDTA	QLVISE				2366
(SEQ ID NO: 6)						

(SEQ ID NO: 6)

EXAMPLE

Endpoints:

Example 1. Safety and Immunogenicity Study of a 20-Valent Pneumococcal Conjugate Vaccine (20vPnC) when Coadministered with an mRNA Vaccine to Prevent Infection with SARS-CoV-2 (BNT162b2)

Primary (Safety):

[0430] Because of the current lack of safety and immunogenicity data on COVID-19 vaccines coadministered with other vaccines, current US Advisory Committee on Immunization Practices (ACIP) guidance is that COVID-19 vaccines should be administered alone, with a minimum interval of 14 days before or after administration of any other vaccine.

- [0435] Prompted local reactions at each injection site (redness, swelling, and pain at the injection site)
- [0436] Prompted systemic events (fever, headache, chills, fatigue, muscle pain, and joint pain)
- [0437] Adverse Events
- [0438] Serious Adverse Events

Secondary:

[0431] A clinical study has been designed to describe the safety and immunogenicity of a 20-valent Pneumococcal Conjugate Vaccine (20vPnC) and a booster dose of BNT162b2 (an mRNA vaccine to prevent infection with SARS-CoV-2) when administered together at the same visit compared to each of the vaccines given alone in adults 65 years of age, as shown in FIG. 1.

- [0439] Pneumococcal Immunogenicity: Pneumococcal OPA titers OPA GMTs approximately 1 month (e.g., about 21 through 35 days) after vaccination
- [0440] BNT162b2 Immunogenicity: Full-length S-binding IgG levels

Estimands:

Objectives:

Primary (Safety):

Primary Objective:

- [0432] To describe the safety profile of 20vPnC and a booster dose of BNT162b2 when coadministered or administered alone.

[0441] In participants receiving at least 1 dose of study intervention and having safety follow-up after vaccination, the percentage of participants reporting:

- [0442] Prompted local reactions at each injection site for up to 10 days following vaccination
- [0443] Prompted systemic events for up to 7 days following vaccination
- [0444] AEs from vaccination at Visit 1 through approximately 1 month after vaccination
- [0445] SAEs from vaccination at Visit 1 through 6 months after vaccination

Secondary Objectives:

Secondary:

- [0433] To describe the immune response elicited by 20vPnC when coadministered with a booster dose of BNT162b2 or when administered alone.
- [0434] To describe the immune response elicited by a booster dose of BNT162b2 when coadministered with 20vPnC or when administered alone.

- [0446] Pneumococcal Immunogenicity: In participants in compliance with the key protocol criteria (evaluable participants): OPA GMTs (Geometric Mean Titers) approximately 1 month after vaccination

[0447] BNT162b2 Immunogenicity: In evaluable participants: GMCs (Geometric Mean Concentration) of full-length S-binding IgG levels approximately 1 month after vaccination, GMFR (Geometric Mean Fold Rise) in full-length S-binding IgG levels from before to approximately 1 month after vaccination

Tertiary/Exploratory:

[0448] Pneumococcal Immunogenicity: In evaluable participants: GMFR in OPA titers from before to approximately 1 month after vaccination, The percentage of participants with a ≥ 4 -fold rise in OPA titers from before to approximately 1 month after vaccination, The percentage of participants with OPA titers $\geq$ LLOQ before vaccination and approximately 1 month after vaccination.

[0449] BNT162b2 Immunogenicity: In evaluable participants: GMTs of SARS-CoV-2 reference-strain neutralizing titers approximately 1 month after vaccination, GMFRs in SARS-CoV-2 reference-strain neutralizing titers from before to approximately 1 month after vaccination

Overall Design

[0450] This is a Phase 3, multicenter, randomized, double-blind study conducted at investigator sites in the US. The purpose of this study is to describe the safety and immunogenicity of 20vPnC and a booster dose of BNT162b2 when administered together at the same visit compared to each of the vaccines given alone in adults >65 years of age, as shown in FIG. 1.

[0451] Approximately 600 participants from the US, 65 years of age and older who received 2 doses of 30 μ g BNT162b2, are stratified by prior pneumococcal vaccine status (no previous pneumococcal vaccine [naïve] or receipt of at least 1 dose of a pneumococcal vaccine [experienced]) and randomized at a 1:1:1 ratio to 1 of 3 vaccine groups.

[0452] At Visit 1 (Day 1), the Coadministration group (20vPnC+BNT162b2) receives 20vPnC and a booster dose of BNT162b2, the 20vPnC-only group (20vPnC+saline) receives 20vPnC and saline, and the BNT162b2-only group (BNT162b2+saline) receives a booster dose of BNT162b2 and saline.

[0453] Participants from all groups have blood drawn at Visit 1 prior to vaccination, and at Visit 2, approximately 1 month after vaccination, for immunogenicity assessments and serological testing for prior COVID-19 infection.

Number of Participants

[0454] Approximately 600 participants (200 per group) are randomly assigned to study intervention.

Intervention Groups and Duration

[0455] Participants are randomized at a 1:1:1 ratio to 1 of 3 vaccine groups. At Visit 1 (Day 1), the Coadministration group receives 20vPnC and a booster dose of BNT162b2, the 20vPnC-only group receives 20vPnC and saline, and the BNT162b2-only group receives a booster dose of BNT162b2 and saline. Study intervention will be administered by an unblinded administrator via intramuscular injection to the upper deltoid muscle of each arm.

[0456] The duration of the study for each participant is approximately 6 months.

Statistical Methods

[0457] Safety is evaluated by descriptive summary statistics (including counts and percentages of participants and the associated 2-sided 95% CIs) for local reactions at each injection site, systemic events, AEs, and SAEs for each vaccine group.

[0458] Pneumococcal immunogenicity is evaluated descriptively by OPA GMTs approximately 1 month after 20vPnC. Other assessments of immune response including OPA GMFRs from before to approximately 1 month after 20vPnC, percentages of participants with >4 -fold rises in OPA titers from before to approximately 1 month after 20vPnC, and percentages of participants with OPA titers LLOQ approximately 1 month after 20vPnC is also described, each with corresponding 95% CIs in evaluable participants from the Coadministration and 20vPnC-only groups.

[0459] BNT162b2 immunogenicity is evaluated descriptively using GMCs of full-length S-binding IgG levels approximately 1 month after BNT162b2, and GMFR from before to approximately 1 month after BNT162b2, each with corresponding 2-sided 95% CIs in evaluable participants from the Coadministration and BNT162b2-only groups.

Dose

[0460] The 20vPnC candidate contains capsular polysaccharides from pneumococcal serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 22F, 23F, and 33F individually conjugated to CRM₁₉₇. The vaccine formulation contains 2.2 μ g of each saccharide, except for 4.4 μ g of 6B, per 0.5-mL dose (Intramuscular injection). In adults, administration of 1 dose of pneumococcal conjugate vaccine induces immune responses. The modRNA BNT162b2 vaccine candidate is administered at a dose of 30 μ g per 0.3-mL dose (Intramuscular injection). This is the dose that has shown to be efficacious and has been authorized for conditional or emergency use and is anticipated to be licensed in the future in the US.

[0461] For these products the term “dose” refers to an injection of a vaccine.

Study Population

Inclusion Criteria

[0462] Participants are eligible to be included in the study only if all of the following criteria apply:

Age and Sex:

[0463] 1. Male or female participants 65 years of age at the time of consent.

Type of Participant and Disease Characteristics:

[0464] 2. Participating or participated in Study C4591001, received 2 doses of 30 μ g BNT162b2 with the second dose given 6 months prior to the first vaccination in this study, and have not received a third dose of BNT162b2.

[0465] 3. Participants who are willing and able to comply with all scheduled visits, treatment plan, laboratory tests, lifestyle considerations, and other study procedures.

- [0466] 4. Participants who are determined by medical history, physical examination (if required), and clinical judgment of the investigator to be eligible for inclusion in the study.
- [0467] 5. Expected to be available for the duration of the study and can be contacted by telephone during study participation.
- [0468] 6. Male participant who is able to father children and willing to use an acceptable method of contraception; or female participant not of childbearing potential; or male participant not able to father children.
- [0469] 7. Adults who have no history of ever receiving a pneumococcal vaccine (ie, pneumococcal vaccine-naïve), or received a licensed pneumococcal vaccination ≥ 12 months prior to the first vaccination in this study.

Informed Consent:

- [0470] 8. Capable of giving signed informed consent.

Exclusion Criteria

[0471] Participants are excluded from the study if any of the following criteria apply: Medical Conditions:

- [0472] 1. History of severe adverse reaction associated with a vaccine and/or severe allergic reaction (eg, anaphylaxis) to any component of the study intervention(s) or any other diphtheria toxoid-containing vaccine.
- [0473] 2. Serious chronic disorder, including metastatic malignancy, severe COPD requiring supplemental oxygen, end-stage renal disease with or without dialysis, cirrhosis of the liver, clinically unstable cardiac disease, or any other disorder that in the investigator's opinion would make the participant inappropriate for entry into the study.
- [0474] 3. History of microbiologically proven invasive disease caused by *S pneumoniae*.
- [0475] 4. Previous clinical or microbiological diagnosis of COVID-19.
- [0476] 5. Known or suspected immunodeficiency (aside from stable HIV) or other conditions associated with immunosuppression, including, but not limited to, immunoglobulin class/subclass deficiencies, generalized malignancy, leukemia, lymphoma, or organ or bone marrow transplant.
- [0477] 6. Bleeding diathesis or condition associated with prolonged bleeding that would, in the opinion of the investigator, contraindicate intramuscular injection.
- [0478] 7. Congenital, functional, or surgical asplenia.
- [0479] 8. Current febrile illness (body temperature 100.4° F. [$>38.0^{\circ}$ C.]) or other acute illness within 48 hours before study intervention administration.
- [0480] 9. Other medical or psychiatric condition including recent (within the past year) or active suicidal ideation/behavior or laboratory abnormality that may increase the risk of study participation or, in the investigator's judgment, make the participant inappropriate for the study.

Prior/Concomitant Therapy:

- [0481] 10. Previous vaccination with any investigational pneumococcal vaccine, or planned receipt of any

licensed or investigational pneumococcal vaccine through study participation.

- [0482] 11. Previous vaccination with any coronavirus vaccine, other than those received in Study C4591001.
- [0483] 12. Currently receives treatment with immunosuppressive therapy, including cytotoxic agents or systemic corticosteroids, receipt of short-term (<14 days) systemic corticosteroids for treatment of an acute illness in the 28 days before administration of study intervention, or planned receipt through the last blood draw (Visit 2). Inhaled/nebulized, intra-articular, intrabursal, or topical (skin, eyes, or ears) corticosteroids are permitted.
- [0484] 13. Receipt of blood/plasma products, immunoglobulin, or monoclonal antibodies from 60 days before administration of study intervention, or receipt of any passive antibody therapy specific to COVID-19 from 90 days before administration of study intervention, or planned receipt through Visit 2.
- [0485] 14. Receipt of any inactivated or otherwise non-live vaccine within 14 days or any live vaccine within 28 days before administration of study intervention.

Prior/Concurrent Clinical Study Experience:

- [0486] 15. Participation in other studies involving investigational drugs, investigational vaccines, or investigational devices within 28 days prior to study entry and/or during study participation other than Study C4591001. Participation in purely observational studies is acceptable.
- [0487] 16. Previous participation in studies other than C4591001 involving study intervention containing LNPs.

Administration

- [0488] Participants receives study intervention at Visit 1.
- [0489] At Visit 1, participants receives a single 0.5-mL dose of either 20vPnC or saline injected intramuscularly into the right deltoid, and a single 0.3-mL dose of either BNT162b2 or saline injected intramuscularly into the left deltoid.

Immunogenicity Assessments

- [0490] Blood samples are collected from all participants at Visits 1 and 2.

Pneumococcal Responses

- [0491] OPA titers for the 20vPnC serotypes are measured in sera collected at Visits 1 and 2 from the Coadministration and 20vPnC-only groups.

BNT162b2 Responses

- [0492] IgG levels are measured in the SARS-CoV-2 full-length S-binding assay in sera collected at Visits 1 and 2 from the Coadministration and BNT162b2-only groups.
- [0493] SARS-CoV-2 reference-strain neutralizing titers may be measured in a subset of sera collected at Visits 1 and 2 from the Coadministration and BNT162b2-only groups.
- [0494] Blood samples taken at Visits 1 and 2 are also measured for the N-binding antibody.

Example 2: Safety, Tolerability, and
Immunogenicity of a Booster Dose of BNT162b2
COVID-19 Vaccine Coadministered with 20-Valent
Pneumococcal Conjugate Vaccine (PCV20) in
Adults 65 Years of Age and Above

[0495] Introduction. Adults 65 years of age are at increased risk of morbidity and mortality from COVID-19 and from pneumococcal disease. The efficacy and safety of 2 doses (30 µg/dose administered 21 days apart) of the BNT162b2 mRNA COVID-19 vaccine (Comirnaty) in preventing Covid-19 were established in a global, randomised, placebo-controlled phase 2/3 trial (Study C4591001, NCT04368728) of individuals 16 years of age, with 94.7% efficacy in those 65 years of age. The 20-valent pneumococcal conjugate vaccine (PCV20), recently approved in the United States and Europe for the prevention of invasive pneumococcal disease and pneumonia due to vaccine serotypes in adults, contains the components of the 13-valent pneumococcal conjugate vaccine (PCV13) plus the conjugated polysaccharides of 7 additional serotypes; PCV13 has demonstrated efficacy and safety against pneumococcal pneumonia, including nonbacteremic pneumonia, in randomised controlled trials in adults 65 years of age. A booster dose of COVID-19 vaccine is now recommended for adults in many countries; thus, the BNT162b2 vaccine may be used in the same population as PCV20, as their target populations overlap.

Study Design and Participants.

[0496] This phase 3, multicentre, randomised, double-blind study (ClinicalTrials.gov NCT04887948) was conducted in the United States from 20 May 2021 to

[0497] 8 Dec. 2021.

[0498] Participants were adults 65 years of age who had received 2 doses of 30 µg BNT162b2 in the pivotal efficacy study (C4591001), with the second dose given ≥6 months before vaccination in this study, and who had not received a booster dose of any COVID-19 vaccine.

[0499] Participants were randomised 1:1:1 into 1 of 3 groups, stratified by prior pneumococcal vaccine status (naïve or experienced), as follows: Coadministration group (both vaccines administered in opposite participant arms at the same visit), PCV20-only group, and BNT162b2-only group. In the control groups, saline was administered in the opposite participant arm to maintain blinding.

[0500] Three visits were performed in the study.

[0501] At Visit 1 (Day 1), participants were screened and enrolled, blood was drawn for immunogenicity testing, and vaccine was administered.

[0502] At Visit 2 (21-35 days after Visit 1), blood was drawn for immunogenicity testing, and safety data were collected.

[0503] At Visit 3 (approximately 6 months after Visit 1), participants were contacted by telephone to collect safety data.

Safety:

[0504] Prompted local reactions (redness, swelling, pain at injection site) at each injection site and systemic events (fever, fatigue, headache, chills, muscle pain,

joint pain) were assessed using an e-diary for 10 and 7 days after vaccination, respectively.

[0505] Adverse events (AEs) and serious AEs (SAEs) were collected for 1 month and 6 months after vaccination, respectively.

Immunogenicity:

[0506] Serotype-specific opsonophagocytic activity (OPA) titres for the PCV20 serotypes were measured in the Pfizer OPA assay before and 1 month after vaccination in the 2 groups that received PCV20.

[0507] SARS-CoV-2 full-length S-binding IgG concentrations were measured before and 1 month after vaccination in the 2 groups that received BNT162b2.

Statistical Analyses:

[0508] The statistical analysis was descriptive, with no hypothesis testing.

[0509] Safety results were descriptively summarised for the safety population, which included all participants who received any study vaccination and had safety follow-up.

[0510] Immunogenicity results were descriptively summarised for the evaluable immunogenicity population, which included participants who were vaccinated as randomised, had at least one OPA titre or SARS-CoV-2 full-length S-binding IgG concentration from a blood sample collected 1 month after vaccination, and had no major protocol deviations as determined by the clinician. Additionally, participants with clinically documented SARS-CoV-2 infection occurring between vaccination and 1 month after BNT162b2 vaccination were excluded from the SARS-CoV2 IgG analysis.

[0511] Serotype-specific OPA geometric mean titres (GMTs) and SARS-CoV-2 full-length S-binding IgG geometric mean concentrations (GMCs) and the corresponding 2-sided 95% CIs were calculated by exponentiating the mean logarithm of the titres or concentrations and the corresponding confidence intervals (Cis; based on the Student's t distribution).

[0512] Geometric mean fold rises (GMFRs) in full-length S-binding IgG levels from before to approximately 1 month after vaccination (secondary endpoint) were calculated as the mean of the difference of logarithmically transformed assay results (later minus earlier) and exponentiated back to the original units. The associated 2-sided 95% CIs were computed by exponentiating the Cis using Student's t distribution for the mean difference on the natural log scale.

[0513] A post hoc analysis using a linear regression model was performed to compare serotype-specific OPA titres 1 month after vaccination in the Coadministration and PCV20-only groups. A similar post hoc analysis evaluated full-length S-binding concentrations 1 month after vaccination in the Coadministration group compared to the BNT162b2-only group.

Study Population:

[0514] A total of 570 participants were randomised.

[0515] 559 participants were vaccinated and comprise the safety population (Coadministration, n=187; PPV20-only, n=187; BNT162b2-only, n=185).

[0516] Demographic characteristics were similar among the vaccine groups (Table 1).

[0517] (The average time elapsed since the second dose of BNT162b2 was 8.1 months, and the average elapsed time since the most recent pneumococcal vaccination was 3.5 years (Table 1).

[0523] The third BNT162b2 dose also elicited robust immune IgG responses to the SARS-CoV-2 full-length S-binding protein, that were similar whether BNT162b2 was coadministered with PCV20 or given alone (FIG. 5).

[0524] The observed GMFR in full-length S-binding IgG levels from before to 1 month after the booster

TABLE 1

Demographic and Baseline Characteristics (Safety Population)				
	Coadministration (PCV20 + BNT162b2) (N = 187)	PCV20 Only (PCV20 + Saline) (N = 187)	BNT162b2 Only (BNT162b2 + Saline) (N = 185)	Total (N = 559)
Sex, n (%)				
Male	98 (52.4)	112 (59.9)	108 (58.4)	318 (56.8)
Female	89 (47.6)	75 (40.1)	77 (41.6)	241 (43.1)
Race, n (%)				
White	164 (87.7)	172 (92.0)	165 (89.2)	501 (89.6)
Black or African American	8 (4.3)	7 (3.7)	10 (5.4)	25 (4.5)
Asian	10 (5.3)	5 (2.7)	9 (4.9)	24 (4.3)
Other	5 (2.7)	3 (1.6)	1 (0.5)	9 (1.6)
Ethnicity, n (%)				
Hispanic/Latino	28 (15.0)	26 (13.9)	25 (13.5)	79 (14.1)
Non-Hispanic/non-Latino	159 (85.0)	161 (86.1)	160 (86.5)	480 (85.9)
Age group, n (%), y				
65 to 69	76 (40.6)	69 (36.9)	72 (38.9)	217 (38.9)
70 to 74	73 (39.0)	69 (36.9)	59 (31.9)	201 (36.0)
75 to 79	29 (15.5)	34 (18.2)	41 (22.2)	104 (18.6)
≥80	9 (4.8)	15 (8.0)	13 (7.0)	37 (6.6)
Age at vaccination, y				
Mean (SD) Elapsed time since second dose of BNT162b2, months	71.0 (4.22)	71.8 (4.94)	71.8 (4.95)	71.5 (4.72)
Mean (SD) Elapsed time since prior pneumococcal vaccination, years	8.2 (0.76)	8.2 (0.66)	8.0 (0.75)	8.1 (0.73)
Mean (SD)	3.3 (1.44)	3.7 (1.70)	3.6 (1.55)	3.5 (1.57)

PCV20 = 20-valent pneumococcal conjugate vaccine; SD = standard deviation.

*Includes American Indian or Alaska Native, multiracial or not reported.

[0518] Safety. The percentages of participants who reported local reactions within 10 days at each of the PCV20 and BNT162b2 injection sites were similar regardless of whether the vaccines were given together or alone, and reactions were generally mild or moderate in severity. The rates of systemic events were similar in the Coadministration group and the BNT162b2-only group and were generally higher than those in the PCV20-only group; events were generally mild to moderate in severity.

[0519] Fatigue was the most frequently reported systemic event for all vaccine groups.

[0520] The rates of AEs within 1 month after vaccination and serious AEs (SAEs) through 6 months after vaccination were low and similar across all groups.

[0521] No SAEs were considered related to vaccine, and there was one death during the study (duodenal perforation, not considered related to vaccine).

[0522] PCV20 elicited robust immune responses to all 20 serotypes that were similar when PCV20 was coadministered with BNT162b2 or given alone (FIG. 4).

dose of BNT162b2 was similar in the coadministration and BNT162b2-only groups (35.5 and 39.0, respectively).

[0525] Based on post hoc analyses, the OPA GMRs of the Coadministration group to the PCV20-only group 1 month after PCV20 ranged from 0.77 (serotypes 8, 19F, and 23F) to 1.11 (serotype 19A; FIG. 6)

[0526] Full-length S-binding IgG GMR of the Coadministration group to the BNT162b2-only group 1 month after the booster dose of BNT162b2 was 1.06 (95% CI, 0.91, 1.23).

[0527] The responses would be statistically noninferior, with the lower bound of the 95% CI of the pneumococcal OPA GMR of coadministration to PCV20-only >0.5 for each serotype and >0.67 for the IgG GMR of coadministration to BNT162b2-only to the SARS-CoV-2 full-length S-binding protein; 0.5 and 0.67 correspond to standard 2-fold and 1.5-fold noninferiority criteria, respectively, for these endpoints.

Conclusion

- [0528] Coadministration of PCV20 and BNT162b2 was well tolerated, with an overall safety profile that was similar to BNT162b2 administered alone.
- [0529] Robust immune responses were observed regardless of concomitant or separate administration of PCV20 and BNT162b2.
- [0530] All publications and patent applications mentioned in the specification are indicative of the level of those skilled in the art to which this invention pertains. All publications and patent applications are hereby incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference.
- [0531] Although the foregoing invention has been described in some detail by way of illustration and example for purposes of clarity of understanding, certain changes and modifications may be practiced within the scope of the appended claims.

EMBODIMENTS

- [0532] 1. A method for eliciting an immunoprotective response in a human subject against *S. pneumoniae* and Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), the method comprising co-administering to the human subject an effective dose of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2.
- [0533] 2. The method of Embodiment 1 wherein said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are administered concurrently or concomitantly.
- [0534] 3. The method of Embodiment 1 wherein said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are administered concurrently.
- [0535] 4. The method of Embodiment 1 wherein said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are administered concomitantly.
- [0536] 5. The method of any one of Embodiments 1 to 4 wherein one dose of each of the vaccines is administered.
- [0537] 6. The method of any one of Embodiments 1 to 4 wherein at least 2 doses of said mRNA vaccine against SARS-CoV-2 is administered.
- [0538] 7. The method of any one of Embodiments 1 to 4 wherein at least 3 doses of said mRNA vaccine against SARS-CoV-2 is administered.
- [0539] 8. The method of any one of Embodiments 1 to 4 wherein at least 4 doses of said mRNA vaccine against SARS-CoV-2 is administered.
- [0540] 9. The method of any one of Embodiments 1 to 4 wherein 2 doses of said mRNA vaccine against SARS-CoV-2 is administered.
- [0541] 10. The method of any one of Embodiments 1 to 4 wherein 3 doses of said mRNA vaccine against SARS-CoV-2 is administered.
- [0542] 11. The method of any one of Embodiments 1 to 4 wherein 4 doses of said mRNA vaccine against SARS-CoV-2 is administered.
- [0543] 12. The method of any one of Embodiments 6 to 11 wherein said doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 12 months.
- [0544] 13. The method of any one of Embodiments 1 to 12 wherein one dose of said pneumococcal conjugate vaccine is administered.
- [0545] 14. The method of any one of Embodiments 1 to 12 wherein at least 2 doses of said pneumococcal conjugate vaccine is administered.
- [0546] 15. The method of any one of Embodiments 1 to 12 wherein at least 3 doses of said pneumococcal conjugate vaccine is administered.
- [0547] 16. The method of any one of Embodiments 1 to 12 wherein at least 4 doses of said pneumococcal conjugate vaccine is administered.
- [0548] 17. The method of any one of Embodiments 1 to 12 wherein 2 doses of said pneumococcal conjugate vaccine is administered.
- [0549] 18. The method of any one of Embodiments 1 to 12 wherein 3 doses of said pneumococcal conjugate vaccine is administered.
- [0550] 19. The method of any one of Embodiments 1 to 12 wherein 4 doses of said pneumococcal conjugate vaccine is administered.
- [0551] 20. The method of any one of Embodiments 14 to 19 wherein said doses of pneumococcal conjugate vaccine are separated by an interval of about 2 weeks to about 12 months.
- [0552] 21. The method of any one of Embodiments 1 to 4 wherein 2 doses of mRNA vaccine against SARS-CoV-2 and one dose of pneumococcal conjugate vaccine are administered.
- [0553] 22. The method of Embodiment 21 wherein said pneumococcal conjugate vaccine is co-administered with the first dose of mRNA vaccine against SARS-CoV-2.
- [0554] 23. The method of Embodiment 21 wherein said pneumococcal conjugate vaccine is co-administered with the second dose of mRNA vaccine against SARS-CoV-2.
- [0555] 24. The method of Embodiment 21 wherein said pneumococcal conjugate vaccine is concomitantly administered with the first dose of mRNA vaccine against SARS-CoV-2.
- [0556] 25. The method of Embodiment 21 wherein said pneumococcal conjugate vaccine is concomitantly administered with the second dose of mRNA vaccine against SARS-CoV-2.
- [0557] 26. The method of Embodiment 21 wherein said pneumococcal conjugate vaccine is concurrently administered with the first dose of mRNA vaccine against SARS-CoV-2.
- [0558] 27. The method of Embodiment 21 wherein said pneumococcal conjugate vaccine is concurrently administered with the second dose of mRNA vaccine against SARS-CoV-2.
- [0559] 28. The method of any one of Embodiments 1 to 4 wherein 3 doses of mRNA vaccine against SARS-CoV-2 and one dose of pneumococcal conjugate vaccine are administered.
- [0560] 29. The method of Embodiment 28 wherein said pneumococcal conjugate vaccine is co-administered with the first dose of mRNA vaccine against SARS-CoV-2.
- [0561] 30. The method of Embodiment 28 wherein said pneumococcal conjugate vaccine is co-administered with the second dose of mRNA vaccine against SARS-CoV-2.
- [0562] 31. The method of Embodiment 28 wherein said pneumococcal conjugate vaccine is co-administered with the third dose of mRNA vaccine against SARS-CoV-2.

- [0563] 32. The method of Embodiment 28 wherein said pneumococcal conjugate vaccine is concurrently administered with the first dose of mRNA vaccine against SARS-CoV-2.
- [0564] 33. The method of Embodiment 28 wherein said pneumococcal conjugate vaccine is concurrently administered with the second dose of mRNA vaccine against SARS-CoV-2.
- [0565] 34. The method of Embodiment 28 wherein said pneumococcal conjugate vaccine is concurrently administered with the third dose of mRNA vaccine against SARS-CoV-2.
- [0566] 35. The method of Embodiment 28 wherein said pneumococcal conjugate vaccine is concomitantly administered with the first dose of mRNA vaccine against SARS-CoV-2.
- [0567] 36. The method of Embodiment 28 wherein said pneumococcal conjugate vaccine is concomitantly administered with the second dose of mRNA vaccine against SARS-CoV-2.
- [0568] 37. The method of Embodiment 28 wherein said pneumococcal conjugate vaccine is concomitantly administered with the third dose of mRNA vaccine against SARS-CoV-2.
- [0569] 38. The method of any one of Embodiments 1 to 4 wherein 4 doses of mRNA vaccine against SARS-CoV-2 and one dose of pneumococcal conjugate vaccine are administered.
- [0570] 39. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is co-administered with the first dose of mRNA vaccine against SARS-CoV-2.
- [0571] 40. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is co-administered with the second dose of mRNA vaccine against SARS-CoV-2.
- [0572] 41. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is co-administered with the third dose of mRNA vaccine against SARS-CoV-2.
- [0573] 42. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is co-administered with the fourth dose of mRNA vaccine against SARS-CoV-2.
- [0574] 43. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is concurrently administered with the first dose of mRNA vaccine against SARS-CoV-2.
- [0575] 44. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is concurrently administered with the second dose of mRNA vaccine against SARS-CoV-2.
- [0576] 45. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is concurrently administered with the third dose of mRNA vaccine against SARS-CoV-2.
- [0577] 46. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is concurrently administered with the fourth dose of mRNA vaccine against SARS-CoV-2.
- [0578] 47. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is concomitantly administered with the first dose of mRNA vaccine against SARS-CoV-2.
- [0579] 48. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is concomitantly administered with the second dose of mRNA vaccine against SARS-CoV-2.
- [0580] 49. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is concomitantly administered with the third dose of mRNA vaccine against SARS-CoV-2.
- [0581] 50. The method of Embodiment 38 wherein said pneumococcal conjugate vaccine is concomitantly administered with the fourth dose of mRNA vaccine against SARS-CoV-2.
- [0582] 51. The method of any one of Embodiments 1 to 4 wherein 2 doses of mRNA vaccine against SARS-CoV-2 and 2 doses of pneumococcal conjugate vaccine are administered.
- [0583] 52. The method of any one of Embodiments 1 to 4 wherein 3 doses of mRNA vaccine against SARS-CoV-2 and 2 doses of pneumococcal conjugate vaccine are administered.
- [0584] 53. The method of any one of Embodiments 1 to 4 wherein 4 doses of mRNA vaccine against SARS-CoV-2 and 2 doses of pneumococcal conjugate vaccine are administered.
- [0585] 54. The method of any one of Embodiments 21 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 6 months.
- [0586] 55. The method of any one of Embodiments 21 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 2 months.
- [0587] 56. The method of any one of Embodiments 21 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 6 weeks.
- [0588] 57. The method of any one of Embodiments 21 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 4 months.
- [0589] 58. The method of any one of Embodiments 21 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 3 weeks.
- [0590] 59. The method of any one of Embodiments 28 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 6 months and the third dose is separated from the second dose by an interval of at least about 6 months.
- [0591] 60. The method of any one of Embodiments 28 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 2 months and the third dose is separated from the second dose by an interval of at least about 6 months.
- [0592] 61. The method of any one of Embodiments 28 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 6 weeks and the third dose is separated from the second dose by an interval of at least about 6 months.
- [0593] 62. The method of any one of Embodiments 28 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 4 months and the third dose is separated from the second dose by an interval of at least about 6 months.
- [0594] 63. The method of any one of Embodiments 28 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 3 weeks and the third dose is separated from the second dose by an interval of at least about 6 months.

[0595] 64. The method of any one of Embodiments 28 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 6 months and the third dose is separated from the second dose by an interval of at least about a year.

[0596] 65. The method of any one of Embodiments 28 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 2 months and the third dose is separated from the second dose by an interval of at least about a year.

[0597] 66. The method of any one of Embodiments 28 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 6 weeks and the third dose is separated from the second dose by an interval of at least about a year.

[0598] 67. The method of any one of Embodiments 28 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 4 months and the third dose is separated from the second dose by an interval of at least about a year.

[0599] 68. The method of any one of Embodiments 28 to 53 wherein the first 2 doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 3 weeks and the third dose is separated from the second dose by an interval of at least about a year.

[0600] 69. The method of any one of Embodiments 1 to 4 wherein said human subject has already received at least one mRNA vaccine dose against SARS-CoV-2 prior to said co-administration.

[0601] 70. The method of Embodiment 69 wherein said at least one mRNA vaccine dose against SARS-CoV-2 has been administered at least about 3 weeks prior to said co-administration.

[0602] 71. The method of Embodiment 69 wherein said at least one mRNA vaccine dose against SARS-CoV-2 has been administered at least about 2 months prior to said co-administration.

[0603] 72. The method of Embodiment 69 wherein said at least one mRNA vaccine dose against SARS-CoV-2 has been administered at least about 6 months prior to said co-administration.

[0604] 73. The method of Embodiment 69 wherein said at least one mRNA vaccine dose against SARS-CoV-2 has been administered at least about one year prior to said co-administration.

[0605] 74. The method of Embodiment 69 wherein said at least one mRNA vaccine dose against SARS-CoV-2 has been administered at least about two years prior to said co-administration.

[0606] 75. The method of any one of Embodiments 1 to 4 wherein said human subject has already received one mRNA vaccine dose against SARS-CoV-2 prior to said co-administration.

[0607] 76. The method of Embodiment 75 wherein said one mRNA vaccine dose against SARS-CoV-2 has been administered at least about 3 weeks prior to said co-administration.

[0608] 77. The method of Embodiment 75 wherein said one mRNA vaccine dose against SARS-CoV-2 has been administered at least about 2 months prior to said co-administration.

[0609] 78. The method of Embodiment 75 wherein said one mRNA vaccine dose against SARS-CoV-2 has been administered at least about 6 months prior to said co-administration.

[0610] 79. The method of Embodiment 75 wherein said one mRNA vaccine dose against SARS-CoV-2 has been administered at least about one year prior to said co-administration.

[0611] 80. The method of Embodiment 75 wherein said one mRNA vaccine dose against SARS-CoV-2 has been administered at least about two years prior to said co-administration.

[0612] 81. The method of any one of Embodiments 1 to 4 wherein said human subject has already received at least two mRNA vaccine doses against SARS-CoV-2 prior to said co-administration.

[0613] 82. The method of Embodiment 81 wherein the last of said at least two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 3 weeks prior to said co-administration.

[0614] 83. The method of Embodiment 81 wherein the last of said at least two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 2 months prior to said co-administration.

[0615] 84. The method of Embodiment 81 wherein the last of said at least two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 6 months prior to said co-administration.

[0616] 85. The method of Embodiment 81 wherein the last of said at least two mRNA vaccine doses against SARS-CoV-2 has been administered at least about one year prior to said co-administration.

[0617] 86. The method of Embodiment 81 wherein the last of said at least two mRNA vaccine doses against SARS-CoV-2 has been administered at least about two years prior to said co-administration.

[0618] 87. The method of any one of Embodiments 1 to 4 wherein said human subject has already received two mRNA vaccine doses against SARS-CoV-2 prior to said co-administration.

[0619] 88. The method of Embodiment 87 wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 3 weeks prior to said co-administration.

[0620] 89. The method of Embodiment 87 wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 2 months prior to said co-administration.

[0621] 90. The method of Embodiment 87 wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 6 months prior to said co-administration.

[0622] 91. The method of Embodiment 87 wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about one year prior to said co-administration.

[0623] 92. The method of Embodiment 87 wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about two years prior to said co-administration.

[0624] 93. The method of any one of Embodiments 1 to 4 wherein said co-administration is a booster dose of said mRNA vaccine against SARS-CoV-2.

[0625] 94. A pneumococcal conjugate vaccine and an mRNA vaccine against SARS-CoV-2 for use in a method for eliciting an immunoprotective response in a human subject against *S. pneumoniae* and Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), said method comprising co-administering to the human subject said vaccines.

[0626] 95. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 94 wherein said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are administered concurrently or concomitantly.

[0627] 96. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 94 wherein said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are administered concurrently.

[0628] 97. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 94 wherein said pneumococcal conjugate vaccine and said mRNA vaccine against SARS-CoV-2 are administered concomitantly.

[0629] 98. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein one dose of each of the vaccines is administered.

[0630] 99. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein at least 2 doses of said mRNA vaccine against SARS-CoV-2 is administered.

[0631] 100. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein at least 3 doses of said mRNA vaccine against SARS-CoV-2 is administered.

[0632] 101. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein at least 4 doses of said mRNA vaccine against SARS-CoV-2 is administered.

[0633] 102. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein 2 doses of said mRNA vaccine against SARS-CoV-2 is administered.

[0634] 103. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein 3 doses of said mRNA vaccine against SARS-CoV-2 is administered.

[0635] 104. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein 4 doses of said mRNA vaccine against SARS-CoV-2 is administered.

[0636] 105. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 99 to 104 wherein said doses of mRNA vaccine against SARS-CoV-2 are separated by an interval of about 2 weeks to about 12 months.

[0637] 106. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 105 wherein one dose of said pneumococcal conjugate vaccine is administered.

[0638] 107. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 105 wherein at least 2 doses of said pneumococcal conjugate vaccine is administered.

[0639] 108. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of

Embodiments 94 to 105 wherein at least 3 doses of said pneumococcal conjugate vaccine is administered.

[0640] 109. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 105 wherein at least 4 doses of said pneumococcal conjugate vaccine is administered.

[0641] 110. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 105 wherein 2 doses of said pneumococcal conjugate vaccine is administered.

[0642] 111. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 105 wherein 3 doses of said pneumococcal conjugate vaccine is administered.

[0643] 112. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 105 wherein 4 doses of said pneumococcal conjugate vaccine is administered.

[0644] 113. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 107 to 112 wherein said doses of pneumococcal conjugate vaccine are separated by an interval of about 2 weeks to about 12 months.

[0645] 114. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein 2 doses of mRNA vaccine against SARS-CoV-2 and one dose of pneumococcal conjugate vaccine are administered.

[0646] 115. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 114 wherein said pneumococcal conjugate vaccine is co-administered with the first dose of mRNA vaccine against SARS-CoV-2.

[0647] 116. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 114 wherein said pneumococcal conjugate vaccine is co-administered with the second dose of mRNA vaccine against SARS-CoV-2.

[0648] 117. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 114 wherein said pneumococcal conjugate vaccine is concomitantly administered with the first dose of mRNA vaccine against SARS-CoV-2.

[0649] 118. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 114 wherein said pneumococcal conjugate vaccine is concomitantly administered with the second dose of mRNA vaccine against SARS-CoV-2.

[0650] 119. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 114 wherein said pneumococcal conjugate vaccine is concurrently administered with the first dose of mRNA vaccine against SARS-CoV-2.

[0651] 120. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 114 wherein said pneumococcal conjugate vaccine is concurrently administered with the second dose of mRNA vaccine against SARS-CoV-2.

[0652] 121. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein 3 doses of mRNA vaccine against SARS-CoV-2 and one dose of pneumococcal conjugate vaccine are administered.

already received one mRNA vaccine dose against SARS-CoV-2 prior to said co-administration.

[0700] 169. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 168 wherein said one mRNA vaccine dose against SARS-CoV-2 has been administered at least about 3 weeks prior to said co-administration.

[0701] 170. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 168 wherein said one mRNA vaccine dose against SARS-CoV-2 has been administered at least about 2 months prior to said co-administration.

[0702] 171. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 168 wherein said one mRNA vaccine dose against SARS-CoV-2 has been administered at least about 6 months prior to said co-administration.

[0703] 172. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 168 wherein said one mRNA vaccine dose against SARS-CoV-2 has been administered at least about one year prior to said co-administration.

[0704] 173. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 168 wherein said one mRNA vaccine dose against SARS-CoV-2 has been administered at least about two years prior to said co-administration.

[0705] 174. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein said human subject has already received at least two mRNA vaccine doses against SARS-CoV-2 prior to said co-administration.

[0706] 175. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 174 wherein the last of said at least two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 3 weeks prior to said co-administration.

[0707] 176. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 174 wherein the last of said at least two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 2 months prior to said co-administration.

[0708] 177. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 174 wherein the last of said at least two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 6 months prior to said co-administration.

[0709] 178. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 174 wherein the last of said at least two mRNA vaccine doses against SARS-CoV-2 has been administered at least about one year prior to said co-administration.

[0710] 179. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 174 wherein the last of said at least two mRNA vaccine doses against SARS-CoV-2 has been administered at least about two years prior to said co-administration.

[0711] 180. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein said human subject has already received two mRNA vaccine doses against SARS-CoV-2 prior to said co-administration.

[0712] 181. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodi-

ment 180 wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 3 weeks prior to said co-administration.

[0713] 182. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 180 wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 2 months prior to said co-administration.

[0714] 183. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 180 wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about 6 months prior to said co-administration.

[0715] 184. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 180 wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about one year prior to said co-administration.

[0716] 185. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of Embodiment 180 wherein the last of said two mRNA vaccine doses against SARS-CoV-2 has been administered at least about two years prior to said co-administration.

[0717] 186. The pneumococcal conjugate vaccine and mRNA vaccine against SARS-CoV-2 for use of any one of Embodiments 94 to 97 wherein said co-administration is a booster dose of said mRNA vaccine against SARS-CoV-2.

[0718] 187. Use of the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 as a booster dose of an mRNA vaccine against SARS-CoV-2.

[0719] 188. Use of the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 as a booster dose of said mRNA vaccine against SARS-CoV-2.

[0720] 189. Use of the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for boosting an mRNA vaccine against SARS-CoV-2.

[0721] 190. Use of the co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for boosting said mRNA vaccine against SARS-CoV-2.

[0722] 191. The co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for use as a booster dose of an mRNA vaccine against SARS-CoV-2.

[0723] 192. The co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for use as a booster dose of said mRNA vaccine against SARS-CoV-2.

[0724] 193. The co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for use in a method of boosting an mRNA vaccine against SARS-CoV-2.

[0725] 194. The co-administration of a pneumococcal conjugate vaccine and of an mRNA vaccine against SARS-CoV-2 for use in a method of boosting said mRNA vaccine against SARS-CoV-2.

[0726] 195. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is a 13-valent pneumococcal vaccine.

[0727] 196. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is a 14-valent pneumococcal vaccine.

[0728] 197. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is a 15-valent pneumococcal vaccine.

[0729] 198. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is a 16-valent pneumococcal vaccine.

[0730] 199. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is a 20-valent pneumococcal vaccine

[0731] 200. The method of any one of Embodiments 1 to 199 wherein the glycoconjugates from *S. pneumoniae* are all individually conjugated to CRM₁₉₇.

[0732] 201. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine comprises 13 glycoconjugates from a *Streptococcus pneumoniae* serotype selected from the group consisting of serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 23F, 22F and 33F.

[0733] 202. The method Embodiment 201 wherein said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0734] 203. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is a 13-valent pneumococcal conjugate vaccine wherein said 13 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F.

[0735] 204. The method Embodiment 203 wherein said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0736] 205. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is a 15-valent pneumococcal conjugate vaccine wherein said 15 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F, 22F and 33F.

[0737] 206. The method Embodiment 205 wherein said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0738] 207. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is a 20-valent pneumococcal conjugate vaccine wherein said 20 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 23F, 22F and 33F.

[0739] 208. The method Embodiment 207 wherein said glycoconjugates are all individually conjugated to CRM₁₉₇.

[0740] 209. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is PRE-VNAR 13®/

[0741] 210. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is V114.

[0742] 211. The method of any one of Embodiments 1 to 194 wherein said pneumococcal conjugate vaccine is 20vPnC.

[0743] 212. The method of any one of Embodiments 1 to 211 wherein said mRNA vaccine against SARS-CoV-2 comprises nucleoside-modified mRNA.

[0744] 213. The method of any one of Embodiments 1 to 211 wherein said mRNA vaccine is BNT162b2 (Comirnaty®).

[0745] 214. The method of any one of Embodiments 1 to 212 wherein said mRNA vaccine against SARS-CoV-2 comprises a sequence having residues 1-102 of SEQ ID NO:

1 and residues 103-4284 of SEQ ID NO: 1, wherein the sequence for the SARS-CoV-2 antigen of SEQ ID NO: 1 is replaced with SARS-CoV-2 antigen of a variant strain.

[0746] 215. The method of any one of Embodiments 1 to 212 wherein said mRNA vaccine is vaccine "mRNA-1273".

[0747] 216. The method of any one of Embodiments 1 to 212 wherein said mRNA vaccine against SARS-CoV-2 comprises a mRNA which includes a first region of linked nucleosides encoding a SARS-CoV-2 antigen (e.g., S protein), a first flanking region located at the 5'-terminus of the first region (e.g., a 5'-UTR), a second flanking region located at the 3'-terminus of the first region (e.g., a 3'-UTR), at least one 5'-cap region, and a 3'-stabilizing region.

[0748] 217. The method of any one of Embodiments 1 to 212 wherein said mRNA vaccine against SARS-CoV-2 comprises a mRNA which includes a first region of linked nucleosides encoding a mutated viral spike (S) glycoprotein of SARS-CoV-2, a first flanking region located at the 5'-terminus of the first region (e.g., a 5'-UTR), a second flanking region located at the 3'-terminus of the first region (e.g., a 3'-UTR), at least one

Embodiments Continued

[0749] 1. A method for eliciting an immune response in a human subject against an infectious disease-causing bacterium and a betacoronavirus, the method comprising co-administering to the human subject an effective dose of a first immunogenic composition comprising an antigen derived from the bacterium and an immunogenic composition comprising mRNA encoding an antigen derived from the betacoronavirus.

[0750] 2. The method according to clause 1, wherein a dose of the first immunogenic composition and a dose of the immunogenic composition comprising mRNA are administered concurrently.

[0751] 3. The method according to any one of clauses 1-2, wherein a dose of the first immunogenic composition and a dose of the immunogenic composition comprising mRNA are administered concomitantly.

[0752] 4. The method according to any one of clauses 1-3, wherein a dose of the first immunogenic composition and a dose of the immunogenic composition comprising mRNA are administered within 24 hours of each respective dose.

[0753] 5. The method according to any one of clauses 1-4, wherein a dose of the first immunogenic composition and a dose of the immunogenic composition comprising mRNA are administered within 48 hours of each respective dose.

[0754] 6. The method according to any one of clauses 1-5, wherein a first dose of the first immunogenic composition and a second dose of the first immunogenic composition are separated by an interval of about 2 weeks to about 12 months.

[0755] 7. The method according to any one of clauses 1-6, wherein a first dose of the immunogenic composition comprising mRNA and a second dose of the immunogenic composition comprising mRNA are separated by an interval of about 2 weeks to about 12 months.

[0756] 8. The method according to clause 1, wherein the bacterium is not *S. pneumoniae*.

[0757] 9. The method according to clause 1, wherein the bacterium is *N. meningitidis*.

[0758] 10. The method according to clause 1, wherein the bacterium is *C. difficile*.

- [0759] 11. The method according to clause 1, wherein the bacterium is *E. coli*.
- [0760] 12. The method according to clause 1, wherein the betacoronavirus is SARS-CoV-2.
- [0761] 13. The method according to any one of clauses 1-12, wherein the human subject has already received at least one dose of the immunogenic composition comprising mRNA against betacoronavirus prior to said co-administration.
- [0762] 14. The method according to any one of clauses 1-13, wherein said at least one dose of the immunogenic composition comprising mRNA against betacoronavirus has been administered at least about 3 weeks prior to said co-administration.
- [0763] 15. The method according to any one of clauses 1-13, wherein said at least one dose of the immunogenic composition comprising mRNA against betacoronavirus has been administered at least about 2 months prior to said co-administration.
- [0764] 16. The method according to any one of clauses 1-13, wherein said at least one dose of the immunogenic composition comprising mRNA against betacoronavirus has been administered at least about 6 months prior to said co-administration.
- [0765] 17. The method according to any one of clauses 1-13, wherein said at least one dose of the immunogenic composition comprising mRNA against betacoronavirus has been administered at least about one year prior to said co-administration.
- [0766] 18. The method according to any one of clauses 1-13, wherein said at least one dose of the immunogenic composition comprising mRNA against betacoronavirus has been administered at least about two years prior to said co-administration.
- [0767] 19. The method according to any one of clauses 1-18, wherein said human subject has already received at least two dose of the immunogenic composition comprising mRNA against betacoronavirus prior to said co-administration.
- [0768] 20. The method according to any one of clauses 1-19, wherein the last of said at least two dose of the immunogenic composition comprising mRNA against Betacoronavirus has been administered at least about 3 weeks prior to said co-administration.
- [0769] 21. The method according to any one of clauses 1-20, wherein the last of said at least two dose of the immunogenic composition comprising mRNA against Betacoronavirus has been administered at least about 2 months prior to said co-administration.
- [0770] 22. The method according to any one of clauses 1-21, wherein the last of said at least two dose of the immunogenic composition comprising mRNA against Betacoronavirus has been administered at least about 6 months prior to said co-administration.
- [0771] 23. The method according to any one of clauses 1-22, wherein the last of said at least two dose of the immunogenic composition comprising mRNA against Betacoronavirus has been administered at least about one year prior to said co-administration.
- [0772] 24. The method according to any one of clauses 1-23, wherein the last of said at least two dose of the immunogenic composition comprising mRNA against Betacoronavirus has been administered at least about two years prior to said co-administration.
- [0773] 25. The method according to any one of clauses 1-24, wherein said human subject has already received two dose of the immunogenic composition comprising mRNA against Betacoronavirus prior to said co-administration.
- [0774] 26. The method according to clause 8, wherein the first composition is a 13-valent pneumococcal vaccine.
- [0775] 27. The method according to clause 8, wherein the first composition is a 14-valent pneumococcal vaccine.
- [0776] 28. The method according to clause 8, wherein the first composition is a 15-valent pneumococcal vaccine.
- [0777] 29. The method according to clause 8, wherein the first composition is a 16-valent pneumococcal vaccine.
- [0778] 30. The method according to clause 8, wherein the first composition is a 20-valent pneumococcal vaccine.
- [0779] 31. The method according to clause 8, wherein the first composition comprises glycoconjugates from *S. pneumoniae*, said glycoconjugates are all individually conjugated to CRM₁₉₇.
- [0780] 32. The method according to clause 8, wherein the first composition comprises 13 glycoconjugates from a *Streptococcus pneumoniae* serotype selected from the group consisting of serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 23F, 22F and 33F.
- [0781] 33. The method according to clause 32, wherein said glycoconjugates are all individually conjugated to CRM₁₉₇.
- [0782] 34. The method according to clause 8, wherein the first composition is a 13-valent pneumococcal conjugate vaccine wherein said 13 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F.
- [0783] 35. The method according to clause 34, wherein said glycoconjugates are all individually conjugated to CRM₁₉₇.
- [0784] 36. The method according to clause 8, wherein the first composition is a 15-valent pneumococcal conjugate vaccine wherein said 15 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 9V, 14, 18C, 19A, 19F, 23F, 22F and 33F.
- [0785] 37. The method according to clause 36, wherein said glycoconjugates are all individually conjugated to CRM₁₉₇.
- [0786] 38. The method according to clause 8, wherein the first composition is a 20-valent pneumococcal conjugate vaccine wherein said 20 conjugates consists of glycoconjugates from *S. pneumoniae* serotypes 1, 3, 4, 5, 6A, 6B, 7F, 8, 9V, 10A, 11A, 12F, 14, 15B, 18C, 19A, 19F, 23F, 22F and 33F.
- [0787] 39. The method according to clause 38, wherein said glycoconjugates are all individually conjugated to CRM₁₉₇.
- [0788] 40. The method according to clause 8, wherein the first composition is PREVNAR 13®.
- [0789] 41. The method according to clause 8, wherein the first composition is V114.
- [0790] 42. The method according to clause 8, wherein the first composition is 20vPnC.
- [0791] 43. The method according to clause 9, wherein the first composition comprises a polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 1 nm.
- [0792] 44. The method according to clause 43, wherein the polypeptide is lipidated.
- [0793] 45. The method according to clause 43, wherein the polypeptide is non-lipidated.

- [0794] 46. The method according to clause 43, wherein the polypeptide is immunogenic.
- [0795] 47. The method according to clause 43, wherein the first composition further comprises an adjuvant.
- [0796] 48. The method according to clause 9, wherein the first composition comprises a) a first lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 1, and b) a second lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 2.
- [0797] 49. The method according to clause 43, wherein the first composition further comprises polysorbate-80, aluminum, histidine, and sodium chloride.
- [0798] 50. The method according to clause 48, wherein the first composition further comprises polysorbate-80, aluminum, histidine, and sodium chloride.
- [0799] 51. The method according to clause 9, wherein the first composition comprises TRUMENBA®.
- [0800] 52. The method according to clause 9, wherein the first composition comprises fHBP antigens.
- [0801] 53. The method according to clause 9, wherein the first composition comprises polysaccharides derived from *N. meningitidis*.
- [0802] 54. The method according to clause 9, wherein the first composition comprises a) a liquid composition comprising (i) a first lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 1 nm; and (ii) a second lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 2 nm and aluminum; and b) a lyophilized composition comprising i) a *Neisseria meningitidis* serogroup A (MenA) capsular saccharide conjugated to an adipic acid dihydrazide (ADH) linker by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, wherein the linker is conjugated to tetanus toxoid (TT) by carbodiimide chemistry (MenAAH-TT conjugate); ii) a *Neisseria meningitidis* serogroup C (MenC) capsular saccharide conjugated to an ADH linker by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, wherein the linker is conjugated to tetanus toxoid (TT) by carbodiimide chemistry (MenCAH-TT conjugate); iii) a *Neisseria meningitidis* serogroup W135 (MenW) capsular saccharide directly conjugated to tetanus toxoid (TT) by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, in the absence of a linker (MenW-TT conjugate); and iv) a *Neisseria meningitidis* serogroup Y (MenY) capsular saccharide directly conjugated to tetanus toxoid (TT) by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, in the absence of a linker (MenY-TT conjugate) wherein the lyophilized composition is reconstituted with the liquid composition.
- [0803] 55. The method according to clause 54, wherein the aluminum comprises aluminum phosphate.
- [0804] 56. The method according to clause 54, wherein the first composition further comprises polysorbate-80.
- [0805] 57. The method according to clause 54, wherein the first composition further comprises Tris-HCl; sodium chloride; sucrose; histidine; polysorbate 80; and aluminum phosphate.
- [0806] 58. The method according to clause 9, wherein the first composition is administered to a human aged 10 to 25 years old.
- [0807] 59. The method according to clause 9, wherein the first composition is administered to a human aged 10 to 26 years old.
- [0808] 60. The method according to clause 10, wherein the first composition comprises a toxoid.
- [0809] 61. The method according to clause 10, wherein the first composition comprises a fusion polypeptide.
- [0810] 62. The method according to clause 10, wherein the first composition is lyophilized.
- [0811] 63. The method according to clause 10, wherein the first composition comprises a cryoprotectant.
- [0812] 64. The method according to clause 10, wherein the first composition comprises sucrose.
- [0813] 65. The method according to clause 10, wherein the first composition comprises trehalose.
- [0814] 66. The method according to clause 10, wherein the first composition does not further comprise formaldehyde.
- [0815] 67. The method according to clause 10, wherein the first composition comprises a polypeptide that comprises the C-terminal domain of a wild-type *C. difficile* toxin A.
- [0816] 68. The method according to clause 10, wherein the first composition comprises a polypeptide that comprises the C-terminal domain of a wild-type *C. difficile* toxin B.
- [0817] 69. The method according to clause 10, wherein the first composition comprises a polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 4cd, wherein a side chain of a lysine residue of the polypeptide is crosslinked to a beta-alanine moiety.
- [0818] 70. The method according to clause 10, wherein the first composition comprises a polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 6cd, wherein a side chain of a lysine residue of the polypeptide is crosslinked to a beta-alanine moiety.
- [0819] 71. The method according to clause 10, wherein the first composition is administered to a human aged 55 and older.
- [0820] 72. The method according to clause 12, wherein said immunogenic composition comprising mRNA comprises nucleoside-modified mRNA.
- [0821] 73. The method according to clause 12, wherein said immunogenic composition comprising mRNA is BNT162b2 (Comirnaty®).
- [0822] 74. The method according to clause 73, wherein said immunogenic composition comprising mRNA comprises a sequence having residues 1-102 of SEQ ID NO: 1 and residues 103-4284 of SEQ ID NO: 1, wherein the sequence for the SARS-CoV-2 antigen of SEQ ID NO: 1 is replaced with SARS-CoV-2 antigen of a variant strain.
- [0823] 75. The method according to clause 12, wherein said immunogenic composition comprising mRNA is vaccine "mRNA-1273".
- [0824] 76. The method according to clause 74, wherein said immunogenic composition comprising mRNA comprises a mRNA which includes a first region of linked nucleosides encoding a SARS-CoV-2 antigen (e.g., S protein), a first flanking region located at the 5'-terminus of the first region (e.g., a 5'-UTR), a second flanking region located at the 3'-terminus of the first region (e.g., a 3'-UTR), at least one 5'-cap region, and a 3'-stabilizing region.
- [0825] The method according to clause 74, wherein said immunogenic composition comprising mRNA-2 comprises a mRNA which includes a first region of linked nucleosides encoding a mutated viral spike (S) glycoprotein of SARS-CoV-2, a first flanking region located at the 5'-terminus of the first region (e.g., a 5'-UTR), a second flanking region located at the 3'-terminus of the first region (e.g., a 3'-UTR), at least one 5'-cap region, and a 3'-stabilizing region.

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 12

<210> SEQ ID NO 1

<211> LENGTH: 4284

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 1

gagaauaaac uaguauucuu cuggucccca cagacucaga gagaacccgc caccauguuc 60
guguuccugg ugcugcugcc ucuggugucc agccagugug ugaaccugac caccagaaca 120
cagcugccuc cagccuacac caacagcuuu accagaggcg uguacuacc cgacaaggug 180
uucagaucca gcgugcugca cucuaccag gaccuguucc ugccuuucuu cagcaacgug 240
accugguucc acgccaucca cguguccggc accauggca ccaagagauu cgacaacccc 300
gugcugcccu ucaacgacgg gguguacuuu gccagcacgg agaaguccaa caucaucaga 360
ggcuggaucu ucggcaccac acuggacagc aagaccaga gccugcugau cgugaacaac 420
gccaccaacg uggucaucaa agugugcgag uuccaguucu gcaacgaccc cuuccugggc 480
gucuacuacc acaagaacaa caagagcugg auggaaagcg aguuccgggu guacagcagc 540
gccacaacuu gcaccuucga guacgugucc cagccuuucc ugauggaccu ggaaggcaag 600
cagggaacuu ucaagaaccu gcgcgaguuc guguuuaaga acaucgacgg cuacuuaag 660
aucuacagca agcacacccc uaucaaccuc gugcgggauc ugccucaggg cuucucugcu 720
cuggaacccc ugguggaucu gcccacggc aucaacauc cccgguuua gacacugcug 780
gccucgaca gaagcuaccu gacaccuggc gauagcagca gcggauggac agcuggugcc 840
gccgcuuacu augugggcuu ccugcagccu agaaccuucc ugcugaagua caacgagaac 900
ggcaccuaua ccgacgccgu ggauugugcu cuggauccuc ugagcgagac aaagugcacc 960
cugaaguccu ucaccgugga aaagggauc uaccagacca gcaacuuccg ggugcagccc 1020
accgaaucca ucgugcgguu ccccaauauc accaauucgu gcccucucgg cagguguuu 1080
aaugccacca gauucgccuc uguguacgcc uggaaccgga agcggauacg caauugcgug 1140
gccgacuacu ccgugcugua caacuccgcc agcuucagca ccuucagug cuacggcgug 1200
uucccuacca agcugaacga ccugugcuuc acaaacgugu acgcccagac cuucgugauc 1260
cggggagaug aagugcgga gauugcccu ggacagacag gcaagaucgc cgacuacaac 1320
uacaagcugc ccgacgacuu caccggcugu gugauugccu ggaacagcaa caaccuggac 1380
uccaaagucg gcggcaacua caauuaccug uaccggcugu uccggaaguc caaucugaag 1440
cccuucgagc gggacauuc caccgagauc uaucaggccg gcagacccc uaguaacggc 1500
guggaaggcu ucaacugcuu cuucccacug caguccuacg gcuuucagcc cacaauaggc 1560
gugggcuauc agcccuacag agugguggug cugagcuucg aacugcugca ugcccugcc 1620
acagugugcg gccuuaagaa aagaccaau cucgugaaga acaauugcgu gaacuuaac 1680
uucacgggcc ugaccggcac cggcgugcug acagagagca acaagaaguu ccugccaauu 1740
cagcaguuuu gccgggauau cggcgauacc acagacggcg uuagagaucc ccagacacug 1800
gaaaucugc acauacccc uugcagcuuc ggcgagugu cugugaucac ccuggcacc 1860
aacaccagca aucagguggc agugcuguac caggacguga acuguaccga agugcccgug 1920

-continued

gccauucacg	ccgaucagcu	gacaccuaca	uggcgggugu	acuccaccgg	cagcaaugug	1980
uuucagacca	gagccggcug	ucugaucgga	gccgagcacg	ugaacaauag	cuacgagugc	2040
gacaucccca	ucggcgucug	aaucugcgcc	agcuaccaga	cacagacaaa	cagcccucgg	2100
agagccagaa	gcguggccag	ccagagcauc	auugccuaca	caaugucucu	ggcgcccgag	2160
aacagcgugg	ccuacuccaa	caacucuauc	gcuaucacca	ccaacuucac	caucagcgug	2220
accacagaga	uccugccugu	guccaugacc	aagaccagcg	uggacugcac	cauguacauc	2280
ugcgcgcauu	ccaccgagug	cuccaaccug	cugcugcagu	acggcagcuu	cugcaccag	2340
cugaauagag	cccugacagg	gaucgcccug	gaacaggaca	agaacacca	agagguguuc	2400
gcccaguga	agcagaucua	caagaccccu	ccuaucagg	acuucggcgg	cuucauuuc	2460
agccagauuc	ugcccgauc	uagcaagccc	agcaagcgga	gcuucaucga	ggaccugcug	2520
uucaacaaag	ugacacuggc	cgacgcccgc	uucaucaagc	aguuggcgga	uugucugggc	2580
gacauugccg	ccagggaucu	gauuugcgcc	cagaaguuaa	acggacugac	agugcugccu	2640
ccucugcuga	ccgaugagau	gaucgcccag	uacacaucug	cccugcuggc	cggcacaauc	2700
acaagcgguu	ggacauuugg	agcaggcgcc	gcucugcaga	uccccuuugc	uauagcagaug	2760
gccuaccggu	ucaacggcau	cggagugacc	cagaauuguc	uguacgagaa	ccagaagcug	2820
aucgccaacc	aguuaacag	cgccaucggc	aagauccagg	acagccugag	cagcacagca	2880
agcgcccugg	gaaagcugca	ggacgugguc	aaccagaau	cccaggcacu	gaacaccucg	2940
gucaagcagc	uguccuccaa	cuucggcgcc	aucagcucug	ugcugaacga	uaucugagc	3000
agacuggacc	cuccugaggc	cgaggugcag	aucgacagac	ugaucacagg	cagacugcag	3060
agccuccaga	cauacgugac	ccagcagcug	aucagagccg	ccgagauuag	agccucugcc	3120
aaucugggcg	ccaccaagau	gucugagugu	gugcuggggc	agagcaagag	aguggacuua	3180
ugcggaagg	gcuaccaccu	gaugagcuuc	ccucagucug	cccucacgg	cgugguguuu	3240
cugcagcuga	cauauugcc	cgcucaagag	aagaauuua	ccaccgcucc	agccaucugc	3300
cacgacggca	aagcccacuu	uccuagagaa	ggcguguucg	uguccaacgg	caccuauugg	3360
uucgugacac	agcggaacuu	cuacgagccc	cagaucuaa	ccaccgacaa	caccuucgug	3420
ucuggcaacu	gcgacgucgu	gaucggcauu	gugaacaaua	ccguguacga	cccucugcag	3480
cccgagcugg	acagcuucaa	agaggaaucg	gacaaguacu	uuuagaacca	cacaagcccc	3540
gacguggacc	ugggcgcauu	cagcggauc	aaugccagcg	ucgugaacau	ccagaaagag	3600
aucgaccggc	ugaacgaggu	ggccaagaau	cugaacgaga	gccugaucga	ccugcaagaa	3660
cugggggaagu	acgagcagua	caucaagugg	cccugguaca	ucuggcuggg	cuuuaucgcc	3720
ggacugauug	ccaucgugau	ggucacaauc	augcuguguu	gcaugaccag	cugcuguagc	3780
ugccugaagg	gcuguuguag	cuguggcagc	ugcugcaagu	ucgacgagga	cgauucugag	3840
cccugcuga	agggcgugaa	acugcacuac	acaugaugac	ucgagcuggu	acugcaugca	3900
cgcaaugcua	gcugccccc	ucccguccug	gguaccccga	gucuccccc	accucggguc	3960
ccagguaugc	ucccaccucc	accugcccca	cucaccaccu	cugcuaguuc	cagacaccuc	4020
ccaagcacgc	agcaaugcag	cuaaaaacgc	uuagccuagc	cacaccccca	cgggaaacag	4080
cagugauuaa	ccuuuagcaa	uaaacgaaag	uuuaacuaag	cuauacuaac	cccagggguu	4140
gucauuuucg	ugccagccac	accugggagc	uagcaaaaaa	aaaaaaaaaa	aaaaaaaaaa	4200

-continued

aaaagcauu gacuaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa aaaaaaaaaa	4260
aaaaaaaaa aaaaaaaaaa aaaa	4284

<210> SEQ ID NO 2
 <211> LENGTH: 4001
 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 2

gggaaauaag agagaaaaga agaguaagaa gaaauauaag accccggcgc cgccaccaug	60
uucguguucc uggugcugcu gccccuggug agcagccagu gcgugaaccu gaccaccgg	120
accagcugc caccagccua caccaacagc uucacccggg gcgucuacua ccccgacaag	180
guguuccgga gcagcugccu gcacagcacc caggaccugu uccugcccuu cuucagcaac	240
gugaccuggu uccacgcca ccacgugagc ggcaccaacg gcaccaagcg guucgacaac	300
cccgugcugc ccuuaacga cggcguguac uucgccagca ccgagaagag caacaucauc	360
cggggcugga ucuucggcac caccucggac agcaagaccc agagccugcu gaucgugaau	420
aacgccacca acguggugau caaggugugc gaguuccagu ucugcaacga ccccuuccug	480
ggcguguacu accacaagaa caacaagagc uggauggaga gcgaguuccg gguguacagc	540
agcgccaaca acugcaccuu cgaguacgua gagccagccc uuccugaugg accuggaggg	600
caagcagggc aacuucaaga accugcggga guucguguuc aagaacaucg acggcuacuu	660
caagaucuac agcaagcaca ccccaucaa ccuggugcgg gaucugcccc agggcuucuc	720
agccucggag cccucggugg accugcccau cggcaucaac aucaccgggu uccagaccu	780
gcugggccug caccggagcu accugacccc aggcgacagc agcagcgggu ggacagcagg	840
cgcggcugcu uacuacgugg gcuaccugca gccccggacc uuccugcuga aguacaacga	900
gaacggcacc aucaccgacg ccguggacug cggccuggac ccucugagcg agaccaagug	960
caccugaag agcuuacccg uggagaaggg caucuaccag accagcaacu uccgggugca	1020
gcccaccgag agcaucgugc gguuucccaa cauacccaac cugugcccu ucggcgaggu	1080
guucaacgcc acccgguucg ccagcgugua cggcuggaac cggaagcgga ucagcaacug	1140
cguggccgac uacagcgugc uguacaacag cgcagcuuc agcaccuua agugcuacgg	1200
cgugagcccc accaagcuga acgaccuguc uucaccaacg uguacggcga caguucguga	1260
uccguggcga cgaggugcgg cagaucgcac cggccagac aggaagauc gccgacuaca	1320
acuacaagcu gcccgacgac uucaccggcu gcgugaucgc cuggaacagc aacaaccucg	1380
acagcaaggu gggcggaac uacaacuacc uguaccggcu guuccggaag agcaaccuga	1440
agccuucga cggggacauc agcaccgaga ucuaccaagc cggcuccacc ccuugcaacg	1500
gcguggaggg cuucaacugc uacuuccuc ugcaagcua cggcuuccag cccaccaacg	1560
gcgugggcu ccagcccuac cggguggugg ugcugagcu cgagcugcug cagccccag	1620
ccaccgugug uggcccaag aagagacca accugguga gaacaagugc gugaacuuca	1680
acuacaacgg ccuuaaccgg accggcgugc ugaccgagag caacaagaaa uuccugccu	1740
uucagcagu cggccgggac aucgccgaca ccaccgacg ugugcgggau cccagaccc	1800
uggagaucuu ggacaucacc ccuugcagcu ucggcgggcu gagcgugauc accccaggca	1860

-continued

ccaacaccag	caaccaggug	gccgugcugu	accaggacgu	gaacugcacc	gaggugccc	1920
uggccaacca	cgccgaccag	cugacacca	ccugcggggu	cuacagcacc	ggcagcaac	1980
uguuccagac	ccgggcccgu	ugccugaucg	gcgcccagca	cgugaacaac	agcuacgagu	2040
gcgacauccc	caucggcgcc	ggcaucugug	ccagcuacca	gacccagacc	aaauaccccc	2100
ggagggaag	gagcguggcc	agccagagca	ucaucgcua	caccaugagc	cugggcccgc	2160
agaacagcgu	ggccuacagc	aacaacagca	ucgccauccc	caccaacuuc	accaucagcg	2220
ugaccaccga	gauucugccc	gugagcauga	ccaagaccag	cguggacugc	accauguaca	2280
ucugcgggca	cagcaccgag	ugcagcaacc	ugcugcugca	guacggcagc	uucugacccc	2340
agcugaaccg	ggcccugacc	ggcaucgcgg	uggagcagga	caagaacacc	caggaggugu	2400
ucgcccaggu	gaagcagauc	uacaagaccc	cucccaucaa	ggacuucggc	ggcuucauu	2460
cagccagauc	cugcccagcc	ccagcaagcc	cagcaagcgg	agcuucaucg	aggaccugcu	2520
guucaacaag	gugaccuag	ccgacgccc	cuucaucaag	caguacggcg	acugccucgg	2580
cgacauagcc	gcccgggacc	ugaucugcgc	ccagaaguuc	aacggccuga	ccgugcugcc	2640
ucccugcug	accgacgaga	ugaucgccc	guacaccagc	gccuguuagc	cggaaccauc	2700
accagcgccu	ggacuucgg	cgucggagcc	gcucugcaga	uccccuucgc	caugcagaug	2760
gccuaccggu	ucaacggcgu	cgcgugaccc	cagaacgugc	uguacgagaa	ccagaagcug	2820
aucgccaacc	aguuaacag	cgccaucggc	aagauccagg	acagccugag	cagcaccgcu	2880
agcgcccug	gcaagcugca	ggagcuggg	aaccagaacg	cccaggcccu	gaacaccucg	2940
gugaagcagc	ugagcagcaa	cuucggcgcc	aucagcagcg	ugcugaacga	cauccugagc	3000
cggcuggacc	cucccgaggc	cgaggugcag	aucgaccggc	ugaucacugg	ccggcugcag	3060
agccugcaga	ccuacugagc	ccagcagcug	auccgggccc	ccgagauucg	ggccagcgcc	3120
aaaccuggcc	ccaccaagau	gagcgagugc	gugcuggggc	agagcaagcg	gguggacuuc	3180
ugcggaagg	gcuaccaccu	gaugagcuuu	cccagagcgg	caccccacgg	agugguguuc	3240
cugcagcuga	ccuacugucc	cgcccaggag	aagaacuua	ccaccgccc	agccaucugc	3300
cacgacggca	agggccacuu	uccccgggag	ggcguguucg	ugagcaacgg	caccacacug	3360
uucgugaccc	agcggaacuu	cuacgagccc	cagaucua	ccaccgacaa	caccuucgug	3420
agcggaacuu	gcgacguggu	gaucggcauc	gugaacaaca	ccguguacga	ucccugcag	3480
cccagcugg	acagcuucaa	ggaggagcug	gacaaguacu	ucaagaauca	caccagcccc	3540
gacguggacc	ugggcgacau	cagcggcac	aacgcccagc	uggugaacau	ccagaaggag	3600
aucgaucggc	ugaacgaggu	ggccaagaac	cugaacgaga	gccugaucga	ccugcaggag	3660
cugggcaagu	acgagcagua	caucaagugg	cccugguaca	ucuggcuggg	cuucaucgcc	3720
ggccugaucg	ccaucgugau	ggugaccauc	augcugugcu	gcaugaccag	cugcugcagc	3780
ugccugaagg	gcuguugcag	cugcggcagc	ugcugcaagu	ucgacgagga	cgacagcgag	3840
cccugcuga	agggcgugaa	gcugcacuac	accugauaa	aggcuggagc	cucgguggcc	3900
uagcuucuu	ccccuugggc	cucuccccag	ccccuccucc	ccuuccugca	cccguacccc	3960
cguggucuuu	gaauaaaguc	ugaguggggc	gcaaaaaaaa	a		4001

<210> SEQ ID NO 3

<211> LENGTH: 258

<212> TYPE: PRT

-continued

```

<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 3

Cys Gly Ser Ser Gly Gly Gly Gly Val Ala Ala Asp Ile Gly Thr Gly
1          5          10          15

Leu Ala Asp Ala Leu Thr Ala Pro Leu Asp His Lys Asp Lys Gly Leu
20          25          30

Lys Ser Leu Thr Leu Glu Asp Ser Ile Ser Gln Asn Gly Thr Leu Thr
35          40          45

Leu Ser Ala Gln Gly Ala Glu Lys Thr Phe Lys Val Gly Asp Lys Asp
50          55          60

Asn Ser Leu Asn Thr Gly Lys Leu Lys Asn Asp Lys Ile Ser Arg Phe
65          70          75          80

Asp Phe Val Gln Lys Ile Glu Val Asp Gly Gln Thr Ile Thr Leu Ala
85          90          95

Ser Gly Glu Phe Gln Ile Tyr Lys Gln Asp His Ser Ala Val Val Ala
100         105         110

Leu Gln Ile Glu Lys Ile Asn Asn Pro Asp Lys Ile Asp Ser Leu Ile
115         120         125

Asn Gln Arg Ser Phe Leu Val Ser Gly Leu Gly Gly Glu His Thr Ala
130         135         140

Phe Asn Gln Leu Pro Ser Gly Lys Ala Glu Tyr His Gly Lys Ala Phe
145         150         155         160

Ser Ser Asp Asp Ala Gly Gly Lys Leu Thr Tyr Thr Ile Asp Phe Ala
165         170         175

Ala Lys Gln Gly His Gly Lys Ile Glu His Leu Lys Thr Pro Glu Gln
180         185         190

Asn Val Glu Leu Ala Ser Ala Glu Leu Lys Ala Asp Glu Lys Ser His
195         200         205

Ala Val Ile Leu Gly Asp Thr Arg Tyr Gly Ser Glu Glu Lys Gly Thr
210         215         220

Tyr His Leu Ala Leu Phe Gly Asp Arg Ala Gln Glu Ile Ala Gly Ser
225         230         235         240

Ala Thr Val Lys Ile Arg Glu Lys Val His Glu Ile Gly Ile Ala Gly
245         250         255

Lys Gln

```

```

<210> SEQ ID NO 4
<211> LENGTH: 261
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthetic construct

```

```

<400> SEQUENCE: 4

Cys Gly Ser Ser Gly Gly Gly Gly Ser Gly Gly Gly Gly Val Thr Ala
1          5          10          15

Asp Ile Gly Thr Gly Leu Ala Asp Ala Leu Thr Ala Pro Leu Asp His
20          25          30

Lys Asp Lys Gly Leu Lys Ser Leu Thr Leu Glu Asp Ser Ile Ser Gln
35          40          45

Asn Gly Thr Leu Thr Leu Ser Ala Gln Gly Ala Glu Lys Thr Tyr Gly

```

-continued

50	55	60
Asn Gly Asp Ser Leu	Asn Thr Gly Lys Leu Lys	Asn Asp Lys Val Ser
65	70	75
Arg Phe Asp Phe Ile Arg	Gln Ile Glu Val Asp Gly Gln Leu Ile Thr	
	85	90
Leu Glu Ser Gly Glu Phe	Gln Val Tyr Lys Gln Ser His Ser Ala Leu	
	100	105
Thr Ala Leu Gln Thr Glu Gln	Glu Gln Asp Pro Glu His Ser Glu Lys	
	115	120
Met Val Ala Lys Arg Arg	Phe Arg Ile Gly Asp Ile Ala Gly Glu His	
	130	135
Thr Ser Phe Asp Lys Leu Pro	Lys Asp Val Met Ala Thr Tyr Arg Gly	
	145	150
Thr Ala Phe Gly Ser Asp Asp	Ala Gly Gly Lys Leu Thr Tyr Thr Ile	
	165	170
Asp Phe Ala Ala Lys Gln Gly	His Gly Lys Ile Glu His Leu Lys Ser	
	180	185
Pro Glu Leu Asn Val Asp Leu	Ala Val Ala Tyr Ile Lys Pro Asp Glu	
	195	200
Lys His His Ala Val Ile Ser	Gly Ser Val Leu Tyr Asn Gln Asp Glu	
	210	215
Lys Gly Ser Tyr Ser Leu Gly	Ile Phe Gly Glu Lys Ala Gln Glu Val	
	225	230
Ala Gly Ser Ala Glu Val Glu	Thr Ala Asn Gly Ile His His Ile Gly	
	245	250
Leu Ala Ala Lys Gln		255
	260	

<210> SEQ ID NO 5

<211> LENGTH: 2699

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 5

Ser Leu Ile Ser Lys Glu Glu Leu Ile Lys Leu Ala Tyr Ser Ile Arg
1 5 10 15
Pro Arg Glu Asn Glu Tyr Lys Thr Ile Leu Thr Asn Leu Asp Glu Tyr
20 25 30
Asn Lys Leu Thr Thr Asn Asn Asn Glu Asn Lys Tyr Leu Gln Leu Lys
35 40 45
Lys Leu Asn Glu Ser Ile Asp Val Phe Met Asn Lys Tyr Lys Thr Ser
50 55 60
Ser Arg Asn Arg Ala Leu Ser Asn Leu Lys Lys Asp Ile Leu Lys Glu
65 70 75 80
Val Ile Leu Ile Lys Asn Ser Asn Thr Ser Pro Val Glu Lys Asn Leu
85 90 95
His Phe Val Trp Ile Gly Gly Glu Val Ser Asp Ile Ala Leu Glu Tyr
100 105 110
Ile Lys Gln Trp Ala Asp Ile Asn Ala Glu Tyr Asn Ile Lys Leu Trp
115 120 125
Tyr Asp Ser Glu Ala Phe Leu Val Asn Thr Leu Lys Lys Ala Ile Val

-continued

130					135					140					
Glu 145	Ser	Ser	Thr	Thr	Glu 150	Ala	Leu	Gln	Leu	Leu 155	Glu	Glu	Glu	Ile	Gln 160
Asn	Pro	Gln	Phe	Asp 165	Asn	Met	Lys	Phe	Tyr 170	Lys	Lys	Arg	Met	Glu 175	Phe
Ile	Tyr	Asp	Arg	Gln	Lys	Arg	Phe	Ile	Asn	Tyr	Tyr	Lys	Ser	Gln	Ile
Asn	Lys	Pro	Thr	Val	Pro	Thr	Ile	Asp	Asp	Ile	Ile	Lys	Ser	His	Leu
Val	Ser	Glu	Tyr	Asn	Arg	Asp 215	Glu	Thr	Val	Leu	Glu	Ser	Tyr	Arg	Thr
Asn	Ser	Leu	Arg	Lys	Ile	Asn	Ser	Asn	His	Gly 235	Ile	Asp	Ile	Arg	Ala 240
Asn	Ser	Leu	Phe	Thr	Glu	Gln	Glu	Leu	Leu	Asn	Ile	Tyr	Ser	Gln	Glu
Leu	Leu	Asn	Arg	Gly	Asn	Leu	Ala	Ala	Ala	Ser	Asp	Ile	Val	Arg	Leu
Leu	Ala	Leu	Lys	Asn	Phe	Gly	Gly	Val	Tyr	Leu	Ala	Val	Ala	Met	Leu
Pro	Gly	Ile	His	Ser	Asp	Leu	Phe	Lys	Thr	Ile	Ser	Arg	Pro	Ser	Ser
Ile	Gly	Leu	Asp	Arg	Trp	Glu	Met	Ile	Lys	Leu	Glu	Ala	Ile	Met	Lys 320
Tyr	Lys	Lys	Tyr	Ile	Asn	Asn	Tyr	Thr	Ser	Glu	Asn	Phe	Asp	Lys	Leu
Asp	Gln	Gln	Leu	Lys	Asp	Asn	Phe	Lys	Leu	Ile	Ile	Glu	Ser	Lys	Ser
Glu	Lys	Ser	Glu	Ile	Phe	Ser	Lys	Leu	Glu	Asn	Leu	Asn	Val	Ser	Asp
Leu	Glu	Ile	Lys	Ile	Ala	Phe	Ala	Leu	Gly	Ser	Val	Ile	Asn	Gln	Ala
Leu	Ile	Ser	Lys	Gln	Gly	Ser	Tyr	Leu	Thr	Asn	Leu	Val	Ile	Glu	Gln
Val	Lys	Asn	Arg	Tyr	Gln	Phe	Leu	Asn	Gln	His	Leu	Asn	Pro	Ala	Ile
Glu	Ser	Asp	Asn	Asn	Phe	Thr	Asp	Thr	Thr	Lys	Ile	Phe	His	Asp	Ser
Leu	Phe	Asn	Ser	Ala	Thr	Ala	Glu	Asn	Ser	Met	Phe	Leu	Thr	Lys	Ile
Ala	Pro	Tyr	Leu	Gln	Val	Gly	Phe	Met	Pro	Glu	Ala	Arg	Ser	Thr	Ile
Ser	Leu	Ser	Gly	Pro	Gly	Ala	Tyr	Ala	Ser	Ala	Tyr	Tyr	Asp	Phe	Ile
Asn	Leu	Gln	Glu	Asn	Thr	Ile	Glu	Lys	Thr	Leu	Lys	Ala	Ser	Asp	Leu
Ile	Glu	Phe	Lys	Phe	Pro	Glu	Asn	Asn	Leu	Ser	Gln	Leu	Thr	Glu	Gln
Glu	Ile	Asn	Ser	Leu	Trp	Ser	Phe	Asp	Gln	Ala	Ser	Ala	Lys	Tyr	Gln
Phe	Glu	Lys	Tyr	Val	Arg	Asp	Tyr	Thr	Gly	Gly	Ser	Leu	Ser	Glu	Asp

-continued

Asn Gly Val Asp Phe	Asn Lys Asn Thr Ala	Leu Asp Lys Asn Tyr Leu
545	550	555 560
Leu Asn Asn Lys Ile Pro Ser Asn Asn Val Glu Glu Ala Gly Ser Lys		
565	570	575
Asn Tyr Val His Tyr Ile Ile Gln Leu Gln Gly Asp Asp Ile Ser Tyr		
580	585	590
Glu Ala Thr Cys Asn Leu Phe Ser Lys Asn Pro Lys Asn Ser Ile Ile		
595	600	605
Ile Gln Arg Asn Met Asn Glu Ser Ala Lys Ser Tyr Phe Leu Ser Asp		
610	615	620
Asp Gly Glu Ser Ile Leu Glu Leu Asn Lys Tyr Arg Ile Pro Glu Arg		
625	630	635 640
Leu Lys Asn Lys Glu Lys Val Lys Val Thr Phe Ile Gly His Gly Lys		
645	650	655
Asp Glu Phe Asn Thr Ser Glu Phe Ala Arg Leu Ser Val Asp Ser Leu		
660	665	670
Ser Asn Glu Ile Ser Ser Phe Leu Asp Thr Ile Lys Leu Asp Ile Ser		
675	680	685
Pro Lys Asn Val Glu Val Asn Leu Leu Gly Ala Asn Met Phe Ser Tyr		
690	695	700
Asp Phe Asn Val Glu Glu Thr Tyr Pro Gly Lys Leu Leu Leu Ser Ile		
705	710	715 720
Met Asp Lys Ile Thr Ser Thr Leu Pro Asp Val Asn Lys Asn Ser Ile		
725	730	735
Thr Ile Gly Ala Asn Gln Tyr Glu Val Arg Ile Asn Ser Glu Gly Arg		
740	745	750
Lys Glu Leu Leu Ala His Ser Gly Lys Trp Ile Asn Lys Glu Glu Ala		
755	760	765
Ile Met Ser Asp Leu Ser Ser Lys Glu Tyr Ile Phe Phe Asp Ser Ile		
770	775	780
Asp Asn Lys Leu Lys Ala Lys Ser Lys Asn Ile Pro Gly Leu Ala Ser		
785	790	795 800
Ile Ser Glu Asp Ile Lys Thr Leu Leu Leu Asp Ala Ser Val Ser Pro		
805	810	815
Asp Thr Lys Phe Ile Leu Asn Asn Leu Lys Leu Asn Ile Glu Ser Ser		
820	825	830
Ile Gly Asp Tyr Ile Tyr Tyr Glu Lys Leu Glu Pro Val Lys Asn Ile		
835	840	845
Ile His Asn Ser Ile Asp Asp Leu Ile Asp Glu Phe Asn Leu Leu Glu		
850	855	860
Asn Val Ser Asp Glu Leu Tyr Glu Leu Lys Lys Leu Asn Asn Leu Asp		
865	870	875 880
Glu Lys Tyr Leu Ile Ser Phe Glu Asp Ile Ser Lys Asn Asn Ser Thr		
885	890	895
Tyr Ser Val Arg Phe Ile Asn Lys Ser Asn Gly Glu Ser Val Tyr Val		
900	905	910
Glu Thr Glu Lys Glu Ile Phe Ser Lys Tyr Ser Glu His Ile Thr Lys		
915	920	925
Glu Ile Ser Thr Ile Lys Asn Ser Ile Ile Thr Asp Val Asn Gly Asn		
930	935	940

-continued

Leu	Leu	Asp	Asn	Ile	Gln	Leu	Asp	His	Thr	Ser	Gln	Val	Asn	Thr	Leu	945	950	955	960
Asn	Ala	Ala	Phe	Phe	Ile	Gln	Ser	Leu	Ile	Asp	Tyr	Ser	Ser	Asn	Lys	965	970	975	
Asp	Val	Leu	Asn	Asp	Leu	Ser	Thr	Ser	Val	Lys	Val	Gln	Leu	Tyr	Ala	980	985	990	
Gln	Leu	Phe	Ser	Thr	Gly	Leu	Asn	Thr	Ile	Tyr	Asp	Ser	Ile	Gln	Leu	995	1000	1005	
Val	Asn	Leu	Ile	Ser	Asn	Ala	Val	Asn	Asp	Thr	Ile	Asn	Val	Leu		1010	1015	1020	
Pro	Thr	Ile	Thr	Glu	Gly	Ile	Pro	Ile	Val	Ser	Thr	Ile	Leu	Asp		1025	1030	1035	
Gly	Ile	Asn	Leu	Gly	Ala	Ala	Ile	Lys	Glu	Leu	Leu	Asp	Glu	His		1040	1045	1050	
Asp	Pro	Leu	Leu	Lys	Lys	Glu	Leu	Glu	Ala	Lys	Val	Gly	Val	Leu		1055	1060	1065	
Ala	Ile	Asn	Met	Ser	Leu	Ser	Ile	Ala	Ala	Thr	Val	Ala	Ser	Ile		1070	1075	1080	
Val	Gly	Ile	Gly	Ala	Glu	Val	Thr	Ile	Phe	Leu	Leu	Pro	Ile	Ala		1085	1090	1095	
Gly	Ile	Ser	Ala	Gly	Ile	Pro	Ser	Leu	Val	Asn	Asn	Glu	Leu	Ile		1100	1105	1110	
Leu	His	Asp	Lys	Ala	Thr	Ser	Val	Val	Asn	Tyr	Phe	Asn	His	Leu		1115	1120	1125	
Ser	Glu	Ser	Lys	Lys	Tyr	Gly	Pro	Leu	Lys	Thr	Glu	Asp	Asp	Lys		1130	1135	1140	
Ile	Leu	Val	Pro	Ile	Asp	Asp	Leu	Val	Ile	Ser	Glu	Ile	Asp	Phe		1145	1150	1155	
Asn	Asn	Asn	Ser	Ile	Lys	Leu	Gly	Thr	Cys	Asn	Ile	Leu	Ala	Met		1160	1165	1170	
Glu	Gly	Gly	Ser	Gly	His	Thr	Val	Thr	Gly	Asn	Ile	Asp	His	Phe		1175	1180	1185	
Phe	Ser	Ser	Pro	Ser	Ile	Ser	Ser	His	Ile	Pro	Ser	Leu	Ser	Ile		1190	1195	1200	
Tyr	Ser	Ala	Ile	Gly	Ile	Glu	Thr	Glu	Asn	Leu	Asp	Phe	Ser	Lys		1205	1210	1215	
Lys	Ile	Met	Met	Leu	Pro	Asn	Ala	Pro	Ser	Arg	Val	Phe	Trp	Trp		1220	1225	1230	
Glu	Thr	Gly	Ala	Val	Pro	Gly	Leu	Arg	Ser	Leu	Glu	Asn	Asp	Gly		1235	1240	1245	
Thr	Arg	Leu	Leu	Asp	Ser	Ile	Arg	Asp	Leu	Tyr	Pro	Gly	Lys	Phe		1250	1255	1260	
Tyr	Trp	Arg	Phe	Tyr	Ala	Phe	Phe	Asp	Tyr	Ala	Ile	Thr	Thr	Leu		1265	1270	1275	
Lys	Pro	Val	Tyr	Glu	Asp	Thr	Asn	Ile	Lys	Ile	Lys	Leu	Asp	Lys		1280	1285	1290	
Asp	Thr	Arg	Asn	Phe	Ile	Met	Pro	Thr	Ile	Thr	Thr	Asn	Glu	Ile		1295	1300	1305	
Arg	Asn	Lys	Leu	Ser	Tyr	Ser	Phe	Asp	Gly	Ala	Gly	Gly	Thr	Tyr		1310	1315	1320	
Ser	Leu	Leu	Leu	Ser	Ser	Tyr	Pro	Ile	Ser	Thr	Asn	Ile	Asn	Leu					

-continued

1325	1330	1335
Ser Lys Asp Asp Leu Trp Ile Phe Asn Ile Asp Asn Glu Val Arg		
1340	1345	1350
Glu Ile Ser Ile Glu Asn Gly Thr Ile Lys Lys Gly Lys Leu Ile		
1355	1360	1365
Lys Asp Val Leu Ser Lys Ile Asp Ile Asn Lys Asn Lys Leu Ile		
1370	1375	1380
Ile Gly Asn Gln Thr Ile Asp Phe Ser Gly Asp Ile Asp Asn Lys		
1385	1390	1395
Asp Arg Tyr Ile Phe Leu Thr Cys Glu Leu Asp Asp Lys Ile Ser		
1400	1405	1410
Leu Ile Ile Glu Ile Asn Leu Val Ala Lys Ser Tyr Ser Leu Leu		
1415	1420	1425
Leu Ser Gly Asp Lys Asn Tyr Leu Ile Ser Asn Leu Ser Asn Ile		
1430	1435	1440
Ile Glu Lys Ile Asn Thr Leu Gly Leu Asp Ser Lys Asn Ile Ala		
1445	1450	1455
Tyr Asn Tyr Thr Asp Glu Ser Asn Asn Lys Tyr Phe Gly Ala Ile		
1460	1465	1470
Ser Lys Thr Ser Gln Lys Ser Ile Ile His Tyr Lys Lys Asp Ser		
1475	1480	1485
Lys Asn Ile Leu Glu Phe Tyr Asn Asp Ser Thr Leu Glu Phe Asn		
1490	1495	1500
Ser Lys Asp Phe Ile Ala Glu Asp Ile Asn Val Phe Met Lys Asp		
1505	1510	1515
Asp Ile Asn Thr Ile Thr Gly Lys Tyr Tyr Val Asp Asn Asn Thr		
1520	1525	1530
Asp Lys Ser Ile Asp Phe Ser Ile Ser Leu Val Ser Lys Asn Gln		
1535	1540	1545
Val Lys Val Asn Gly Leu Tyr Leu Asn Glu Ser Val Tyr Ser Ser		
1550	1555	1560
Tyr Leu Asp Phe Val Lys Asn Ser Asp Gly His His Asn Thr Ser		
1565	1570	1575
Asn Phe Met Asn Leu Phe Leu Asp Asn Ile Ser Phe Trp Lys Leu		
1580	1585	1590
Phe Gly Phe Glu Asn Ile Asn Phe Val Ile Asp Lys Tyr Phe Thr		
1595	1600	1605
Leu Val Gly Lys Thr Asn Leu Gly Tyr Val Glu Phe Ile Cys Asp		
1610	1615	1620
Asn Asn Lys Asn Ile Asp Ile Tyr Phe Gly Glu Trp Lys Thr Ser		
1625	1630	1635
Ser Ser Lys Ser Thr Ile Phe Ser Gly Asn Gly Arg Asn Val Val		
1640	1645	1650
Val Glu Pro Ile Tyr Asn Pro Asp Thr Gly Glu Asp Ile Ser Thr		
1655	1660	1665
Ser Leu Asp Phe Ser Tyr Glu Pro Leu Tyr Gly Ile Asp Arg Tyr		
1670	1675	1680
Ile Asn Lys Val Leu Ile Ala Pro Asp Leu Tyr Thr Ser Leu Ile		
1685	1690	1695
Asn Ile Asn Thr Asn Tyr Tyr Ser Asn Glu Tyr Tyr Pro Glu Ile		
1700	1705	1710

-continued

Ile Val 1715	Leu Asn Pro Asn Thr 1720	Phe His Lys Lys Val 1725	Asn Ile Asn
Leu Asp 1730	Ser Ser Ser Phe Glu 1735	Tyr Lys Trp Ser Thr 1740	Glu Gly Ser
Asp Phe 1745	Ile Leu Val Arg Tyr 1750	Leu Glu Glu Ser Asn 1755	Lys Lys Ile
Leu Gln 1760	Lys Ile Arg Ile Lys 1765	Gly Ile Leu Ser Asn 1770	Thr Gln Ser
Phe Asn 1775	Lys Met Ser Ile Asp 1780	Phe Lys Asp Ile Lys 1785	Lys Leu Ser
Leu Gly 1790	Tyr Ile Met Ser Asn 1795	Phe Lys Ser Phe Asn 1800	Ser Glu Asn
Glu Leu 1805	Asp Arg Asp His Leu 1810	Gly Phe Lys Ile Ile 1815	Asp Asn Lys
Thr Tyr 1820	Tyr Tyr Asp Glu Asp 1825	Ser Lys Leu Val Lys 1830	Gly Leu Ile
Asn Ile 1835	Asn Asn Ser Leu Phe 1840	Tyr Phe Asp Pro Ile 1845	Glu Phe Asn
Leu Val 1850	Thr Gly Trp Gln Thr 1855	Ile Asn Gly Lys Lys 1860	Tyr Tyr Phe
Asp Ile 1865	Asn Thr Gly Ala Ala 1870	Leu Ile Ser Tyr Lys 1875	Ile Ile Asn
Gly Lys 1880	His Phe Tyr Phe Asn 1885	Asn Asp Gly Val Met 1890	Gln Leu Gly
Val Phe 1895	Lys Gly Pro Asp Gly 1900	Phe Glu Tyr Phe Ala 1905	Pro Ala Asn
Thr Gln 1910	Asn Asn Asn Ile Glu 1915	Gly Gln Ala Ile Val 1920	Tyr Gln Ser
Lys Phe 1925	Leu Thr Leu Asn Gly 1930	Lys Lys Tyr Tyr Phe 1935	Asp Asn Asp
Ser Lys 1940	Ala Val Thr Gly Trp 1945	Arg Ile Ile Asn Asn 1950	Glu Lys Tyr
Tyr Phe 1955	Asn Pro Asn Asn Ala 1960	Ile Ala Ala Val Gly 1965	Leu Gln Val
Ile Asp 1970	Asn Asn Lys Tyr Tyr 1975	Phe Asn Pro Asp Thr 1980	Ala Ile Ile
Ser Lys 1985	Gly Trp Gln Thr Val 1990	Asn Gly Ser Arg Tyr 1995	Tyr Phe Asp
Thr Asp 2000	Thr Ala Ile Ala Phe 2005	Asn Gly Tyr Lys Thr 2010	Ile Asp Gly
Lys His 2015	Phe Tyr Phe Asp Ser 2020	Asp Cys Val Val Lys 2025	Ile Gly Val
Phe Ser 2030	Thr Ser Asn Gly Phe 2035	Glu Tyr Phe Ala Pro 2040	Ala Asn Thr
Tyr Asn 2045	Asn Asn Ile Glu Gly 2050	Gln Ala Ile Val Tyr 2055	Gln Ser Lys
Phe Leu 2060	Thr Leu Asn Gly Lys 2065	Lys Tyr Tyr Phe Asp 2070	Asn Asn Ser
Lys Ala 2075	Val Thr Gly Trp Gln 2080	Thr Ile Asp Ser Lys 2085	Lys Tyr Tyr

-continued

Phe	Asn	Thr	Asn	Thr	Ala	Glu	Ala	Ala	Thr	Gly	Trp	Gln	Thr	Ile
2090						2095					2100			
Asp	Gly	Lys	Lys	Tyr	Tyr	Phe	Asn	Thr	Asn	Thr	Ala	Glu	Ala	Ala
2105						2110					2115			
Thr	Gly	Trp	Gln	Thr	Ile	Asp	Gly	Lys	Lys	Tyr	Tyr	Phe	Asn	Thr
2120						2125					2130			
Asn	Thr	Ala	Ile	Ala	Ser	Thr	Gly	Tyr	Thr	Ile	Ile	Asn	Gly	Lys
2135						2140					2145			
His	Phe	Tyr	Phe	Asn	Thr	Asp	Gly	Ile	Met	Gln	Ile	Gly	Val	Phe
2150						2155					2160			
Lys	Gly	Pro	Asn	Gly	Phe	Glu	Tyr	Phe	Ala	Pro	Ala	Asn	Thr	Asp
2165						2170					2175			
Ala	Asn	Asn	Ile	Glu	Gly	Gln	Ala	Ile	Leu	Tyr	Gln	Asn	Glu	Phe
2180						2185					2190			
Leu	Thr	Leu	Asn	Gly	Lys	Lys	Tyr	Tyr	Phe	Gly	Ser	Asp	Ser	Lys
2195						2200					2205			
Ala	Val	Thr	Gly	Trp	Arg	Ile	Ile	Asn	Asn	Lys	Lys	Tyr	Tyr	Phe
2210						2215					2220			
Asn	Pro	Asn	Asn	Ala	Ile	Ala	Ala	Ile	His	Leu	Cys	Thr	Ile	Asn
2225						2230					2235			
Asn	Asp	Lys	Tyr	Tyr	Phe	Ser	Tyr	Asp	Gly	Ile	Leu	Gln	Asn	Gly
2240						2245					2250			
Tyr	Ile	Thr	Ile	Glu	Arg	Asn	Asn	Phe	Tyr	Phe	Asp	Ala	Asn	Asn
2255						2260					2265			
Glu	Ser	Lys	Met	Val	Thr	Gly	Val	Phe	Lys	Gly	Pro	Asn	Gly	Phe
2270						2275					2280			
Glu	Tyr	Phe	Ala	Pro	Ala	Asn	Thr	His	Asn	Asn	Asn	Ile	Glu	Gly
2285						2290					2295			
Gln	Ala	Ile	Val	Tyr	Gln	Asn	Lys	Phe	Leu	Thr	Leu	Asn	Gly	Lys
2300						2305					2310			
Lys	Tyr	Tyr	Phe	Asp	Asn	Asp	Ser	Lys	Ala	Val	Thr	Gly	Trp	Gln
2315						2320					2325			
Thr	Ile	Asp	Gly	Lys	Lys	Tyr	Tyr	Phe	Asn	Leu	Asn	Thr	Ala	Glu
2330						2335					2340			
Ala	Ala	Thr	Gly	Trp	Gln	Thr	Ile	Asp	Gly	Lys	Lys	Tyr	Tyr	Phe
2345						2350					2355			
Asn	Leu	Asn	Thr	Ala	Glu	Ala	Ala	Thr	Gly	Trp	Gln	Thr	Ile	Asp
2360						2365					2370			
Gly	Lys	Lys	Tyr	Tyr	Phe	Asn	Thr	Asn	Thr	Phe	Ile	Ala	Ser	Thr
2375						2380					2385			
Gly	Tyr	Thr	Ser	Ile	Asn	Gly	Lys	His	Phe	Tyr	Phe	Asn	Thr	Asp
2390						2395					2400			
Gly	Ile	Met	Gln	Ile	Gly	Val	Phe	Lys	Gly	Pro	Asn	Gly	Phe	Glu
2405						2410					2415			
Tyr	Phe	Ala	Pro	Ala	Asn	Thr	His	Asn	Asn	Asn	Ile	Glu	Gly	Gln
2420						2425					2430			
Ala	Ile	Leu	Tyr	Gln	Asn	Lys	Phe	Leu	Thr	Leu	Asn	Gly	Lys	Lys
2435						2440					2445			
Tyr	Tyr	Phe	Gly	Ser	Asp	Ser	Lys	Ala	Val	Thr	Gly	Leu	Arg	Thr
2450						2455					2460			
Ile	Asp	Gly	Lys	Lys	Tyr	Tyr	Phe	Asn	Thr	Asn	Thr	Ala	Val	Ala

-continued

2465	2470	2475
Val Thr Gly Trp Gln Thr Ile Asn Gly Lys Lys Tyr Tyr Phe Asn		
2480	2485	2490
Thr Asn Thr Ser Ile Ala Ser Thr Gly Tyr Thr Ile Ile Ser Gly		
2495	2500	2505
Lys His Phe Tyr Phe Asn Thr Asp Gly Ile Met Gln Ile Gly Val		
2510	2515	2520
Phe Lys Gly Pro Asp Gly Phe Glu Tyr Phe Ala Pro Ala Asn Thr		
2525	2530	2535
Asp Ala Asn Asn Ile Glu Gly Gln Ala Ile Arg Tyr Gln Asn Arg		
2540	2545	2550
Phe Leu Tyr Leu His Asp Asn Ile Tyr Tyr Phe Gly Asn Asn Ser		
2555	2560	2565
Lys Ala Ala Thr Gly Trp Val Thr Ile Asp Gly Asn Arg Tyr Tyr		
2570	2575	2580
Phe Glu Pro Asn Thr Ala Met Gly Ala Asn Gly Tyr Lys Thr Ile		
2585	2590	2595
Asp Asn Lys Asn Phe Tyr Phe Arg Asn Gly Leu Pro Gln Ile Gly		
2600	2605	2610
Val Phe Lys Gly Ser Asn Gly Phe Glu Tyr Phe Ala Pro Ala Asn		
2615	2620	2625
Thr Asp Ala Asn Asn Ile Glu Gly Gln Ala Ile Arg Tyr Gln Asn		
2630	2635	2640
Arg Phe Leu His Leu Leu Gly Lys Ile Tyr Tyr Phe Gly Asn Asn		
2645	2650	2655
Ser Lys Ala Val Thr Gly Trp Gln Thr Ile Asn Gly Lys Val Tyr		
2660	2665	2670
Tyr Phe Met Pro Asp Thr Ala Met Ala Ala Ala Gly Gly Leu Phe		
2675	2680	2685
Glu Ile Asp Gly Val Ile Tyr Phe Phe Gly Val		
2690	2695	

<210> SEQ ID NO 6

<211> LENGTH: 2365

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 6

Ser Leu Val Asn Arg Lys Gln Leu Glu Lys Met Ala Asn Val Arg Phe		
1	5	10 15
Arg Thr Gln Glu Asp Glu Tyr Val Ala Ile Leu Asp Ala Leu Glu Glu		
20	25	30
Tyr His Asn Met Ser Glu Asn Thr Val Val Glu Lys Tyr Leu Lys Leu		
35	40	45
Lys Asp Ile Asn Ser Leu Thr Asp Ile Tyr Ile Asp Thr Tyr Lys Lys		
50	55	60
Ser Gly Arg Asn Lys Ala Leu Lys Lys Phe Lys Glu Tyr Leu Val Thr		
65	70	75 80
Glu Val Leu Glu Leu Lys Asn Asn Asn Thr Pro Val Glu Lys Asn		
85	90	95
Leu His Phe Val Trp Ile Gly Gly Gln Ile Asn Asp Thr Ala Ile Asn		

-continued

100							105					110				
Tyr	Ile	Asn	Gln	Trp	Lys	Asp	Val	Asn	Ser	Asp	Tyr	Asn	Val	Asn	Val	
115							120					125				
Phe	Tyr	Asp	Ser	Asn	Ala	Phe	Leu	Ile	Asn	Thr	Leu	Lys	Lys	Thr	Val	
130							135					140				
Val	Glu	Ser	Ala	Ile	Asn	Asp	Thr	Leu	Glu	Ser	Phe	Arg	Glu	Asn	Leu	
145	150							155					160			
Asn	Asp	Pro	Arg	Phe	Asp	Tyr	Asn	Lys	Phe	Phe	Arg	Lys	Arg	Met	Glu	
165							170					175				
Ile	Ile	Tyr	Asp	Lys	Gln	Lys	Asn	Phe	Ile	Asn	Tyr	Tyr	Lys	Ala	Gln	
180							185					190				
Arg	Glu	Glu	Asn	Pro	Glu	Leu	Ile	Ile	Asp	Asp	Ile	Val	Lys	Thr	Tyr	
195							200					205				
Leu	Ser	Asn	Glu	Tyr	Ser	Lys	Glu	Ile	Asp	Glu	Leu	Asn	Thr	Tyr	Ile	
210							215					220				
Glu	Glu	Ser	Leu	Asn	Lys	Ile	Thr	Gln	Asn	Ser	Gly	Asn	Asp	Val	Arg	
225	230							235					240			
Asn	Phe	Glu	Glu	Phe	Lys	Asn	Gly	Glu	Ser	Phe	Asn	Leu	Tyr	Glu	Gln	
245							250					255				
Glu	Leu	Val	Glu	Arg	Trp	Asn	Leu	Ala	Ala	Ala	Ser	Asp	Ile	Leu	Arg	
260							265					270				
Ile	Ser	Ala	Leu	Lys	Glu	Ile	Gly	Gly	Met	Tyr	Leu	Ala	Val	Ala	Met	
275							280					285				
Leu	Pro	Gly	Ile	Gln	Pro	Asp	Leu	Phe	Glu	Ser	Ile	Glu	Lys	Pro	Ser	
290							295					300				
Ser	Val	Thr	Val	Asp	Phe	Trp	Glu	Met	Thr	Lys	Leu	Glu	Ala	Ile	Met	
305	310							315					320			
Lys	Tyr	Lys	Glu	Tyr	Ile	Pro	Glu	Tyr	Thr	Ser	Glu	His	Phe	Asp	Met	
325							330					335				
Leu	Asp	Glu	Glu	Val	Gln	Ser	Ser	Phe	Glu	Ser	Val	Leu	Ala	Ser	Lys	
340							345					350				
Ser	Asp	Lys	Ser	Glu	Ile	Phe	Ser	Ser	Leu	Gly	Asp	Met	Glu	Ala	Ser	
355							360					365				
Pro	Leu	Glu	Val	Lys	Ile	Ala	Phe	Asn	Ser	Lys	Gly	Ile	Ile	Asn	Gln	
370							375					380				
Gly	Leu	Ile	Ser	Val	Lys	Asp	Ser	Tyr	Cys	Ser	Asn	Leu	Ile	Val	Lys	
385	390							395					400			
Gln	Ile	Glu	Asn	Arg	Tyr	Lys	Ile	Leu	Asn	Asn	Ser	Leu	Asn	Pro	Ala	
405							410					415				
Ile	Ser	Glu	Asp	Asn	Asp	Phe	Asn	Thr	Thr	Thr	Asn	Thr	Phe	Ile	Asp	
420							425					430				
Ser	Ile	Met	Ala	Glu	Ala	Asn	Ala	Asp	Asn	Gly	Arg	Phe	Met	Met	Glu	
435							440					445				
Leu	Gly	Lys	Tyr	Leu	Arg	Val	Gly	Phe	Phe	Pro	Asp	Val	Lys	Thr	Thr	
450							455					460				
Ile	Asn	Leu	Ser	Gly	Pro	Glu	Ala	Tyr	Ala	Ala	Ala	Tyr	Gln	Asp	Leu	
465	470							475					480			
Leu	Met	Phe	Lys	Glu	Gly	Ser	Met	Asn	Ile	His	Leu	Ile	Glu	Ala	Asp	
485							490					495				
Leu	Arg	Asn	Phe	Glu	Ile	Ser	Lys	Thr	Asn	Ile	Ser	Gln	Ser	Thr	Glu	
500							505					510				

-continued

Gln	Glu	Met	Ala	Ser	Leu	Trp	Ser	Phe	Asp	Asp	Ala	Arg	Ala	Lys	Ala
	515						520					525			
Gln	Phe	Glu	Glu	Tyr	Lys	Arg	Asn	Tyr	Phe	Glu	Gly	Ser	Leu	Gly	Glu
	530					535					540				
Asp	Asp	Asn	Leu	Asp	Phe	Ser	Gln	Asn	Ile	Val	Val	Asp	Lys	Glu	Tyr
545					550					555					560
Leu	Leu	Glu	Lys	Ile	Ser	Ser	Leu	Ala	Arg	Ser	Ser	Glu	Arg	Gly	Tyr
			565						570					575	
Ile	His	Tyr	Ile	Val	Gln	Leu	Gln	Gly	Asp	Lys	Ile	Ser	Tyr	Glu	Ala
			580					585					590		
Ala	Cys	Asn	Leu	Phe	Ala	Lys	Thr	Pro	Tyr	Asp	Ser	Val	Leu	Phe	Gln
		595					600					605			
Lys	Asn	Ile	Glu	Asp	Ser	Glu	Ile	Ala	Tyr	Tyr	Tyr	Asn	Pro	Gly	Asp
	610					615					620				
Gly	Glu	Ile	Gln	Glu	Ile	Asp	Lys	Tyr	Lys	Ile	Pro	Ser	Ile	Ile	Ser
625					630					635					640
Asp	Arg	Pro	Lys	Ile	Lys	Leu	Thr	Phe	Ile	Gly	His	Gly	Lys	Asp	Glu
			645					650						655	
Phe	Asn	Thr	Asp	Ile	Phe	Ala	Gly	Phe	Asp	Val	Asp	Ser	Leu	Ser	Thr
			660					665					670		
Glu	Ile	Glu	Ala	Ala	Ile	Asp	Leu	Ala	Lys	Glu	Asp	Ile	Ser	Pro	Lys
		675					680				685				
Ser	Ile	Glu	Ile	Asn	Leu	Leu	Gly	Ala	Asn	Met	Phe	Ser	Tyr	Ser	Ile
	690				695					700					
Asn	Val	Glu	Glu	Thr	Tyr	Pro	Gly	Lys	Leu	Leu	Leu	Lys	Val	Lys	Asp
705					710					715					720
Lys	Ile	Ser	Glu	Leu	Met	Pro	Ser	Ile	Ser	Gln	Asp	Ser	Ile	Ile	Val
			725					730						735	
Ser	Ala	Asn	Gln	Tyr	Glu	Val	Arg	Ile	Asn	Ser	Glu	Gly	Arg	Arg	Glu
			740				745						750		
Leu	Leu	Asp	His	Ser	Gly	Glu	Trp	Ile	Asn	Lys	Glu	Glu	Ser	Ile	Ile
		755				760					765				
Lys	Asp	Ile	Ser	Ser	Lys	Glu	Tyr	Ile	Ser	Phe	Asn	Pro	Lys	Glu	Asn
	770				775					780					
Lys	Ile	Thr	Val	Lys	Ser	Lys	Asn	Leu	Pro	Glu	Leu	Ser	Thr	Leu	Leu
785					790				795						800
Gln	Glu	Ile	Arg	Asn	Asn	Ser	Asn	Ser	Ser	Asp	Ile	Glu	Leu	Glu	Glu
			805					810						815	
Lys	Val	Met	Leu	Thr	Glu	Cys	Glu	Ile	Asn	Val	Ile	Ser	Asn	Ile	Asp
		820					825						830		
Thr	Gln	Ile	Val	Glu	Glu	Arg	Ile	Glu	Glu	Ala	Lys	Asn	Leu	Thr	Ser
	835					840						845			
Asp	Ser	Ile	Asn	Tyr	Ile	Lys	Asp	Glu	Phe	Lys	Leu	Ile	Glu	Ser	Ile
	850				855						860				
Ser	Asp	Ala	Leu	Cys	Asp	Leu	Lys	Gln	Gln	Asn	Glu	Leu	Glu	Asp	Ser
865					870					875					880
His	Phe	Ile	Ser	Phe	Glu	Asp	Ile	Ser	Glu	Thr	Asp	Glu	Gly	Phe	Ser
			885					890						895	
Ile	Arg	Phe	Ile	Asn	Lys	Glu	Thr	Gly	Glu	Ser	Ile	Phe	Val	Glu	Thr
			900					905					910		

-continued

Glu	Lys	Thr	Ile	Phe	Ser	Glu	Tyr	Ala	Asn	His	Ile	Thr	Glu	Glu	Ile
	915						920					925			
Ser	Lys	Ile	Lys	Gly	Thr	Ile	Phe	Asp	Thr	Val	Asn	Gly	Lys	Leu	Val
	930					935					940				
Lys	Lys	Val	Asn	Leu	Asp	Thr	Thr	His	Glu	Val	Asn	Thr	Leu	Asn	Ala
945				950					955					960	
Ala	Phe	Phe	Ile	Gln	Ser	Leu	Ile	Glu	Tyr	Asn	Ser	Ser	Lys	Glu	Ser
			965						970					975	
Leu	Ser	Asn	Leu	Ser	Val	Ala	Met	Lys	Val	Gln	Val	Tyr	Ala	Gln	Leu
		980						985						990	
Phe	Ser	Thr	Gly	Leu	Asn	Thr	Ile	Thr	Asp	Ala	Ala	Lys	Val	Val	Glu
	995						1000					1005			
Leu	Val	Ser	Thr	Ala	Leu	Asp	Glu	Thr	Ile	Asp	Leu	Leu	Pro	Thr	
	1010					1015					1020				
Leu	Ser	Glu	Gly	Leu	Pro	Ile	Ile	Ala	Thr	Ile	Ile	Asp	Gly	Val	
	1025					1030					1035				
Ser	Leu	Gly	Ala	Ala	Ile	Lys	Glu	Leu	Ser	Glu	Thr	Ser	Asp	Pro	
	1040					1045					1050				
Leu	Leu	Arg	Gln	Glu	Ile	Glu	Ala	Lys	Ile	Gly	Ile	Met	Ala	Val	
	1055					1060					1065				
Asn	Leu	Thr	Thr	Ala	Thr	Thr	Ala	Ile	Ile	Thr	Ser	Ser	Leu	Gly	
	1070					1075					1080				
Ile	Ala	Ser	Gly	Phe	Ser	Ile	Leu	Leu	Val	Pro	Leu	Ala	Gly	Ile	
	1085					1090					1095				
Ser	Ala	Gly	Ile	Pro	Ser	Leu	Val	Asn	Asn	Glu	Leu	Val	Leu	Arg	
	1100					1105					1110				
Asp	Lys	Ala	Thr	Lys	Val	Val	Asp	Tyr	Phe	Lys	His	Val	Ser	Leu	
	1115					1120					1125				
Val	Glu	Thr	Glu	Gly	Val	Phe	Thr	Leu	Leu	Asp	Asp	Lys	Ile	Met	
	1130					1135					1140				
Met	Pro	Gln	Asp	Asp	Leu	Val	Ile	Ser	Glu	Ile	Asp	Phe	Asn	Asn	
	1145					1150					1155				
Asn	Ser	Ile	Val	Leu	Gly	Lys	Cys	Glu	Ile	Trp	Arg	Met	Glu	Gly	
	1160					1165					1170				
Gly	Ser	Gly	His	Thr	Val	Thr	Asp	Asp	Ile	Asp	His	Phe	Phe	Ser	
	1175					1180					1185				
Ala	Pro	Ser	Ile	Thr	Tyr	Arg	Glu	Pro	His	Leu	Ser	Ile	Tyr	Asp	
	1190					1195					1200				
Val	Leu	Glu	Val	Gln	Lys	Glu	Glu	Leu	Asp	Leu	Ser	Lys	Asp	Leu	
	1205					1210					1215				
Met	Val	Leu	Pro	Asn	Ala	Pro	Asn	Arg	Val	Phe	Ala	Trp	Glu	Thr	
	1220					1225					1230				
Gly	Trp	Thr	Pro	Gly	Leu	Arg	Ser	Leu	Glu	Asn	Asp	Gly	Thr	Lys	
	1235					1240					1245				
Leu	Leu	Asp	Arg	Ile	Arg	Asp	Asn	Tyr	Glu	Gly	Glu	Phe	Tyr	Trp	
	1250					1255					1260				
Arg	Tyr	Phe	Ala	Phe	Ile	Ala	Asp	Ala	Leu	Ile	Thr	Thr	Leu	Lys	
	1265					1270					1275				
Pro	Arg	Tyr	Glu	Asp	Thr	Asn	Ile	Arg	Ile	Asn	Leu	Asp	Ser	Asn	
	1280					1285					1290				
Thr	Arg	Ser	Phe	Ile	Val	Pro	Ile	Ile	Thr	Thr	Glu	Tyr	Ile	Arg	

-continued

1295	1300	1305
Glu Lys Leu Ser Tyr Ser Phe Tyr Gly Ser Gly Gly Thr Tyr Ala 1310 1315 1320		
Leu Ser Leu Ser Gln Tyr Asn Met Gly Ile Asn Ile Glu Leu Ser 1325 1330 1335		
Glu Ser Asp Val Trp Ile Ile Asp Val Asp Asn Val Val Arg Asp 1340 1345 1350		
Val Thr Ile Glu Ser Asp Lys Ile Lys Lys Gly Asp Leu Ile Glu 1355 1360 1365		
Gly Ile Leu Ser Thr Leu Ser Ile Glu Glu Asn Lys Ile Ile Leu 1370 1375 1380		
Asn Ser His Glu Ile Asn Phe Ser Gly Glu Val Asn Gly Ser Asn 1385 1390 1395		
Gly Phe Val Ser Leu Thr Phe Ser Ile Leu Glu Gly Ile Asn Ala 1400 1405 1410		
Ile Ile Glu Val Asp Leu Leu Ser Lys Ser Tyr Lys Leu Leu Ile 1415 1420 1425		
Ser Gly Glu Leu Lys Ile Leu Met Leu Asn Ser Asn His Ile Gln 1430 1435 1440		
Gln Lys Ile Asp Tyr Ile Gly Phe Asn Ser Glu Leu Gln Lys Asn 1445 1450 1455		
Ile Pro Tyr Ser Phe Val Asp Ser Glu Gly Lys Glu Asn Gly Phe 1460 1465 1470		
Ile Asn Gly Ser Thr Lys Glu Gly Leu Phe Val Ser Glu Leu Pro 1475 1480 1485		
Asp Val Val Leu Ile Ser Lys Val Tyr Met Asp Asp Ser Lys Pro 1490 1495 1500		
Ser Phe Gly Tyr Tyr Ser Asn Asn Leu Lys Asp Val Lys Val Ile 1505 1510 1515		
Thr Lys Asp Asn Val Asn Ile Leu Thr Gly Tyr Tyr Leu Lys Asp 1520 1525 1530		
Asp Ile Lys Ile Ser Leu Ser Leu Thr Leu Gln Asp Glu Lys Thr 1535 1540 1545		
Ile Lys Leu Asn Ser Val His Leu Asp Glu Ser Gly Val Ala Glu 1550 1555 1560		
Ile Leu Lys Phe Met Asn Arg Lys Gly Asn Thr Asn Thr Ser Asp 1565 1570 1575		
Ser Leu Met Ser Phe Leu Glu Ser Met Asn Ile Lys Ser Ile Phe 1580 1585 1590		
Val Asn Phe Leu Gln Ser Asn Ile Lys Phe Ile Leu Asp Ala Asn 1595 1600 1605		
Phe Ile Ile Ser Gly Thr Thr Ser Ile Gly Gln Phe Glu Phe Ile 1610 1615 1620		
Cys Asp Glu Asn Asp Asn Ile Gln Pro Tyr Phe Ile Lys Phe Asn 1625 1630 1635		
Thr Leu Glu Thr Asn Tyr Thr Leu Tyr Val Gly Asn Arg Gln Asn 1640 1645 1650		
Met Ile Val Glu Pro Asn Tyr Asp Leu Asp Asp Ser Gly Asp Ile 1655 1660 1665		
Ser Ser Thr Val Ile Asn Phe Ser Gln Lys Tyr Leu Tyr Gly Ile 1670 1675 1680		

-continued

Asp Ser Cys Val Asn Lys Val Val Ile Ser Pro Asn Ile Tyr Thr 1685 1690 1695
Asp Glu Ile Asn Ile Thr Pro Val Tyr Glu Thr Asn Asn Thr Tyr 1700 1705 1710
Pro Glu Val Ile Val Leu Asp Ala Asn Tyr Ile Asn Glu Lys Ile 1715 1720 1725
Asn Val Asn Ile Asn Asp Leu Ser Ile Arg Tyr Val Trp Ser Asn 1730 1735 1740
Asp Gly Asn Asp Phe Ile Leu Met Ser Thr Ser Glu Glu Asn Lys 1745 1750 1755
Val Ser Gln Val Lys Ile Arg Phe Val Asn Val Phe Lys Asp Lys 1760 1765 1770
Thr Leu Ala Asn Lys Leu Ser Phe Asn Phe Ser Asp Lys Gln Asp 1775 1780 1785
Val Pro Val Ser Glu Ile Ile Leu Ser Phe Thr Pro Ser Tyr Tyr 1790 1795 1800
Glu Asp Gly Leu Ile Gly Tyr Asp Leu Gly Leu Val Ser Leu Tyr 1805 1810 1815
Asn Glu Lys Phe Tyr Ile Asn Asn Phe Gly Met Met Val Ser Gly 1820 1825 1830
Leu Ile Tyr Ile Asn Asp Ser Leu Tyr Tyr Phe Lys Pro Pro Val 1835 1840 1845
Asn Asn Leu Ile Thr Gly Phe Val Thr Val Gly Asp Asp Lys Tyr 1850 1855 1860
Tyr Phe Asn Pro Ile Asn Gly Gly Ala Ala Ser Ile Gly Glu Thr 1865 1870 1875
Ile Ile Asp Asp Lys Asn Tyr Tyr Phe Asn Gln Ser Gly Val Leu 1880 1885 1890
Gln Thr Gly Val Phe Ser Thr Glu Asp Gly Phe Lys Tyr Phe Ala 1895 1900 1905
Pro Ala Asn Thr Leu Asp Glu Asn Leu Glu Gly Glu Ala Ile Asp 1910 1915 1920
Phe Thr Gly Lys Leu Ile Ile Asp Glu Asn Ile Tyr Tyr Phe Asp 1925 1930 1935
Asp Asn Tyr Arg Gly Ala Val Glu Trp Lys Glu Leu Asp Gly Glu 1940 1945 1950
Met His Tyr Phe Ser Pro Glu Thr Gly Lys Ala Phe Lys Gly Leu 1955 1960 1965
Asn Gln Ile Gly Asp Tyr Lys Tyr Tyr Phe Asn Ser Asp Gly Val 1970 1975 1980
Met Gln Lys Gly Phe Val Ser Ile Asn Asp Asn Lys His Tyr Phe 1985 1990 1995
Asp Asp Ser Gly Val Met Lys Val Gly Tyr Thr Glu Ile Asp Gly 2000 2005 2010
Lys His Phe Tyr Phe Ala Glu Asn Gly Glu Met Gln Ile Gly Val 2015 2020 2025
Phe Asn Thr Glu Asp Gly Phe Lys Tyr Phe Ala His His Asn Glu 2030 2035 2040
Asp Leu Gly Asn Glu Glu Gly Glu Glu Ile Ser Tyr Ser Gly Ile 2045 2050 2055

-continued

Leu	Asn	Phe	Asn	Asn	Lys	Ile	Tyr	Tyr	Phe	Asp	Asp	Ser	Phe	Thr
2060						2065					2070			
Ala	Val	Val	Gly	Trp	Lys	Asp	Leu	Glu	Asp	Gly	Ser	Lys	Tyr	Tyr
2075						2080					2085			
Phe	Asp	Glu	Asp	Thr	Ala	Glu	Ala	Tyr	Ile	Gly	Leu	Ser	Leu	Ile
2090						2095					2100			
Asn	Asp	Gly	Gln	Tyr	Tyr	Phe	Asn	Asp	Asp	Gly	Ile	Met	Gln	Val
2105						2110					2115			
Gly	Phe	Val	Thr	Ile	Asn	Asp	Lys	Val	Phe	Tyr	Phe	Ser	Asp	Ser
2120						2125					2130			
Gly	Ile	Ile	Glu	Ser	Gly	Val	Gln	Asn	Ile	Asp	Asp	Asn	Tyr	Phe
2135						2140					2145			
Tyr	Ile	Asp	Asp	Asn	Gly	Ile	Val	Gln	Ile	Gly	Val	Phe	Asp	Thr
2150						2155					2160			
Ser	Asp	Gly	Tyr	Lys	Tyr	Phe	Ala	Pro	Ala	Asn	Thr	Val	Asn	Asp
2165						2170					2175			
Asn	Ile	Tyr	Gly	Gln	Ala	Val	Glu	Tyr	Ser	Gly	Leu	Val	Arg	Val
2180						2185					2190			
Gly	Glu	Asp	Val	Tyr	Tyr	Phe	Gly	Glu	Thr	Tyr	Thr	Ile	Glu	Thr
2195						2200					2205			
Gly	Trp	Ile	Tyr	Asp	Met	Glu	Asn	Glu	Ser	Asp	Lys	Tyr	Tyr	Phe
2210						2215					2220			
Asn	Pro	Glu	Thr	Lys	Lys	Ala	Cys	Lys	Gly	Ile	Asn	Leu	Ile	Asp
2225						2230					2235			
Asp	Ile	Lys	Tyr	Tyr	Phe	Asp	Glu	Lys	Gly	Ile	Met	Arg	Thr	Gly
2240						2245					2250			
Leu	Ile	Ser	Phe	Glu	Asn	Asn	Asn	Tyr	Tyr	Phe	Asn	Glu	Asn	Gly
2255						2260					2265			
Glu	Met	Gln	Phe	Gly	Tyr	Ile	Asn	Ile	Glu	Asp	Lys	Met	Phe	Tyr
2270						2275					2280			
Phe	Gly	Glu	Asp	Gly	Val	Met	Gln	Ile	Gly	Val	Phe	Asn	Thr	Pro
2285						2290					2295			
Asp	Gly	Phe	Lys	Tyr	Phe	Ala	His	Gln	Asn	Thr	Leu	Asp	Glu	Asn
2300						2305					2310			
Phe	Glu	Gly	Glu	Ser	Ile	Asn	Tyr	Thr	Gly	Trp	Leu	Asp	Leu	Asp
2315						2320					2325			
Glu	Lys	Arg	Tyr	Tyr	Phe	Thr	Asp	Glu	Tyr	Ile	Ala	Ala	Thr	Gly
2330						2335					2340			
Ser	Val	Ile	Ile	Asp	Gly	Glu	Glu	Tyr	Tyr	Phe	Asp	Pro	Asp	Thr
2345						2350					2355			
Ala	Gln	Leu	Val	Ile	Ser	Glu								
2360						2365								

<210> SEQ ID NO 7

<211> LENGTH: 574

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 7

Met	Glu	Leu	Leu	Ile	Leu	Lys	Ala	Asn	Ala	Ile	Thr	Thr	Ile	Leu	Thr
1						5					10				15

-continued

Ala	Val	Thr	Phe	Cys	Phe	Ala	Ser	Gly	Gln	Asn	Ile	Thr	Glu	Glu	Phe
			20					25					30		
Tyr	Gln	Ser	Thr	Cys	Ser	Ala	Val	Ser	Lys	Gly	Tyr	Leu	Ser	Ala	Leu
	35						40					45			
Arg	Thr	Gly	Trp	Tyr	Thr	Ser	Val	Ile	Thr	Ile	Glu	Leu	Ser	Asn	Ile
	50					55					60				
Lys	Glu	Asn	Lys	Cys	Asn	Gly	Thr	Asp	Ala	Lys	Val	Lys	Leu	Ile	Lys
65					70					75					80
Gln	Glu	Leu	Asp	Lys	Tyr	Lys	Asn	Ala	Val	Thr	Glu	Leu	Gln	Leu	Leu
				85					90					95	
Met	Gln	Ser	Thr	Pro	Pro	Thr	Asn	Asn	Arg	Ala	Arg	Arg	Glu	Leu	Pro
			100					105					110		
Arg	Phe	Met	Asn	Tyr	Thr	Leu	Asn	Asn	Ala	Lys	Lys	Thr	Asn	Val	Thr
		115					120					125			
Leu	Ser	Lys	Lys	Arg	Lys	Arg	Arg	Phe	Leu	Gly	Phe	Leu	Leu	Gly	Val
	130					135					140				
Gly	Ser	Ala	Ile	Ala	Ser	Gly	Val	Ala	Val	Ser	Lys	Val	Leu	His	Leu
145					150					155					160
Glu	Gly	Glu	Val	Asn	Lys	Ile	Lys	Ser	Ala	Leu	Leu	Ser	Thr	Asn	Lys
				165					170					175	
Ala	Val	Val	Ser	Leu	Ser	Asn	Gly	Val	Ser	Val	Leu	Thr	Ser	Lys	Val
			180					185					190		
Leu	Asp	Leu	Lys	Asn	Tyr	Ile	Asp	Lys	Gln	Leu	Leu	Pro	Ile	Val	Asn
	195						200					205			
Lys	Gln	Ser	Cys	Ser	Ile	Ser	Asn	Ile	Glu	Thr	Val	Ile	Glu	Phe	Gln
	210					215					220				
Gln	Lys	Asn	Asn	Arg	Leu	Leu	Glu	Ile	Thr	Arg	Glu	Phe	Ser	Val	Asn
225					230					235					240
Ala	Gly	Val	Thr	Thr	Pro	Val	Ser	Thr	Tyr	Met	Leu	Thr	Asn	Ser	Glu
				245					250					255	
Leu	Leu	Ser	Leu	Ile	Asn	Asp	Met	Pro	Ile	Thr	Asn	Asp	Gln	Lys	Lys
			260				265						270		
Leu	Met	Ser	Asn	Asn	Val	Gln	Ile	Val	Arg	Gln	Gln	Ser	Tyr	Ser	Ile
	275						280					285			
Met	Ser	Ile	Ile	Lys	Glu	Glu	Val	Leu	Ala	Tyr	Val	Val	Gln	Leu	Pro
	290					295					300				
Leu	Tyr	Gly	Val	Ile	Asp	Thr	Pro	Cys	Trp	Lys	Leu	His	Thr	Ser	Pro
305					310					315					320
Leu	Cys	Thr	Thr	Asn	Thr	Lys	Glu	Gly	Ser	Asn	Ile	Cys	Leu	Thr	Arg
				325					330					335	
Thr	Asp	Arg	Gly	Trp	Tyr	Cys	Asp	Asn	Ala	Gly	Ser	Val	Ser	Phe	Phe
			340					345					350		
Pro	Gln	Ala	Glu	Thr	Cys	Lys	Val	Gln	Ser	Asn	Arg	Val	Phe	Cys	Asp
	355						360					365			
Thr	Met	Asn	Ser	Leu	Thr	Leu	Pro	Ser	Glu	Ile	Asn	Leu	Cys	Asn	Val
	370					375					380				
Asp	Ile	Phe	Asn	Pro	Lys	Tyr	Asp	Cys	Lys	Ile	Met	Thr	Ser	Lys	Thr
385					390					395					400
Asp	Val	Ser	Ser	Ser	Val	Ile	Thr	Ser	Leu	Gly	Ala	Ile	Val	Ser	Cys
				405					410				415		
Tyr	Gly	Lys	Thr	Lys	Cys	Thr	Ala	Ser	Asn	Lys	Asn	Arg	Gly	Ile	Ile

-continued

420	425	430
Lys Thr Phe Ser Asn Gly Cys Asp Tyr Val Ser Asn Lys Gly Met Asp		
435	440	445
Thr Val Ser Val Gly Asn Thr Leu Tyr Tyr Val Asn Lys Gln Glu Gly		
450	455	460
Lys Ser Leu Tyr Val Lys Gly Glu Pro Ile Ile Asn Phe Tyr Asp Pro		
465	470	475
Leu Val Phe Pro Ser Asp Glu Phe Asp Ala Ser Ile Ser Gln Val Asn		
485	490	495
Glu Lys Ile Asn Gln Ser Leu Ala Phe Ile Arg Lys Ser Asp Glu Leu		
500	505	510
Leu His Asn Val Asn Ala Gly Lys Ser Thr Thr Asn Ile Met Ile Thr		
515	520	525
Thr Ile Ile Ile Val Ile Ile Val Ile Leu Leu Ser Leu Ile Ala Val		
530	535	540
Gly Leu Leu Leu Tyr Cys Lys Ala Arg Ser Thr Pro Val Thr Leu Ser		
545	550	555
Lys Asp Gln Leu Ser Gly Ile Asn Asn Ile Ala Phe Ser Asn		
565	570	

<210> SEQ ID NO 8

<211> LENGTH: 574

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 8

Met Glu Leu Leu Ile His Arg Ser Ser Ala Ile Phe Leu Thr Leu Ala		
1	5	10
Val Asn Ala Leu Tyr Leu Thr Ser Ser Gln Asn Ile Thr Glu Glu Phe		
20	25	30
Tyr Gln Ser Thr Cys Ser Ala Val Ser Arg Gly Tyr Phe Ser Ala Leu		
35	40	45
Arg Thr Gly Trp Tyr Thr Ser Val Ile Thr Ile Glu Leu Ser Asn Ile		
50	55	60
Lys Glu Thr Lys Cys Asn Gly Thr Asp Thr Lys Val Lys Leu Ile Lys		
65	70	75
Gln Glu Leu Asp Lys Tyr Lys Asn Ala Val Thr Glu Leu Gln Leu Leu		
85	90	95
Met Gln Asn Thr Pro Ala Ala Asn Asn Arg Ala Arg Arg Glu Ala Pro		
100	105	110
Gln Tyr Met Asn Tyr Thr Ile Asn Thr Thr Lys Asn Leu Asn Val Ser		
115	120	125
Ile Ser Lys Lys Arg Lys Arg Arg Phe Leu Gly Phe Leu Leu Gly Val		
130	135	140
Gly Ser Ala Ile Ala Ser Gly Ile Ala Val Ser Lys Val Leu His Leu		
145	150	155
Glu Gly Glu Val Asn Lys Ile Lys Asn Ala Leu Leu Ser Thr Asn Lys		
165	170	175
Ala Val Val Ser Leu Ser Asn Gly Val Ser Val Leu Thr Ser Lys Val		
180	185	190
Leu Asp Leu Lys Asn Tyr Ile Asn Asn Arg Leu Leu Pro Ile Val Asn		

-continued

195				200				205							
Gln	Gln	Ser	Cys	Arg	Ile	Ser	Asn	Ile	Glu	Thr	Val	Ile	Glu	Phe	Gln
210						215					220				
Gln	Met	Asn	Ser	Arg	Leu	Leu	Glu	Ile	Thr	Arg	Glu	Phe	Ser	Val	Asn
225				230						235					240
Ala	Gly	Val	Thr	Thr	Pro	Leu	Ser	Thr	Tyr	Met	Leu	Thr	Asn	Ser	Glu
				245						250					255
Leu	Leu	Ser	Leu	Ile	Asn	Asp	Met	Pro	Ile	Thr	Asn	Asp	Gln	Lys	Lys
			260					265							270
Leu	Met	Ser	Ser	Asn	Val	Gln	Ile	Val	Arg	Gln	Gln	Ser	Tyr	Ser	Ile
			275					280							285
Met	Ser	Ile	Ile	Lys	Glu	Glu	Val	Leu	Ala	Tyr	Val	Val	Gln	Leu	Pro
			290					295							300
Ile	Tyr	Gly	Val	Ile	Asp	Thr	Pro	Cys	Trp	Lys	Leu	His	Thr	Ser	Pro
305					310					315					320
Leu	Cys	Thr	Thr	Asn	Ile	Lys	Glu	Gly	Ser	Asn	Ile	Cys	Leu	Thr	Arg
				325						330					335
Thr	Asp	Arg	Gly	Trp	Tyr	Cys	Asp	Asn	Ala	Gly	Ser	Val	Ser	Phe	Phe
			340					345							350
Pro	Gln	Ala	Asp	Thr	Cys	Lys	Val	Gln	Ser	Asn	Arg	Val	Phe	Cys	Asp
			355					360							365
Thr	Met	Asn	Ser	Leu	Thr	Leu	Pro	Ser	Glu	Val	Ser	Leu	Cys	Asn	Thr
			370					375							380
Asp	Ile	Phe	Asn	Ser	Lys	Tyr	Asp	Cys	Lys	Ile	Met	Thr	Ser	Lys	Thr
385					390					395					400
Asp	Ile	Ser	Ser	Ser	Val	Ile	Thr	Ser	Leu	Gly	Ala	Ile	Val	Ser	Cys
				405						410					415
Tyr	Gly	Lys	Thr	Lys	Cys	Thr	Ala	Ser	Asn	Lys	Asn	Arg	Gly	Ile	Ile
			420					425							430
Lys	Thr	Phe	Ser	Asn	Gly	Cys	Asp	Tyr	Val	Ser	Asn	Lys	Gly	Val	Asp
			435					440							445
Thr	Val	Ser	Val	Gly	Asn	Thr	Leu	Tyr	Tyr	Val	Asn	Lys	Leu	Glu	Gly
			450					455							460
Lys	Asn	Leu	Tyr	Val	Lys	Gly	Glu	Pro	Ile	Ile	Asn	Tyr	Tyr	Asp	Pro
465					470						475				480
Leu	Val	Phe	Pro	Ser	Asp	Glu	Phe	Asp	Ala	Ser	Ile	Ser	Gln	Val	Asn
				485						490					495
Glu	Lys	Ile	Asn	Gln	Ser	Leu	Ala	Phe	Ile	Arg	Arg	Ser	Asp	Glu	Leu
			500					505							510
Leu	His	Asn	Val	Asn	Thr	Gly	Lys	Ser	Thr	Thr	Asn	Ile	Met	Ile	Thr
			515					520							525
Thr	Ile	Ile	Ile	Val	Ile	Ile	Val	Val	Leu	Leu	Ser	Leu	Ile	Ala	Ile
			530					535							540
Gly	Leu	Leu	Leu	Tyr	Cys	Lys	Ala	Lys	Asn	Thr	Pro	Val	Thr	Leu	Ser
545					550					555					560
Lys	Asp	Gln	Leu	Ser	Gly	Ile	Asn	Asn	Ile	Ala	Phe	Ser	Lys		
				565						570					

<210> SEQ ID NO 9

<211> LENGTH: 84

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 9

Gln Asn Ile Thr Glu Glu Phe Tyr Gln Ser Thr Cys Ser Ala Val Ser
 1 5 10 15

Lys Gly Tyr Leu Ser Ala Leu Arg Thr Gly Trp Tyr Thr Ser Val Ile
 20 25 30

Thr Ile Glu Leu Ser Asn Ile Lys Glu Asn Lys Cys Asn Gly Thr Asp
 35 40 45

Ala Lys Val Lys Leu Ile Lys Gln Glu Leu Asp Lys Tyr Lys Asn Ala
 50 55 60

Val Thr Glu Leu Gln Leu Leu Met Gln Ser Thr Pro Ala Cys Asn Ser
 65 70 75 80

Arg Ala Arg Arg

<210> SEQ ID NO 10

<211> LENGTH: 377

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 10

Phe Leu Gly Phe Leu Leu Gly Val Gly Ser Ala Cys Ala Ser Gly Ile
 1 5 10 15

Ala Val Ser Lys Val Leu His Leu Glu Gly Glu Val Asn Lys Ile Lys
 20 25 30

Ser Ala Leu Leu Ser Thr Asn Lys Ala Val Val Ser Leu Ser Asn Gly
 35 40 45

Val Ser Val Leu Thr Ile Lys Val Leu Asp Leu Lys Asn Tyr Ile Asp
 50 55 60

Lys Gln Leu Leu Pro Ile Val Asn Lys Gln Ser Cys Ser Ile Ser Asn
 65 70 75 80

Ile Glu Thr Val Ile Glu Phe Gln Gln Lys Asn Asn Arg Leu Leu Glu
 85 90 95

Ile Thr Arg Glu Phe Ser Val Asn Ala Gly Val Thr Thr Pro Val Ser
 100 105 110

Thr Tyr Met Leu Thr Asn Ser Glu Leu Leu Ser Leu Ile Asn Asp Met
 115 120 125

Pro Ile Thr Asn Asp Gln Lys Lys Leu Met Ser Ser Asn Val Gln Ile
 130 135 140

Val Arg Gln Gln Ser Tyr Ser Ile Met Ser Ile Ile Lys Glu Glu Val
 145 150 155 160

Leu Ala Tyr Val Val Gln Leu Pro Leu Tyr Gly Val Ile Asp Thr Pro
 165 170 175

Cys Trp Lys Leu His Thr Ser Pro Leu Cys Thr Thr Asn Thr Lys Glu
 180 185 190

Gly Ser Asn Ile Cys Leu Thr Arg Thr Asp Arg Gly Trp Tyr Cys Asp
 195 200 205

Asn Ala Gly Ser Val Ser Phe Phe Pro Gln Ala Glu Thr Cys Lys Val
 210 215 220

Gln Ser Asn Arg Val Phe Cys Asp Thr Met Asn Ser Leu Thr Leu Pro
 225 230 235 240

-continued

Ser Glu Val Asn Leu Cys Asn Ile Asp Ile Phe Asn Pro Lys Tyr Asp
245 250 255

Cys Lys Ile Met Thr Ser Lys Thr Asp Val Ser Ser Ser Val Ile Thr
260 265 270

Ser Leu Gly Ala Ile Val Ser Cys Tyr Gly Lys Thr Lys Cys Thr Ala
275 280 285

Ser Asn Lys Asn Arg Gly Ile Ile Lys Thr Phe Ser Asn Gly Cys Asp
290 295 300

Tyr Val Ser Asn Lys Gly Val Asp Thr Val Ser Val Gly Asn Thr Leu
305 310 315 320

Tyr Tyr Val Asn Lys Gln Glu Gly Lys Ser Leu Tyr Val Lys Gly Glu
325 330 335

Pro Ile Ile Asn Phe Tyr Asp Pro Leu Val Phe Pro Ser Ser Glu Phe
340 345 350

Asp Ala Ser Ile Ser Gln Val Asn Glu Lys Ile Asn Gln Ser Leu Ala
355 360 365

Phe Ile Arg Lys Ser Asp Glu Leu Leu
370 375

<210> SEQ ID NO 11
<211> LENGTH: 84
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 11

Gln Asn Ile Thr Glu Glu Phe Tyr Gln Ser Thr Cys Ser Ala Val Ser
1 5 10 15

Arg Gly Tyr Phe Ser Ala Leu Arg Thr Gly Trp Tyr Thr Ser Val Ile
20 25 30

Thr Ile Glu Leu Ser Asn Ile Lys Glu Thr Lys Cys Asn Gly Thr Asp
35 40 45

Thr Lys Val Lys Leu Ile Lys Gln Glu Leu Asp Lys Tyr Lys Asn Ala
50 55 60

Val Thr Glu Leu Gln Leu Leu Met Gln Asn Thr Pro Ala Cys Asn Asn
65 70 75 80

Arg Ala Arg Arg

<210> SEQ ID NO 12
<211> LENGTH: 377
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: synthetic construct

<400> SEQUENCE: 12

Phe Leu Gly Phe Leu Leu Gly Val Gly Ser Ala Cys Ala Ser Gly Ile
1 5 10 15

Ala Val Ser Lys Val Leu His Leu Glu Gly Glu Val Asn Lys Ile Lys
20 25 30

-continued

Asn	Ala	Leu	Leu	Ser	Thr	Asn	Lys	Ala	Val	Val	Ser	Leu	Ser	Asn	Gly
	35						40					45			
Val	Ser	Val	Leu	Thr	Ile	Lys	Val	Leu	Asp	Leu	Lys	Asn	Tyr	Ile	Asn
	50					55					60				
Asn	Gln	Leu	Leu	Pro	Ile	Val	Asn	Gln	Gln	Ser	Cys	Arg	Ile	Ser	Asn
65					70					75					80
Ile	Glu	Thr	Val	Ile	Glu	Phe	Gln	Gln	Lys	Asn	Ser	Arg	Leu	Leu	Glu
			85						90					95	
Ile	Thr	Arg	Glu	Phe	Ser	Val	Asn	Ala	Gly	Val	Thr	Thr	Pro	Leu	Ser
			100					105					110		
Thr	Tyr	Met	Leu	Thr	Asn	Ser	Glu	Leu	Leu	Ser	Leu	Ile	Asn	Asp	Met
		115					120					125			
Pro	Ile	Thr	Asn	Asp	Gln	Lys	Lys	Leu	Met	Ser	Ser	Asn	Val	Gln	Ile
	130					135					140				
Val	Arg	Gln	Gln	Ser	Tyr	Ser	Ile	Met	Ser	Ile	Ile	Lys	Glu	Glu	Val
145					150					155					160
Leu	Ala	Tyr	Val	Val	Gln	Leu	Pro	Ile	Tyr	Gly	Val	Ile	Asp	Thr	Pro
			165					170						175	
Cys	Trp	Lys	Leu	His	Thr	Ser	Pro	Leu	Cys	Thr	Thr	Asn	Ile	Lys	Glu
		180						185					190		
Gly	Ser	Asn	Ile	Cys	Leu	Thr	Arg	Thr	Asp	Arg	Gly	Trp	Tyr	Cys	Asp
		195					200					205			
Asn	Ala	Gly	Ser	Val	Ser	Phe	Phe	Pro	Gln	Ala	Asp	Thr	Cys	Lys	Val
	210					215					220				
Gln	Ser	Asn	Arg	Val	Phe	Cys	Asp	Thr	Met	Asn	Ser	Leu	Thr	Leu	Pro
225					230					235					240
Ser	Glu	Val	Ser	Leu	Cys	Asn	Thr	Asp	Ile	Phe	Asn	Ser	Lys	Tyr	Asp
			245					250						255	
Cys	Lys	Ile	Met	Thr	Ser	Lys	Thr	Asp	Ile	Ser	Ser	Ser	Val	Ile	Thr
		260						265					270		
Ser	Leu	Gly	Ala	Ile	Val	Ser	Cys	Tyr	Gly	Lys	Thr	Lys	Cys	Thr	Ala
	275						280					285			
Ser	Asn	Lys	Asn	Arg	Gly	Ile	Ile	Lys	Thr	Phe	Ser	Asn	Gly	Cys	Asp
	290				295					300					
Tyr	Val	Ser	Asn	Lys	Gly	Val	Asp	Thr	Val	Ser	Val	Gly	Asn	Thr	Leu
305					310					315					320
Tyr	Tyr	Val	Asn	Lys	Leu	Glu	Gly	Lys	Asn	Leu	Tyr	Val	Lys	Gly	Glu
			325					330						335	
Pro	Ile	Ile	Asn	Tyr	Tyr	Asp	Pro	Leu	Val	Phe	Pro	Ser	Ser	Glu	Phe
			340					345					350		
Asp	Ala	Ser	Ile	Ser	Gln	Val	Asn	Glu	Lys	Ile	Asn	Gln	Ser	Leu	Ala
	355					360						365			
Phe	Ile	Arg	Arg	Ser	Asp	Glu	Leu	Leu							
	370					375									

What is claimed is:

1. A method for eliciting an immune response in a human subject against an infectious disease-causing bacterium and a betacoronavirus, the method comprising co-administering to the human subject an effective dose of a first immunogenic composition comprising an antigen derived from the

bacterium and an immunogenic composition comprising mRNA encoding an antigen derived from the betacoronavirus.

2. The method according to claim 1, wherein a dose of the first immunogenic composition and a dose of the immunogenic composition comprising mRNA are administered concurrently.

3. (canceled)
4. The method according to claim 1, wherein a dose of the first immunogenic composition and a dose of the immunogenic composition comprising mRNA are administered within 24 hours of each respective dose.
5. (canceled)
6. (canceled)
7. (canceled)
8. The method according to claim 1, wherein the beta-coronavirus is SARS-CoV-2.
9. The method according to claim 1, wherein the bacterium is *N. meningitidis*.
10. The method according to claim 1, wherein the first composition comprises a polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 1.
11. (canceled)
12. (canceled)
13. (canceled)
14. The method according to claim 1, wherein the first immunogenic composition comprises a) a first lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 1, and b) a second lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 2.
15. The method according to claim 1, wherein the first immunogenic composition further comprises polysorbate-80, aluminum, histidine, and sodium chloride.
16. The method according to claim 9, wherein the first immunogenic composition comprises TRUMENBA®.
17. The method according to claim 1, wherein the first immunogenic composition comprises polysaccharides derived from *N. meningitidis*.
18. The method according to claim 1, wherein the first immunogenic composition comprises a) a liquid composition comprising (i) a first lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 3; and (ii) a second lipidated polypeptide comprising the amino acid sequence set forth in SEQ ID NO: 4 and aluminum; and b) a lyophilized composition comprising i) a *Neisseria meningitidis* serogroup A (MenA) capsular saccharide conjugated to an adipic acid dihydrazide (ADH) linker by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, wherein the linker is conjugated to tetanus toxoid (TT) by carbodiimide chemistry (MenAAH-TT conjugate); ii) a *Neisseria meningitidis* serogroup C (MenC) capsular saccharide conjugated to an ADH linker by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, wherein the linker is conjugated to tetanus toxoid (TT) by carbodiimide chemistry (MenCAH-TT conjugate); iii) a *Neisseria meningitidis* serogroup W135 (MenW) capsular saccharide directly conjugated to tetanus

toxoid (TT) by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, in the absence of a linker (MenW-TT conjugate); and iv) a *Neisseria meningitidis* serogroup Y (MenY) capsular saccharide directly conjugated to tetanus toxoid (TT) by 1-cyano-4-dimethylamino pyridinium tetrafluoroborate, in the absence of a linker (MenY-TT conjugate) wherein the lyophilized composition is reconstituted with the liquid composition.

19. The method according to claim 1, wherein the bacterium is *C. difficile*.
20. (canceled)
21. The method according to claim 19, wherein the first immunogenic composition comprises a fusion polypeptide.
22. (canceled)
23. (canceled)
24. (canceled)
25. (canceled)
26. (canceled)
27. The method according to claim 19, wherein the first composition comprises a polypeptide selected from the group consisting of a C-terminal domain of a wild-type *C. difficile* toxin A and a C-terminal domain of a wild-type *C. difficile* toxin B.
28. (canceled)
29. The method according to claim 19, wherein the first immunogenic composition is administered to a human aged 55 and older.
30. The method according to claim 1, wherein the bacterium is *E. coli*.
31. The method according to claim 1, wherein said immunogenic composition comprising mRNA comprises nucleoside-modified mRNA.
32. The method according to claim 1, wherein said immunogenic composition comprising mRNA is BNT162b2 (Cominmaty®).
33. The method according to claim 1, wherein said immunogenic composition comprising mRNA comprises a sequence having residues 1-102 of SEQ ID NO:1 and residues 103-4284 of SEQ ID NO:1, wherein the sequence for the SARS-CoV-2 antigen of SEQ ID NO:1 is replaced with SARS-CoV-2 antigen of a variant strain.
34. The method according to claim 1, wherein said immunogenic composition comprising mRNA comprises a mRNA which includes a first region of linked nucleosides encoding a SARS-CoV-2 antigen, a first flanking region located at the 5'-terminus of the first region, a second flanking region located at the 3'-terminus of the first region, at least one 5'-cap region, and a 3'-stabilizing region.

* * * * *